REFERENCE HANDBOOK

Polynomial Arithmetic Atlas

Polynomial irreducibility and Galois groups

Results, methods, examples, and references.

37 articles / 232 references

Complete offline edition / Literature checked through 12 November 2026

This edition includes the articles, references, questions, and reading guides from the website. The bibliography is not exhaustive.

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Introduction

This handbook describes methods for proving polynomial irreducibility and determining Galois groups. It covers the rationals, number fields, and finite fields, with related material on specialization, iteration, and absolute irreducibility in several variables.

The edition contains 37 articles and 232 annotated references. Each result includes its hypotheses and source. References give theorem or section numbers when these have been checked, together with links and publication details. Central statements were compared with the cited sources, and selected examples were checked by exact computation. Not every cited proof has been checked independently.

Publication status

Classical or published result. The statement appears in a published work or an established exposition.

Conditional result. An additional assumption is part of the statement.

Preprint. The entry reports the cited version. Limits on access and verification are recorded in the reference entry.

Open in a cited source. The question was open in that source; later work may have resolved it.

Research direction. A suggested question for further study, not necessarily a new conjecture or an open problem.

Example. A computation concerns the stated polynomial. A finite sample does not prove a result for an entire family.

Editions

The literature review is dated 12 November 2026. The earlier 188-source bibliography is a separate fixed edition. The current handbook includes later additions and corrections. Linked books and papers remain with their authors or publishers; this bibliography provides publication details, annotations, and links.

Using the methods

1. State the field, degree, separability, coefficient assumptions, and any factor removed by normalization.
2. Establish irreducibility or list the possible factor degrees.
3. Find elements of the Galois group or a subgroup containing it.
4. Apply group-theoretic criteria or resolvents to determine the group. Check the hypotheses before using the discriminant criterion.
5. Record the computations and theorem references used in the proof.

Reading guide

From a polynomial to its Galois group

Prove irreducibility and determine the Galois group from its permutation action.

Background: A first course in field theory

1. **Fields, irreducibility, and Galois actions**. Fix the ground field, distinguish irreducibility from separability, and relate factorization to the action on roots.
2. **Reduction modulo primes**. Use one finite-field factorization or intersect the factor degrees permitted by several primes. Examine why this method can be inconclusive for an irreducible polynomial.
3. **Frobenius and factorization modulo primes**. At a prime of good reduction, a squarefree factorization gives a cycle type in the global Galois group. Group-theoretic criteria then restrict the possible groups.
4. **Transitivity, primitivity, and cycle criteria**. Apply block systems, transpositions, and long-cycle criteria to permutations obtained from local arithmetic.
5. **Discriminants, parity, and ramification**. The discriminant detects repeated roots. For a separable polynomial in characteristic zero, its square class determines the sign character of the root action, but does not determine the group.

   References: Milne2022, Cox2012, ConradIrredQ, ConradModP, ConradSnAn
6. **Exercise**. Give a proof of $\mathrm{Gal}(x^5-x-1/\mathbb{Q}) = S_5$ and state exactly what each prime contributes.

From valuations to large groups

Use coefficient divisibility to restrict factor degrees and obtain long prime cycles.

Background: Readers comfortable with valuations and finite permutation groups

1. **Schönemann-Eisenstein and Dumas criteria**. Coefficient divisibility can exclude every proper factor degree. Translation and Newton-polygon slopes extend the elementary criterion.
2. **Newton polygons and Galois groups**. Determine p-adic root valuations from coefficients, intersect the permitted factor degrees, and apply group criteria to the resulting slope denominators.
3. **Transitivity, primitivity, and cycle criteria**. Apply block systems, transpositions, and long-cycle criteria to permutations obtained from local arithmetic.
4. **Taylor polynomials**. Compare exponential truncations, with alternating or symmetric groups, and geometric truncations, whose factors are cyclotomic.
5. **Generalized Laguerre polynomials**. Use factorial coefficient products to relate local slopes to Galois-group constraints. Distinguish fixed parameter with increasing degree from fixed degree with varying parameter.

   References: Serre1979, dumas1906, Coleman1987, Hajir2005, FilasetaLam2002
6. **Exercise**. Separate the irreducibility input, the Newton-polygon input, Jordan's theorem, and the discriminant step.

Irreducibility and Galois groups in families

Identify forced factors and symmetries, then determine the ambient permutation group.

Background: Researchers moving from examples to a family theorem

1. **Reciprocal polynomials**. Reduce a reciprocal polynomial to a polynomial of half the degree, then study the remaining quadratic extensions to determine irreducibility and the ambient Galois group.
2. **Sparse polynomials and factor reversal**. Separate persistent reciprocal factors from the nonreciprocal part and use coefficient sparsity to restrict possible factorizations.
3. **Trinomials**. The polynomials $x^n - x - 1$ realize S_n in every degree at least two. Modular reductions give certificates for individual examples; inertia gives a uniform proof.
4. **Eulerian polynomials**. Derive reciprocal symmetry from descent statistics, apply Eisenstein after a shift at prime indices, and compute the Galois group in a quartic example.
5. **Fekete polynomials**. Quadratic characters impose rational factors and reciprocal symmetry. Trace polynomials relate irreducibility and Galois groups to special values and square classes.
6. **Hermite, Bessel, and Legendre polynomials**. Study the factorization problems of Hermite, Bessel, and Legendre polynomials through parity reduction and coefficient products, accounting for the normalization of each family.

   References: Schinzel2000, Prasolov2004, sawinshustermanstoll2020, MinacNguyenTan2023, CullinanHajir2014
7. **Exercise**. State the normalization, the ambient permutation group, a proved result, and an unresolved question with its source and stated range.

Models and asymptotic regimes for random polynomials

Specify the asymptotic parameter, coefficient law, conditioning, and analytic hypotheses.

Background: Researchers reading counting and random-polynomial papers

1. **Height bounds for exceptional Galois groups**. At fixed degree, almost all integer polynomials have full symmetric group. Study the exceptional counts and the changes imposed by reciprocal symmetry.
2. **Random polynomials: models and methods**. Specify the coefficient law and limiting parameter, then distinguish irreducibility, alternating-group containment, and full symmetric-group conclusions.
3. **Littlewood polynomials**. Independent sign coefficients impose congruence constraints. The examples also show why reduction modulo two does not resolve the all-degree probabilistic problem.
4. **Random reciprocal polynomials of growing degree**. For reciprocal polynomials with independent free coefficients, roots occur in inverse pairs and the Galois group lies in a signed-permutation group. Study this model through the associated trace polynomial.
5. **Characteristic polynomials of random matrices**. Independent matrix entries give dependent characteristic-polynomial coefficients. Local rank estimates provide input specific to these models.

   References: Bhargava2025, BreuillardVarju2019, BarySorokerKoukoulopoulosKozma2023, BarySoroker-HokkenKozmaPoonen2025, Eberhard2022

6. **Exercise**. Explain why a fixed-degree height theorem does not imply a growing-degree fair-sign theorem.

From generic families to arithmetic specializations

Relate generic monodromy to arithmetic specialization, with the quantifiers in each conclusion stated explicitly.

Background: Graduate readers interested in arithmetic geometry and inverse Galois theory

1. **Critical points and polynomial monodromy**. Use critical points and critical values to determine the generic group, then compare it with the groups of four rational fibers in a cubic example.

   References: Serre2007Topics
2. **Specialization and Hilbert irreducibility**. Specialize a polynomial over a rational function field and study the exceptional parameter values.
3. **Inverse Galois theory and generic polynomials**. Construct polynomials with prescribed groups and distinguish individual realizations, regular families, and universal parametrizations.
4. **Composition, iteration, and arboreal groups**. Study repeated polynomial composition through extension towers and rooted trees. Distinguish stability from maximality of the Galois action.
5. **Generalized Laguerre polynomials**. Use factorial coefficient products to relate local slopes to Galois-group constraints. Distinguish fixed parameter with increasing degree from fixed degree with varying parameter.

   References: Serre2007Topics, FriedJarden2023, HajirWong2006, JensenLedetYui2002, Jones2013Arboreal
6. **Exercise**. Distinguish infinitely many good specializations, all but finitely many, density one, and an infinite tower.

Exact computation and certificates

Study factorization and Galois-group algorithms and identify the steps that certify their outputs.

Background: Readers using computer algebra in proofs

1. **Irreducibility over finite fields**. Use Frobenius-power tests to certify irreducibility, count irreducibles, and distinguish irreducible from primitive polynomials.
2. **Polynomial factorization algorithms**. Study Frobenius and linear algebra in finite-field factorization, followed by p-adic lifting and lattice reduction for factorization over the rationals.
3. **Resolvents and subgroup tests**. A resolvent records the group action on selected expressions in the roots. Rational factors can detect an invariant partition or restrict the group to a smaller subgroup.
4. **Algorithms for Galois groups**. Combine exact arithmetic, subgroup constraints, and certification, recording the inputs and certificates needed to reproduce the proof.

   References: LidlNiederreiter1997, GathenGerhard2013, LLL1982, FiekerKlueners2014, MagmaGaloisDocumentation, PARIDocumentation
5. **Exercise**. Record the exact input, software version, field, factorization or resolvent certificate, and certification step.

Books

Fields and Galois Theory / Milne, James S.

Field extensions, separability, and the fundamental theorem of Galois theory; Chapter 4 gives explicit Galois-group computations. The frozen course notes v5.10 and the 2022 book have different pagination. Cite the version actually used.

Introductory text

Galois Theory / Cox, David A.

Explicit polynomials, resolvents, radicals, and permutation actions, developed alongside the general theory.

Examples and constructions

Polynomials / Prasolov, Victor V.

Irreducible polynomials, special forms, root properties, and Galois theory. The linked electronic edition carries a 2009 release date; the English book is ©2004.

Polynomial theory

Polynomials with Special Regard to Reducibility / Schinzel, Andrzej

Structured, sparse, and multivariable polynomials, with references to the origins of the criteria. The publisher excludes results valid only over finite fields, local fields, or Q from the monograph's scope; these require separate field-specific references.

Research reference on reducibility

Local Fields / Serre, Jean-Pierre

Valuations, ramification, and local extensions underlying Newton-polygon and inertia arguments.

Local fields

Permutation Groups / Dixon, John D. and Mortimer, Brian

Transitivity, blocks, primitivity, and cycle criteria for determining permutation groups.

Permutation groups

Finite Fields / Lidl, Rudolf and Niederreiter, Harald

Irreducible and primitive polynomials, extension fields, counting, constructions, and factorization.

Finite fields

Modern Computer Algebra / von zur Gathen, Joachim and Gerhard, Jürgen

Polynomial arithmetic, finite-field algorithms, Hensel lifting, lattice reduction, and factorization over the rationals.

Algorithms and complexity

A Course in Computational Algebraic Number Theory / Cohen, Henri

Defining polynomials, number fields, integral bases, discriminants, and explicit arithmetic.

Computational number theory

Topics in Galois Theory / Serre, Jean-Pierre

Hilbert irreducibility, thin sets, regular realizations, and sieve arguments. The electronic edition has copyright 2007; the physical second edition is also catalogued as 2008.

Specialization and inverse Galois theory

Field Arithmetic / Fried, Michael D. and Jarden, Moshe

Hilbertian fields, PAC fields, absolute Galois groups, and specialization in the fourth edition.

Field arithmetic

Inverse Galois Theory / Malle, Gunter and Matzat, B. Heinrich

Regular realizations, rigidity, embedding problems, and explicit construction methods.

Inverse Galois theory

Methods

Is a given polynomial irreducible over Q?

Given. Factorization modulo one or more small primes

Method. Degree-preserving reduction and intersection of possible factor degrees.

Conclusion. An irreducibility certificate if every proper factor degree is excluded

Limitations. An irreducible polynomial need not have an irreducible reduction at any prime.

Can coefficient divisibility prove irreducibility?

Given. One prime divides all lower coefficients, or the valuations have a simple pattern

Method. Schönemann-Eisenstein and Dumas/Newton polygons.

Conclusion. Irreducibility over a local field, hence over Q, or restrictions on factor degrees

Limitations. Failure of a sufficient criterion does not imply reducibility.

Can several local tests work together?

Given. Different primes exclude different factor degrees

Method. Intersect Newton-polygon degree constraints.

Conclusion. A global irreducibility proof even when each individual prime permits factors

Limitations. The degree restrictions from different primes may still have a common proper degree.

Can root sizes rule out factors?

Given. A monic integer polynomial satisfying Perron's coefficient inequality, or a root bound at a prime-valued argument

Method. Perron's criterion and archimedean factor bounds.

Conclusion. Irreducibility from root location and integral constant terms or factor evaluations

Limitations. Perron's test needs monicity and a nonzero constant term; the displayed bounds are strict.

Can integer evaluations rule out every factor?

Given. Many distinct integer arguments with small nonzero values, or at least 2n+1 unit or prime values for a primitive degree-n polynomial

Method. Schur unit-value arguments, Pólya interpolation, and prime-value counting.

Conclusion. An irreducibility certificate from restrictions on the values of possible integral factors

Limitations. Use the stated degree-dependent threshold and distinct arguments; one prime value alone is insufficient.

Is a composition or binomial irreducible?

Given. An irreducible outer polynomial F, a nonconstant inner polynomial g, or a binomial in characteristic zero

Method. Capelli root-field reduction and the exact binomial power criterion.

Conclusion. A lower-degree irreducibility question over a root field; for binomials, necessary and sufficient power exclusions

Limitations. Irreducibility of both inputs is insufficient. Binomials with exponent divisible by 4 require the exceptional negative-four-times-a-fourth-power exclusion.

What happens when a sparse polynomial has a large gap?

Given. A family x^N c(x^{-1})+d(x) with fixed c and d

Method. Nonreciprocal-part theorems and Capelli exclusions.

Conclusion. Irreducibility of a specified nonreciprocal part for sufficiently large N

Limitations. The conclusion requires separate treatment of reciprocal factors and Capellian exceptions.

Which permutations occur in the Galois group?

Given. Squarefree factorization at a prime of good reduction

Method. Frobenius cycle types.

Conclusion. An element of the specified cycle type

Limitations. The squarefree-reduction argument does not apply at bad primes; a missing cycle type need not be absent from the group.

When do cycle types force A_n or S_n?

Given. Transitivity, primitivity, and an appropriate prime cycle or transposition

Method. Permutation-group recognition.

Conclusion. A proved group containment or equality

Limitations. A transitive subgroup containing a transposition need not be S_n.

How can A_n be distinguished from S_n?

Given. Containment of A_n has already been proved

Method. Discriminant square class or an odd permutation.

Conclusion. Equality with A_n or S_n

Limitations. A nonsquare discriminant alone does not prove S_n.

Does a root symmetry force a smaller ambient group?

Given. Reciprocal, even, or compositional structure

Method. Invariant blocks and quotient variables.

Conclusion. A wreath-product or other structural upper bound

Limitations. Root symmetry can exclude the full symmetric group S_n.

How can remaining candidate groups be separated?

Given. Several subgroups fit the observed Frobenius types

Method. Resolvents, relative invariants, and subgroup descent.

Conclusion. Additional exact constraints or a certified group

Limitations. A numerical approximation or a candidate group alone does not certify the result.

Do generic results survive parameter specialization?

Given. An irreducible family over Q(t), or over K(t) with K Hilbertian

Method. Hilbert irreducibility and specialized geometric criteria.

Conclusion. Irreducible specializations, with the exceptional set described under the stated hypotheses

Limitations. Infinitely many, density one, and all but finitely many are different conclusions.

Are all iterates irreducible?

Given. A polynomial f and the tower of its iterates

Method. Capelli-type steps, critical orbits, and arboreal groups.

Conclusion. Irreducibility at finitely many levels or at every level, under the corresponding hypotheses

Limitations. Specializing each finite level does not select one specialization valid at every level.

Can an irreducible polynomial over F_q be constructed or certified?

Given. Specified degree and possibly prescribed coefficients

Method. Frobenius congruences, gcd tests, and finite-field constructions.

Conclusion. A finite-field irreducibility certificate or an existence theorem

Limitations. Primitive is stronger than irreducible; an irreducible degree-n polynomial has cyclic splitting-field group.

Is a multivariate polynomial absolutely irreducible?

Given. Sparse exponent support or an effective specialization problem

Method. Newton polytopes and absolute-irreducibility criteria.

Conclusion. Irreducibility after extending the coefficient field to an algebraic closure

Limitations. Exclude monomial factors; this differs from ordinary irreducibility over the ground field.

Which coefficient models does a probabilistic theorem cover?

Given. A coefficient law and a specified limiting parameter

Method. Local mixing, sieve methods, and factor exclusion.

Conclusion. A probability estimate for the exact model in the theorem

Limitations. State the conditioning, the distinction between A_n and S_n, and any RH or GRH hypothesis.

What is the generic Galois group of F(X)−T?

Given. Simple critical points with pairwise distinct critical values in characteristic zero

Method. Morse criterion: finite branch inertia generates transpositions.

Conclusion. Geometric and arithmetic group S_n; a regular splitting field

Limitations. A full generic group does not determine every rational fiber. Check specialization separately.

Part I

Foundations

CHAPTER 1

Fields, irreducibility, and Galois actions

foundation / Start here / general fields / Classical theory

Irreducible factors correspond to Galois orbits. The ground field and separability determine the action on roots.

Overview

- What does irreducibility depend on?
- How do irreducible factors correspond to Galois orbits?
- Why does irreducibility not determine the Galois group?

1.1 Irreducibility and the ground field

A nonconstant polynomial $f \in K[x]$ is irreducible over a field K if a factorization $f = gh$ in $K[x]$ forces one factor to be constant. Irreducibility depends on the field: $x^2 - 2$ is irreducible over $\mathbb{Q}$, but splits over $\mathbb{Q}(\sqrt{2})$. Over an algebraic closure, every polynomial of degree greater than one factors. Thus a univariate polynomial is absolutely irreducible only in degree one. Polynomials in several variables can be absolutely irreducible in higher degree.

For nonzero $f \in \mathbb{Z}[x]$, the content is the greatest common divisor of its coefficients; dividing by it gives a primitive polynomial. Gauss's lemma says that a primitive polynomial has a factorization of positive degrees over $\mathbb{Q}$ exactly when it has one over $\mathbb{Z}$. Integral divisibility and coefficient bounds therefore give criteria for irreducibility over the rationals. A constant factor such as the 2 in $2x + 2$ is a unit over $\mathbb{Q}$, but is not a unit over $\mathbb{Z}$.

References

[Lang2002]. Polynomial rings, content, and irreducibility over fields and unique factorization domains. Source.

[ConradIrredQ] / §2. A proof of Gauss's lemma and its use in rational irreducibility tests. Source.

1.2 Separability

A polynomial is *separable* if all its roots in an algebraic closure are distinct. For nonzero f, this is equivalent to $\gcd(f, f') = 1$. Every irreducible polynomial over a field of characteristic zero is separable. In positive characteristic, $x^p - t \in \mathbb{F}_p(t)[x]$ is irreducible, yet has only one distinct root in its splitting field.

For a polynomial of degree n, the action on that many distinct roots requires separability. Removing a specified factor changes the set on which the group acts. The group of the original polynomial and the group acting on a selected factor need not be the same permutation representation.

References

[Milne2022] / Chapter 2, Propositions 2.21 and 2.24. Separable polynomials, the derivative criterion, and perfect fields; numbering refers to the free version 5.10. Author's book page.

1.3 Irreducible factors and Galois orbits

Theorem / factor-orbit correspondence

Let $f \in K[x]$ be separable of degree n, let L be its splitting field, and let $G = \mathrm{Gal}(L/K)$. The action of G on the roots is faithful. Its orbits are exactly the root sets of the irreducible factors of f over K. In particular, f is irreducible if and only if $G \leq S_n$ is transitive.

Proof sketch

An automorphism fixing K carries a root to another root of its minimal polynomial. Conversely, the isomorphism between the simple extensions generated by two roots of the same irreducible polynomial extends to an automorphism of the normal splitting field. These are exactly the roots in one orbit. Faithfulness follows because the roots generate L.

Irreducibility establishes transitivity. Further permutations or invariants are generally needed to determine the group.

References

[Milne2022] / Proposition 4.5. Irreducibility is equivalent to transitivity for a separable polynomial. Author's book page.

1.4 Example

Example / $x^3 - 2$

Eisenstein at 2 proves irreducibility over $\mathbb{Q}$. Write $\alpha = \sqrt[3]{2}$ and let ζ_3 be a primitive cube root of unity. Its splitting field is $L = \mathbb{Q}(\alpha, \zeta_3)$. The real field $\mathbb{Q}(\alpha)$ has degree 3, and adjoining the nonreal element ζ_3 has degree 2. Hence $[L : \mathbb{Q}] = 6$, so the faithful subgroup of S_3 is all of S_3. Over $\mathbb{Q}(\zeta_3)$, the same splitting field has degree 3, and the group is cyclic of order 3. Over $\mathbb{Q}(\alpha)$, the polynomial already has a linear factor and the splitting-field group has order 2. The splitting-field group therefore depends on the ground field.

References

[Cox2012]. Field extensions, splitting fields, and explicit low-degree Galois theory; the displayed degree calculation is supplied here. Source.

1.5 Related questions and terminology

- **Factorization:** is f irreducible, or what are its irreducible factor degrees?
- **Permutation group:** which subgroup of S_n acts on the roots?
- **Field realization:** which extension of K is the splitting field, including its ramification?
- **Uniform family statement:** does the conclusion hold for every parameter, outside an explicit exceptional set, or with a limiting probability?

The word *primitive* also has several meanings: relatively prime integer coefficients; a permutation action with no nontrivial block system; or a finite-field element generating the multiplicative group. These are distinct uses of the same term.

References

[Schinzel2000]. The research reference for reducibility questions over different fields and within special polynomial families. Source.

[DixonMortimer1996]. Transitive and primitive permutation actions. Source.

Questions

Does irreducibility imply a symmetric Galois group?

Settled in stated scope / Separable polynomials over Q

No. The polynomial $x^4 + 1 = \Phi_8(x)$ is irreducible, but its splitting field is $\mathbb{Q}(\zeta_8)$ and its group is $(\mathbb{Z}/8\mathbb{Z})^\times \cong C_2 \times C_2$, acting transitively on four roots. Its transitive group is a proper subgroup of the symmetric group.

References

[Milne2022] / Chapter 5. Cyclotomic extensions and their automorphism groups. Author's book page.

Related topics

Applies to. Reduction modulo primes. Reduction of integral factors gives restrictions on rational factor degrees.

Applies to. Frobenius and factorization modulo primes. At good primes, irreducible factor degrees are Frobenius cycle lengths.

Applies to. Transitivity, primitivity, and cycle criteria. Additional group-theoretic hypotheses can strengthen transitivity to alternating or symmetric containment.

Contrasts with. Absolute irreducibility and Newton polytopes. In several variables, absolute irreducibility means irreducibility over the algebraic closure.

Part II

Irreducibility methods

CHAPTER 2

Reduction modulo primes

method / Start here / Q / Classical theory

Reduction modulo primes restricts the degrees of rational factors. These restrictions can prove irreducibility, but need not detect every irreducible polynomial.

Overview

- When does irreducibility modulo p prove irreducibility over Q?
- How can several reducible reductions prove rational irreducibility?
- Does reduction detect every irreducible polynomial?

Prerequisites: Fields, irreducibility, and Galois actions

2.1 The reduction criterion

Theorem / reduction criterion

Let $f \in \mathbb{Z}[x]$ be primitive of degree $n \geq 2$. If a prime p does not divide its leading coefficient and $\bar{f} \in \mathbb{F}_p[x]$ is irreducible, then f is irreducible over $\mathbb{Q}$.

The leading-coefficient condition is essential. For example, $2x^2 + 3x + 1 = (2x+1)(x+1)$ becomes the irreducible linear polynomial $x+1$ modulo 2. Here reduction lowers the degree, so the criterion does not apply. No squarefreeness hypothesis is separately needed for the reduction criterion: an irreducible polynomial over a finite field is already separable.

Proof

Gauss's lemma turns a rational factorization into $f = gh$ in $\mathbb{Z}[x]$. Since the product of the leading coefficients is not divisible by p, both factors retain their positive degrees after reduction. Their reductions would factor $\bar{f}$, a contradiction.

References

[ConradIrredQ] / §3 and Appendix A. Reduction modulo a prime and the Gauss-lemma proof. Source.

2.2 Factor-degree restrictions

Suppose the irreducible factors of $\bar{f}$, listed with multiplicity, have degrees $d_1, \dots, d_r$. Every rational factor degree must lie in

$$D_p = \left\{ \sum_{i \in I} d_i : I \subseteq \{1, \dots, r\} \right\} \cap \{1, \dots, n-1\}.$$

A rational factor reduces to a product of some of these finite-field factors. Therefore $\bigcap_{p \in P} D_p = \emptyset$, for any finite set of degree-preserving primes P, proves irreducibility. In computation one can retain only degrees at most $n/2$, because a reducible polynomial has a factor in that range.

This degree argument remains valid at a repeated reduction: multiplicities must be included. By contrast, reading the same factor degrees as a Frobenius cycle type requires a squarefree reduction.

References

[Prasolov2004] / Chapter 2. Reduction and coefficient methods for rational irreducibility; the subset-sum formulation follows directly from reduction of an integral factorization. Source.

2.3 Example

Example / $f = x^4 + x^3 + x + 2$

$$f \bmod 2 = x(x^3 + x^2 + 1),$$
$$f \bmod 3 = (x^2 + 1)(x^2 + x - 1).$$

The cubic has no root in $\mathbb{F}_2$, so it is irreducible there. The two quadratics have no roots in $\mathbb{F}_3$, so they are irreducible there. Modulo 2, a proper rational factor could have degree 1 or 3. Modulo 3, it could only have degree 2. The sets are disjoint; hence f is irreducible over $\mathbb{Q}$.

References

[ConradIrredQ] / §3. The reduction principle underlying the certificate; these two displayed factorizations are calculated for this entry. Source.

2.4 A polynomial reducible modulo every prime

The polynomial x^4+1 is irreducible over $\mathbb{Q}$: translating by 1 gives $x^4+4x^3+6x^2+4x+2$, Eisenstein at 2. Nevertheless it is reducible modulo every prime. At 2 it is $(x+1)^4$. For odd p, its roots are primitive eighth roots of unity; since $p^2 \equiv 1 \pmod 8$, they lie in $\mathbb{F}_{p^2}$, so all irreducible factors have degree at most two.

In fact degree 2 remains compatible with every reduction, so even intersecting all the sets D_p does not prove irreducibility here. Indeed, $C_2 \times C_2$, the Galois group, contains no 4-cycle. For a separable irreducible polynomial over $\mathbb{Q}$, Chebotarev and the Frobenius factorization theorem show that an irreducible good reduction exists exactly when the group contains an n-cycle.

References

[ConradAllPrimes]. A complementary quartic example and a proof that reduction at every prime can fail. Source.

[StevenhagenLenstra1996]. The relation between Frobenius conjugacy classes, factorization types, and Chebotarev density. Source.

2.5 Combining local information

The intersections of the possible factor degrees can prove irreducibility even when each reduction is reducible. If these intersections remain nonempty, coefficient valuations may give further restrictions through Newton polygons. Reciprocal or compositional structure can also exclude particular cycle types.

A finite prime search can be inconclusive. Proving that the group contains no n-cycle establishes that every good reduction is reducible.

References

[ConradModP]. Good-prime factorizations as Galois-group information. Source.

Questions

Does every irreducible rational polynomial have an irreducible reduction?

Settled in stated scope / Separable irreducible polynomials over Q

No. The precise condition is that the Galois group contain a full-length cycle. The polynomial $x^4 + 1$ is a counterexample.

References

[StevenhagenLenstra1996]. Chebotarev gives primes in each available conjugacy class. Source.

[ConradAllPrimes]. An explicit failure of the converse. Source.

Related topics

Uses. Fields, irreducibility, and Galois actions. Gauss's lemma gives integral factors from rational factors.

Contrasts with. Newton polygons and Galois groups. Valuations give further restrictions when reduction has repeated factors.

Supplies examples for. Frobenius and factorization modulo primes. Squarefree modular factorizations give Frobenius cycle types.

Uses. Irreducibility over finite fields. Finite-field irreducibility tests verify the modular factors.

CHAPTER 3

Schönemann-Eisenstein and Dumas criteria

method / Start here / Q / Classical theory

Eisenstein and the one-edge Dumas criterion prove local irreducibility from coefficient valuations. Translation can make these criteria applicable.

Overview

- Why does divisibility by p, but not p^2, prove irreducibility?
- How can a change of variable make an irreducibility criterion applicable?
- How does the one-edge Dumas criterion extend Eisenstein?

Prerequisites: Fields, irreducibility, and Galois actions

3.1 The Schönemann-Eisenstein criterion

Theorem / Schönemann-Eisenstein

Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$, with $n \geq 1$. Suppose a prime p satisfies $p \nmid a_n$, $p \mid a_i$ for $0 \leq i < n$, and $p^2 \nmid a_0$. Then f is irreducible over $\mathbb{Q}$.

Proof sketch

A factorization into positive degrees can be taken in $\mathbb{Z}[x]$. Reducing it modulo p gives a factorization of the monomial $a_n x^n$. Both reduced factors are monomials of positive degree, because their leading coefficients remain nonzero. Thus both original constant terms are divisible by p, which would make a_0 divisible by p^2.

Schönemann's work of 1846 precedes Eisenstein's 1850 paper. Cox's historical account explains their different motivations and the connection with finite fields and the division of the lemniscate.

References

[schoenemann1846]. Original historical source. Source.
[eisenstein1850]. Eisenstein's original treatment. Source.
[cox2011eisenstein] / pp. 3-4. Historical attribution and the elementary form of the criterion. Source.

3.2 Changes of variable

For $a \in K^\times$ and $b \in K$, substitution $x \mapsto ax + b$ is an automorphism of $K[x]$ and preserves irreducibility. It may change the coefficient divisibility required by the criterion. Reversing coefficients, $f^*(x) = x^{\deg f} f(1/x)$, also preserves irreducibility when $f(0) \neq 0$.

Example

$$(x+1)^4+1 = x^4+4x^3+6x^2+4x+2.$$

The right side is Eisenstein at 2, proving irreducibility of x^4+1. More generally, if p is prime,

$$\Phi_p(x+1) = \frac{(x+1)^p-1}{x} = \sum_{j=1}^{p} \binom{p}{j} x^{j-1}$$

is Eisenstein at p: all intermediate binomial coefficients are divisible by p, while the constant coefficient is exactly p.

References

[cox2011eisenstein] / p. 4. The prime cyclotomic example after translation. Source.

3.3 Schönemann's residue-power criterion

The following form allows a higher-degree irreducible factor in the reduction. Let $\phi \in \mathbb{Z}[x]$ be monic with irreducible reduction modulo p, let $e \geq 2$, and let $M \in \mathbb{Z}[x]$ satisfy $\deg M < e \deg \phi$ and $\gcd(\bar{\phi}, \bar{M}) = 1$. Then

$$f = \phi^e + pM$$

is irreducible over $\mathbb{Q}$. Indeed, a monic integral factorization would reduce to $\bar{g} = \bar{\phi}^a$, $\bar{h} = \bar{\phi}^b$, where $a, b > 0$ and $a + b = e$. Write $g = \phi^a + pg_1$, $h = \phi^b + ph_1$. Comparing with f, then dividing by p and reducing modulo p, forces $\bar{\phi} \mid \bar{M}$, a contradiction.

For example, $(x^2+x+1)^2+2 = x^4+2x^3+3x^2+2x+3$ is irreducible: x^2+x+1 is irreducible over $\mathbb{F}_2$, and $M = 1$. The reduction is a power of an irreducible quadratic.

References

[schoenemann1846]. Historical source of the residue-power criterion. Source.

[bonciocat2015multiprime] / Introduction. Modern statement of Schönemann's criterion. Source.

3.4 The Dumas criterion

Theorem / one-edge Dumas criterion

Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Q}_p[x]$, with $a_0 a_n \neq 0$, $v_p(a_n) = 0$, and $v_p(a_0) = k > 0$. Suppose $\gcd(n, k) = 1$ and

$$v_p(a_i) \geq k\left(1 - \frac{i}{n}\right) \qquad (0 < i < n),$$

where $v_p(0) = +\infty$. Then f is irreducible over $\mathbb{Q}_p$. For a rational polynomial, this also proves irreducibility over $\mathbb{Q}$.

The Newton polygon has one segment of slope $-k/n$, so every root has valuation k/n. A monic factor of degree d over $\mathbb{Q}_p$ would have constant coefficient of valuation dk/n, an integer. Coprimality forces $n \mid d$. Hence no proper positive factor degree is possible. Eisenstein is the case $k = 1$.

References

[dumas1906]. Classical source for the Newton-polygon criterion. Source.

[bonciocat2015multiprime] / Introduction. Introduction states Dumas's criterion in reversed-coefficient form; the form here follows by coefficient reversal. Source.

[Hajir2005] / Theorem 2.1. Roots and slopes over a discretely valued field. Source.

3.5 Example

Example / $f = x^5 + 4x + 4$

At 2, the nonzero coefficient points are $(0, 2), (1, 2), (5, 0)$. The middle point lies above the segment joining $(0, 2)$ to $(5, 0)$. Its slope is $-2/5$, and $\gcd(2, 5) = 1$. Thus f is irreducible over $\mathbb{Q}_2$, hence over $\mathbb{Q}$.

The ordinary criterion at 2 does not apply to the displayed polynomial, since $4 \mid f(0)$. The Newton polygon uses the denominator of the root valuation, without requiring the constant term to have valuation one.

References

[bonciocat2015multiprime] / Introduction. The one-edge criterion used here; the valuation diagram and polynomial are an explicit worked computation. Source.

3.6 Limitations and translation

A polygon with several edges, or one whose endpoint increments have a common divisor, need not imply reducibility. The resulting restrictions may leave several possible factor degrees. Other primes, residual polynomials, or a change of variable can give further information. Local irreducibility alone does not determine the global Galois group.

Even allowing every integral translation does not make the elementary Eisenstein test complete. Conrad proves that $x^4 - 10x^2 + 1$ is irreducible over $\mathbb{Q}$, but $(x + c)^4 - 10(x + c)^2 + 1$ is never Eisenstein for any $c \in \mathbb{Z}$. Thus translated Eisenstein criteria do not characterize irreducibility.

The stronger one-edge criterion succeeds after translation: at $c = 1$ the polynomial becomes $x^4 + 4x^3 - 4x^2 - 16x - 8$. Its 2-adic coefficient valuations are $3, 4, 2, 2, 0$, and the lower polygon is a single edge of slope $-3/4$. Dumas therefore proves irreducibility even over $\mathbb{Q}_2$.

References

[ConradAllPrimes] / Theorems 1 and 2. Irreducibility and failure of every integral Eisenstein shift. Source.

[dumas1906]. The one-edge criterion applies to the translated quartic, whose coefficients and valuations are calculated here. Source.

Questions

Can an integral translation always make an irreducible polynomial Eisenstein?

Settled in stated scope / Translations x↦x+c with c∈Z

No. The irreducible polynomial $x^4 - 10x^2 + 1$ is a counterexample. The counterexample concerns integral translations, and does not exclude other changes of variable or valuation criteria.

References

[ConradAllPrimes] / Theorems 1 and 2. Explicit counterexample. Source.

Related topics

Uses. Fields, irreducibility, and Galois actions. Gauss's lemma gives an integral factorization from rational reducibility.

Uses. Newton polygons and Galois groups. The Dumas criterion is the one-edge case of the Newton-polygon argument.

Contrasts with. Reduction modulo primes. An Eisenstein polynomial has repeated reduction at the prime used.

Applies to. Taylor polynomials. Factorial valuations determine Newton polygons of exponential truncations.

CHAPTER 4

Newton polygons and Galois groups

method / Intermediate / Q / Classical theory

Newton slopes restrict factor degrees and ramification indices. Residual polynomials refine these restrictions, while suitable prime divisors imply large Galois groups.

Overview

- How do Newton slopes determine root valuations?
- How can several primes prove irreducibility?
- How do residual polynomials refine a single Newton edge?
- When does a local slope force containment of the alternating group?

Prerequisites: Fields, irreducibility, and Galois actions; Schönemann-Eisenstein and Dumas criteria

4.1 Newton polygons and root valuations

Write $f(x) = \sum_{i=0}^{n} a_i x^i \in \mathbb{Q}_p[x]$ with $a_0 a_n \neq 0$, and normalize $v_p(p) = 1$. The Newton polygon is the lower convex hull of the points $(i, v_p(a_i))$ with $a_i \neq 0$. Zero coefficients have infinite valuation and contribute no finite points.

> **Theorem / slopes and roots**
>
> A segment of slope $-u/v$, in lowest terms with $v > 0$, and horizontal length ℓ, corresponds to exactly ℓ roots of valuation u/v, counted with multiplicity. There is a factor over $\mathbb{Q}_p$ consisting of all roots with each given valuation.

For a factor over $\mathbb{Q}_p$, the number of its roots belonging to that slope must be a multiple of v. Indeed, the constant coefficient of its corresponding monic slope factor has integer valuation. Thus the possible factor degrees are sums of multiples of the slope denominators, bounded by the lengths available on the respective segments.

References

[Hajir2005] / §2.1, Theorem 2.1. Definition and root-slope theorem. Source.

4.2 Combining primes

A rational factor remains a factor over every $\mathbb{Q}_p$. Consequently its degree must satisfy the restrictions from every prime used. These local restrictions can be intersected, as with factor degrees modulo primes. This can be useful even when every root has the same reduction modulo p.

The polygon-product rule states that the segments of the product polygon are assembled from those of the factor polygons, ordered by slope. Integral lattice lengths constrain how much of a segment a factor can use. Dumas already combined such restrictions at several primes; Bonciocat develops explicit criteria using arbitrarily many primes.

References

[dumas1906] / p. 238, item 4. Historical source for the polygon-product and several-prime method. Source.

[bonciocat2015multiprime] / §1, Theorem A (arXiv:1304.0874v2, p. 3). Systematic several-prime irreducibility criteria. Source.

4.3 Residual polynomials

The coefficients on a Newton edge also define a residual polynomial. It can exclude factor degrees permitted by the slope denominator.

Theorem / residual-polynomial criterion

Let $f(x) = \sum_{i=0}^{ed} a_i x^i \in \mathbb{Z}_p[x]$ be monic. Suppose its Newton polygon is the single segment from $(0, hd)$ to $(ed, 0)$, where h, e, d are positive integers and $\gcd(h, e) = 1$. Define

$$R_f(Y) = \sum_{j=0}^{d} \overline{\frac{a_{ej}}{p^{h(d-j)}}}\, Y^j \in \mathbb{F}_p[Y].$$

The polygon hypothesis makes every displayed quotient integral. If R_f is irreducible, then f is irreducible over $\mathbb{Q}_p$.

Proof

In a monic factorization $f = g_1 g_2$ over $\mathbb{Q}_p$, the factors have integral coefficients. All roots have valuation h/e, so the constant coefficients show that the factor degrees are ed_1, ed_2, with $d_1 + d_2 = d$. The residual-polynomial product rule gives $R_f = R_{g_1} R_{g_2}$, with degrees d_1, d_2. This rule follows directly from coefficient multiplication: terms on the edge survive after normalization and reduction; terms above it vanish. If two indices summing to a multiple of e are individually not multiples of e, their integral valuations put their product strictly above the edge. Irreducibility of R_f therefore forces one factor degree to be zero.

Example

For $f = x^4 + 2x^2 + 4$ at $p = 2$, the points $(0, 2), (2, 1), (4, 0)$ lie on one edge of slope $-1/2$. Its denominator leaves a possible quadratic factor. Here $h = 1, e = 2, d = 2$, and

$$R_f(Y) = 1 + Y + Y^2.$$

This quadratic has no root in $\mathbb{F}_2$. Thus f is irreducible over $\mathbb{Q}_2$, and hence over $\mathbb{Q}$, although its ordinary reduction is x^4. The criterion does not assume tame ramification. When $d = 1$, it recovers the one-edge Dumas criterion.

References

[GuardiaMontesNart2012Newton] / §1.3-1.4: Theorem 1.13 and Corollary 1.20 in arXiv:0807.2620v2. The product rule and residual-polynomial irreducibility criterion. The elementary specialization and quartic calculation are worked out here. Source.

[GuardiaMontesNart2011Montes] / §2.1-2.2, Theorems 2.1-2.2. The same criterion is encoded by a complete type; the source explains its ramification and residue-degree information. Source.

4.4 Example

Example

$$F(x) = 6!\sum_{j=0}^{6} \frac{x^j}{j!} = 720 + 720x + 360x^2 + 120x^3 + 30x^4 + 6x^5 + x^6.$$

The coefficient valuations, from constant to leading term, are

i	0	1	2	3	4	5	6
$v_2(a_i)$	4	4	3	3	1	1	0
$v_3(a_i)$	2	2	2	1	1	1	0

.

At 2, the vertices are $(0,4),(4,1),(6,0)$. The slopes are $-3/4, -1/2$, with lengths $4, 2$. The only proper factor degrees permitted are $2, 4$. At 3, there is one segment from $(0,2)$ to $(6,0)$, with slope $-1/3$; the only proper factor degree permitted is 3. Hence F is irreducible over $\mathbb{Q}$. Coleman's general treatment of exponential Taylor polynomials uses the same Newton-polygon method.

References

[Coleman1987]. Newton-polygon treatment of exponential Taylor polynomials; the degree-six valuation table is calculated here. Source.

[Hajir2005] / Theorem 2.1. The local theorem justifying the factor-degree restrictions. Source.

4.5 Alternating-group containment

Theorem / a form of Hajir's Newton-index criterion

Let $f \in \mathbb{Q}[x]$ be irreducible of degree n. Suppose a prime q divides the denominator, in lowest terms, of some Newton slope of f at a finite prime. If

$$\frac{n}{2} < q \leq n - 3,$$

then $\mathrm{Gal}(f/\mathbb{Q})$ contains A_n.

The root valuation forces q to divide a ramification index in the local splitting field, and hence the order of the global group. Cauchy's theorem gives an element of order q; because $2q > n$, this element is a single q-cycle. Transitivity comes from irreducibility. A nontrivial block system would have both its block size and its number of blocks smaller than q; an element of order q could act nontrivially on neither, which is impossible. Thus the action is primitive. Jordan's theorem then gives A_n.

The discriminant decides between A_n and S_n only after this containment has been proved. A nonsquare discriminant alone does not imply the full symmetric group.

References

[Hajir2005] / Theorem 2.2. Newton index divides the Galois-group order and a prime in the stated interval forces alternating-group containment. Source.

[Jordan1871]. Classical primitive-group theorem used in the final step. Source.

[DixonMortimer1996]. Permutation-group background for primitivity and long prime cycles. Source.

4.6 Example

Example / $f = x^8 + 2x^5 + 6$ has group S_8

Eisenstein at 2 proves irreducibility. At 3, the Newton polygon has vertices $(0,1),(5,0),(8,0)$, hence slopes $-1/5, 0$ of lengths $5, 3$. The denominator forces 5 to divide the Galois-group order. Since $8 < 2 \cdot 5$, an element of order 5 is a single 5-cycle. Irreducibility supplies transitivity, and $8/2 < 5 \leq 8 - 3$ satisfies the criterion above. Thus the group contains A_8.
The exact discriminant is

$$\operatorname{disc}(f) = 2^{12}3^9 \cdot 79 \cdot 733.$$

It is nonsquare, so the group is S_8. The proof uses Eisenstein for transitivity, a Newton slope for a prime cycle, and the discriminant for parity. The useful primes $2, 3$ have highly repeated reductions: x^8 and $x^5(x-1)^3$. Reading these multiplicities as Frobenius cycle lengths would be invalid.

References

[Hajir2005] / Theorem 2.2. The Newton-index theorem proves alternating containment for the displayed octic. The valuations and resultant were computed exactly for this entry. Source.

[Milne2022]. The discriminant-square criterion completes the group determination. Author's book page.

4.7 Slope denominators and cycle lengths

A slope denominator is an arithmetic constraint, not automatically a cycle of that length. The prime-order argument above uses both the denominator and the interval $n/2 < q \leq n - 3$. If the relevant prime is smaller, an element of order q may be a product of several disjoint q-cycles. If irreducibility is unavailable, transitivity cannot be assumed.

Counterexample

The polynomial $x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2)$ has one Newton slope $-1/2$ at 2. Its roots are $1 \pm i, -1 \pm i$, and its splitting field is $\mathbb{Q}(i)$. The only nonidentity automorphism acts as a product of two transpositions, never a single transposition. Its discriminant 2^{14} is a square, consistently with this even action. Capelli's exceptional fourth-power case explains the factorization.

The same phenomenon occurs for irreducible polynomials. The irreducible quartic $x^4 + 2x^2 + 4$ certified above also has slope denominator two. Its roots are $(\pm\sqrt{2} \pm \sqrt{-6})/2$, with independent sign choices, and its splitting field is $\mathbb{Q}(\sqrt{2}, \sqrt{-6})$. Its Galois group is V_4 in the transitive degree-four action: all three nonidentity elements are double transpositions. Its discriminant $2^{10}3^2$ is a square.

A common argument first uses factor-degree restrictions to establish irreducibility. Ramification then gives suitable permutations. Once alternating containment is known, the discriminant distinguishes the alternating and symmetric groups. Explicit coefficient formulas make these steps accessible for Laguerre and Taylor families, under the hypotheses of the corresponding results.

References

[Hajir2005] / Theorem 2.2. The irreducibility input and prime interval are explicit hypotheses. Source.

[capelli1901binomials]. The fourth-power exception explains the explicitly factored quartic. Source.

[GuardiaMontesNart2012Newton] / Corollary 1.20 in arXiv:0807.2620v2. The earlier residual-polynomial criterion proves irreducibility of the second counterexample; its radicals, field, permutation action, and discriminant are calculated here. Source.

Questions

How can one refine Newton-polygon restrictions on factor degrees?

Settled in stated scope / Local factorization of a specified separable polynomial over a p-adic field

Residual polynomials give the first refinement, illustrated above. If a residual factor is repeated, that stage may leave irreducibility undecided; higher Newton polygons refine the associated valuation and residue data. The Guàrdia-Montes-Nart theory gives a terminating local factorization procedure for separable polynomials. Finding shorter irreducibility proofs for particular families remains a separate question.

References

[GuardiaMontesNart2012Newton] / Introduction; §3 and Theorem 4.18 in arXiv:0807.2620v2. Source.
[GuardiaMontesNart2011Montes] / §2.2-3. Source.

Related topics

Generalizes. Schönemann-Eisenstein and Dumas criteria. Eisenstein corresponds to one edge of slope $-1/n$.
Contrasts with. Reduction modulo primes. Newton polygons apply at repeated reductions, where the squarefree Frobenius test does not apply.
Uses. Transitivity, primitivity, and cycle criteria. Cauchy's and Jordan's theorems turn suitable ramification divisors into alternating containment.
Uses. Discriminants, parity, and ramification. After alternating containment, the discriminant determines whether the group is alternating or symmetric.
Supplies examples for. Capelli's theorem and polynomial composition. The binomials x^6-72 and x^4+4 exhibit power obstructions and valuation constraints.
Applies to. Generalized Laguerre polynomials. Factorial and binomial coefficient formulas determine useful Newton slopes.
Develops into. Polynomial factorization algorithms. Higher Newton polygons and residual polynomials define the types used by the Montes algorithm.

CHAPTER 5

Root bounds and prime-value criteria

method / Intermediate / Q / Classical theory

Root bounds and integer evaluations give irreducibility criteria through restrictions on the constant terms and values of possible factors.

Overview

- How can one large coefficient prove irreducibility?
- When does a single prime value suffice?
- What changes when coefficients are digits of a prime?

Prerequisites: Fields, irreducibility, and Galois actions

5.1 Perron's criterion

Theorem / Perron

Let $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$, where $n \geq 2$ and $a_0 \neq 0$. If

$$|a_{n-1}| > 1 + \sum_{i=0}^{n-2} |a_i|,$$

then f is irreducible over $\mathbb{Q}$.

Proof sketch

On $|z| = 1$, the term $a_{n-1}z^{n-1}$ strictly dominates the sum of all other terms. Rouché's theorem gives exactly $n-1$ roots inside the unit circle and one outside. If the monic polynomial factored over $\mathbb{Z}$, one positive-degree monic factor would have all roots strictly inside. Its constant coefficient would be a nonzero integer whose absolute value is the product of numbers smaller than one, an impossibility.

The nonzero constant term prevents a zero root; monicity identifies a factor's constant coefficient with a product of its roots, up to sign. The strict inequality controls roots on the unit circle.

References

[perron1907] / Theorem I. Original source for coefficient-dominance irreducibility. Source.
[Prasolov2004] / Theorem 2.2.5. A modern presentation of Perron's criterion. Source.

5.2 Example

Example / $x^5 + 3x + 1$

Its reciprocal polynomial is

$$f^*(x) = x^5 f(1/x) = x^5 + 3x^4 + 1.$$

Now $3 > 1 + 1$, so Perron's criterion proves f^* irreducible. Since the constant term of f is nonzero, coefficient reversal preserves factorizations and proves f irreducible. The same argument works for $x^n + 3x + 1$ for every $n \geq 2$.

The coefficient inequality proves the required root bound without numerical approximations.

References

[perron1907]. Coefficient-dominance theorem; the reversal and worked family are direct applications. Source.

5.3 A prime-value criterion

Theorem / prime-value criterion

Let $f \in \mathbb{Z}[x]$ be nonconstant. Suppose an integer m satisfies $|f(m)|$ prime and $|m - \alpha| > 1$ for every complex root α of f. Then f is irreducible over $\mathbb{Q}$.

In an integral factorization $f = gh$, each positive-degree factor has an integer leading coefficient $b \neq 0$, so

$$|g(m)| = |b| \prod_{g(\alpha)=0} |m - \alpha| > 1.$$

Both factors would therefore give integer divisors of $f(m)$ larger than one in absolute value. A practical sufficient condition is a certified root bound $|\alpha| \leq R$ and $|m| > R + 1$. More general prime-value criteria are available in the cited literature.

For example, the roots of $x^4 - 10x^2 + 1$ are $\pm\sqrt{2} \pm \sqrt{3}$, all of absolute value less than 4. At $m = 6$, the value is 937, a prime. Every distance is greater than 2, so the criterion proves irreducibility. Primality can be checked by the primes at most $\sqrt{937} < 31$.

References

[murty2002primevalues]. Prime evaluations together with root or coefficient bounds; the displayed distance criterion is proved in the text. Source.

[ConradAllPrimes]. The roots of the quartic used in the example. Source.

5.4 Cohn's criterion

Cohn's decimal theorem says that if $p = a_n 10^n + \cdots + a_0$ is prime, with $n \geq 1$, $0 \leq a_i \leq 9$, and $a_n \neq 0$, then $a_n x^n + \cdots + a_0$ is irreducible over $\mathbb{Q}$. Brillhart, Filaseta, and Odlyzko extend the result to arbitrary integer bases $b \geq 2$, again assuming a nonconstant digit polynomial. The digit restrictions are essential to this statement; a prime value alone does not imply irreducibility.

For instance, 101 is prime, so its decimal digits give the irreducible polynomial $x^2 + 1$. The unrestricted prime-value inference fails for $(x-1)(x+1)$, which takes the prime value 3 at $x = 2$, and is reducible.

References

[brillhartfilasetaodlyzko1981] / Theorem 1 (p. 1055) and Corollary 2 (p. 1058). Cohn's theorem, prime-value root separation, and extension beyond decimal digits. Source.

[murty2002primevalues]. A further exposition and proof of the arbitrary-base extension. Source.

5.5 Limitations

In Perron's criterion, root location contradicts integrality of a factor's constant coefficient. In the prime-value criterion, the root bound forces both nonconstant factors to have values of absolute value greater than one. Real-rootedness alone gives neither contradiction and is compatible with a large Galois group.

A complete factorization algorithm decides any one rational polynomial, but a uniform theorem for a sparse or parameterized family may require a new inequality or a different method.

References

[Schinzel2000]. General reducibility theory and criteria for special forms. Source.

[GathenGerhard2013]. Exact polynomial factorization algorithms. Source.

Questions

Does one prime value prove irreducibility?

Settled in stated scope / Integer polynomials without additional coefficient or root restrictions

No: $(x-1)(x+1)$ takes the prime value 3 at $x = 2$. A distance-to-roots bound, a sufficiently large evaluation, or digit restrictions can turn a prime value into an irreducibility criterion.

References

[murty2002primevalues]. Precise additional hypotheses for prime-value arguments. Source.

Related topics

Contrasts with. Reduction modulo primes. Root bounds can prove irreducibility when every prime reduction is reducible.

Uses. Reciprocal polynomials. Coefficient reversal can put the dominant coefficient in the required position.

Applies to. Sparse polynomials and factor reversal. Sparse support simplifies root and coefficient-norm estimates.

Contrasts with. Polynomial factorization algorithms. A family criterion can give a short proof without general factorization.

CHAPTER 6

Irreducibility from integer values

method / Intermediate / Q / Classical theory

Unit, small, and prime values restrict the possible integral factors of a polynomial. Schur's and Pólya's criteria provide explicit examples.

Overview

- Why is a product of distinct linear integer factors minus one irreducible?
- How small must many values be to force irreducibility?
- How many prime or unit values suffice?

Prerequisites: Fields, irreducibility, and Galois actions

6.1 Schur's unit-value problem

Theorem / the minus-one case of Schur's problem

For distinct integers $a_1, \dots, a_m$, with $m \geq 1$, the polynomial

$$P(x) = \prod_{i=1}^{m} (x - a_i) - 1$$

is irreducible over $\mathbb{Q}$.

Proof

A nontrivial factorization gives monic $g, h \in \mathbb{Z}[x]$ of positive degrees with $P = gh$. At each a_i, the integers $g(a_i), h(a_i)$ multiply to -1, so $g(a_i) + h(a_i) = 0$. The polynomial $g + h$ has degree less than m and at least m distinct roots. It must vanish identically, contradicting the monicity of both factors.

For the plus-one polynomial, two exceptions are

$$x(x-2) + 1 = (x-1)^2, \qquad x(x-1)(x-2)(x-3) + 1 = (x^2 - 3x + 1)^2.$$

Westlund and Flügel's classical solution, recalled in Győry-Hajdu-Tijdeman, shows that these are the only reducible plus-one cases, up to an integer translation of the variable. Schur posed the question in 1908.

References

[schur1908problem226]. The original problem, rather than an attribution of the later solution. Historical citation verification.

[gyoryhajdutijdeman2011] / §1, p. 416. The minus-one theorem, the complete plus-one exception list, and attribution of the 1909 solution. Source.

6.2 Pólya's criterion

Theorem / Pólya's criterion

Let $P \in \mathbb{Z}[x]$ have degree $n \geq 2$, and put $N = \lceil n/2 \rceil$. If there are n distinct integers a such that

$$0 < |P(a)| < \frac{N!}{2^N},$$

then P is irreducible over $\mathbb{Q}$.

Proof

If $P = QR$ with positive-degree integral factors, choose the larger degree $d = \deg Q$, so $N \leq d \leq n-1$. At every selected integer, $|Q(a)| \leq |P(a)|$, because the other integral factor is nonzero. Choose $d+1$ of those arguments and order them as $b_0 < \cdots < b_d$. Their integer spacing gives $\prod_{i \neq j} |b_j - b_i| \geq j!(d-j)!$. Writing $M = \max_j |Q(b_j)|$, Lagrange interpolation gives

$$1 \leq |\operatorname{lc}(Q)| \leq M \sum_{j=0}^{d} \frac{1}{j!(d-j)!} = M \frac{2^d}{d!}.$$

Consequently $M \geq d!/2^d \geq N!/2^N$, contradicting the strict bound. Gauss's lemma justifies the integral factorization even when P is not monic.

Example

For $P(x) = \prod_{j=0}^{9}(x-j) + 2$, the ten values at $0, \ldots, 9$ are all 2. Since $2 < 5!/2^5 = 15/4$, Pólya proves irreducibility without expanding the polynomial. Nonzero values are essential: an integer zero would itself give a rational linear factor.

References

[polya1919]. The classical small-value irreducibility criterion. Source.

[gyoryhajdutijdeman2011] / §1, p. 416. Confirms the ceiling in N and the strict factorial threshold; the displayed interpolation proof is supplied here. Source.

6.3 A prime-value criterion

Theorem / prime and unit values

Let $f \in \mathbb{Z}[x]$ be primitive of degree $n \geq 1$. If $|f(a)|$ is either 1 or a prime at $2n+1$ distinct integers a, then f is irreducible over $\mathbb{Q}$.

Proof

For a positive-degree integral factorization $f = gh$, every selected argument satisfies $g(a) = 1$, $g(a) = -1$, $h(a) = 1$, or $h(a) = -1$. Those four equations have at most $2\deg g + 2\deg h = 2n$ distinct solutions in total, a contradiction.

Example

At $a = 0, 1, 2, 3, 4$, the values of $x^2 + x + 41$ are $41, 43, 47, 53, 61$, all prime. Thus the criterion proves irreducibility. It says nothing about primality at all later arguments; at 40 the value is 41^2.

References

[ConradIrredQ] / Theorem 5.1. The finite prime-or-unit-value criterion and its elementary counting proof. Source.

6.4 Limitations

Remark

A single prime value is insufficient: $(x - 1)(x + 1)$ takes the prime value 3 at 2. Conversely, $x^2 + x + 4$ is irreducible because its discriminant is -15, yet every integer value is even and at least 4. The preceding criterion does not apply.

A single prime value can suffice when extra information controls the factors: Cohn's criterion restricts the coefficients to base digits, while root bounds force nonconstant factors to have large absolute values at a chosen integer. For more general small-value criteria, Levit, Tverberg, and Győry-Hajdu-Tijdeman develop sharper bounds and extensions. The number and sizes of the values are essential hypotheses in these results.

References

[ConradIrredQ] / §5, p. 5. The irreducible polynomial with no prime or unit evaluations. Source.

[levit1968]. Further criteria based on low absolute values. Source.

[tverberg1973smallvalues]. Refinements of the small-value method. Source.

[gyoryhajdutijdeman2011]. Extensions involving Schur-type and Pólya-type families. Source.

Questions

Must an irreducible integer polynomial take a prime or unit value?

Settled in stated scope / Primitive polynomials in Z[x], with integer arguments and without changing variables

No. The primitive irreducible polynomial $x^2 + x + 4$ has only positive even values at integers, all at least 4. Thus the prime-value criterion is not necessary for irreducibility.

References

[ConradIrredQ] / §5, p. 5. The same explicit obstruction is recorded after the prime-value criterion. Source.

Related topics

Uses. Fields, irreducibility, and Galois actions. Gauss's lemma allows integer evaluations to constrain rational factors.

Contrasts with. Root bounds and prime-value criteria. One-value criteria require additional coefficient or root bounds.

Applies to. Capelli's theorem and polynomial composition. Root-field criteria can also apply to compositions arising from Schur-type constructions.

Contrasts with. Polynomial factorization algorithms. Evaluation criteria can prove irreducibility without computing a factorization.

CHAPTER 7

Sparse polynomials and factor reversal

method / Research / Q / Published results

Factor reversal preserves reciprocal products and coefficient norms. These identities constrain nonreciprocal factors of sparse polynomials.

Overview

- Why do large-gap theorems remove reciprocal factors?
- What does factor reversal preserve?
- What do sufficiently large gaps imply about factorization?

Prerequisites: Fields, irreducibility, and Galois actions; Reciprocal polynomials

7.1 Large gaps and reciprocal factors

Consider the family

$$f_N(x) = x^N c(x^{-1}) + d(x), \qquad N > \deg c + \deg d,$$

where $c, d \in \mathbb{Z}[x]$ are fixed and $c(0)d(0) \neq 0$. The high and low blocks of coefficients are separated by a gap. Write $h^*(x) = x^{\deg h} h(1/x)$. A nonconstant irreducible factor is reciprocal when $h^* = \pm h$; the *nonreciprocal part* is obtained by removing all such irreducible factors, including their multiplicities.

This removal is different from removing only cyclotomic factors. Every cyclotomic polynomial is reciprocal up to sign, but a reciprocal polynomial need not be cyclotomic. Irreducibility of the nonreciprocal part therefore does not imply irreducibility of the original polynomial.

References

[Schinzel2000]. Lacunary polynomial factorization and reciprocal factors. Source.
[sawinshustermanstoll2020] / §1. Large-gap notation and definitions. Source.

7.2 Schinzel's theorem

Call (c, d) *Capellian* if $-d(x)/c(x^{-1})$ is a p-th power in $\mathbb{Q}(x)$ for some prime p, or if $d(x)/c(x^{-1})$ is four times a fourth power. These conditions explain systematic binomial-type factorizations.

> **Theorem / Schinzel, in large-gap form**
> Assume $c(0)d(0) \neq 0$, $\gcd_{\mathbb{Z}[x]}(c^*, d) = 1$, $c \neq \pm d$, and that (c, d) is not Capellian. Then there is an effective bound N_0, depending on c, d, such that for $N > N_0$ the nonreciprocal part of f_N is irreducible over $\mathbb{Q}$.

The cited formulation is Theorem 1.2 of Sawin-Shusterman-Stoll, attributed there to Schinzel. Filaseta-Ford-Konyagin improve explicit thresholds. Under an additional robustness condition, Sawin-Shusterman-Stoll obtain substantially more practical bounds and algorithms; their bounds require that condition.

References

[Schinzel2000] / Theorem 74, as identified in Sawin-Shusterman-Stoll. The general lacunary result behind this formulation. Source.

[filasetafordkonyagin2000]. Improved effective bounds with explicit exclusions. Source.

[sawinshustermanstoll2020] / Theorems 1.2, 1.5, and 1.6. Exact hypotheses of the large-gap theorem and the extra robustness assumption for its improved bound. Source.

7.3 Factor reversal

For $f(0) \neq 0$ and a factorization $f = gh$, form $G = gh^*$. The reciprocal product does not change:

$$GG^* = ff^*.$$

Equivalently, as Laurent polynomials, one may use $g(x)h(x^{-1})$; multiplication by a power of x only changes the presentation. On the unit circle, a real-coefficient polynomial satisfies $|h^*(z)| = |h(z)|$. Consequently G and f have the same squared coefficient norm, by Parseval's identity:

$$\sum_j |[x^j]G|^2 = \frac{1}{2\pi}\int_0^{2\pi} |G(e^{it})|^2\,dt = \sum_j |[x^j]f|^2.$$

For a sparse polynomial with small integral coefficients, the right side is small. Only finitely many distributions of that integer norm among nonzero coefficients are possible. Comparing the full identity $GG^* = ff^*$ can then force the reversal to be trivial or expose a reciprocal factor. Ljunggren's method and its later extensions use this restriction.

References

[Ljunggren1960]. Original factor-reversal approach to sparse trinomials and quadrinomials. Source.

[filaseta1999smallnorm]. Small Euclidean coefficient norm as a factorization constraint. Source.

[sawinshustermanstoll2020] / §§3-4. Reversal identity and the later algorithmic development. Source.

7.4 Example

Example / $x^N + x + 1$

Let ζ be a primitive cube root of unity. Since $\zeta + 1 = -\zeta^2$,

$$\zeta^N + \zeta + 1 = 0 \quad\Longleftrightarrow\quad N \equiv 2 \pmod 3.$$

Thus $x^2 + x + 1$ divides $x^N + x + 1$ exactly in that residue class. In degree 5, for example,

$$x^5 + x + 1 = (x^2 + x + 1)(x^3 - x^2 + 1).$$

The cubic has no rational root, hence is irreducible over $\mathbb{Q}$. Thus infinitely many members are reducible because of a recurring cyclotomic factor.

The periodicity is general for a fixed cyclotomic factor: evaluation at a root of unity depends on N modulo its order. The large-gap literature combines this elementary observation with bounds that control other factors.

References

[Ljunggren1960]. Sparse trinomial factorization and reciprocal obstructions; the displayed root-of-unity calculation is given here. Source.

[sawinshustermanstoll2020] / Corollary 1.4. Effective eventual description by finitely many exceptional exponents and arithmetic progressions. Source.

7.5 A family with an exceptional degree

Sawin-Shusterman-Stoll prove that $x^N - 7x^2 + 1$ is irreducible for every $N \geq 5$, and also for $N = 3$. The degree 4 is exceptional:

$$x^4 - 7x^2 + 1 = (x^2 - 3x + 1)(x^2 + 3x + 1).$$

Their general theory supplies an eventual threshold, and a finite check handles the remaining exponents in this example. The conclusion combines the uniform bound with exact calculations at the remaining degrees.

References

[sawinshustermanstoll2020] / Example 7.1. The complete k=7 example and its exceptional quartic. Source.

Questions

How small can a uniform large-gap threshold be?

Open in cited source / Pairs satisfying the hypotheses of the large-gap theorem; degree sum δ and squared coefficient weight at most w

Sawin-Shusterman-Stoll define $N_{\mathrm{opt}}(\delta, w)$ as the smallest uniform threshold for their Theorem 1.2 over pairs with $\deg c + \deg d = \delta$ and $\|c\|_2^2 + \|d\|_2^2 \leq w$. Their Question 1.7 asks, for fixed $\delta > 1$, whether this is $O_\delta(\log w)$; Question 1.8 asks for the uniform bound $O(\delta \log w)$. This records the questions as posed in that paper; it is not a claim that all later work has been excluded.

References

[sawinshustermanstoll2020] / Questions 1.7 and 1.8. Precise definition and proposed threshold questions. Source.

Related topics

Uses. Reciprocal polynomials. Large-gap theorems concern the nonreciprocal part.

Applies to. Trinomials. Trinomials give examples of factor reversal and recurring cyclotomic factors.

Contrasts with. Root bounds and prime-value criteria. Coefficient norms and root bounds supplement coefficient-dominance criteria.

Uses. Capelli's theorem and polynomial composition. Capellian obstructions are binomial power obstructions over a rational function field.

Contrasts with. Composition, iteration, and arboreal groups. Large-gap families and iterated compositions have different parameter structures.

CHAPTER 8

Capelli's theorem and polynomial composition

method / Intermediate / general fields; characteristic zero for binomials / Classical theory

Capelli's theorem reduces irreducibility of a composition to irreducibility over a root field. Power classes and norms then give criteria for binomials and iterates.

Overview

- When is a composition of polynomials irreducible?
- Exactly when is a binomial irreducible in characteristic zero?
- How can one prove irreducibility of every iterate?

Prerequisites: Fields, irreducibility, and Galois actions

8.1 Capelli's composition theorem

Theorem / Capelli's composition lemma

Let K be any field, let $F \in K[x]$ be irreducible of degree $m \geq 1$, and let $g \in K[x]$ have degree $d \geq 1$. For a root α of F, the composition $F(g(x))$ is irreducible over K if and only if $g(x) - \alpha$ is irreducible over $K(\alpha)$.

Proof

Choose a root β of $g(x) - \alpha$. Since $\alpha = g(\beta)$, we have $K(\alpha) \subseteq K(\beta)$, and

$$[K(\beta) : K] = m\,[K(\beta) : K(\alpha)].$$

The polynomial $F(g(x))$, of degree md, has root β. It is irreducible exactly when the left side is md, equivalently when the minimal polynomial of β over $K(\alpha)$ has degree d. No separability assumption is needed for this argument.

The composition problem is thus reduced to irreducibility over a root field. When $g = x^r + c$, the new question is whether $x^r - (\alpha - c)$ factors; power classes and norms give useful criteria.

References

[AyadMcQuillan2000] / Lemma 1, p. 87. Capelli's composition lemma, stated for arbitrary fields. Source.
[Schinzel2000]. Composition and reducibility in extension fields. Source.

8.2 Example

Example

Both $F = x^2 - 8$ and $g = x^2 - 3$ are irreducible over $\mathbb{Q}$, but

$$F(g(x)) = x^4 - 6x^2 + 1 = (x^2 - 2x - 1)(x^2 + 2x - 1).$$

For $\alpha = 2\sqrt{2}$, we have $3 + \alpha = (1 + \sqrt{2})^2$, so $g(x) - \alpha$ splits in $\mathbb{Q}(\sqrt{2})[x]$. Irreducibility of the two inputs over $\mathbb{Q}$ does not exclude this square in the root field.

References

[AyadMcQuillan2000] / Lemma 1. The root-field criterion explains the explicitly verified example. Source.

8.3 Capelli's binomial criterion

Theorem / Capelli's binomial criterion

Let K have characteristic zero, $a \in K^\times$, and $n \geq 2$. Then $x^n - a$ is irreducible over K exactly when

- $a \notin K^p$ for every prime $p \mid n$;
- if $4 \mid n$, then $a \notin -4K^4$.

Here $K^p = \{b^p : b \in K\}$. This statement is restricted to characteristic zero; finite-field questions are treated separately.

The first obstruction gives a difference-of-powers factorization. The exceptional fourth-power obstruction comes from

$$x^{4r} + 4b^4 = (x^{2r} - 2bx^r + 2b^2)(x^{2r} + 2bx^r + 2b^2).$$

These prove necessity. The proof of sufficiency uses successive prime-power extensions and excludes powers in the intermediate root fields. Odd-prime steps use norms; the power of two requires precisely the additional obstruction above.

Example

The polynomial $x^4 + 4$ factors by the displayed identity even though -4 is not a rational square. In contrast, $x^6 - 72$ is irreducible: $72 = 2^3 3^2$ is neither a square nor a cube. In fact $x^n - 72$ is irreducible for every $n \geq 2$, because no prime divides both valuation exponents $3, 2$, and a positive rational number cannot be -4 times a rational fourth power.

References

[capelli1901binomials]. The classical binomial reducibility theorem, including the exceptional fourth-power case. Source.

[Lang2002]. The field-theoretic treatment of irreducible binomials and the norm argument. Source.

8.4 Example

Example

For $h = x^6 - 72$, the Newton polygon at 2 has one slope $-3/6 = -1/2$, allowing proper factor degrees $2, 4$. At 3, its slope is $-2/6 = -1/3$, allowing only degree 3. The empty intersection gives a second proof of irreducibility, complementing Capelli. Neither polygon individually forces the full degree.

Let the positive root be $\alpha = \sqrt{2}\sqrt[3]{3}$. The identities

$$\sqrt{2} = \alpha^3/6, \qquad \sqrt[3]{3} = \alpha^4/12$$

show that the splitting field is $L = \mathbb{Q}(\sqrt{2}, \sqrt[3]{3}, \zeta_3)$. The splitting field M of $x^3 - 3$ has group S_3: the cubic is Eisenstein at 3, and its discriminant is -243. Its unique quadratic subfield is $\mathbb{Q}(\sqrt{-3})$. Thus $M \cap \mathbb{Q}(\sqrt{2}) = \mathbb{Q}$, and restriction gives

$$\mathrm{Gal}(h/\mathbb{Q}) \cong S_3 \times C_2, \qquad |\mathrm{Gal}(h/\mathbb{Q})| = 12.$$

The discriminant $2^{21}3^{16}$ is nonsquare, but the group is much smaller than S_6. The group calculation requires information beyond either irreducibility proof or the discriminant square class.

References

[dumas1906]. The slope-denominator principle applied to the explicitly computed two-prime certificate. Source.

[Milne2022]. Galois correspondence, composita, and the quadratic subfield determined by a discriminant; the displayed sextic calculation is supplied here. Author's book page.

[capelli1901binomials]. The complementary binomial criterion. Source.

8.5 Irreducibility of quadratic iterates

Example / every iterate of $f = x^2 + 1$ is irreducible over $\mathbb{Q}$

Put $F_r = f^{\circ r}$ and $a_r = F_r(0)$, with $a_0 = 0$. Then $a_{r+1} = a_r^2 + 1$, beginning $0, 1, 2, 5, 26, 677$. The base case $F_1 = x^2 + 1$ is irreducible. Suppose F_r is irreducible and let α be a root. Capelli reduces the next step to showing that $\alpha - 1$ is not a square in $\mathbb{Q}(\alpha)$.

Since F_r is monic of even degree,

$$N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(\alpha - 1) = F_r(1) = a_{r+1} = a_r^2 + 1.$$

For $r \geq 1$, this integer lies strictly between a_r^2 and $(a_r + 1)^2$. It is therefore not a rational square. A square in $\mathbb{Q}(\alpha)$ would have square norm, giving the contradiction and completing induction.

Remark

A nonsquare norm certifies a nonsquare element; a square norm does not certify a square element. The argument proves stability, meaning irreducibility of every iterate. Determining every Galois group in the resulting tree requires further information. The correction to Ayad-McQuillan withdraws a claimed converse in a broader stability criterion.

References

[AyadMcQuillan2000] / Lemma 1. Capelli's lemma underlying the supplied norm induction. Source.

[AyadMcQuillan2001Correction] / Theorem A. The corrected sufficient stability criterion and the failure of the original converse. Source.

[Jones2013Arboreal] / §5. Stability and its relation to arboreal Galois representations. Source.

Questions

Does composition preserve irreducibility?

Settled in stated scope / Nonconstant polynomials over Q

No. The pair x^2-8, x^2-3 above is a counterexample. Capelli identifies the missing condition: the inner polynomial minus a root of the outer one must remain irreducible over that root field.

References

[AyadMcQuillan2000] / Lemma 1. The exact criterion, applied here to an independently expanded example. Source.

Related topics

Uses. Fields, irreducibility, and Galois actions. The tower formula turns polynomial irreducibility into an equality of extension degrees.

Applies to. Composition, iteration, and arboreal groups. Stability gives transitivity at every level of an arboreal action.

Supplies examples for. Newton polygons and Galois groups. The sextic x^6-72 has both a Capelli proof and a proof using Newton polygons at two primes.

Uses. Discriminants, parity, and ramification. The sextic example has nonsquare discriminant and a Galois group of order 12.

Applies to. Irreducibility from integer values. Restrictions on integer values give irreducibility criteria for other composed families.

Contrasts with. Irreducibility over finite fields. Finite-field irreducibility tests use the field size and Frobenius.

Part III

Galois group detection

CHAPTER 9

Frobenius and factorization modulo primes

method / Intermediate / Q and number fields / Classical theory

Squarefree factorization modulo a prime determines a cycle type in the Galois group. Group-theoretic criteria then convert these cycle types into group containments.

Overview

- What does a squarefree reduction say about the Galois group?
- When can irreducibility be witnessed by one prime?
- How do two primes prove that a quintic has group S5?

Prerequisites: Fields, irreducibility, and Galois actions; Reduction modulo primes

9.1 The Dedekind-Frobenius theorem

Theorem / Dedekind-Frobenius

Let $f \in \mathbb{Z}[x]$ be monic and separable, and let G be the Galois group of its splitting field over $\mathbb{Q}$, acting on its n roots. If $p \nmid \operatorname{disc}(f)$ and

$$\bar{f} = f_1 \cdots f_r \quad \text{in } \mathbb{F}_p[x]$$

is its factorization into distinct monic irreducibles, then G contains a permutation whose cycle lengths are $\deg f_1, \dots, \deg f_r$.

The cycle statement does not require irreducibility of f. Irreducibility is the additional condition that makes G transitive. For nonmonic integral polynomials, also exclude primes dividing the leading coefficient.

References

[ConradModP] / Theorem 1 and its proof. The note treats irreducible monic polynomials; the same reduction argument gives the separable form stated here. Source.

[Milne2022] / Chapter 4, "Computing Galois groups over Q". Author's book page.

9.2 Frobenius orbits

Choose a prime $\mathfrak{p}$ of the splitting field above p. The discriminant condition ensures that the roots remain distinct after reduction. Frobenius acts on the residue field by $z \mapsto z^p$. A root of an irreducible degree-d factor has an orbit of length exactly d under this map. Lifting that action to the roots gives the required permutation.

A different choice of $\mathfrak{p}$ conjugates Frobenius. Thus factorization determines the cycle type. It does not identify the root labels, so permutations obtained from different primes have no specified common labeling.

References

[ConradModP] / Proof of Theorem 1. Source.

9.3 Example

Example: S_5

Take $f = x^5 - x - 1$. Modulo 3, it has no root. The only monic irreducible quadratics over $\mathbb{F}_3$ are

$$x^2 + 1, \quad x^2 + x + 2, \quad x^2 + 2x + 2.$$

The remainders on division of f by these are $-1, x - 1, x - 1$, respectively. Every reducible quintic has a factor of degree at most two, so $\bar{f}$ is irreducible over $\mathbb{F}_3$. Hence f is irreducible over $\mathbb{Q}$, and its group contains a 5-cycle.

Modulo 2,

$$f = (x^2 + x + 1)(x^3 + x^2 + 1),$$

with distinct irreducible factors. This gives an element of type $(2, 3)$, whose cube is a transposition.

A transitive action of prime degree is primitive. A primitive group containing a transposition is the full symmetric group, as proved in the permutation-group entry. Therefore $\mathrm{Gal}(f/\mathbb{Q}) = S_5$.

References

[ConradSnAn] / Theorem 2.1. Recognition criterion used after the explicit finite-field computations. Source.

[Selmer1956]. Classical irreducibility of the broader trinomial family; the present degree-five proof is self-contained. Source.

9.4 Chebotarev density

By Chebotarev, every conjugacy class of G occurs at infinitely many unramified primes. The density for a class C is $|C|/|G|$; a cycle type may comprise several classes of G, in which case their densities add.

Consequently, an irreducible polynomial of degree n has an irreducible reduction at some good prime if and only if G contains an n-cycle. Transitivity alone does not guarantee this. For $x^4 + 1$, the group is the Klein four-group and no good prime yields an irreducible quartic reduction.

References

[StevenhagenLenstra1996]. Chebotarev density and Frobenius interpretation. Source.

[ConradAllPrimes]. Source.

9.5 Limits of modular tests

Remark

The theorem requires squarefree reduction. Failure to observe a cycle type in a finite search does not exclude it. A finite list of factorizations determines the group only if a recognition theorem or a complete subgroup analysis excludes every other compatible group.

At primes of repeated reduction, ramification and Newton polygons may give further permutations. Resolvents can distinguish groups with the same observed cycle types.

References

[ConradSnAn] / Section 1. Source.

Questions

Does every irreducible rational polynomial remain irreducible modulo some prime?

Settled in stated scope / Monic irreducible polynomials over Q

No. Such a reduction exists precisely when the Galois group contains a full cycle. The polynomial $x^4 + 1$ is a counterexample.

References

[ConradAllPrimes]. Source.

[StevenhagenLenstra1996]. Source.

Related topics

Uses. Transitivity, primitivity, and cycle criteria. Permutation-group criteria determine which groups contain the observed cycle types.

Uses. Discriminants, parity, and ramification. The discriminant identifies good primes and the parity character.

Contrasts with. Newton polygons and Galois groups. Ramification and Newton polygons apply at primes where reduction is not squarefree.

Applies to. Algorithms for Galois groups. Modular factorizations restrict the candidate groups in an exact computation.

CHAPTER 10

Discriminants, parity, and ramification

method / Intermediate / Fields of characteristic not 2; Q / Classical theory

The discriminant vanishes precisely when roots repeat. For a separable polynomial in characteristic different from two, its square class determines the sign character of the Galois group.

Overview

- Why does a square discriminant put the group inside A_n?
- What distinguishes polynomial and number-field discriminants?
- When does parity finish a Galois-group proof?

Prerequisites: Fields, irreducibility, and Galois actions

10.1 Definition

For $f(x) = a_n \prod_{i=1}^{n} (x - \alpha_i)$ of degree n, the polynomial discriminant is

$$\Delta(f) = a_n^{2n-2} \prod_{i<j} (\alpha_i - \alpha_j)^2 = (-1)^{n(n-1)/2} a_n^{-1} \operatorname{Res}(f, f').$$

It is nonzero exactly when the roots are distinct. The resultant formula computes it from coefficients without computing the roots. Multiplying f by a nonzero scalar changes its discriminant by an even power, so its square class is unchanged.

References

[Milne2022] / Chapter 4 and its appendix on resultants. Author's book page.

10.2 The discriminant criterion

Theorem / discriminant criterion

If $f \in K[x]$ is separable and $\operatorname{char} K \neq 2$, then

$$\operatorname{Gal}(f/K) \subseteq A_n \quad \Longleftrightarrow \quad \Delta(f) \in K^{\times 2}.$$

Proof

Set $D = a_n^{n-1} \prod_{i<j} (\alpha_i - \alpha_j)$, so $D^2 = \Delta(f)$. An automorphism σ sends D to $\operatorname{sgn}(\sigma) D$. Thus all automorphisms are even exactly when D belongs to the ground field. Conversely, if $D^2 = b^2$ with $b \in K$, then $D = \pm b$, so $D \in K$. Characteristic two is excluded because the signs coincide.

If the group is not contained in A_n, the quadratic field $K(\sqrt{\Delta(f)})$ is the fixed field of the kernel of the sign character.

References

[ConradCubicQuartic] / Theorem 1.3. Source.
[Milne2022] / Chapter 4, “When is G_f contained in A_n?”. Author’s book page.

10.3 Cubics

Example

A separable irreducible cubic has a transitive subgroup of S_3, hence group A_3 or S_3. For $x^3 + ax + b$, the discriminant is $-4a^3 - 27b^2$. Therefore

f	$\Delta(f)$	$\mathrm{Gal}(f/\mathbb{Q})$
$x^3 - x - 1$	-23	S_3
$x^3 - 3x - 1$	81	C_3

The rational-root test proves both polynomials irreducible: neither 1 nor -1 is a root. Irreducibility is required to apply the cubic classification.

References

[ConradCubicQuartic] / Theorem 2.1 and Example 2.2. Source.

10.4 Quartic examples

The polynomial $x^4 + 1$ is irreducible over $\mathbb{Q}$, since replacing x by $x + 1$ makes it Eisenstein at 2. Its discriminant is 256, but its splitting field is $\mathbb{Q}(\zeta_8)$, whose group is $(\mathbb{Z}/8\mathbb{Z})^\times \simeq C_2 \times C_2$, strictly smaller than A_4.

For a nonsquare example, $x^4 - 2$ is Eisenstein at 2 and has nonsquare discriminant -2048. Its splitting field is $\mathbb{Q}(\sqrt[4]{2}, i)$, of degree eight: the real field $\mathbb{Q}(\sqrt[4]{2})$ has degree four and does not contain i. The group is dihedral of order eight, not S_4.

Remark

If $A_n \subseteq G$ is known, a square discriminant gives $G = A_n$, and a nonsquare discriminant gives $G = S_n$. The containment is essential.

References

[Cox2012]. Cyclotomic and radical extensions; discriminant interpretation. Source.
[ConradCubicQuartic] / Section 3. Source.

10.5 Polynomial and field discriminants

If f is the monic minimal polynomial of an algebraic integer α, and $K = \mathbb{Q}(\alpha)$, then

$$\Delta(f) = \mathrm{disc}(K)\,[\mathcal{O}_K : \mathbb{Z}[\alpha]]^2.$$

The two discriminants have the same square class, but can have different prime divisors. A prime dividing $\Delta(f)$ may come from the index rather than ramification in K.

For $f = x^2 - 5$, the polynomial discriminant is 20, whereas $\mathrm{disc}(\mathbb{Q}(\sqrt{5})) = 5$. Reduction modulo 2 is repeated, although 2 is unramified in the quadratic field. The integral generator $(1 + \sqrt{5})/2$, with polynomial $x^2 - x - 1$, removes this index obstruction.

Over $\mathbb{Q}$, complex conjugation also determines the sign of the discriminant: with r nonreal conjugate pairs it has type $2^r 1^{n-2r}$, and $\mathrm{sgn}\,\Delta(f) = (-1)^r$.

References

[Neukirch1999]. Discriminants of orders, indices, and ramification. Source.
[ConradCubicQuartic] / Remark 3.11 and the discussion of real roots. Source.

Questions

When can a discriminant determine the full Galois group?

Settled in stated scope / Separable irreducible cubics; groups already known to contain A_n.

It determines the group for irreducible cubics, and whenever an independent argument has proved $A_n \subseteq G$. In general it determines only whether $G \subseteq A_n$.

References

[ConradCubicQuartic] / Theorems 1.3 and 2.1. Source.

Related topics

Applies to. Frobenius and factorization modulo primes. Primes not dividing the polynomial discriminant have squarefree reduction.
Applies to. Resolvents and subgroup tests. The cubic resolvent and the discriminant together distinguish several quartic groups.
Uses. Transitivity, primitivity, and cycle criteria. Containment of A_n and an odd permutation imply equality with S_n.
Applies to. Random polynomials: models and methods. Alternating-group containment and a nonsquare discriminant require separate probability estimates.

CHAPTER 11

Ramification and inertia

method / Intermediate / Q and local fields / Classical theory

Inertia describes the action on roots at ramified primes. For a monic integral polynomial, an odd prime dividing its discriminant exactly once gives a transposition.

Overview

- How does inertia act at a ramified prime?
- When does a discriminant valuation force a transposition?
- Can repeated reduction occur without ramification?

Prerequisites: Fields, irreducibility, and Galois actions; Discriminants, parity, and ramification; Frobenius and factorization modulo primes

11.1 Decomposition and inertia groups

Let $L/\mathbb{Q}$ be a finite Galois extension with group G, and choose a prime $\mathfrak{P}$ above p. Its **decomposition group** $D_{\mathfrak{P}}$ consists of automorphisms preserving $\mathfrak{P}$. The **inertia group** $I_{\mathfrak{P}}$ consists of those acting trivially on the residue field $k_{\mathfrak{P}} = \mathcal{O}_L/\mathfrak{P}$. There is an exact sequence

$$1 \longrightarrow I_{\mathfrak{P}} \longrightarrow D_{\mathfrak{P}} \longrightarrow \mathrm{Gal}(k_{\mathfrak{P}}/\mathbb{F}_p) \longrightarrow 1.$$

Also $D_{\mathfrak{P}} \cong \mathrm{Gal}(L_{\mathfrak{P}}/\mathbb{Q}_p)$. The local group is therefore a subgroup of the global Galois group. When p is unramified, inertia is trivial and residue Frobenius lifts uniquely inside $D_{\mathfrak{P}}$. At a ramified prime it determines a coset modulo inertia. Inertia fixes the reductions of integral roots, while possibly permuting roots that collide.

References

[Milne2020ANT] / Theorem 7.58; Proposition 8.10. Exact sequence, maximal unramified subfield, and identification with the local Galois group. Source.

11.2 Tame inertia

In residue characteristic p, inertia has a normal wild p-subgroup and a cyclic quotient of order prime to p. If the extension is **tame**, wild inertia is trivial, so inertia itself is cyclic.

For irreducible f, put $K = \mathbb{Q}(\alpha)$. If its splitting field is tame at p, write $p\mathcal{O}_K = \prod_i \mathfrak{p}_i^{e_i}$, with residue degrees f_i. A generator of inertia acts on the roots with an e_i-cycle repeated f_i times for each i. Indeed, the decomposition orbit for $\mathfrak{p}_i$ has size $e_i f_i$, and splits into f_i inertia orbits of size e_i; a cyclic generator traverses each orbit. These are ramification indices in the ring of integers, not automatically exponents in f mod p.

For example, $x^3 - 5$ is Eisenstein over $\mathbb{Q}_5$. Its root has valuation $1/3$, forcing ramification index three. Adjoining a primitive cube root of unity is unramified at 5, since $x^2 + x + 1$ has squarefree reduction. The splitting field therefore has cyclic inertia of order three, acting as a three-cycle. Here $\Delta(x^3 - 5) = -675$ has valuation two at 5: tame inertia gives a 3-cycle although the discriminant valuation is not one.

References

[Serre1979] / Ramification Groups, pp. 61-79. Background for tame and wild inertia; the cycle description is the orbit consequence explained here. Source.

[Milne2020ANT] / Corollary 7.59 and the following paragraph. Cyclicity of the tame quotient and the criterion for tame ramification. Source.

11.3 A simple discriminant prime

Theorem / simple discriminant prime

Let $f \in \mathbb{Z}[x]$ be monic and separable, with splitting field L. If p is odd and $v_p(\Delta(f)) = 1$, then inertia at every prime of L above p has order two, and its nonidentity element acts as a transposition on the roots. Irreducibility is not required.

Proof

Multiplication by f' on the free $\mathbb{Z}_p$-module $\mathbb{Z}_p[x]/(f)$ has determinant $\operatorname{Res}(f, f') = \pm\Delta(f)$. Valuation one means its Smith normal form has exactly one nonunit diagonal entry, so reduction modulo p has cokernel dimension one. This cokernel is $\mathbb{F}_p[x]/(\bar{f}, \bar{f}')$; hence $\deg \gcd(\bar{f}, \bar{f}') = 1$.

Since p is odd, necessarily $\bar{f} = (x-a)^2\bar{h}$, where $a \in \mathbb{F}_p$, $\bar{h}$ is squarefree, and $\bar{h}(a) \neq 0$. Hensel's lemma lifts these coprime factors to $f = qh$ over $\mathbb{Z}_p$, with q quadratic. The identity

$$\Delta(f) = \Delta(q)\Delta(h)\operatorname{Res}(q, h)^2$$

and the unit factors $\Delta(h)$ and $\operatorname{Res}(q, h)$ give $v_p(\Delta(q)) = 1$. The splitting field $\mathbb{Q}_p(\sqrt{\Delta(q)})$ is ramified quadratic; the squarefree reduction of h makes its splitting field unramified. Their compositum therefore has inertia of order two, swapping precisely the two roots of q.

References

[Milne2020ANT] / Theorem 7.33; Proposition 7.50. Hensel lifting and unramified extensions used in the explicit proof; the valuation-one deduction is supplied here. Source.

[Serre1979] / Discriminant and Different, pp. 47-60; Ramification Groups, pp. 61-79. Discriminants and local ramification framework. Source.

11.4 Example

Example / $x^3 - x - 1$

The rational-root test proves $f = x^3 - x - 1$ irreducible: neither 1 nor -1 is a root. Its discriminant and bad-prime reduction are

$$\Delta(f) = -4(-1)^3 - 27(-1)^2 = -23, \qquad f \equiv (x-10)^2(x-3) \pmod{23}.$$

Indeed, $f - (x-10)^2(x-3) = 23(x^2 - 7x + 13)$. The theorem supplies an inertia transposition at 23. Since $G \leq S_3$ is transitive, $3 \mid |G|$; the transposition gives $2 \mid |G|$, so $G = S_3$. The discriminant valuation proves nontrivial inertia; the repeated reduction alone would not suffice.

References

[Serre1979] / Ramification Groups, pp. 61-79. Local-group background; all numerical identities and the elementary group argument are verified directly here. Source.

11.5 The index of an order

Remark

For monic irreducible f with root α and $K = \mathbb{Q}(\alpha)$,

$$\Delta(f) = \operatorname{disc}(K)\,[\mathcal{O}_K : \mathbb{Z}[\alpha]]^2.$$

Consider $x^2 - 18$, with root $3\sqrt{2}$. Modulo 3 it becomes x^2, yet $K = \mathbb{Q}(\sqrt{2})$ has discriminant 8, so 3 is unramified and its inertia group is trivial. The polynomial discriminant is $72 = 8 \cdot 3^2$: the repeated reduction comes from the index 3. Conversely, valuation one excludes any index contribution at that prime.

A primitive subgroup containing a transposition is S_n. Transitivity alone does not suffice.

References

[Milne2020ANT] / Remark 2.25; Theorem 3.35. Index-squared discriminant identity and the ramification-discriminant criterion. Source.

Questions

Does a repeated root modulo p imply nontrivial inertia?

Settled in stated scope / Monic integral polynomials over Q

No: $x^2 - 18$ modulo 3 is a counterexample, because its root field is unramified at 3. The order index can account for repeated reduction; the valuation-one theorem excludes this obstruction.

References

[Milne2020ANT] / Remark 2.25. The index-squared discriminant identity explains the obstruction. Source.

Related topics

Contrasts with. Frobenius and factorization modulo primes. Squarefree reduction gives Frobenius cycle types; repeated reduction requires ramification and index information.

Uses. Discriminants, parity, and ramification. A simple discriminant prime isolates a ramified quadratic factor.

Applies to. Transitivity, primitivity, and cycle criteria. An inertia transposition and primitivity imply the full symmetric group.

Contrasts with. Newton polygons and Galois groups. Newton polygons constrain local factor degrees; inertia describes ramification permutations.

CHAPTER 12

Transitivity, primitivity, and cycle criteria

method / Intermediate / Permutation groups; polynomial Galois groups / Classical theory

Blocks, transpositions, and long cycles give criteria for alternating-group containment and equality with the symmetric group.

Overview

- How can primitivity be proved?
- Why does a primitive group with a transposition equal S_n?
- Which single cycles force containment of A_n?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

12.1 Blocks and primitivity

Let $G \leq S_n$ act on Ω. A nonempty subset $B \subseteq \Omega$ is a block if each translate gB is either equal to B or disjoint from it. A transitive action is primitive if its only blocks have size 1 or n.

In a transitive action, translates of a block partition Ω into subsets of equal size. Hence a transitive action of prime degree is primitive. For an irreducible separable polynomial with root α, the action is primitive exactly when the root field $K(\alpha)$ has no proper intermediate field between K and $K(\alpha)$. This follows by applying the Galois correspondence to subgroups between a root stabilizer and the whole group.

For a polynomial map, decomposition of $f(X)$ is related to primitivity of the monodromy action of $f(X) - t$ over a function field. This is a different action from the Galois action on the roots of a fixed polynomial.

References

[DixonMortimer1996]. Blocks, stabilizers, and primitive actions. Source.

[Cox2012]. Galois correspondence and the action on conjugates. Source.

12.2 Transpositions

Theorem

A primitive subgroup $G \leq S_n$ containing a transposition is S_n.

Proof

Form a graph on Ω, joining a to b when $(a\ b) \in G$. Conjugation by G preserves this graph, so its connected components form a block system. There is an edge, and primitivity forces the graph to be connected. The transpositions associated with the edges of a connected graph generate every transposition: along a path, conjugating successive edge transpositions moves an endpoint step by step. They therefore generate S_n, which is contained in G.

Primitivity cannot be replaced by transitivity. The dihedral group $\langle (1234), (13) \rangle$ has order eight and contains a transposition, while preserving the block system $\{\{1,3\},\{2,4\}\}$. It is a proper subgroup of S_4.

References

[DixonMortimer1996]. Primitive permutation groups and generation by transpositions. Source.

[ConradSnAn] / Theorem 2.1. A useful sufficient condition for the required primitivity. Source.

12.3 Long prime cycles

Suppose G is transitive and contains a q-cycle, with q prime and $q > n/2$. If a nontrivial block system had a blocks of size b, then $a, b \leq n/2 < q$. The cycle's induced action on the blocks has order dividing q; since $a < q$, it must fix every block. Its restriction to each block is also trivial because $b < q$. This contradicts the presence of a q-cycle. Thus G is primitive.

Taking powers can isolate a cycle under a coprimality condition. If one cycle has length m coprime to every other cycle length, raising the permutation to the least common multiple of the other lengths eliminates those cycles and preserves an m-cycle. An element of type $(3,3)$ does not yield a single 3-cycle by this argument.

References

[ConradSnAn] / Theorem 2.1. Source.

[Jones2014Cycles] / Remarks 1.4 and 1.6 (published version, p. 160). Source.

12.4 Double transitivity

Theorem

For $n \geq 3$, a transitive subgroup $G \leq S_n$ containing an $(n-1)$-cycle is doubly transitive. If it also contains a transposition, then $G = S_n$.

Proof

If the cycle fixes a, its powers act transitively on the other $n-1$ points. Thus the stabilizer G_a is transitive there. Transitivity of G now moves any ordered pair of distinct points to any other: first move the first point, then use its stabilizer to move the second. This is double transitivity. Conjugating one transposition by these elements supplies every transposition, which generates S_n.

No primality condition on $n-1$ is needed. In arithmetic applications, a good reduction of factor degrees $(1, n-1)$ supplies the long cycle. Irreducibility supplies transitivity. A second squarefree pattern with one 2-cycle and all other cycles of odd lengths supplies a transposition by taking an odd power. These conditions depend only on cycle types and therefore require no common labeling of roots across primes.

References

[DixonMortimer1996] / Chapter 1, The Basic Ideas. The stabilizer characterization of double transitivity; the short proof is included here. Source.

[ConradSnAn] / §1 and Example 2.3. Arithmetic cycle-type witnesses and extracting a transposition by taking a power. Source.

12.5 Jordan's theorem and Jones's extension

Theorem (Jordan)

If a primitive subgroup $G \leq S_n$ contains a q-cycle with q prime and $q \leq n-3$, then $A_n \subseteq G$.

Combining this with the preceding argument gives the following criterion: transitivity plus a prime cycle with $n/2 < q \leq n-3$ forces A_n. Newton polygons often produce just such a prime divisor of the group order; the inequality $q > n/2$ ensures an element of order q is one cycle.

Theorem (Jones)

A primitive subgroup containing a nontrivial single cycle that fixes at least three points contains A_n. The cycle length need not be prime.

This stronger result follows from the classification in Jones's paper and uses the classification of finite simple groups. The classical prime-cycle theorem does not require that classification. Both theorems assume primitivity.

References

[Jordan1871]. Historical source for primitive-group criteria. Source.
[Jones2014Cycles] / Theorem 1.1, Theorem 1.2, and Corollary 1.3. Source.

12.6 Example

Example / $x^5 - 4x + 2$

This polynomial is Eisenstein at 2, hence irreducible. Its action has prime degree five and is primitive. Put $a = (4/5)^{1/4}$. The derivative $5x^4 - 4$ vanishes only at $-a, a$; the local maximum has value $2 + 16a/5 > 0$, and the local minimum has value $2 - 16a/5 < 0$, since $a > 5/8$. Monotonicity on the three intervals gives exactly three real roots.
The remaining two roots form one nonreal conjugate pair. Complex conjugation is therefore a transposition. The transposition criterion proves $\mathrm{Gal}(x^5 - 4x + 2\,/\mathbb{Q}) = S_5$.

References

[Milne2022] / Chapter 4, "Examples of polynomials with S_p as Galois group over Q". Author's book page.
[ConradSnAn] / Theorem 2.1. Source.

Questions

Can the prime-length assumption in Jordan's cycle criterion be removed?

Settled in stated scope / Primitive finite permutation groups containing a nontrivial cycle with at least three fixed points

Yes: Jones's Corollary 1.3 gives containment of A_n without that assumption. The stronger theorem uses the classification of finite simple groups; the primitivity and fixed-point conditions remain essential.

References

[Jones2014Cycles] / Corollary 1.3. Source.

Related topics

Uses. Newton polygons and Galois groups. Suitable Newton-slope denominators force long prime cycles.

Uses. Discriminants, parity, and ramification. Once A_n is contained, the discriminant decides between A_n and S_n.

Contrasts with. Composition, iteration, and arboreal groups. Polynomial decomposition concerns function-field monodromy; primitivity of a root action concerns intermediate fields.

Contrasts with. Reciprocal polynomials. Inverse pairs form blocks for the Galois action.

CHAPTER 13

Resolvents and subgroup tests

method / Intermediate / Q; fields of characteristic not 2 for quartics / Classical theory

Resolvent factorization describes Galois orbits of expressions in the roots. A simple rational root detects containment in a conjugate of the corresponding stabilizer.

Overview

- How does a polynomial in the roots detect a subgroup?
- How is the cubic resolvent of a quartic constructed?
- What distinguishes S4, A4, V4, and the C4/D4 cases?

Prerequisites: Fields, irreducibility, and Galois actions; Discriminants, parity, and ramification

13.1 Invariant polynomials and coset actions

Let $f \in K[x]$ be separable, with roots $\alpha_1, \dots, \alpha_n$; suppose its Galois group is contained in $G_0 \leq S_n$. Choose $\Phi \in K[X_1, \dots, X_n]$ with formal stabilizer H in G_0. Evaluate its orbit to obtain values $\beta_1, \dots, \beta_m$, where $m = [G_0 : H]$, and form

$$R_\Phi(T) = \prod_{j=1}^{m} (T - \beta_j).$$

The actual Galois group permutes these values, so R_Φ has coefficients in the ground field. If the values are distinct, its irreducible factor degrees are the orbit sizes for this new action. A root in the ground field gives a fixed coset, hence containment of the Galois group in a conjugate of H.

Distinctness is essential: equal evaluated invariants can otherwise give a false subgroup containment. A nonzero resolvent discriminant certifies distinctness; if it vanishes, different root expressions may be needed.

References

[FiekerKlueners2014] / Section 3, Lemma 3.1 and Theorem 3.2. Source.

[Stauduhar1973]. The subgroup-descent principle. Source.

13.2 The cubic resolvent

For $f = x^4 + ax^3 + bx^2 + cx + d$, the three partitions of its four roots into two unordered pairs give

$$\beta_1 = \alpha_1\alpha_2 + \alpha_3\alpha_4, \quad \beta_2 = \alpha_1\alpha_3 + \alpha_2\alpha_4, \quad \beta_3 = \alpha_1\alpha_4 + \alpha_2\alpha_3.$$

Expanding $\prod(T - \beta_i)$ in elementary symmetric functions yields

$$R_f(T) = T^3 - bT^2 + (ac - 4d)T + 4bd - a^2d - c^2.$$

The action $S_4 \to S_3$ on the three pairings has kernel the normal Klein four-group. Moreover, $\beta_1 - \beta_2 = (\alpha_1 - \alpha_4)(\alpha_2 - \alpha_3)$, with analogous formulas for the other differences. Their squared product proves $\operatorname{disc}(R_f) = \operatorname{disc}(f)$, so this resolvent is separable whenever f is.

References

[ConradCubicQuartic] / Definition 3.1, equation (3.7), and Theorem 3.4. Source.

[Cox2012]. Resolvents and their relation to permutation actions. Source.

13.3 Quartic Galois groups

Assume $\operatorname{char} K \neq 2$ and f is a separable irreducible quartic over K. Then:

Resolvent over K	**Discriminant of f**	**Galois group**
Irreducible	Nonsquare	S_4
Irreducible	Square	A_4
Splits completely	Square	V_4
One linear factor and one irreducible quadratic	Nonsquare	C_4 or D_4

Here D_4 denotes the dihedral group of order eight. The last row requires an additional test. If $r \in K$ is the unique root of R_f, they are separated by the following test: the group is C_4 precisely when both $T^2 + aT + (b - r)$ and $T^2 - rT + d$ split over $K(\sqrt{\Delta(f)})$; otherwise it is D_4.

References

[KappeWarren1989Quartic] / pp. 133-137. Original published quartic-group test; Conrad Theorem 4.1 gives the exact formulation used here. Source.

[ConradCubicQuartic] / Theorem 3.6, Corollary 3.8, and Theorem 4.1 (Kappe-Warren). Source.

13.4 Example

Example / $f = x^4 + 2x + 2$

Eisenstein at 2 proves irreducibility. The resolvent is $R_f = T^3 - 8T - 4$. By the rational-root test a rational root would be one of $\pm 1, \pm 2, \pm 4$; the six values, in the order $-4, -2, -1, 1, 2, 4$, are $-36, 4, 3, -11, -12, 28$. None is zero, so R_f is irreducible.

The discriminant is

$$256 \cdot 2^3 - 27 \cdot 2^4 = 1616 = 16 \cdot 101.$$

It is nonsquare. The first line of the classification gives $\operatorname{Gal}(f/\mathbb{Q}) = S_4$.

References

[ConradCubicQuartic] / Table 5. The example also appears in the source; the arithmetic is displayed here in full. Source.

13.5 Cyclic and dihedral examples

Example (cyclic group)

For $f = x^4 + x^3 + x^2 + x + 1$, translation gives $f(x + 1) = x^4 + 5x^3 + 10x^2 + 10x + 5$, which is Eisenstein at 5. Here

$$\Delta(f) = 125, \qquad R_f(T) = (T - 2)(T^2 + T - 1).$$

The unique rational resolvent root is $r = 2$, and the discriminant field is $\mathbb{Q}(\sqrt{5})$. The two test polynomials are $T^2 + T - 1$ and $(T - 1)^2$. Both split over that field, so the group is C_4.

Example (dihedral group)

For $f = x^4 - 2$, Eisenstein at 2 proves irreducibility. Now

$$\Delta(f) = -2048, \qquad R_f(T) = T(T^2 + 8).$$

Thus $r = 0$, the discriminant field is $\mathbb{Q}(\sqrt{-2})$, and the test polynomials are T^2 and $T^2 - 2$. The latter does not split there: a real element of this imaginary quadratic field is rational, while $\sqrt{2}$ is not. Hence the group is D_4, of order eight.

Both resolvents have the same factor degrees $(1, 2)$. The extra quadratic-field test distinguishes their groups.

References

[KappeWarren1989Quartic] / pp. 133-137. Original group-separation method. Source.

[ConradCubicQuartic] / Theorem 4.1 and Corollary 4.3. Exact test used in these independently calculated examples. Source.

13.6 Relative resolvents

Factorization modulo a good prime determines a cycle type. Resolvent factorization describes orbits in another permutation action. The two tests give complementary restrictions; a rational resolvent root can identify a preserved partition that the observed cycle types do not determine.

In larger degrees, an absolute resolvent can be expensive to compute. Stauduhar's method descends through successive subgroup candidates using relative invariants. Fieker and Klüners explain how to construct invariants from group structure and how to certify the relevant factors. A numerical approximation to an integer must be verified before it can certify a factorization.

References

[Stauduhar1973]. Source.

[FiekerKlueners2014] / Sections 3 and 7. Source.

Questions

How are cyclic and dihedral quartic groups distinguished?

Settled in stated scope / Separable irreducible quartics over fields of characteristic not 2

For a reducible cubic resolvent, the discriminant first distinguishes the Klein-four case. In the remaining case, the Kappe-Warren criterion tests whether two specified quadratics split over the discriminant quadratic extension.

References

[ConradCubicQuartic] / Theorem 4.1. Source.

[KappeWarren1989Quartic] / pp. 133-137. Source.

Related topics

Uses. Discriminants, parity, and ramification. The discriminant distinguishes the alternating and symmetric cases in the quartic classification.

Applies to. Algorithms for Galois groups. Relative resolvents test subgroup containment.

Uses. Transitivity, primitivity, and cycle criteria. The factorization of a resolvent describes orbits in a coset action.

Supplies examples for. Reciprocal polynomials. The partition into inverse pairs gives invariant expressions for resolvent construction.

CHAPTER 14

Reciprocal polynomials

method / Intermediate / Characteristic different from 2 / Published results

The trace substitution halves the degree of a reciprocal polynomial. Quadratic extensions determine its irreducibility and the kernel of its Galois action on inverse pairs.

Overview

- How can a reciprocal polynomial be reduced to half its degree?
- Why is irreducibility of the trace polynomial insufficient?
- When is the Galois group the full wreath product?

Prerequisites: Fields, irreducibility, and Galois actions

14.1 Trace polynomial

Let K have characteristic different from 2. A monic polynomial f of degree $2m$ is reciprocal if $x^{2m} f(x^{-1}) = f(x)$. Its constant term is then 1. There is a unique monic $q \in K[y]$ of degree m such that

$$f(x) = x^m q(x + x^{-1}).$$

To construct it, express the symmetric Laurent polynomial $x^{-m} f(x)$ in the basis $1, x^j + x^{-j}$. The recurrence $s_{j+1} = y s_j - s_{j-1}$, with $s_0 = 2, s_1 = y$, expresses every $s_j = x^j + x^{-j}$ as a polynomial in y. The roots of q record the inverse pairs of roots of f.

References

[AndersonBertelliODorney2026] / §2. Reciprocal polynomials and the trace substitution. Source.

14.2 Irreducibility criterion

Criterion

Assume q is irreducible and separable, and let y be one of its roots. Then $x^m q(x + x^{-1})$ is irreducible over K if and only if $y^2 - 4$ is not a square in $K(y)$.

A root α above y satisfies $\alpha^2 - y\alpha + 1 = 0$, and $y = \alpha + \alpha^{-1}$. Thus $[K(\alpha) : K] = m[K(\alpha) : K(y)]$. The second factor is two exactly when the quadratic discriminant is a nonsquare. This proves the criterion by comparing the degree with $2m$. Conversely, irreducibility of f forces irreducibility of q.

References

[MinacNguyenTan2023] / §4. The quadratic-lift method for reciprocal polynomials; the degree argument is included here. Source.

14.3 Galois group

Suppose f is separable and $f(1)f(-1) \neq 0$. Let M be the splitting field of q, with roots $y_1, \dots, y_m$. The splitting field of f is

$$L = M(\sqrt{y_1^2 - 4}, \dots, \sqrt{y_m^2 - 4}).$$

Every Galois automorphism permutes the inverse pairs and may exchange the two roots in each pair. Hence $\mathrm{Gal}(f/K) \leq C_2 \wr S_m = C_2^m \rtimes S_m$. Its quotient on pairs is $\mathrm{Gal}(q/K)$; its kernel has order 2^r, where r is the dimension of the span of the square classes $[y_i^2 - 4]$ in $M^\times / M^{\times 2}$. In particular, the full wreath product occurs exactly when the quotient is S_m and all m classes are independent.

References

[AndersonBertelliODorney2026] / §2. Signed permutations are the natural ambient group. Source.

[Cox2012]. Quadratic extensions and Galois actions; the square-class formulation is standard Kummer theory. Source.

14.4 Examples

Example

For $q(y) = y^2 - 5$, the lift is $f = x^4 - 3x^2 + 1 = (x^2 + x - 1)(x^2 - x - 1)$. The polynomial q is irreducible, but $y^2 - 4 = 1$ in $\mathbb{Q}(y)$, so the lifting criterion fails.

Example

For $q(y) = y^2 - 3$, the lift is $x^4 - x^2 + 1 = \Phi_{12}(x)$. Its splitting field is $\mathbb{Q}(\sqrt{3}, i)$, of degree four, and its group is $C_2 \times C_2$. Both radicands equal -1, so their square classes span only one dimension. The group is a proper subgroup of the order-eight wreath product.

References

[Cox2012]. Cyclotomic splitting fields and quadratic extensions; the two factorizations are checked directly. Source.

14.5 Discriminant

For monic q, the quadratic-factor decomposition gives

$$\mathrm{disc}(f) = q(2)q(-2)\,\mathrm{disc}(q)^2 = (-1)^m f(1)f(-1)\,\mathrm{disc}(q)^2.$$

Indeed, the factors $x^2 - y_i x + 1$ have discriminants $y_i^2 - 4$, and the resultant of the factors indexed by i, j is $(y_i - y_j)^2$. Multiplying the discriminants and the squares of these resultants proves the identity. It determines the parity of the action on all $2m$ roots, but does not determine the entire kernel.

The same construction applies to even polynomials $q(x^2)$, with pairs $\{\alpha, -\alpha\}$ and radicands y_i. Affine reflection $x \mapsto 1 - x$ becomes this situation after centering at $1/2$.

References

[AndersonBertelliODorney2026] / §2. Discriminants in the reciprocal setting. Source.

[CullinanHajir2014] / §1. The analogous signed action for even Legendre polynomials. Source.

Questions

When are the quadratic extensions independent?

Research direction / A specified reciprocal family over a field of characteristic different from 2.

After determining the Galois group of the trace polynomial, the remaining problem is to find relations among the radicands in its splitting field. These square-class relations determine the kernel of the action on inverse pairs.

References

[AndersonBertelliODorney2026] / §§2-3. Source.

Related topics

Applies to. Eulerian polynomials. Eulerian reciprocity produces inverse pairs after the forced factor is removed.
Applies to. Fekete polynomials. Character identities produce reciprocal normalizations and special-value tests.
Uses. Discriminants, parity, and ramification. The discriminant detects one parity character of the full action.
Generalizes. Hermite, Bessel, and Legendre polynomials. Even polynomials have an analogous action on sign pairs.

Part IV

Polynomial families

CHAPTER 15

Trinomials

family / Intermediate / Q / Published results

The polynomials x^n-x-1 have Galois group S_n over Q for every $n\geq 2$. Ramification proves this uniformly; power compositions require other group classifications.

Overview

- Which trinomial has group S_n in every degree $n\geq 2$?
- How can two primes certify the quintic case?
- Why does the all-degree proof use ramification?
- How can a degree-twelve trinomial be classified using lower-degree polynomials?

Prerequisites: Fields, irreducibility, and Galois actions

15.1 The family x^n-x-1

Theorem (Selmer; Nart-Vila)

For every integer $n \geq 2$, $f_n(x) = x^n - x - 1$ is irreducible over $\mathbb{Q}$, and its splitting field has Galois group S_n.

Selmer proved irreducibility in 1956. Nart and Vila first proved the all-degree group statement in 1979; Osada obtained it independently as a consequence of a broader trinomial criterion.

References

[Selmer1956]. Irreducibility of x^n−x−1. Source.

[NartVila1979]. The first all-degree symmetric-group result. Source.

[Osada1987] / Corollary 3. Independent proof through a general trinomial criterion. Source.

15.2 Prime-degree irreducibility

Let p be prime and let α satisfy $\alpha^p - \alpha - 1 = 0$ over $\overline{\mathbb{F}}_p$. Then $\alpha^{p^k} = \alpha + k$, where k is read in $\mathbb{F}_p$. The least positive k for which this equals α is p. Thus the Frobenius orbit has length p, and $x^p - x - 1$ is irreducible over $\mathbb{F}_p$. Reduction then proves irreducibility over $\mathbb{Q}$.

This Artin-Schreier argument uses the equality of the degree and the residue characteristic. It does not cover composite degrees.

References

[LidlNiederreiter1997]. Frobenius orbits and irreducible polynomials over finite fields; the orbit computation is explicit. Source.

15.3 Example

Example

For $x^5 - x - 1$, reduction modulo 5 gives a 5-cycle. Modulo 2,

$$x^5 + x + 1 = (x^2 + x + 1)(x^3 + x^2 + 1),$$

where the two factors are irreducible. The group therefore contains an element of type $(2, 3)$, whose cube is a transposition. Label the roots by $\mathbb{Z}/5\mathbb{Z}$ so that the 5-cycle adds one. Conjugating the transposition by its powers gives all edges with a fixed nonzero difference. These edges form a connected graph. Edge transpositions of a connected graph generate S_5, proving the claim.

The conclusion uses this connectivity argument. A transitive subgroup containing one transposition need not be symmetric in general.

References

[ConradModP]. Factorization patterns give Frobenius cycle types. Source.

[ConradSnAn]. Permutation-group criteria; the graph proof is included here. Source.

15.4 Ramification

For f_n, the simultaneous equations $f_n(a) = f_n'(a) = 0$ modulo a prime give $na^{n-1} = 1$ and $(n-1)a = -n$. They force a unique possible multiple root. Such a prime divides neither n nor $n-1$, and $f_n''(a) = n(n-1)a^{n-2} \neq 0$, so that root is double. Every nontrivial inertia element therefore exchanges just the two roots that coalesce.

The normal subgroup generated by the finite inertia groups fixes an extension of $\mathbb{Q}$ unramified at all finite primes. Minkowski's discriminant bound implies that this extension is trivial. Hence the whole Galois group is generated by transpositions. Selmer's irreducibility makes its action transitive, and the connected-graph argument now gives S_n.

References

[Osada1987] / Corollary 3 and its proof mechanism. Ramification and generation by transpositions. Source.

[Neukirch1999]. Minkowski's discriminant bound and the absence of nontrivial everywhere-finitely-unramified extensions of Q. Source.

15.5 General trinomials

For $x^n + ax^r + b$, a common divisor of the exponents may give a power composition. Cyclotomic factors and the nonreciprocal part require separate analysis. Common roots of the polynomial and its derivative determine whether a bad prime produces a single double root or a more complicated multiplicity pattern. The latter may give inertia elements other than transpositions. Selmer, Ljunggren, and Cohen-Movahhedi-Salinier study factorization and Galois groups in these broader families.

References

[Ljunggren1960]. Factorization phenomena for trinomials and quadrinomials. Source.

[CohenMovahhediSalinier1999]. Broader trinomial Galois-group analysis. Source.

15.6 Degree twelve

Suppose $f = x^{12} + ax^6 + b \in \mathbb{Q}[x]$ is irreducible. Chen classifies all 16 possible Galois groups. The proof uses the decompositions

$$f = g_4(x^3) = g_6(x^2), \qquad g_4 = x^4 + ax^2 + b, \quad g_6 = x^6 + ax^3 + b.$$

The groups of g_4, g_6, followed by at most two additional rational-square tests, distinguish the possibilities. Theorem 3.8 gives the complete algorithm, including tests for rational cubes and reducibility of $r = x^3 - 3bx + ab$. Irreducibility of f is an input hypothesis.

Example

For $f = x^{12} + 4x^6 + 2$, Eisenstein at 2 proves irreducibility. Here $b(a^2 - 4b) = 16$, $b = 2$ is not a rational cube, and $r = x^3 - 6x + 8$ has no rational root. The first branch of the algorithm gives $12T39$, the transitive-group catalogue label.

References

[Chen2026Dodecic] / Theorem 3.8; Table 2. Source.

15.7 Cubic power compositions

For $g = x^3 + ax^2 + bx + c \in \mathbb{Q}[x]$ and irreducible $f = g(x^3)$, Awtrey-Patane-Toone classify 11 possible transitive groups in degree nine. They use the discriminants of f and g, together with two further resolvents. Their work also addresses the converse problem: whether a given number field admits a defining polynomial of this compositional shape. These polynomials usually have four terms. Their relation to trinomials is the root partition induced by power composition.

References

[AwtreyPataneToone2025] / Abstract and Introduction. Source.

Questions

Which sparse families admit a uniform ramification argument?

Research direction / A specified family $x^n + a_n x^{r_n} + b_n$ over $\mathbb{Q}$.

The relevant data are the primes at which roots coincide and the resulting multiplicities. A uniform proof requires transitivity and sufficient inertia generators; modular cycle types computed in individual degrees do not suffice.

References

[Osada1987]. Source.
[CohenMovahhediSalinier1999]. Source.

Related topics

Uses. Frobenius and factorization modulo primes. Small good primes provide finite certificates in individual degrees.
Uses. Transitivity, primitivity, and cycle criteria. A transitive group generated by transpositions is symmetric.
Generalizes. Sparse polynomials and factor reversal. Sparse families may require separating reciprocal and nonreciprocal factors.
Supplies examples for. Inverse Galois theory and generic polynomials. The family realizes every symmetric group over Q.
Uses. Capelli's theorem and polynomial composition. The factorization of a power composition is controlled over a root field; irreducibility must precede the classification.

CHAPTER 16

Taylor polynomials

family / Intermediate / Q / Published results

Exponential truncations have alternating or symmetric Galois groups. Geometric truncations factor into cyclotomic polynomials. Their proofs use coefficient valuations, discriminants, and explicit factorization.

Overview

- What are the Galois groups of exponential truncations?
- How does the discriminant computation work?
- Why do geometric truncations behave differently?

Prerequisites: Fields, irreducibility, and Galois actions

16.1 Definition

For $F(x) = \sum_{j\geq 0} a_j x^j \in \mathbb{Q}[[x]]$, let $F_{\leq n} = \sum_{j=0}^{n} a_j x^j$. Its degree is n only when $a_n \neq 0$. Multiplying by a nonzero rational scalar clears denominators without changing the roots or splitting field. Analytic convergence and the limiting location of complex roots provide useful structure, but do not alone prove irreducibility over $\mathbb{Q}$.

Katz and Rivin ask about the frequencies of Galois groups in such sequences. Rabarison, Pazuki, and Molin give a later survey and extend the experimental viewpoint to Padé approximants. Truncations and rational approximants require distinct definitions and normalizations.

References

[KatzRivinSpeculations] / §§1-2. Limiting-frequency questions and explicitly labeled computational evidence. Source.

[RabarisonPazukiMolin2024]. Survey of truncations and experiments with Padé approximants. Source.

16.2 Schur's theorem

Theorem (Schur)

For every $n \geq 2$, $T_n(x) = \sum_{j=0}^{n} x^j/j!$ is irreducible over $\mathbb{Q}$, and

$$\mathrm{Gal}(T_n/\mathbb{Q}) = \begin{cases} A_n, & 4 \mid n, \\ S_n, & 4 \nmid n. \end{cases}$$

The proof excludes every proper factor degree, proves containment of A_n, and computes the discriminant square class. Coleman's Newton-polygon proof illustrates the use of factorial valuations.

References

[Schur1930]. Original all-degree group theorem. Historical citation verification.

[Coleman1987]. Newton polygons and the exact alternating/symmetric conclusion. Source.

16.3 Discriminant

Set $P_n = n!T_n$, which is monic. Since $P_n' = P_n - x^n$, each root r satisfies $P_n'(r) = -r^n$. The constant term is $n!$, so multiplying these identities over the roots gives

$$\operatorname{disc}(P_n) = (-1)^{n(n-1)/2}(n!)^n.$$

For even n, this is a rational square exactly when $4 \mid n$: its magnitude is already a square and its sign decides the issue. For odd $n \geq 3$, a prime between $n/2$ and n occurs exactly once in $n!$, so the discriminant is a nonsquare. This step determines the group only after A_n-containment has been proved.

References

[Coleman1987]. The discriminant square test; the resultant computation is included explicitly. Source.

16.4 Example

Example

$$P_6 = x^6 + 6x^5 + 30x^4 + 120x^3 + 360x^2 + 720x + 720.$$

At 2, its lower Newton polygon has vertices $(0,4), (4,1), (6,0)$; a proper rational factor can therefore have only degree 2 or 4. At 3, the polygon is the segment from $(0,2)$ to $(6,0)$, so a proper factor can have only degree 3. The allowed degrees are disjoint, so the polynomial is irreducible.

References

[Coleman1987]. The Newton-polygon mechanism; the displayed degree-six coefficients and vertices are checked directly. Source.

16.5 Geometric truncations

The degree-n truncation of $(1-x)^{-1}$ is

$$1 + x + \cdots + x^n = \prod_{\substack{d \mid n+1 \\ d>1}} \Phi_d(x).$$

For $n \geq 1$, it is irreducible exactly when $n+1$ is prime; its splitting field is $\mathbb{Q}(\zeta_{n+1})$, with group $(\mathbb{Z}/(n+1)\mathbb{Z})^\times$. The groups are therefore abelian.

In other truncation families, parity, vanishing coefficients, or rational-function identities may force factors. The remaining factors can be studied through uniform valuation patterns, specialization theorems, or permutation-group arguments. Numerical frequencies alone do not establish a limit.

References

[Cox2012]. Cyclotomic factorization and cyclotomic Galois groups. Source.

[KatzRivinSpeculations] / §§1-4. Frequency questions, variants, and shifts. Source.

Questions

When does a truncation family have a limiting Galois-group frequency?

Research direction / A specified rational power series, a stated normalization, and n tending to infinity.

The exponential and geometric families have different Galois-group distributions. Katz and Rivin ask about limiting frequencies in more general truncation families. Such a distribution depends on the factors removed and the permutation action chosen.

References

[KatzRivinSpeculations] / §§1-4. Source.

Related topics

Uses. Newton polygons and Galois groups. Factorial valuations provide incompatible local factor degrees.

Uses. Discriminants, parity, and ramification. A resultant identity determines the sign character after large-group containment.

Generalizes. Generalized Laguerre polynomials. Exponential truncations occur as a varying-parameter Laguerre specialization.

Contrasts with. Specialization and Hilbert irreducibility. A family indexed by degree is different from specialization of one fixed-degree cover.

CHAPTER 17

Generalized Laguerre polynomials

family / Intermediate / Q / Published results

Arithmetic-progression products determine coefficient valuations and constrain Galois groups. The main theorems distinguish a fixed rational parameter from a fixed degree.

Overview

- What changes when the parameter is fixed instead of the degree?
- Why do primes in arithmetic progressions appear?
- Which negative integer parameters force reducibility?
- Which small rational parameters admit an all-degree irreducibility theorem?

Prerequisites: Fields, irreducibility, and Galois actions

17.1 Definition and normalization

For $n \geq 1$ and $\alpha \in \mathbb{Q}$, set

$$L_n^{(\alpha)}(x) = \sum_{j=0}^{n} \binom{n+\alpha}{n-j} \frac{(-x)^j}{j!}.$$

The generalized binomial coefficient is a polynomial in α. For real $\alpha > -1$, these polynomials form a classical orthogonal family; the arithmetic question also makes sense outside that range. Their monic normalization is $M_n^{(\alpha)} = (-1)^n n! L_n^{(\alpha)}$.

If $\alpha = a/b$ in lowest terms, $b > 0$, then an integral monic normalization with the same splitting field is

$$b^n n! L_n^{(a/b)}(-x/b) = \sum_{j=0}^{n} \binom{n}{j} \Big(\prod_{k=j+1}^{n} (bk + a) \Big) x^j.$$

The product permits valuation estimates using an arithmetic progression.

References

[Hajir2005] / §§1 and 3. Definition and the integral coefficient-product normalization. Source.

17.2 Fixed parameter

> **Theorem (Filaseta-Lam; Hajir)**
> Fix $\alpha \in \mathbb{Q} \setminus \mathbb{Z}_{<0}$. For every sufficiently large n, depending on α, the polynomial $L_n^{(\alpha)}$ is irreducible over $\mathbb{Q}$ and its group contains A_n. It equals A_n or S_n according to whether its discriminant is a rational square.

Filaseta-Lam supply eventual irreducibility. Hajir obtains a large prime divisor of a Newton-slope denominator, then uses a permutation-group criterion. The irreducibility input is essential to that group argument.

References

[FilasetaLam2002]. Eventual irreducibility for a fixed rational parameter outside the negative integers. Source.

[Hajir2005] / Theorems 1.1 and 2.2. Eventual alternating containment and the Newton-index criterion. Source.

17.3 Small rational parameters

Theorem (Laishram-Shorey)

Let $q \in \{\pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{1}{4}, \pm\frac{3}{4}\}$. For every $n \geq 1$, $L_n^{(q)}(x)$ is irreducible over $\mathbb{Q}$, except when $(q, n) = (\frac{1}{4}, 2)$.

The proof combines Newton polygons with large prime divisors of arithmetic-progression products.

Example

$$L_2^{(1/4)}(x) = \frac{(4x-3)(4x-15)}{32}.$$

The exception is explicit in arXiv:1306.0736v2. The power-substitution statement in the cited theorem requires a low-degree qualification: although $L_1^{(-3/4)}(x) = 1/4 - x$ is irreducible, $L_1^{(-3/4)}(x^4) = (1/2 - x^2)(1/2 + x^2)$.

References

[LaishramShorey2016III] / Theorem 1 and §8; arXiv:1306.0736v2. The ordinary Laguerre assertion only is used; see the explicit boundary example for power substitutions. Source.

17.4 Fixed degree

Fix a number field K and a degree $n \geq 10$. Then all but finitely many $\alpha \in K$ give an irreducible polynomial $L_n^{(\alpha)}$ with Galois group S_n over K. Hajir-Wong prove this using intermediate covers and finiteness of rational points on curves of sufficiently high genus.

This does not assert a common threshold in n for every rational α; nor does ordinary Hilbert irreducibility by itself make the exceptional set finite. A varying sequence α_n requires its own argument.

References

[HajirWong2006] / Theorem 1(a). The finite exceptional set for fixed n≥10 and fixed number field K. Source.

17.5 Discriminant

Schur's product formula for the monic normalization is

$$\operatorname{disc}(M_n^{(\alpha)}) = \prod_{j=2}^{n} j^j (\alpha + j)^{j-1}.$$

Example (α=0, n=3)

Here $M_3 = x^3 - 9x^2 + 18x - 6$. Eisenstein at 3 proves irreducibility. Its discriminant is $2^3 3^5 = 1944$, a nonsquare. An irreducible cubic has group A_3 or S_3; therefore this group is S_3.

The product also determines the parameters with repeated roots: for degree $n \geq 2$, the exceptional values are $-2, -3, \ldots, -n$. Separability and irreducibility are separate properties; at $\alpha = -1$ there is already a simple rational root at zero.

References

[Schur1931]. Classical algebraic properties and discriminants. Source.

[Hajir2005] / §1. Schur's discriminant formula for the monic normalization. Source.

17.6 Negative integer parameters

Remark

For integers $m \geq 1$ and $n \geq m$, coefficient comparison gives

$$L_n^{(-m)}(x) = (-x)^m \frac{(n-m)!}{n!} L_{n-m}^{(m)}(x).$$

For example, $L_3^{(-2)}(x) = x^2(3-x)/6$. Thus a fixed negative integer parameter supplies a forced zero for every sufficiently large degree. This explains the negative-integer exclusion in the fixed-parameter theorem.

The varying parameter $\alpha = -n-1$ gives $L_n^{(-n-1)} = (-1)^n T_n$, the exponential truncation. Here the parameter depends on the degree, so the fixed-parameter obstruction does not apply.

References

[Hajir2005] / §1, Remarks 1 and 3. The negative-integer exclusion and the exponential specialization. Source.

17.7 Alternating groups

For every $n > 2$ with $n \equiv 2 \pmod 4$, the polynomial $L_n^{(n)}$ is irreducible and has group A_n. Together with Schur's other Laguerre specializations, this realizes every A_n in this family. At $n = 2$, $L_2^{(2)} = (x-2)(x-6)/2$ is split and its group is the trivial group A_2; irreducibility is not claimed there.

For integer parameters, Banerjee-Filaseta-Finch-Leidy classify nonzero square discriminants. Banerjee-Bera subsequently study the actual alternating Galois groups for integer $\alpha \neq n$, up to finitely many exceptions. The square-discriminant classification and the group classification answer different questions.

References

[FilasetaKiddTrifonov2012] / Theorem 1 and Corollary 1. Source.

[BanerjeeFilasetaFinchLeidy2013]. Classification of nonzero square discriminants. Source.

[BanerjeeBera2022] / Abstract and Introduction. The $\alpha \neq n$ restriction is essential to the stated classification scope. Source.

Questions

Which explicit varying-parameter sequences have irreducible Laguerre polynomials?

Research direction / $L_n^{(\alpha_n)}$ over Q, with the sequence α_n specified in advance.

Eventual irreducibility for each fixed parameter does not imply irreducibility when the parameter changes with the degree. A sequence of parameters requires uniform control of factor degrees or a separate argument using its coefficient products.

References

[Hajir2009]. Source.
[FilasetaKiddTrifonov2012]. Source.

Are there only finitely many exceptional parameters in fixed degree $n \geq 10$?

Settled in stated scope / Any fixed number field K and fixed $n \geq 10$.

Yes: the parameters in K for which the polynomial is reducible or has Galois group different from S_n form a finite set. The theorem does not automatically provide an effective list of those parameters.

References

[HajirWong2006] / Theorem 1(a) and Remark 1. Source.

Related topics

Uses. Newton polygons and Galois groups. Arithmetic-progression products determine coefficient valuations and Newton slopes.

Uses. Specialization and Hilbert irreducibility. Fixed-degree parameter families are studied through covers and exceptional specializations.

Supplies examples for. Taylor polynomials. The parameter $-n-1$ gives the exponential truncation.

Generalizes. Hermite, Bessel, and Legendre polynomials. Hermite and Bessel identities connect to Laguerre parameters and substitutions.

Connects to. Capelli's theorem and polynomial composition. Power substitutions can create factors even when the original Laguerre polynomial is irreducible.

CHAPTER 18

Hermite, Bessel, and Legendre polynomials

family / Intermediate / Q / Research overview

Parity and coefficient products constrain irreducibility and Galois groups in these classical families. The results depend on the polynomial normalization and parameter range.

Overview

- Why must odd Hermite and Legendre polynomials be normalized?
- Which all-degree result is known for Bessel polynomials?
- Which parts of a local proof can transfer to another classical family?

Prerequisites: Fields, irreducibility, and Galois actions

18.1 Arithmetic properties

Orthogonal polynomials occur in approximation, differential equations, and spectral problems. Their recurrence and differential identities often give exact coefficients and simple real roots. Neither property alone forces irreducibility over $\mathbb{Q}$: a polynomial can have distinct real algebraic roots and still factor rationally.

Schur's work on Laguerre and Hermite polynomials established local arithmetic methods for such families. The known results differ by family: Bessel irreducibility has an all-degree theorem, whereas the cited Legendre paper treats the general factorization statement as a conjecture.

References

[Schur1931]. Foundational algebraic study of Laguerre and Hermite families. Source.

[FilasetaTrifonov2002]. All-degree Bessel irreducibility. Source.

[CullinanHajir2014] / Introduction. Stieltjes' conjecture and the remaining Legendre problem in this source. Source.

18.2 Hermite polynomials

For the physicists' Hermite convention $H_n(x) = (-1)^n e^{x^2}(d/dx)^n e^{-x^2}$, one has

$$H_{2m}(x) = (-1)^m 2^{2m} m! L_m^{(-1/2)}(x^2),$$

$$H_{2m+1}(x) = (-1)^m 2^{2m+1} m! x L_m^{(1/2)}(x^2).$$

The odd polynomial has a forced factor x; all remaining roots occur in opposite pairs. A theorem concerning $L_m^{(\pm 1/2)}$ alone does not automatically determine the quadratic extensions obtained when square roots of its roots are adjoined.

Example

Here $H_4 = 16x^4 - 48x^2 + 12$. Its pair polynomial is $4y^2 - 12y + 3$, with roots $(3 \pm \sqrt{6})/2$. The pair field has degree two; the full splitting field also requires square roots of these roots. The same construction applies in higher degree.

Theorem (Schur)

For every $m \geq 1$, the physicists' polynomials $H_{2m}(x)$ and $H_{2m+1}(x)/x$ are irreducible over $\mathbb{Q}$. This uses the normalization above. Rescaling the variable by an irrational number need not preserve irreducibility over $\mathbb{Q}$: for example, $H_2 = 4x^2 - 2$ is irreducible, whereas the probabilists' polynomial $\mathrm{He}_2 = x^2 - 1$ is reducible.

References

[CullinanHajirSell2009] / §1. The two Hermite-Laguerre identities. Source.

[Schur1931]. The classical arithmetic context. Source.

[LaishramShorey2016III] / Introduction. Restates Schur's theorem for L_n^(±1/2)(x^2) and the physicists' Hermite normalization. Source.

18.3 Hermite Wronskians

Define the monic probabilists' polynomials by $e^{xt - t^2/2} = \sum_{n \geq 0} \mathrm{He}_n(x) t^n / n!$. For a partition $\lambda = (\lambda_1 \geq \cdots \geq \lambda_r > 0)$, put $n_i = \lambda_{r-i+1} + i - 1$ and

$$W_\lambda(x) = \frac{\mathrm{Wr}(\mathrm{He}_{n_1}, \ldots, \mathrm{He}_{n_r})}{\prod_{i<j}(n_j - n_i)}.$$

Write $W_\lambda = x^{e_\lambda} R_\lambda$, where e_λ is the multiplicity at zero. Grosu-Grosu prove that $R_{(n,1)}$ is irreducible for $n = 1$ or $n \geq 3$, $R_{(n,2)}$ for $n \geq 2$, and $R_{(n,n)}$ when $n + 1$ is a square.

At $n = 2$, $W_{(2,1)} = x^3$, so $R_{(2,1)} = 1$, a unit, hence not irreducible. Their methods combine Newton-slope bounds with Schur-type congruences. Irreducibility implies simple nonzero roots, connecting arithmetic to differential equations. Irreducibility does not hold for every partition: $R_{(6,3,2,1)} = x^2 - 9$. This concerns the displayed probabilists' normalization.

References

[GrosuGrosu2021] / Definition 1, Theorems 1-3 and 7-8, and the discussion after Proposition 6 (arXiv:2007.00065v1). Source.

18.4 Bessel polynomials

Theorem (Filaseta-Trifonov)

Let $n \geq 1$, and let $a_0, \dots, a_n \in \mathbb{Z}$ satisfy $|a_0| = |a_n| = 1$. Then

$$\sum_{j=0}^{n} a_j \frac{(n+j)!}{2^j (n-j)! j!} x^j$$

is irreducible over $\mathbb{Q}$. Taking every $a_j = 1$ proves the irreducibility of the Bessel polynomial y_n in every positive degree.

The method bounds possible factor degrees by Newton polygons and prime divisors of factorial products. The valuation bounds allow the stated integral multipliers on the intermediate coefficients.

Example

The cubic Bessel polynomial is $15x^3 + 15x^2 + 6x + 1$. Modulo 2 it is $x^3 + x^2 + 1$, which has no root and is therefore irreducible. Its discriminant is -135, a nonsquare, so its Galois group is S_3.

References

[FilasetaTrifonov2002] / Main theorem and §2. All-degree coefficient-perturbation theorem and the Newton-polygon mechanism. Source.

18.5 Legendre polynomials

Define $P_n(x) = \frac{1}{2^n n!}(d/dx)^n (x^2 - 1)^n$. Since $P_n(-x) = (-1)^n P_n(x)$, the factor x is forced at odd indices. Stieltjes' conjecture, as presented by Cullinan-Hajir, asks whether P_{2m} and P_{2m+1}/x are irreducible over $\mathbb{Q}$ for every $m \geq 1$. Their Galois-group questions concern these even polynomials and the corresponding signed-permutation action.

Example

After scaling, $8P_4 = 35x^4 - 30x^2 + 3$, which is Eisenstein at 3. Its pair polynomial $35y^2 - 30y + 3$ has discriminant 480. Thus its pair field is $\mathbb{Q}(\sqrt{30})$. This calculation proves irreducibility in degree four only.

References

[CullinanHajir2014] / §1. Stieltjes' irreducibility conjecture and parity-reduced Galois groups. Source.

18.6 A claimed proof of Stieltjes' conjecture

Preprint claim

Zikang Deng's September 2026 preprint claims a proof of Stieltjes' irreducibility conjecture in every degree. Version 2 uses local and real estimates for a resultant. The complete proof has not been independently checked here, and no peer-reviewed publication has been verified. The claim is distinct from the published 2014 account.

References

[Deng2026Legendre] / Theorem 1.1, arXiv:2609.22336v2 (27 September 2026). A preprint claims a complete resolution; this atlas has not independently verified the full proof. Source.

18.7 Methods and parameter ranges

The main arithmetic tools are explicit coefficient products, discriminant formulas, Newton slopes, and the action on root pairs. Their application depends on the polynomial convention, scalar normalization, affine change of variable, removed factors, and base field. Differential equations alone do not determine these arithmetic properties.

For Jacobi families with a parameter, the results distinguish a fixed parameter with degree growing from a fixed degree with parameter varying. Those regimes lead to different arithmetic and geometric methods.

References

[CullinanHajirSell2009]. A fixed-degree one-parameter Jacobi family. Source.

[HajirWong2006]. Geometric specialization methods and the distinction between parameter regimes. Source.

Questions

Are P_{2m} and P_{2m+1}/x always irreducible over $\mathbb{Q}$?

Resolution claimed in preprint / The standard Legendre polynomials and all integers m≥1.

This is Stieltjes' conjecture in the cited 2014 paper. Deng's September 2026 preprint claims a complete proof. The proof has not been independently verified here, so the result is recorded as a preprint claim.

References

[CullinanHajir2014] / §1. Source.

[Deng2026Legendre] / Theorem 1.1, arXiv:2609.22336v2 (27 September 2026). A preprint claims a complete resolution; this atlas has not independently verified the full proof. Source.

Are the Bessel polynomials irreducible in every positive degree?

Settled in stated scope / $y_n(x) = \sum_{j=0}^{n} (n+j)!x^j/(2^j(n-j)!j!)$ over Q, n≥1.

Yes. The Filaseta-Trifonov theorem proves this and permits arbitrary integral intermediate coefficient multipliers with endpoint multipliers ± 1.

References

[FilasetaTrifonov2002] / Main theorem. Source.

Related topics

Uses. Generalized Laguerre polynomials. Hermite identities reduce the pair polynomials to Laguerre parameters $\pm 1/2$.

Uses. Newton polygons and Galois groups. Factorial products constrain factor degrees.

Generalizes. Reciprocal polynomials. The actions on sign pairs and inverse pairs lie in analogous wreath products.

Applies to. Specialization and Hilbert irreducibility. Parameterized Jacobi families connect to fixed-degree specialization theorems.

CHAPTER 19

Eulerian polynomials

family / Intermediate / Q / Published results

Eulerian polynomials are reciprocal after removal of a forced linear factor. Shifted Eisenstein and Newton-polygon arguments prove irreducibility at prime and prime-plus-one indices and bound the degree of a largest irreducible factor.

Overview

- Why is x+1 removed at even indices?
- Why does an Eisenstein shift work at prime indices?
- Why is the natural group a wreath product rather than S_(n−1)?

Prerequisites: Fields, irreducibility, and Galois actions

19.1 Definition and normalization

For $n \geq 1$, define $A_n(x) = \sum_{\sigma \in S_n} x^{\mathrm{des}(\sigma)}$, where a descent is an index i with $\sigma(i) > \sigma(i+1)$. Their generating function is

$$\sum_{k \geq 1} k^n x^{k-1} = \frac{A_n(x)}{(1-x)^{n+1}}.$$

Complementing the values of a permutation gives $x^{n-1} A_n(x^{-1}) = A_n(x)$. Set $\widetilde{A}_n = A_n/(x+1)$ for even n, and $\widetilde{A}_n = A_n$ for odd n. This notation distinguishes the normalized Eulerian polynomial from the Appell Euler polynomial E_n. For $n \geq 3$, $\widetilde{A}_n$ is monic, reciprocal, and has positive even degree.

References

[FoataSchutzenberger1970] / Chapters II and IV. Permutation statistics and generating functions. Authorized digital retypesetting.

19.2 Roots and reciprocity

The recurrence

$$A_n = (1 + (n-1)x) A_{n-1} + x(1-x) A'_{n-1}, \qquad A_1 = 1,$$

gives simple negative roots by induction. At each negative root of A_{n-1}, the factor $x(1-x)$ reverses the sign of its derivative. Together with the signs at 0 and at negative infinity, this places one root of A_n in each intervening interval and one on each outside interval. Reciprocity then pairs roots by inversion. Since $A_n(1) = n!$, the only possible fixed point of inversion is -1. It occurs exactly when n is even, and simplicity makes its multiplicity one.

References

[FoataSchutzenberger1970] / Chapters II, IV, and V. Eulerian identities and the value at -1; the interlacing argument is explained here. Authorized digital retypesetting.

19.3 Limiting zero distribution

Let $\rho_{n,1}, \dots, \rho_{n,n-1} < 0$ be the roots of A_n. Then

$$\frac{1}{n-1}\sum_{j=1}^{n-1}\delta_{-\rho_{n,j}} \implies \frac{\mathbf{1}_{t>0}}{t(\pi^2+\log^2 t)}\,dt.$$

Melotti gives a combinatorial proof of this log-Cauchy limit, using exact stabilized moments after the transformation $u = (1-\rho)^{-1}$. His convention uses $xA_n(x)$, including a zero root; deleting that one atom and changing the normalization leaves the weak limit unchanged. The limiting distribution does not determine the Galois orbits or settle the irreducibility conjecture.

References

[Melotti2025Eulerian] / Theorems 1-2; arXiv:2507.15908v2. The limiting law is older; the paper supplies a new proof via stabilized moments. Source.

19.4 Prime indices

Theorem

For every odd prime p, A_p is irreducible over $\mathbb{Q}$.

Fermat's congruence in the generating identity gives

$$\frac{A_p(x)}{(1-x)^{p+1}} \equiv \sum_{k\geq 1} kx^{k-1} = \frac{1}{(1-x)^2} \pmod p.$$

Therefore $A_p(1+t) \equiv t^{p-1} \pmod p$. The shifted polynomial is monic, while its constant term is $A_p(1) = p!$, divisible by p but not p^2. Eisenstein applies. This is the $r = 1$ instance of Sharma-Straus, Theorem 6.

References

[SharmaStraus1974] / Theorem 6, r=1. The classical prime-index case within the Euler-Frobenius family. Source.

19.5 A large irreducible factor

Theorem

For $n \geq 3$, let p be the largest prime at most n. The normalized polynomial $\widetilde{A}_n$ has an irreducible factor over $\mathbb{Q}$ of degree at least $p-1$. In particular, $\widetilde{A}_p$ and $\widetilde{A}_{p+1}$ are irreducible for every odd prime p.

Put $r = n-p+1$. Bertrand's postulate gives $1 \leq r < p$ and $n < 2p$. Fermat's congruence and the generating identity imply

$$A_n(1+t) \equiv t^{p-1}A_r(1+t) \pmod p.$$

Since $A_r(1) = r!$ is a p-adic unit, the coefficient of t^{p-1} is a unit and every preceding coefficient is divisible by p. The constant term $n!$ has valuation one. The first Newton edge is therefore from $(0,1)$ to $(p-1,0)$. It gives an irreducible factor of degree $p-1$ over $\mathbb{Q}_p$. Some rational

irreducible factor must contain it, hence has at least this degree. At even n, removing $x+1$ removes only a local linear factor with slope zero, because p is odd.

This bound is associated with Heidrich's work. It does not exclude an additional smaller factor at every composite index.

References

[Heidrich1984Eulerian]. The large-factor result; original full text was not accessible in this review, so no theorem number is asserted. Source.

[FoataSchutzenberger1970] / Chapter IV. Generating identity used in the displayed proof. Authorized digital retypesetting.

19.6 Example

Example

$$A_5 = x^4 + 26x^3 + 66x^2 + 26x + 1 = x^2 q(x + x^{-1}), \quad q(y) = y^2 + 26y + 64.$$

The splitting field of q is $M = \mathbb{Q}(\sqrt{105})$. If its roots are y_1, y_2, then

$$(y_1^2 - 4)(y_2^2 - 4) = q(2)q(-2) = 1920 = 64 \cdot 30.$$

Each radicand has norm 1920, a nonsquare in $\mathbb{Q}$, so neither is a square in M. Their product is also a nonsquare in M: a rational number is a square in $\mathbb{Q}(\sqrt{105})$ only if it or its quotient by 105 is a rational square; neither condition holds for 1920. The two classes are independent, so the splitting field has degree $2 \cdot 4 = 8$. Thus $\mathrm{Gal}(A_5/\mathbb{Q}) = C_2 \wr S_2$, the dihedral group of order eight.

References

[AndersonBertelliODorney2026] / §2. The reciprocal group framework; this index-five calculation is worked out directly. Source.

19.7 General indices

The prime and prime-plus-one cases prove transitivity on individual roots. They do not determine the group on inverse pairs or the quadratic kernel. For arbitrary indices, the largest-prime bound supplies a large factor, while the 1974 source presents broader irreducibility conjectures for its Euler-Frobenius family. Finite computations do not settle those conjectures.

The relevant arithmetic data are $\widetilde{A}_n$, its trace polynomial, good-prime factorization patterns, discriminant square classes, and exact certificates.

References

[SharmaStraus1974] / p. 266 and Theorem 6. Conjectural scope in the original paper and one proven range. Source.

Questions

Is the normalized Eulerian polynomial irreducible for every index n≥3?

Conjecture in cited source / The classical r=1 Euler-Frobenius specialization over Q; remove x+1 at even indices.

The prime and prime-plus-one cases and the general large-factor bound are proved above. The all-index statement is a conjecture in the cited 1974 source; this status does not exclude subsequent advances.

References

[SharmaStraus1974] / p. 266 and Theorem 6. Source.
[Heidrich1984Eulerian]. General large-factor results. Source.

For which indices is the group the full signed-permutation group?

Research direction / The separable reciprocal polynomial $\widetilde{A}_n$, of degree 2m.

The group is determined by the action on the degree-m trace polynomial and the rank of the square classes of the quadratic radicands. The index-five example computes both; this computation does not give a uniform argument.

References

[AndersonBertelliODorney2026] / §2. Source.

Related topics

Uses. Schönemann-Eisenstein and Dumas criteria. After the shift x↦1+x, a prime divides every nonleading coefficient.
Uses. Reciprocal polynomials. The trace polynomial separates the pair action from its quadratic kernel.
Contrasts with. Bernoulli and Euler polynomials. Appell Euler polynomials use a different generating function and have different forced factors.
Uses. Frobenius and factorization modulo primes. Good-prime factorizations give further cycle constraints.

CHAPTER 20

Bernoulli and Euler polynomials

family / Intermediate / Q / Published results

Reflection symmetry constrains the Galois action. Bernoulli denominators yield infinite irreducible subfamilies; multiple roots are classified in both families. Even Bernoulli groups contain fixed-point-free elements without an irreducibility assumption.

Overview

- How do Euler, Bernoulli, and Eulerian polynomials differ?
- Which rational factors are forced by the generating identities?
- Why must repeated factors be checked before describing a root action?
- Which infinite Bernoulli subfamilies are known to be irreducible?
- Is the multiple-root problem settled?
- What can one prove about even Bernoulli Galois groups without assuming irreducibility?

Prerequisites: Fields, irreducibility, and Galois actions

20.1 Definitions

Define the standard Appell families by

$$\frac{te^{xt}}{e^t-1}=\sum_{n\ge 0}B_n(x)\frac{t^n}{n!},\qquad \frac{2e^{xt}}{e^t+1}=\sum_{n\ge 0}E_n(x)\frac{t^n}{n!}.$$

The Bernoulli difference identity $B_n(x+1)-B_n(x)=nx^{n-1}$ explains their role in sums of powers. The Euler identity $E_n(x+1)+E_n(x)=2x^n$ is the corresponding alternating relation. Both families satisfy the Appell derivative rule $P_n'=nP_{n-1}$. The derivative identity relates multiple roots to common roots of adjacent polynomials.

The notation E_n here always denotes the Euler polynomial defined above, not a normalized Eulerian descent polynomial.

References

[DLMFBernoulliEuler] / §§24.2 and 24.4. Generating functions, difference equations, and derivative identities. Source.

20.2 Reflection symmetry

Identity

$$B_n(1-x) = (-1)^n B_n(x), \qquad E_n(1-x) = (-1)^n E_n(x).$$

Replacing x by $z+1/2$ makes each polynomial even or odd according to n. Hence for odd n, $x-1/2$ is forced. After removing that factor, the polynomial is a polynomial in $(x-1/2)^2$. Its nonfixed roots come in pairs $\{\alpha, 1-\alpha\}$. For a separable remaining degree $2m$, the Galois action embeds in $C_2 \wr S_m$, just as for sign pairs of an even polynomial.

The irreducibility problem and the permutation action depend on which forced and repeated factors have been removed.

References

[DLMFBernoulliEuler] / §24.4(ii). Reflection identities; the group restriction follows from the displayed pairing. Source.

[Cox2012]. Galois actions preserving a root partition. Source.

20.3 Rational factors

For odd $n \geq 3$, the Bernoulli values at 0 and 1 vanish, so $x(x-1)(x-1/2)$ divides $B_n(x)$. For even $n \geq 2$, the Euler polynomial vanishes at 0 and 1, so $x(x-1)$ divides $E_n(x)$. These statements follow directly from the generating functions and reflection. Their removal need not leave an irreducible polynomial.

Carlitz and Brillhart are classical arithmetic sources. Their results concern these Appell polynomials, which differ from the Euler-Frobenius and Eulerian families.

References

[DLMFBernoulliEuler] / §§24.2 and 24.4. Special values and reflection. Source.

[Carlitz1952BernoulliEuler]. Classical irreducibility investigations in the Appell families. Source.

[Brillhart1969]. Factorization and roots of Euler and Bernoulli polynomials. Source.

20.4 Carlitz's irreducibility criterion

Theorem (Carlitz)

Let p be prime and $n = k(p-1)p^r$ be even, with $1 \leq k < p$ and $r \geq 0$. Then $B_n(x)$ is irreducible over $\mathbb{Q}$. For $p = 2$, the evenness condition means $n = 2^r$ with $r \geq 1$.

Proof sketch

Write $B_j = B_j(0)$. Von Staudt-Clausen says that for even $j \geq 2$, the denominator of B_j contains p exactly when $p-1 \mid j$, and then only to the first power. The base-p digits of $n = (p-k)p^r + (k-1)p^{r+1}$ have sum $p-1$. If $0 < j < n$ and $p-1 \mid j$, Lucas's congruence gives $p \mid \binom{n}{j}$: otherwise each digit of j would be bounded by the corresponding digit of n, forcing its positive digit sum, a multiple of $p-1$, to equal $p-1$, and hence $j = n$.

Consequently every nonconstant coefficient of $pB_n(x)$ is divisible by p in $\mathbb{Z}_p$, whereas its constant coefficient pB_n is a unit. Its leading coefficient is exactly p. The exceptional Bernoulli

number $B_1 = -1/2$ causes no problem: for $p = 2$, its coefficient is $-n$, which is even. Thus $px^n B_n(1/x)$ is Eisenstein over $\mathbb{Z}_p$. Taking reciprocals preserves irreducibility.

Example (n=4, p=5)

$$30B_4(x) = 30x^4 - 60x^3 + 30x^2 - 1.$$

Its reciprocal $-x^4 + 30x^2 - 60x + 30$ is Eisenstein at 5. Hence the polynomial is irreducible over $\mathbb{Q}$.

References

[Carlitz1952BernoulliEuler]. Original irreducibility source. The stated scope is explicitly restated in Kimura's report; no unchecked original theorem number is claimed. Source.

[Kimura2003Bernoulli] / §2, p. 3, theorem attributed to Carlitz (1952). The report uses the reversed coefficient convention for a p-Eisenstein polynomial; the reciprocal formulation above uses the usual convention. Source.

[DLMFBernoulliEuler] / §24.10(i), equation (24.10.1). Bernoulli denominators; the displayed proof supplies the digit argument explicitly. Source.

20.5 Examples

Example

$$B_5(x) = x(x-1)(x-\tfrac{1}{2})(x^2 - x - \tfrac{1}{3}).$$

After removing the three rational factors, the quadratic has discriminant $7/3$, so its splitting field is $\mathbb{Q}(\sqrt{21})$.

Example

$$E_5(x) = (x-\tfrac{1}{2})(x^2 - x - 1)^2.$$

The residual polynomial is a square. Its distinct roots generate only $\mathbb{Q}(\sqrt{5})$. Thus removing the rational factor alone need not leave an irreducible or separable polynomial. The splitting field and its Galois action are defined using the distinct roots.

References

[Brillhart1969]. The Euler/Bernoulli factorization literature; the displayed identities are verified by expansion. Source.

[DLMFBernoulliEuler] / §24.12(iv). The repeated-factor example E_5. Source.

20.6 Multiple roots

Theorem (Brillhart; Dilcher)

Every $B_n(x)$, $n \geq 1$, is squarefree. Among the Euler polynomials, only $E_5(x)$ has multiple roots; its two irrational roots each have multiplicity two.

Brillhart proved the odd Bernoulli case and restricted any possible even-degree multiple root to a root of $x^2 - x - b$, with b a positive odd integer. Dilcher's Theorem 2 eliminates these remaining possibilities. The method combines centered expansions, Bernoulli denominator congruences, and the requirement that a multiple root also annihilate the adjacent Appell polynomial. Brillhart's Euler classification is also recorded in Dilcher's §4.

The classification of multiple roots is distinct from irreducibility of the normalized factors and determination of their Galois groups.

References

[Dilcher2008MultipleZeros] / Theorem 2, p. 150; proof in §3; Euler discussion in §4. Complete Bernoulli squarefreeness theorem and the earlier Euler classification. Source.

[Brillhart1969]. Original Euler classification and earlier Bernoulli restrictions. Source.

[DLMFBernoulliEuler] / §24.12(iv). Authoritative concise statement of both complete multiplicity classifications. Source.

20.7 Fixed-point-free automorphisms

Theorem (Patel-Siksek)

For every even $n \geq 2$, $\mathrm{Gal}(B_n/\mathbb{Q})$ contains an element fixing none of the roots. Also $B_n(x)$ has no root in $\mathbb{Q}_2$.

The proof uses the 2-adic Newton polygon, the pairing $\alpha \leftrightarrow 1 - \alpha$, and a decomposition-group argument. Chebotarev then supplies a positive density of primes at which B_n has no root. This yields obstructions to equations involving sums of consecutive powers. A permutation without fixed points does not by itself imply that the entire Galois action is transitive.

References

[PatelSiksek2017Powers] / Proposition 3.1, Theorem 4, and §§3.2, 5. The group and local-root conclusions are unconditional, independent of the conjectured irreducibility of every even Bernoulli polynomial. Source.

20.8 Normalized polynomials

For a fixed parity and choice of removed rational factors, one may study irreducibility of the remaining polynomial in x, irreducibility of its polynomial in $(x - 1/2)^2$, or the Galois action on pairs. These problems are distinct. Taking a squarefree part changes only the exceptional Euler member E_5 in the standard families discussed here.

The Appell identity $P_n' = nP_{n-1}$ turns a multiple-root question into a common-root question for adjacent members. The classification above resolves this question for Bernoulli and Euler polynomials. The same reduction to adjacent polynomials is available for other Appell families.

References

[Dilcher2008MultipleZeros] / Theorem 2 and §4. Source.

[DLMFBernoulliEuler] / §§24.4(vii), 24.12(iv). Source.

Questions

Is every even Bernoulli polynomial irreducible over Q?

Conjecture in cited source / The standard Bernoulli polynomial B_n(x), for every even n≥2; explicitly described as a conjecture by Patel-Siksek (2017), §3.

Carlitz's criterion proves irreducibility for infinite sets of indices. The unconditional derangement theorem does not settle the general question. The statement is a conjecture in the cited paper; this status does not exclude later advances.

References

[PatelSiksek2017Powers] / §3, immediately after Proposition 3.1. Source.
[Carlitz1952BernoulliEuler]. Source.

What are the Galois groups of the normalized polynomials?

Research direction / A stated parity class in the Bernoulli or Appell Euler family, with an explicit squarefree normalization.

The degree-five Euler example shows that the choice of normalization matters. The group on pairs and the kernel of the corresponding quadratic extensions are separate parts of the problem.

References

[Carlitz1952BernoulliEuler]. Source.
[Brillhart1969]. Source.

Which Bernoulli or Euler polynomials have repeated roots?

Settled in stated scope / The standard Appell families B_n and E_n over characteristic zero, n≥1.

None of the Bernoulli polynomials does. The unique Euler exception is E_5, with factorization displayed above.

References

[Dilcher2008MultipleZeros] / Theorem 2 and §4. Source.
[Brillhart1969]. Source.
[DLMFBernoulliEuler] / §24.12(iv). Source.

Related topics

Contrasts with. Eulerian polynomials. The similarly named descent polynomials have a different definition and symmetry.
Generalizes. Reciprocal polynomials. Reflection becomes sign pairing after an affine change of variable.
Uses. Discriminants, parity, and ramification. Repeated factors make the discriminant zero; the squarefree part has the same splitting field.
Supplies examples for. Composition, iteration, and arboreal groups. Centering gives a quadratic composition that constrains the Galois action.
Uses. Newton polygons and Galois groups. Denominator valuations and slopes yield local irreducible factors and Galois information.
Uses. Frobenius and factorization modulo primes. A derangement produces primes with no linear factor through Chebotarev.

CHAPTER 21

Fekete polynomials

family / Intermediate / Q / Published results

Quadratic characters determine rational factors and reciprocal symmetry. After normalization, trace polynomials and arithmetic special values relate irreducibility to the Galois action on inverse pairs.

Overview

- Which rational factors must be removed?
- Why can half the degree suffice for irreducibility?
- What extra information is needed for the full splitting group?

Prerequisites: Fields, irreducibility, and Galois actions

21.1 Definition

For an odd prime p, let $\chi_p(a) = (a/p)$ be the Legendre symbol and define

$$F_p(x) = \sum_{a=1}^{p-1} \chi_p(a) x^a.$$

The coefficients encode the quadratic residues modulo p. These polynomials entered the study of Dirichlet L-functions; their special values also connect to class numbers and generalized Bernoulli numbers. Their coefficients are deterministic and strongly correlated, so theorems about independent random signs cannot be applied directly.

References

[MinacNguyenTan2023] / Introduction. Historical motivation and arithmetic special values. Source.

21.2 Normalization

The identity $\chi_p(p-a) = \chi_p(-1)\chi_p(a)$ gives $x^p F_p(x^{-1}) = \chi_p(-1) F_p(x)$. Define

$$f_p(x) = \begin{cases} \dfrac{F_p(x)}{x(1-x)}, & p \equiv 3 \pmod 4, \\ \dfrac{F_p(x)}{x(1-x)^2(x+1)}, & p \equiv 1 \pmod 4. \end{cases}$$

The factors at 1 and -1 have precisely the indicated multiplicities. The result is monic and reciprocal. Write $f_p = x^m g_p(x + x^{-1})$, with $m = (p-3)/2$ or $(p-5)/2$, respectively. For $p \geq 7$, irreducibility of f_p over $\mathbb{Q}$ is equivalent to irreducibility of g_p. This equivalence uses arithmetic special values, not reciprocity alone.

References

[MinacNguyenTan2023] / Proposition 2.1, Definition 3.2, and Corollary 4.2. Exact forced factors and irreducibility equivalence. Source.

21.3 Example

Example

The quadratic residues modulo 7 are $1, 2, 4$, so

$$F_7 = x + x^2 - x^3 + x^4 - x^5 - x^6.$$

Dividing by $x(1-x)$ gives

$$f_7 = x^4 + 2x^3 + x^2 + 2x + 1 = x^2((x + x^{-1})^2 + 2(x + x^{-1}) - 1).$$

Thus $g_7(y) = y^2 + 2y - 1$, with roots $-1 \pm \sqrt{2}$. Their field is $M = \mathbb{Q}(\sqrt{2})$, and the quadratic radicands are $-1 \mp 2\sqrt{2}$. Each has norm -7. A square in M has rational norm equal to a rational square, so neither radicand is a square. Their product is -7, which is not a square in the real field M. The two square classes are independent, giving $[L : M] = 4$ and $[L : \mathbb{Q}] = 8$. The group of f_7 is therefore $C_2 \wr S_2$.

The calculation requires both the splitting field of the trace polynomial and independence of the quadratic lifts.

References

[Cox2012]. Quadratic extensions and norms; this conductor-seven calculation is explicit. Source.

21.4 Galois-group conjecture

Miná̌c-Nguyen-Tân conjecture that the normalized prime-conductor family has full group $C_2 \wr S_m$, with $m = \deg g_p$. This is a conjecture in the cited source, supported by structural results and finite computations. The broader generalized-character family can have additional symmetries and a smaller allowed group.

References

[MinacNguyenTan2023] / Conjecture 4.13 in arXiv:2111.05256v3. The full wreath-product prediction. Source.
[MinacNguyenTan2024]. Generalized conductors, cyclotomic factors, and extra symmetry. Source.

21.5 Trace polynomials and square classes

Character identities determine reciprocal or anti-reciprocal symmetry and help identify the cyclotomic factors. After removing those factors, one studies the trace polynomial. Good-prime cycle types and square classes obtained from special values then constrain the full splitting field. A smaller Galois group may result from an additional cyclotomic factor, a reducible trace polynomial, a smaller group on pairs, or a relation among the quadratic radicands.

Thus character sums provide arithmetic information about the Galois group. Zero distributions and finite computations do not, by themselves, prove uniform irreducibility.

References

[MinacNguyenTan2024]. The organization by forced factors, trace polynomials, and modular Galois tests. Source.

21.6 Principal characters

For the principal character modulo $N > 1$, the corresponding polynomial is $F_N^{\mathrm{prin}}(x) = \sum_{1\le a\le N,\,(a,N)=1} x^a$. Möbius inversion gives

$$F_N^{\mathrm{prin}}(x) = \sum_{d\mid N} \mu(d)\,\frac{x^d(1-x^N)}{1-x^d}.$$

Each quotient is a finite geometric sum, so the expression is polynomial. Chidambaram-Ináč-Nguyen-Tân study its cyclotomic factors, remaining factors, and Galois groups. This family has coefficients 0 and 1 and is not covered by the prime quadratic-character conjecture above.

References

[ChidambaramMinacNguyenTan2025] / Introduction. Principal-character family, cyclotomic factors, and associated conjectures. Source.

Questions

Does every normalized prime-conductor Fekete polynomial have the full wreath-product group?

Conjecture in cited source / Odd primes p≥7, the normalization f_p above, and m=deg(g_p).

The cited conjecture predicts $\mathrm{Gal}(f_p/\mathbb{Q}) = C_2 \wr S_m$. This predicts both irreducibility and a maximal quadratic kernel. It remains a conjecture in the cited version; finite computations do not prove the statement for all primes.

References

[MinacNguyenTan2023] / Conjecture 4.13 in arXiv:2111.05256v3. Source.

Related topics

Uses. Reciprocal polynomials. The trace substitution and quadratic kernel organize the root action.

Contrasts with. Eulerian polynomials. Both families are reciprocal after normalization, but their arithmetic inputs differ.

Uses. Frobenius and factorization modulo primes. Good-prime factorizations constrain the group on the remaining roots.

Supplies examples for. Bernoulli and Euler polynomials. Generalized Bernoulli special values connect character sums with arithmetic invariants.

Part V

Counting and probability

CHAPTER 22

Height bounds for exceptional Galois groups

research / Research / Q / Research overview

Counts of polynomials with exceptional Galois group at fixed degree, including sharp bounds for monic and reciprocal families.

Overview

- What is the exceptional proportion in a monic coefficient box?
- Which polynomials give the lower bound?
- What is the corresponding order for reciprocal polynomials?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

22.1 Monic polynomials

Fix $n \geq 3$ and consider the monic polynomials

$$f = x^n + a_1 x^{n-1} + \cdots + a_n, \qquad |a_i| \leq H.$$

There are $(2\lfloor H \rfloor + 1)^n$ choices. Let $E_n(H)$ count those whose Galois group on the roots is not S_n. This count includes reducible polynomials and polynomials with repeated roots. The degree is fixed, and implied constants may depend on it.

Theorem / Bhargava

For fixed $n \geq 3$,

$$E_n(H) = O_n(H^{n-1}).$$

The $(2\lfloor H \rfloor + 1)^{n-1}$ choices with $a_n = 0$ all have the factor x and give a lower bound of the same order. The exceptional proportion therefore has order H^{-1}.

References

[Bhargava2025] / Theorem 1. Sharp upper bound in the fixed-degree box model; the safe stated range here is n≥3. Source.

22.2 Quadratics

Example

Consider $(x+a)(x+b)$ for integers $2 \leq a \leq \sqrt{H}/2$ and $a \leq b \leq H/(2a)$. For sufficiently large H, the coefficient bounds $a + b \leq H$ and $ab \leq H$ hold. Distinct unordered pairs give distinct polynomials. Summing over the available b's for each a gives $\gg H \log H$ reducible quadratics.

Thus the bound $O(H^{n-1})$ for all exceptional polynomials fails at $n = 2$. A count restricted to irreducible polynomials with exceptional transitive group excludes these examples.

22.3 Nonmonic polynomials

Let $R_n(H)$ count degree-n polynomials in $\mathbb{Z}[x]$ that are reducible over $\mathbb{Q}$ and whose coefficients have absolute value at most H. The leading coefficient may be any nonzero integer in the given range. Kuba proves

$$R_n(H) \asymp_n H^n \quad (n \geq 3), \qquad R_2(H) \asymp H^2 \log H.$$

The total count is $2\lfloor H \rfloor (2\lfloor H \rfloor + 1)^n \asymp_n H^{n+1}$, so the reducible proportion has order H^{-1} for $n \geq 3$. The nonmonic count has one additional power of H relative to the monic count.

The proof counts factorizations using height inequalities and lattice-point estimates. For $n \geq 3$, all but $O_n(H^{n-1}(\log H)^2)$ reducible polynomials have an irreducible factor of degree $n-1$ and hence a linear cofactor.

References

[Kuba2009] / Theorems 1, 2 and 4; Lemma 1. The polynomial count is nonmonic and reducibility is over Q, so a nonunit integer content alone does not count as reducibility. Source.

22.4 Sieves and number fields

At good primes, a Galois group restricts the possible factorization types. Gallagher's large sieve bounds the number of coefficient vectors whose reductions satisfy these restrictions at many primes.

Bhargava's sharper bounds use the structure of the exceptional permutation groups and the discriminants of the number fields generated by roots, together with Fourier and geometric-sieve estimates. The count is over coefficient vectors. Many polynomials can define the same number field, so a count by field discriminant answers a different question.

References

[Gallagher1973]. The large sieve in probabilistic Galois theory. Source.

[Bhargava2025] / Introduction and Theorem 2. Group-sensitive bounds and the link with counting number fields. Source.

22.5 Prescribed permutation actions

Write $B(H; G, \Omega)$ for the set of monic separable integer polynomials of degree $|\Omega|$, height at most H, and root action isomorphic to (G, Ω).

Reported preprint result / Ben-Porath (2026)

If $G \leq S_m$ is k-transitive and $[m]_k$ denotes the ordered k-tuples of distinct elements, then for fixed G, m, k and every $\varepsilon > 0$,

$$|B(H; G, [m]_k)| \ll_{G,m,k,\varepsilon} H^{m+1+\varepsilon}.$$

For S_5 on ordered pairs, the degree is 20 and the bound is $O_\varepsilon(H^{6+\varepsilon})$. The natural action has degree five.

The coefficients of $\prod_{\alpha \in A}(x - \alpha)$ generate the field fixed by the setwise stabilizer of the root subset A. The proof chooses families of subsets to recover the splitting field while controlling the heights. The paper also treats subset actions, primitive wreath-product actions, and regular actions.

References

[BenPorath2026RootSubsets] / Theorem 1.3; Lemma 3.1 and Proposition 3.3. The degree is the size of the specified permutation set, not the natural permutation degree m. Source.

22.6 Reciprocal polynomials

A monic reciprocal polynomial of degree $2m$ has m free coefficients. If the polynomial is separable, its Galois group lies in $W_m = C_2 \wr S_m$.

Theorem / Anderson-Bertelli-O'Dorney

For each fixed $m \geq 3$, among separable monic reciprocal polynomials with coefficients bounded by H, the number whose group is a proper subgroup of W_m is

$$\asymp_m H^{m-1} \log H.$$

For nonmonic reciprocal polynomials, the corresponding count is $\asymp_m H^m \log H$ for every $m \geq 1$.

The logarithmic factor arises from a square-value condition. Polynomials whose Galois group lies in the index-two subgroup of even permutations give the main exceptional contribution. The corrected source leaves the monic quartic case $m = 2$ open.

References

[AndersonBertelliODorney2026] / Theorems 1.1 and 1.3; Remark 1.4, corrected v2. Degree ranges, exceptional orders and the quartic boundary. Source.

22.7 Example

For a monic reciprocal quadratic $f = x^2 + ax + 1$, the discriminant is $a^2 - 4$. If this equals an integer square d^2, then $(a-d)(a+d) = 4$. Since the two factors have the same parity, the integer solutions have $a = \pm 2$ and $d = 0$. Every separable polynomial in this family has group S_2, so the exceptional count in monic degree two is zero.

In the fixed-degree ranges of the ordinary and reciprocal counting theorems, the maximal group occurs with density one.

Questions

What is the order of the exceptional count for monic reciprocal quartics?

Open in cited source / Separable monic reciprocal quartics over Z, with coefficient height H tending to infinity.

The corrected source proves the monic bound for half-degree $m \geq 3$ and leaves $m = 2$ open. The nonmonic bound includes quartics.

References

[AndersonBertelliODorney2026] / Remark 1.4 and Remark 1.6, v2. Explicit low-degree exception. Source.

Related topics

Generalizes. Specialization and Hilbert irreducibility. Height counts quantify exceptional specializations.

Uses. Frobenius and factorization modulo primes. Factorization modulo primes gives the local conditions for the sieve.

Uses. Reciprocal polynomials. The inverse pairs determine the ambient permutation group.

Contrasts with. Random polynomials: models and methods. The height tends to infinity at fixed degree.

CHAPTER 23

Random polynomials: models and methods

research / Research / Q / Research overview

Irreducibility and Galois groups in fixed-degree and growing-degree models, with applications of Fourier analysis, sieves, and discriminants.

Overview

- Which parameters are fixed in each model?
- What conditions distinguish irreducibility, A_n-containment, and the full group S_n?
- How do Fourier estimates, sieves, and discriminants enter the proofs?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

23.1 Probability models

We consider three models with different limiting parameters.

Model	Fixed data	Limit
Monic coefficient box	Degree n	Height $H \to \infty$
Independent bounded coefficients	A probability measure μ on $\mathbb{Z}$	Degree $n \to \infty$
Widening coefficient box	A sequence $L(n)$	$n \to \infty$, $L(n) \to \infty$

In the second model, let $F_n = x^n + \sum_{j=0}^{n-1} \xi_j x^j$, with independent coefficients of common law μ. If $\mu(0) > 0$, then $x \mid F_n$ with probability $\mu(0)$. To allow irreducibility with probability tending to one, one may condition on $\xi_0 \neq 0$, fix a nonzero constant term, or choose a law with no atom at zero.

To use a bound for fixed n as the degree grows, one needs control of its dependence on n. A theorem for $L(n) \to \infty$ gives no conclusion for a fixed two-point law.

References

[Bhargava2025] / Theorem 1. Fixed-degree height counting. Source.

[BarySorokerGoldgraber2025]. Full symmetric when the coefficient interval grows, however slowly. Source.

23.2 Small factors and interval laws

Theorem / Bary-Soroker-Koukoulopoulos-Kozma

For every fixed, finitely supported, nonconstant law μ, there is $\theta > 0$ such that, conditional on $\xi_0 \neq 0$, the probability of a nonconstant factor of degree less than θn tends to zero.
If μ is uniform on a fixed interval of at least 35 consecutive integers, then, under the same conditioning,

$$\Pr(F_n \text{ irreducible and } \mathrm{Gal}(F_n/\mathbb{Q}) \supseteq A_n) \longrightarrow 1.$$

The small-factor estimate leaves open the possibility of two factors of comparable degree, for example near $n/2$. In the interval case, the group is A_n or S_n with probability tending to one.

References

[BarySorokerKoukoulopoulosKozma2023] / Theorems 1, 2 and 6. Small-factor exclusion; interval-law irreducibility; Galois-group conclusions. Source.

23.3 Cyclotomic factors and error bounds

For coefficients uniform on a fixed interval of $N \geq 10^8$ integers, Bazin's Theorem 1.1.2 gives

$$\Pr(F_n \text{ reducible}) \leq \Pr(\xi_0 = 0) + O(n^{-1/2}).$$

For each fixed $C > 0$, if the interval is sufficiently long in terms of C, Theorem 1.1.4 gives

$$\Pr(F_n \text{ reducible}) = \Pr(\exists \omega \in U_C : F_n(\omega) = 0) + O_C(n^{-C/2}),$$

where U_C consists of zero and the roots of unity of degree less than C over $\mathbb{Q}$.

References

[Bazin2025] / Theorems 1.1.2 and 1.1.4, v2. The constant-term obstruction and cyclotomic refinement are explicit. Source.

23.4 Repeated roots

Consider $P_n(x) = \sum_{j=0}^{n} \eta_j x^j$, with all coefficients, including the leading coefficient, independent and of a fixed bounded integer law μ with $\max_a \mu(a) \leq 1/2$. Feldheim-Sen prove

$$\Pr(P_n \text{ has a repeated root}) = \Pr(P_n \text{ has a repeated root in } \{0, 1, -1\}) + o(n^{-2}).$$

If $\mu(0) = 0$, this probability is $O(n^{-2})$. If $\mu(0) > 0$, then $\eta_0 = \eta_1 = 0$ has probability $\mu(0)^2$ and forces a repeated root at zero.

The proof separates possible roots by algebraic degree and size and uses anti-concentration for integer evaluations. Applying the theorem to a monic or conditioned ensemble requires a separate justification.

References

[FeldheimSen2017] / Theorem 1.1 and §1.1. The full iid model and the largest-atom condition are essential to the stated theorem. Source.

23.5 Root separation and discriminant size

For independent, identically distributed mean-zero sub-Gaussian coefficients with no atom at zero, Michelen-Yakir prove that the minimum distance between roots, indexed with multiplicity, has scale $n^{-5/4}$ and a nondegenerate limiting law. Their Corollary 1.2 gives a vanishing probability of repeated roots under these hypotheses, without the largest-atom restriction for bounded integer laws. The earlier theorem for Littlewood polynomials gives a sharper quantitative bound in that model.

In a separate preprint, for $P_n(x) = \sum_{j=0}^{n} \eta_j x^j$ under the additional normalization $\mathbb{E}|\eta_j|^2 = 1$, they report

$$\frac{\log|\operatorname{disc}(P_n)| - 2n\log n}{n} \xrightarrow{\Pr} -D_*, \qquad D_* > 0.$$

The proof uses a reciprocal form of the discriminant identity, Mahler measure, and root statistics. Distinguishing A_n from S_n requires an additional arithmetic argument about the square class. The no-zero-atom hypothesis removes the repeated-root obstruction at zero.

References

[MichelenYakir2026RootSeparation] / Theorem 1.1 and Corollary 1.2. The hypotheses here retain mean zero, sub-Gaussian tails and no atom at zero. Source.

[MichelenYakir2025Discriminant] / Theorem 1; §1.3; equation (10). The displayed convergence is a preprint theorem; the no-zero-atom and variance assumptions are explicit in the full statement. Source.

23.6 Fourier estimates and Frobenius

For a nonzero candidate divisor $D \in \mathbb{F}_p[x]$, divisibility is the event that a random sum vanishes in $\mathbb{F}_p[x]/(D)$. Independence gives a product formula for its Fourier transform. Decay of the product controls deviation from uniformity, with exceptions determined by the characters and the structure of the quotient algebra. Possible rational factor degrees must occur among the subset sums of the modular factor degrees at each prime. Combinatorial estimates bound the probability of a common subset sum.

At primes of squarefree reduction, the modular factors give Frobenius cycle types. Probabilistic permutation-group estimates exclude most proper transitive groups. A second method averages root counts over primes and uses effective prime-ideal estimates to study global Galois orbits. Breuillard-Varjú use Riemann hypotheses for Dedekind zeta functions in this step.

References

[BarySorokerKoukoulopoulosKozma2023] / §2. Fourier estimates, factor-degree sieving and permutation groups. Source.

[BreuillardVarju2019]. Conditional prime-ideal and random-walk method. Source.

23.7 The square-discriminant probability

Reported preprint result / Blachar-Breuillard-Kozma, v1

Let $F_n = x^n + \sum_{j=1}^{n-1} \xi_j x^j + 1$, where $\xi_1, \dots, \xi_{n-1}$ are independent with a fixed finitely supported integer law μ. If

$$H_2(\mu) := -\log \sum_{a \in \mathbb{Z}} \mu(a)^2 > 12,$$

the preprint reports $\Pr(\operatorname{disc} F_n \text{ is a square}) = O_\mu(1/\log n)$. This is the constant-term-one case of its stated Theorem 1.1.

Corollary 1.2 states that when all $\xi_0, \dots, \xi_{n-1}$ are independent and uniform on a fixed interval of $N > e^{12}$ integers, $x^n + \sum_{j=0}^{n-1} \xi_j x^j$ is irreducible with group S_n, conditional on $\xi_0 \neq 0$, with probability $1 - O_\mu(1/\log n)$.

Remark on the scope of Theorem 1.1

The unqualified square-discriminant bound in v1 fails for some permitted constant-term laws. For $R_n = x^n + \sum_{j=0}^{n-1} \xi_j x^j$, if ξ_0, ξ_1 are independent with law μ and $\mu(0) > 0$, then

$$\Pr(\operatorname{disc} R_n = 0) \geq \Pr(\xi_0 = \xi_1 = 0) = \mu(0)^2.$$

On this event, $x^2 \mid R_n$ and the discriminant is the square zero. The obstruction occurs even for a uniform interval with $N > e^{12}$ containing zero. The $O(1/n)$ repeated-root estimate invoked at the start of Section 8 cannot hold in that setting. The restrictions in the displayed claims remove this obstruction. The remaining proof has not been independently verified here.

The argument studies double roots modulo primes and their lifts modulo prime squares to bound discriminant valuations. Once A_n-containment is known, a nonsquare discriminant forces S_n. For a uniform law on N values, $H_2 = \log N$. The fair-sign law has $H_2 = \log 2$, below the required threshold.

References

[BlacharBreuillardKozma2026] / Theorem 1.1, Corollary 1.2, and opening of §8, v1. Reported preprint statements; the atlas excludes the zero-constant obstruction and records the scope issue in the original square-discriminant assertion. Source.

23.8 Example

For coefficients uniform on $\{-17, \dots, 17\}$, the 35-point theorem gives irreducibility and A_n-containment with probability tending to one conditional on $\xi_0 \neq 0$. Without conditioning, x is a factor with probability $1/35$.

For coefficients uniform on $\{-1, 1\}$, the constant term is nonzero. The 35-consecutive-integer theorem and the entropy bound both require larger supports. For coefficients uniform on $[-L(n), L(n)] \cap \mathbb{Z}$ with $L(n) \to \infty$, Bary-Soroker-Goldgraber give full S_n with probability tending to one. In this last model the probability of a zero constant term tends to zero.

In numerical experiments, specify the coefficient law, degree, constant-term convention, sample size, and certification method for the group. A frequency of S_n concerns the sampled polynomials.

References

[BarySorokerGoldgraber2025]. The widening-interval theorem. Source.

23.9 Multiplicative coefficients

Choose independent fair signs X_p for primes p, put $X_1 = 1$, and extend completely multiplicatively by $X_n = \prod_{p^a \| n} X_p^a$. Define

$$P_d(x) = \sum_{j=1}^{d} X_j x^j.$$

Assuming RH for the Dedekind zeta functions of all number fields, Varjú-Xu prove that for every $\varepsilon > 0$,

$$\Pr(P_d(x)/x \text{ irreducible over } \mathbb{Q}) \geq 1 - O_\varepsilon(d^{-1/2+\varepsilon}).$$

The theorem also covers the variant with $X_j = 0$ at nonsquarefree integers. In both models one removes the factor x; in the second, the degree of the original polynomial may be less than d. For example, the coefficients satisfy $X_6 = X_2 X_3$. The proof combines finite-field root averages with arithmetic equidistribution. The corresponding deterministic questions for Fekete polynomials are separate.

References

[VarjuXu2026] / Theorem 1.1; §1.1; Problem 1.2. Both multiplicative models are specified in the introduction; the conditional conclusion concerns P_d/x. Source.

23.10 Polynomials over a function field

Fix a prime power q and set

$$f(x,y) = y^n + \sum_{i=0}^{n-1} a_i(x) y^i, \qquad a_i \text{ independent and uniform in } \{a \in \mathbb{F}_q[x] : \deg a \leq d\}.$$

Entin-Popov take the splitting field over $\mathbb{F}_q(x)$. For fixed n, q and $d \to \infty$, the probability of irreducibility, separability, and Galois group S_n tends to one. For fixed $d \geq 1, q$ and $n \to \infty$,

$$\Pr(f \text{ irreducible}) \longrightarrow 1 - q^{-d}, \qquad \Pr(G_f \in \{A_n, S_n\} \mid f \text{ irreducible}) \longrightarrow 1.$$

In odd characteristic, the authors obtain S_n with probability tending to one conditional on irreducibility, under the uniform polynomial Chowla conjecture stated in Conjecture 1.5. In the finite-field setting, an irreducible polynomial over $\mathbb{F}_q$ has cyclic Galois group.

References

[EntinPopov2024] / Theorems 1.1, 1.2, 1.4 and 1.6; Conjecture 1.5. The full symmetric conclusion in the fixed-d regime is conditional, and Theorem 1.6 assumes odd q. Source.

Questions

Can one prove full symmetric groups for small fixed coefficient laws?

Research direction / Independent bounded coefficients over Q, with the zero-constant obstruction excluded.

The entropy hypothesis of the cited preprint excludes small-support laws, including fair signs. For these laws, the stated estimate gives no bound for the square-discriminant probability.

References

[BlacharBreuillardKozma2026] / Introduction after Theorem 1.1. The authors discuss the entropy assumption as a limitation of their method. Source.

Related topics

Applies to. Littlewood polynomials. The fair-sign law has two atoms.

Contrasts with. Factor reversal for random binary polynomials. Binary coefficients admit a reciprocal-product identity.

Contrasts with. Height bounds for exceptional Galois groups. The degree is fixed and the coefficient height tends to infinity.

Uses. Discriminants, parity, and ramification. A nonsquare discriminant distinguishes S_n from A_n once containment is proved.

CHAPTER 24

Littlewood polynomials

family / Research / Q / Research overview

Irreducibility for random sign coefficients, with deterministic criteria modulo two and a theorem in special degrees.

Overview

- When does reduction modulo two prove irreducibility?
- Which degree sequences have unconditional high-probability irreducibility?
- How does a Newton polygon rule out all but small factors?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

24.1 The sign model

A Littlewood polynomial is $L_n(x) = \sum_{j=0}^{n} \varepsilon_j x^j$, with $\varepsilon_j \in \{-1, 1\}$. In the random model, the signs are independent and fair. Multiplying by the leading sign preserves the roots and irreducibility and yields a monic polynomial whose remaining signs are again independent and fair.

By the rational-root theorem, the only possible rational roots are 1 and -1. If n is even, $L_n(1)$ and $L_n(-1)$ are both sums of an odd number of odd integers and therefore do not vanish. For $n = 2m - 1$, counting sign patterns gives

$$\Pr(L_n(1) = 0) = \frac{\binom{2m}{m}}{2^{2m}} \sim \frac{1}{\sqrt{\pi m}}.$$

The event at -1 has the same probability. The two events are dependent. A reducibility estimate must also account for factors of higher degree.

References

[BarySorokerHokkenKozmaPoonen2025] / §1. Definition of the random sign model and the all-degree conjecture. Source.

24.2 Repeated roots

Theorem / Peled-Sen-Zeitouni

For fair independent signs, write $D_n = \Pr(L_n \text{ has a repeated complex root})$. Then

$$D_n = \begin{cases} \dfrac{8\sqrt{3}}{\pi n^2} + o(n^{-2}), & 4 \mid n+1, \\ o(n^{-2}), & 4 \nmid n+1. \end{cases}$$

The main term comes from repeated roots at 1 and -1. The estimate concerns vanishing of the discriminant. A squarefree polynomial may still be reducible, and a nonzero discriminant may still be a square.

The proof separates roots by algebraic degree and distance from the unit circle, then uses anti-concentration and a local central limit calculation for the exceptional roots.

References

[PeledSenZeitouni2016] / Theorems 1.1 and 1.2; §1.1. The displayed constant specializes Theorem 1.2 to mean-zero signs of variance one. Source.

24.3 Reduction modulo two

Example

Modulo 2, all signs reduce to 1, so every degree-four Littlewood polynomial reduces to

$$1 + x + x^2 + x^3 + x^4 = \Phi_5(x).$$

The order of 2 in $(\mathbb{Z}/5\mathbb{Z})^\times$ is 4. Hence Φ_5 is irreducible over $\mathbb{F}_2$, and all degree-four Littlewood polynomials are irreducible over $\mathbb{Q}$.

In general, $\overline{L}_n = (x^{n+1}-1)/(x-1)$. The same argument applies if $n+1$ is an odd prime for which 2 is a primitive root. If $n+1 = 2^r$, then $\overline{L}_n = (x+1)^n$, and the reduction gives no irreducibility certificate.

References

[LidlNiederreiter1997]. Cyclotomic factor degrees over finite fields. Source.

24.4 Irreducibility in special degrees

Theorem / Bary-Soroker-Hokken-Kozma-Poonen

There is an absolute $c > 0$ such that, for $n = 2^r - 1$,

$$\Pr(L_n \text{ irreducible over } \mathbb{Q}) \geq 1 - n^{-c}.$$

In particular, $\limsup_{n\to\infty} \Pr(L_n \text{ irreducible}) = 1$. The paper also treats degrees $n = p^r - 1$ when $p \geq 7$ is prime and 2 generates $(\mathbb{Z}/p^2\mathbb{Z})^\times$.

For $g(x) = L_n(x+1)$, every coefficient below the leading one is even. Their residues modulo 4 are independent and uniform in $\{0, 2\}$. With probability tending to one, a coefficient near the constant term has valuation exactly one. The 2-adic Newton polygon then gives a local irreducible factor of degree close to n. If the polynomial factors over the rationals, a remaining factor must have small degree. The separate small-factor theorem excludes such a factor with probability tending to one.

References

[BarySorokerHokkenKozmaPoonen2025] / Theorem 1.2 and §2. Special degrees and the Newton-polygon argument. Source.

24.5 Example

Let $L(x) = 1 + x + x^2 + x^3 + x^4 + x^5 + x^6 - x^7$. Modulo 2, $L(x+1) = x^7$, so every coefficient below the leading one is even. The constant term $L(1) = 6$ is not divisible by 4. Eisenstein at 2 proves that $L(x+1)$, and hence $L(x)$, is irreducible over $\mathbb{Q}$.

In the preceding probabilistic argument, the first coefficient of valuation one need not be the constant term.

References

[ConradIrredQ]. Eisenstein and translation preserve irreducibility. Source.

24.6 The all-degree problem

Breuillard-Varjú prove high-probability irreducibility along all degrees under Riemann hypotheses for Dedekind zeta functions. The published unconditional result above gives a subsequence. An unconditional all-degree theorem with group S_n requires more than either statement.

Kogler's 2026 preprint concerns monic 0/1 polynomials with constant term 1. In that model, the internal coefficients remain random after reduction modulo 2. Reduction of sign coefficients modulo two is deterministic.

References

[BreuillardVarju2019]. Conditional growing-degree theorem. Source.

[Kogler2026] / Theorem 1.1, v2. The binary model is different. Source.

Questions

Does irreducibility hold with probability tending to one along all degrees?

Open in cited source / Fair independent ± 1 coefficients over Q, with degree n tending to infinity.

The cited Littlewood paper formulates the all-degree conjecture and proves irreducibility with probability tending to one along a subsequence. The all-degree theorem of Breuillard-Varjú assumes Riemann hypotheses for Dedekind zeta functions.

References

[BarySorokerHokkenKozmaPoonen2025] / Equation (1.1), Theorems 1.1-1.2. Source-scoped conjecture and proved subsequence. Source.

Related topics

Uses. Newton polygons and Galois groups. Translation yields a Newton polygon with a large irreducible local factor.
Uses. Irreducibility over finite fields. The order of 2 modulo an odd prime determines the cyclotomic factor degrees.
Applies to. Random polynomials: models and methods. Independent signs give a two-point coefficient law.
Contrasts with. Factor reversal for random binary polynomials. The binary model has random reduction modulo two.

CHAPTER 25

Factor reversal for random binary polynomials

research / Research / Q / Preprint

Reversing a factor of a binary polynomial preserves its reciprocal product. Kogler's preprint bounds the probability of a polynomial with the same product other than the original and its reversal.

Overview

- What does reversing a factor preserve?
- Why do the coefficients remain binary?
- Which additional estimates are needed for irreducibility?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

25.1 The binary model

Let

$$P_n(x) = 1 + \sum_{j=1}^{n-1} \xi_j x^j + x^n, \qquad \Pr(\xi_j = 0) = \Pr(\xi_j = 1) = \tfrac{1}{2},$$

with independent ξ_j.

> **Preprint theorem / Kogler, v2, 14 October 2026**
> Theorem 1.1 states the unconditional asymptotic
>
> $$\Pr(P_n \text{ reducible over } \mathbb{Q}) = \sqrt{\frac{2}{\pi n}} + O(n^{-1}).$$

The leading contribution comes from the factor $x + 1$. The theorem concerns reducibility in this binary model. The Galois group and the fair-sign model are separate questions.

References

[Kogler2026] / Theorem 1.1, v2. Exact model, asymptotic and preprint status. Source.

25.2 The reciprocal product

For a polynomial P with nonzero constant term, let $P^*(x) = x^{\deg P} P(1/x)$. Suppose a binary polynomial with leading and constant coefficients 1 factors as $P = AB$, with $A, B \in \mathbb{Z}[x]$ monic. Put $Q = AB^*$. Then

$$QQ^* = PP^*.$$

The new polynomial is binary.

Proof

The identity follows because reversal is multiplicative and involutive on polynomials with nonzero constant term.
Write $P = \sum p_j x^j$ and $Q = \sum q_j x^j$. The central coefficient of the identity gives $\sum q_j^2 = \sum p_j^2$. Reversal preserves evaluation at 1, so $Q(1) = P(1)$. Since $p_j^2 = p_j$,

$$\sum_j q_j(q_j - 1) = 0.$$

Each summand is a nonnegative integer. Thus each q_j equals 0 or 1, and its leading and constant coefficients equal 1.

To deduce irreducibility with high probability, one must bound the probability of a polynomial with the same reciprocal product other than the original and its reversal. Factorizations that produce only these two polynomials require separate estimates.

References

[Ljunggren1960]. An antecedent of factor-reversal arguments. Source.
[filaseta1999smallnorm]. The Euclidean-norm perspective on factorization. Source.
[Kogler2026] / Lemma 3.1. The binary twin-factorization step. Source.

25.3 Example

Let $A = 1 + x + x^3$ and $B = 1 + x^4 + x^{12}$. The supports of $P = AB$ and $Q = AB^*$ are

$$\operatorname{supp}(P) = \{0, 1, 3, 4, 5, 7, 12, 13, 15\},$$
$$\operatorname{supp}(Q) = \{0, 1, 3, 8, 9, 11, 12, 13, 15\}.$$

In each product, the shifted copies of A have disjoint supports, so the coefficients are binary. The reciprocal products agree. Yet $Q \neq P$ and $Q \neq P^*$, since the support of P^* is $\{0, 2, 3, 8, 10, 11, 12, 14, 15\}$.

Thus the autocorrelation encoded by PP^* can agree for binary polynomials that are distinct even up to reversal.

25.4 The congruence estimate

Kogler's Theorem 1.3 bounds the probability of a nontrivial binary twin even when equality of reciprocal products is replaced by congruence modulo 4. Reduction modulo 2 restricts the possible factors, and the congruence imposes further conditions on opposite coefficient pairs. Cyclotomic factors and reciprocal noncyclotomic factors are treated separately.

The factor-reversal identity also applies in deterministic arguments, including those for polynomials with large gaps.

References

[Kogler2026] / Theorem 1.3 and §3, v2. Probabilistic collision estimate and exceptional factors. Source.
[sawinshustermanstoll2020]. Related deterministic use of Ljunggren-type ideas. Source.

Questions

Does the binary model have full symmetric group with probability tending to one?

Research direction / Monic binary polynomials over Q with constant term 1 and independent fair internal coefficients.

The cited preprint states a high-probability irreducibility theorem. A full-S_n theorem also requires control of the transitive group and the discriminant square class.

References

[Kogler2026] / Theorem 1.1. Scope of the cited result. Source.

Related topics

Applies to. Sparse polynomials and factor reversal. Deterministic large-gap arguments also use reversal and norm identities.

Uses. Reciprocal polynomials. The same reversal defines reciprocal polynomials.

Contrasts with. Littlewood polynomials. Reduction of sign coefficients modulo two is deterministic.

Supplies examples for. Random polynomials: models and methods. The binary proof uses a reciprocal-product congruence.

CHAPTER 26

Characteristic polynomials of random matrices

research / Research / Q / Research overview

Irreducibility and Galois groups of characteristic polynomials of random integer matrices, using finite-field rank estimates.

Overview

- How are the characteristic coefficients dependent?
- Which entry laws admit an unconditional theorem?
- What does irreducibility imply about rational invariant subspaces?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

26.1 The independent-entry model

Let $M_n = (m_{ij})$ have independent entries of common law μ, a fixed finitely supported nonconstant law on $\mathbb{Z}$, and set $\chi_n(t) = \det(tI - M_n)$. As the dimension grows, one asks whether χ_n is irreducible with probability tending to one and whether its Galois group is S_n with probability tending to one.

Example

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$\chi_M(t) = t^2 - (a+d)t + (ad - bc).$$

If a, b, c, d are signs and $a + d = 2$, the determinant is 0 or 2. If $a + d = 0$, it is -2 or 0. For independent fair signs, the conditional laws of the determinant differ, so the trace and determinant are dependent.

The dependence prevents a direct application of the independent-coefficient theorems to χ_n. Symmetric, tridiagonal, and fixed-size height-bounded matrices are separate ensembles.

26.2 Eberhard's theorems

Theorem / Eberhard

In the independent-entry model above:

- If μ is uniform modulo a product of four distinct primes, irreducibility and $\mathrm{Gal}(\chi_n/\mathbb{Q}) \supseteq A_n$ hold with probability tending to one, unconditionally.

- For any fixed finitely supported nonconstant law μ, assume the Riemann hypothesis for Dedekind zeta functions of all number fields. Then $\Pr(\chi_n \text{ reducible}) = O(e^{-cn})$ and $\Pr(\mathrm{Gal}(\chi_n/\mathbb{Q}) \not\supseteq A_n) = O(e^{-c\sqrt{n}})$, with constants depending on μ.

The uniform law on $\{0, 1, \dots, 209\}$ satisfies the first hypothesis, since $210 = 2 \cdot 3 \cdot 5 \cdot 7$. The fair-sign law fails this hypothesis. Both theorems leave the alternatives A_n and S_n.

References

[Eberhard2022] / Theorems 1.2 and 1.3. Unconditional four-prime theorem and conditional general theorem. Source.

26.3 Finite-field eigenvalues and Galois orbits

The equality $\chi_M(a) = 0$ holds exactly when $aI - M$ is singular. Root counts modulo a prime therefore depend on the rank of a random matrix over $\mathbb{F}_p$. Counting tuples of roots requires joint singularity estimates for distinct values of a.

For a squarefree polynomial with root set Ω, the orbit-counting lemma identifies the number of G-orbits on Ω with the average number of fixed roots of elements of G. At good primes, the number of roots modulo the prime equals the number fixed by Frobenius. Effective prime-ideal estimates relate these local counts to the Galois average. Higher root correlations correspond to actions on tuples.

References

[Eberhard2022] / §1.1 and §3. Local root statistics and the effective global-to-local step. Source.

[StevenhagenLenstra1996]. Frobenius and density as the general framework. Source.

26.4 Other matrix ensembles

Ensemble	Result	Source
Symmetric fair-sign matrices	High-probability irreducibility under ERH, using finite-field rank estimates.	Ferber-Jain-Sah-Sawhney
Specified random tridiagonal matrices	Irreducibility with exponentially small failure probability and alternating or symmetric Galois group, under the stated Dedekind RH assumptions.	Bary-Soroker-Garzoni-Sodin
Fixed-size matrices with entries bounded by height	Counts of exceptional groups as the height tends to infinity at fixed size.	Anderson-O'Dorney

The tridiagonal paper also gives a smaller group for a related ensemble. The allowed group can depend on the matrix structure.

References

[FerberJainSahSawhney2023]. Conditional symmetric-matrix irreducibility and rank estimates. Source.

[BarySorokerGarzoniSodin2026]. Conditional tridiagonal results and contrasting ensemble. Source.

[AndersonODorney2026Matrices]. Fixed-size height-counting model. Source.

26.5 Rational invariant subspaces

An irreducible characteristic polynomial excludes nonzero proper rational invariant subspaces.

Proof

If $W \subset \mathbb{Q}^n$ is M-invariant and $0 < \dim W < n$, a basis adapted to W puts M in block upper-triangular form. Then χ_M is the product of the characteristic polynomials on W and $\mathbb{Q}^n/W$, both of positive degree.

The nonsingular matrix $\mathrm{diag}(1, 2)$ has distinct eigenvalues and two rational invariant lines.

Questions

Does irreducibility hold unconditionally for every fixed entry law?

Open in cited source / Independent entries of any fixed nonconstant finitely supported law on Z, with n tending to infinity.

Eberhard's Conjecture 1.1 concerns irreducibility for arbitrary fixed entry laws. His general theorem assumes ERH, and the unconditional theorem requires uniformity modulo four distinct primes. In particular, the latter theorem does not cover fair signs. Full S_n is a further question.

References

[Eberhard2022] / Conjecture 1.1; Theorems 1.2-1.3. Exact conjectural and proved scopes. Source.

Related topics

Contrasts with. Random polynomials: models and methods. Independent matrix entries give dependent characteristic coefficients.

Uses. Frobenius and factorization modulo primes. Root counts modulo primes give fixed-point statistics for Frobenius.

Contrasts with. Height bounds for exceptional Galois groups. The entry height grows at fixed matrix size.

Uses. Discriminants, parity, and ramification. The discriminant square class distinguishes A_n from S_n after containment.

CHAPTER 27

Random reciprocal polynomials of growing degree

research / Research / Q / Preprint

Irreducibility and Galois groups for reciprocal polynomials with independent free coefficients, in the limit of growing degree.

Overview

- Which permutation group acts on the inverse pairs?
- How does the trace change of variables affect coefficient independence?
- Which group alternatives remain when the discriminant is nonsquare?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

27.1 The reciprocal model

For a positive integer m, let

$$F_m(x) = x^{2m} + 1 + a_0 x^m + \sum_{j=1}^{m-1} a_j (x^{m-j} + x^{m+j}),$$

where $a_0, \dots, a_{m-1}$ are independent. The two coefficients at reflected exponents share a single random variable. The ordinary model uses $2m$ independent coefficients.

When F_m is separable, its roots form m inverse pairs. Galois automorphisms permute the pairs and can exchange the two roots within a pair. The group therefore embeds in $W_m = C_2^m \rtimes S_m$, of order $2^m m!$, in its action on $2m$ roots. In degree at least four, this is a proper subgroup of S_{2m}.

References

[HokkenKoukoulopoulos2025] / Equation (1.2) and §14, v2. Model and signed-permutation action. Source.

27.2 Uniform interval laws

Preprint theorem / Hokken-Koukoulopoulos, v2

If the free coefficients are independent and uniform on a fixed interval of at least 35 consecutive integers, then F_m is irreducible with probability tending to one. With probability tending to one, its Galois group is one of

$$W_m, \qquad C_2 \wr A_m, \qquad G_2,$$

where, in signed-permutation coordinates (ε, σ),

$$G_2 = \{(\varepsilon, \sigma) : \operatorname{sgn}(\sigma) \prod_i \varepsilon_i = 1\}.$$

For $m \geq 2$, the second and third groups have index two in W_m. The general theorem permits varying coefficient laws under explicit Fourier hypotheses.

References

[HokkenKoukoulopoulos2025] / Theorem A; Theorem C; equation (14.3), v2. Concrete uniform-interval case, general hypothesis and exact alternatives. Source.

27.3 The trace polynomial

Every monic reciprocal polynomial of degree $2m$ has a unique expression $F(x) = x^m Q(x + x^{-1})$, with Q monic of degree m.

Example

$$F(x) = x^6 + ax^5 + bx^4 + cx^3 + bx^2 + ax + 1,$$
$$Q(y) = y^3 + ay^2 + (b-3)y + (c-2a).$$

This follows from $x^2 + x^{-2} = y^2 - 2$ and $x^3 + x^{-3} = y^3 - 3y$. Even if a, b, c are independent, the coefficients a and $c - 2a$ of Q are generally dependent.

Each root y of Q yields the two solutions of $x^2 - yx + 1 = 0$. The splitting field of Q determines the permutation action on pairs. Splitting the quadratics determines the additional extensions.

References

[HokkenKoukoulopoulos2025] / §5. Trace polynomials and the dependence issue. Source.

27.4 Factorization and the pair action

The irreducible factors of Q describe the Galois orbits on inverse pairs. Even when Q is irreducible, F can split into a factor and its reversal. The proof bounds the probability of this factorization separately. It applies Fourier estimates to the original free coefficients, which are independent.

The group is studied through its action on pairs and the subgroup acting within pairs. A non-square discriminant of F excludes $W_m \cap A_{2m}$. Other index-two subgroups of W_m can contain odd permutations, and the parity test does not distinguish the three groups in the theorem.

References

[HokkenKoukoulopoulos2025] / §2, §7 and §§14-16. Exceptional reciprocal factors and the group analysis. Source.

27.5 Fixed-degree height counting

The fixed-degree height theorem gives an exceptional proportion of order $H^{-1} \log H$, with m fixed and $H \to \infty$, in the stated degree range. In the present theorem, the free-coefficient law is fixed and $m \to \infty$. Both models have the same inverse pairing of roots.

References

[AndersonBertelliODorney2026] / Theorem 1.1. Fixed-degree comparison. Source.

Questions

When does the Galois group equal the full wreath product with high probability?

Research direction / Growing-degree reciprocal model under the hypotheses of Hokken-Koukoulopoulos.

The preprint allows $C_2 \wr A_m$ and G_2 as alternatives. Determining their probabilities or proving that W_m occurs with probability tending to one for particular laws requires further information beyond irreducibility and a nonsquare discriminant of F_m.

References

[HokkenKoukoulopoulos2025] / Theorem A and §14, v2. Exact surviving group alternatives. Source.

Related topics

Uses. Reciprocal polynomials. The trace polynomial describes the action on inverse pairs.
Generalizes. Random polynomials: models and methods. The Fourier estimates use the independent free coefficients.
Contrasts with. Height bounds for exceptional Galois groups. The height tends to infinity at fixed degree.
Generalizes. Twisted reciprocity and square discriminants. The involution x↦b/x gives a corresponding trace construction.

CHAPTER 28

Twisted reciprocity and square discriminants

research / Research / Q / Preprint

A discriminant formula for polynomials whose roots are paired by x↦b/x, and its use in counting exceptional Galois groups.

Overview

- What is the trace variable for x↦b/x?
- How is the discriminant expressed in terms of the trace polynomial?
- Which square-value condition detects even permutations?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

28.1 The trace substitution

Fix an integer $b \neq 0$. A rational polynomial F of degree $2m$ is b-reciprocal if

$$F(x) = \frac{x^{2m}}{b^m} F(b/x).$$

Its nonzero roots are paired by $\alpha \mapsto b/\alpha$. In the trace expression below, a fixed point $\alpha^2 = b$ would be a repeated root. For separable F, the group therefore acts on distinct pairs and embeds in $W_m = C_2 \wr S_m$.

The invariant variable is $y = x + b/x$, and $F(x) = x^m Q(x + b/x)$ for a degree-m polynomial Q. Each trace root y_i corresponds to the quadratic $x^2 - y_i x + b$.

References

[ODorney2026Twisted]. Twisted-reciprocal family and hyperoctahedral ambient group. Source.

28.2 The discriminant formula

Identity

If $F(x) = x^m Q(x + b/x)$, then

$$\operatorname{disc}(F) = b^{m(m-1)} Q(2\sqrt{b}) Q(-2\sqrt{b}) \operatorname{disc}(Q)^2.$$

Proof

Write $Q(y) = a \prod_{i=1}^{m} (y - y_i)$. The quadratic $x^2 - y_i x + b$ has discriminant $y_i^2 - 4b$. For $i \neq j$, the two quadratics have resultant $b(y_i - y_j)^2$. Indeed, evaluating their difference at the two roots of the first quadratic gives a product of $b(y_i - y_j)^2$. The discriminant product formula gives

$$a^{4m-2}\prod_i (y_i^2-4b)\,b^{m(m-1)}\prod_{i<j}(y_i-y_j)^4,$$

which is the right-hand side of the identity.

For a separable polynomial, since $m(m-1)$ is even, its discriminant has the square class of $Q(2\sqrt{b})Q(-2\sqrt{b})$. The product is rational even when $\sqrt{b}\notin\mathbb{Q}$.

References

[Cox2012]. Standard discriminant and resultant product identities. Source.

28.3 Example

Take $Q(y)=y^2-y-1$. Then

$$F(x)=x^2Q(x+2/x)=x^4-x^3+3x^2-2x+4.$$

Here $\operatorname{disc}(Q)=5$, and

$$Q(2\sqrt{2})Q(-2\sqrt{2})=(7-2\sqrt{2})(7+2\sqrt{2})=41.$$

Thus $\operatorname{disc}(F)=2^2\cdot 41\cdot 5^2=4100$ is nonsquare. The Galois group contains an odd permutation, so it is not contained in A_4. This leaves the exact subgroup of W_2 undetermined.

28.4 Exceptional groups at fixed degree

O'Dorney's September 2026 preprint uses an adapted Euclidean height. For fixed $b\neq 0$ and $m\geq 4$, it reports that the number of polynomials with Galois group smaller than W_m is an explicit constant times $H^m\log H$, with error $O_{m,b}(H^m)$. Groups contained in an index-two subgroup supply the main term. The counting convention and constant are those of the linked preprint version.

The discriminant formula reduces the even-permutation condition to a square-value condition on the trace polynomial. Anderson-Bertelli-O'Dorney treat the untwisted case $b=1$ in the published paper.

References

[ODorney2026Twisted] / Abstract, v1. Verified primary indexed abstract; full-text theorem details are not asserted here. Source.

[AndersonBertelliODorney2026]. Published untwisted counting theorem. Source.

Related topics

Generalizes. Reciprocal polynomials. The trace construction extends to x+b/x.

Applies to. Discriminants, parity, and ramification. Two evaluations determine the discriminant square class.

Applies to. Height bounds for exceptional Galois groups. The square-value condition contributes to the exceptional count.

Contrasts with. Random reciprocal polynomials of growing degree. The height grows with the degree fixed.

Part VI

Specialization and inverse Galois

CHAPTER 29

Specialization and Hilbert irreducibility

method / Intermediate / number fields / Classical theory

Hilbert irreducibility preserves the generic Galois group outside a thin set. Geometry and restrictions on the parameters give more precise descriptions of the exceptions.

Overview

- When does a specialization retain the generic Galois group?
- Why can the exceptional set be infinite, and when can it be finite?
- What changes when parameters are restricted to units?
- Why can arithmetic and geometric Galois groups differ?

Prerequisites: Fields, irreducibility, and Galois actions; Frobenius and factorization modulo primes

29.1 Specialization of Galois groups

Fix a number field K and a monic, separable, irreducible polynomial $F(T,x) \in K(T)[x]$. Its *arithmetic* group is $G = \mathrm{Gal}(F/K(T))$, acting on the roots. Exclude the finitely many parameter values where a coefficient has a pole or the discriminant vanishes. At every other $t \in K$, the specialized group G_t is conjugate to a subgroup of G.

This inclusion comes from a decomposition group above the place $T = t$ in the generic splitting field. The roots remain distinct, so the unramified residue action is the action on the specialized roots. The specialized group may be a proper subgroup of the generic group.

References

[Serre2007Topics] / §3.3, polynomial interpretation. Specialization of the root permutation action. Source.

29.2 Arithmetic and geometric groups

Let K have characteristic zero, let T be transcendental, and let $L/K(T)$ be the splitting field of a separable polynomial. Inside a common algebraic closure, put $C = L \cap \overline{K}$, the field of algebraic constants. Define

$$G_{\mathrm{arith}} = \mathrm{Gal}(L/K(T)), \qquad G_{\mathrm{geom}} = \mathrm{Gal}(L\overline{K}/\overline{K}(T)).$$

The constants exact sequence

The extension C/K is finite Galois, and restriction to constants gives

$$1 \longrightarrow G_{\mathrm{geom}} \longrightarrow G_{\mathrm{arith}} \longrightarrow \mathrm{Gal}(C/K) \longrightarrow 1.$$

Normality makes C stable under the arithmetic group, and automorphisms of $C(T)/K(T)$ extend to L. The kernel is $\mathrm{Gal}(L/C(T))$. Since C is relatively algebraically closed in L, the extension is regular over C; extending constants to $\overline{K}$ preserves this kernel. Thus the two groups agree exactly when $C = K$, the regular case. The exact sequence need not split in general.

Example: $X^3 - T$ over $\mathbb{Q}(T)$

Choose $\alpha^3 = T$ and a primitive cube root ζ. The roots are $\alpha, \zeta\alpha, \zeta^2\alpha$, so $L = \mathbb{Q}(\zeta)(\alpha)$. Because α is transcendental, this rational function field has algebraic constants exactly $C = \mathbb{Q}(\zeta) = \mathbb{Q}(\sqrt{-3})$. Eisenstein at T over $C[T]$ gives

$$[L : \mathbb{Q}(T)] = [L : C(T)]\,[C(T) : \mathbb{Q}(T)] = 3 \cdot 2 = 6.$$

The automorphism r sends $\alpha \mapsto \zeta\alpha$ and fixes ζ; the automorphism s fixes α and sends $\zeta \mapsto \zeta^{-1}$. They satisfy $r^3 = s^2 = 1$ and $srs = r^{-1}$. Hence $G_{\mathrm{arith}} \cong S_3$, while $G_{\mathrm{geom}} = \langle r \rangle \cong C_3$. The quotient C_2 acts on the quadratic constant field. The discriminant $-27T^2 = -3(3T)^2$ identifies the same quadratic constant field.

For comparison, $X^3 - X - T$ is irreducible over $\overline{\mathbb{Q}}(T)$: apply Gauss's lemma to its irreducibility as a polynomial linear in T with unit leading coefficient in $\overline{\mathbb{Q}}[X, T]$. Its discriminant $4 - 27T^2$ has two simple zeros, so is not a square in that rational function field. The cubic criterion gives $G_{\mathrm{geom}} = S_3$; consequently $G_{\mathrm{arith}} = S_3$ and $C = \mathbb{Q}$. Thus this family is regular, whereas $X^3 - T$ is not.

References

[Milne2022] / Free notes v5.10, Propositions 7.7 and 7.15. Restriction and extension of Galois automorphisms and change of base field; the constants exact sequence is derived here from these facts. Author's book page.

[JensenLedetYui2002] / §3.3, Proposition 3.3.6 (pp. 74-75). Regular Galois extensions preserve their group after extension of constants. Source.

[ConradCubicQuartic] / Equation (1.1), Theorems 2.1 and 2.8. Cubic discriminant formula, cubic group criterion, and splitting field from one root and the discriminant; the two parameter examples are worked out explicitly here. Source.

29.3 Hilbert irreducibility

Theorem (Hilbert irreducibility)

For F as above, there is a thin set $B \subset K$ such that every $t \in K \setminus B$ is admissible, $F(t, x)$ is irreducible, and $G_t = G$ up to conjugacy. In particular, infinitely many parameters retain the generic group.

Here "thin" means contained in a finite union of finite sets and images of rational points under dominant, generically finite maps from geometrically integral curves to the parameter line of degree at least two. Such a set may be infinite. The fixed fields of proper subgroups of G define resolvent covers. If a specialization has a smaller group, an associated cover has a rational point above that parameter. Hilbert's theorem gives parameters outside the union of these images.

References

[Hilbert1892]. Original theorem; modern formulations use Hilbertian fields. Source.

[Serre2007Topics] / Proposition 3.3.5 and Theorem 3.4.1. Full-group specialization outside a thin set over number fields. Source.

29.4 Example

Example

Let $F(T,x) = x^3 - x - T$. The quotient $\mathbb{Q}[T,x]/(F)$ is $\mathbb{Q}[x]$, with $T = x^3 - x$. Hence F is irreducible in the polynomial ring and, by Gauss's lemma, in $\mathbb{Q}(T)[x]$. Its discriminant is

$$\Delta(T) = 4 - 27T^2.$$

The two zeros of this rational function are simple, so it is not a square in $\mathbb{Q}(T)$. A transitive cubic group with nonsquare discriminant is S_3.
At $t = 1$, the reduction of $x^3 - x - 1$ modulo 2 has no root, and its discriminant is -23, giving S_3 again. But every $t = u^3 - u$, with $u \in \mathbb{Q}$, gives the rational root u. The parameters $t = u^3 - u$ form precisely the reducible-fiber locus, an infinite subset of the full-group exceptional set. There are also irreducible exceptional fibers: at $t = 1/7$, the cubic remains irreducible but its discriminant is $169/49$, giving A_3.

References

[ConradCubicQuartic]. The transitive cubic group and discriminant criterion; calculations here are explicit. Source.

29.5 Finite exceptional sets

In some families, proper-subgroup resolvent covers give a finite exceptional set. If the corresponding curves have genus at least two, each has only finitely many rational points. The required genus bounds depend on the ramification and the subgroup structure.

Theorem (Hajir-Wong)

Fix a number field K. For fixed $n \geq 5$, all but finitely many $\alpha \in K$ give an irreducible generalized Laguerre polynomial $L_n^{(\alpha)}$ whose group contains A_n. For $n \geq 10$, the group is S_n outside a finite set.

Here the degree is fixed and the parameter varies. Fixing α and letting n grow is a different problem. The argument does not generally determine the finite exceptional set effectively.

References

[HajirWong2006] / Theorem 1(a), p.1129; §§2 and 6. The displayed thresholds are ≥5 and ≥10; the PDF page was checked visually. Source.

29.6 Power parameters

Hilbert irreducibility over all rational parameters does not by itself control a prescribed sequence such as 2^n. For $X^2 - t$, exactly the odd integers n make $X^2 - 2^n$ irreducible over $\mathbb{Q}$; their density is $1/2$.

Theorem (Bary-Soroker-Garzoni-Matei, conditional on GRH)

Let $f(\mathbf{t}, X) \in \mathbb{Z}[t_1, \dots, t_r, X]$, with $\deg_X f \geq 1$, satisfy the pullback condition: for every tuple $m_i \geq 1$,

$$f(t_1^{m_1}, \dots, t_r^{m_r}, X) \text{ is irreducible in } \overline{\mathbb{Q}}[\mathbf{t}, X].$$

Fix $a_i \in \mathbb{Z}\setminus\{0,1,-1\}$. Assuming GRH, the proportion of exponent vectors $\mathbf{n} \in ([-N,N]\cap\mathbb{Z})^r$ for which $f(a_1^{n_1},\dots,a_r^{n_r},X)$ is irreducible over $\mathbb{Q}$ tends to one as $N\to\infty$.

The pullback condition excludes reducibility caused by a power substitution. For X^2-t, substituting $t=u^2$ already produces a factorization. The theorem concerns irreducibility. Preservation of the Galois group requires a resolvent-cover argument and its additional hypotheses.

For one parameter there is a stronger unconditional result. Under the same pullback condition with $r=1$, Dèbes proves that $f(a^n,X)$ is irreducible for all but finitely many integers n, without GRH. His Theorem 1.2 shows that infinitely many reducible fibers force a reducible scaled power pullback; geometric (PB) excludes this. The proof uses Siegel's theorem on integral points on curves. The exceptional exponents are finite in number, but the theorem does not give an effective list.

References

[BarySorokerGarzoniMatei2025Powers] / §1, condition (PB), Theorem 1.1, and the following examples. Published theorem; GRH and absolute irreducibility of every power pullback are essential. Source.

[Debes1992Powers] / Theorem 1.2, p. 144; see also Bary-Soroker-Garzoni-Matei, §1 after Theorem 1.1. The original theorem characterizes infinitely many reducible power specializations; geometric (PB) rules out its scaled power-pullback obstruction. Source.

29.7 Unit parameters

Preprint, 3 September 2026. Let $\pi : X \to \mathbb{P}^1$ be a nonconstant morphism from a smooth projective irreducible curve over a number field K, let S be a finite set of non-archimedean places of K, and put $U=\mathcal{O}_S^\times$. Let $\mathrm{Red}(\pi)$ be the set of rational parameters with a reducible fiber, together with all rational branch values. Siksek-Stoll prove

$$U\cap \mathrm{Red}(\pi) = I \cup \bigcup_{j=1}^{k} \gamma_j U^{r_j}, \qquad |I|<\infty, \quad \gamma_j\in U, \quad r_j\ge 2.$$

The power cosets are effectively computable; the proof does not make the finite remainder I effective. This describes the exceptional unit parameters more precisely than thinness alone. For x^2-T, the exceptional unit parameters are simply the squares U^2: if a square in K is an S-unit, its square root is also an S-unit.

References

[SiksekStoll2026] / Theorem 3 and following remark. Version 1; this is a preprint theorem, not a complete effective algorithm. Source.

29.8 Limitations

Irreducibility of one fiber does not imply a uniform theorem in degree or determine all exceptional fibers. The arithmetic group over $K(T)$ can exceed the geometric group over $\overline{K}(T)$. A theorem at each finite iteration depth need not give one parameter valid at every depth. This requires control of an intersection of infinitely many Hilbert sets.

References

[FriedJarden2023]. Hilbertian fields and specialization framework. Source.

[DittmannKadets2022] / Introduction and Theorem 1.2. A concrete warning against interchanging finite-level and infinite-level quantifiers. Source.

Questions

Can the finite exceptional parameters be determined effectively?

Research direction / A specified one-parameter family over a fixed number field

A finiteness theorem need not determine the exceptional parameters effectively. Hajir-Wong discuss this difficulty for their curves; the 2026 unit-specialization theorem also leaves a finite set ineffective. For a given family, the problem reduces to determining rational or integral points on the associated curves.

References

[HajirWong2006] / Remark 1. Source.
[SiksekStoll2026] / Remark following Theorem 3. Source.

Related topics

Applies to. Inverse Galois theory and generic polynomials. A regular realization over K(T) specializes to realizations over K.

Uses. Resolvents and subgroup tests. Proper-subgroup fixed fields describe possible drops of Galois group.

Supplies examples for. Generalized Laguerre polynomials. Laguerre specializations illustrate finite exceptional sets.

Contrasts with. Composition, iteration, and arboreal groups. Separate results at every finite depth need not give a single parameter valid throughout the infinite tower.

Generalizes. Height bounds for exceptional Galois groups. Counting points in thin exceptional sets gives quantitative specialization theorems.

CHAPTER 30

Critical points and polynomial monodromy

method / Advanced / Characteristic zero; specializations over Q / Classical theory

In characteristic zero, simple critical points with distinct critical values give full symmetric monodromy. Specialization can reduce the group.

Overview

- How can derivatives determine a generic Galois group?
- Why do simple critical points with distinct values give transpositions?
- When do rational specializations retain the generic group?

Prerequisites: Fields, irreducibility, and Galois actions; Transitivity, primitivity, and cycle criteria; Ramification and inertia; Specialization and Hilbert irreducibility

30.1 Morse polynomials

Let K have characteristic zero and let $F \in K[X]$ have degree $n \geq 2$. The family $F(X) - T$ is a polynomial over $K(T)$, with T an indeterminate. A **critical point** is a root c of F', and its **critical value** is $F(c)$. We call F Morse when F' has $n-1$ distinct roots over $\overline{K}$ and their critical values are pairwise distinct.

For example, for $F = X^4 - 2X^2$, the critical points $0, 1, -1$ are simple, but $F(1) = F(-1) = -1$. Thus simple critical points need not have distinct critical values. Composition $F = (X^2)^2 - 2X^2$ also gives an intermediate cover and a block system on the four generic roots.

The geometric group is the splitting-field group over $\overline{K}(T)$; the arithmetic group is over $K(T)$. Both act on the same n roots. A theorem about these groups does not assert the same group for every specialization $T = t$.

References

[Serre2007Topics] / §4.4, Theorems 4.4.1 and 4.4.5, pp. 39-41. Morse hypotheses and the geometric construction; the quartic boundary example is computed here. Source.

30.2 The Morse criterion

Theorem / Hilbert's Morse criterion

If $F \in K[X]$ is Morse and $\operatorname{char} K = 0$, then

$$\operatorname{Gal}(F(X) - T/\overline{K}(T)) = S_n,$$
$$\operatorname{Gal}(F(X) - T/K(T)) = S_n.$$

The splitting field over $K(T)$ is regular over K: it introduces no new algebraic constants.

The geometric group is S_n and is a subgroup of the arithmetic group, which already lies in S_n. Equality follows, and the arithmetic/geometric quotient, which measures the constant extension, is trivial.

This is a sufficient criterion. Polynomials failing one of the Morse conditions can still have full symmetric monodromy; this requires a separate ramification argument.

References

[Serre2007Topics] / Theorems 4.4.1 and 4.4.5. The general theorem is applied after base change to the algebraic closure; arithmetic equality and regularity follow. Source.

30.3 Proof sketch

Proof sketch

Over $\overline{K}$, the degree-n cover $X \mapsto F(X)$ is connected, so its monodromy group G is transitive. Equivalently, $F(X) - T$ is irreducible over $\overline{K}(T)$: in $\overline{K}[X, T]$, a factor independent of T would have to divide the coefficient -1 of T.

At a simple critical point c, the local expansion begins $F(c) + a(X - c)^2 + \cdots$, with $a \neq 0$. The distinct-value hypothesis ensures that only these two sheets meet over $F(c)$. Its inertia generator is therefore a single transposition.

Let H be the normal subgroup generated by inertia at all finite branch values. The quotient of the Galois closure by H gives a connected degree-d cover of the projective line, unramified away from infinity. In characteristic zero all ramification is tame. If $r \geq 1$ points lie above infinity, Riemann-Hurwitz reads $2g - 2 = -2d + (d - r) = -d - r$. Since $g \geq 0$, necessarily $d = r = 1$. Thus $H = G$: the finite transpositions generate the whole group.

Consider the graph whose vertices are the roots and whose edges are these transpositions, including their conjugates. Its connected components are the orbits of the generated group. Transitivity makes the graph connected, and edge transpositions of a connected graph generate every permutation. Hence $G = S_n$.

A transitive group containing one transposition need not be symmetric; here the transpositions generate the group.

References

[Serre2007Topics] / Theorem 4.4.5 and Proposition 4.4.6, p. 41. Inertia generation via Riemann-Hurwitz. The graph argument expands the transposition step. Source.

30.4 Example

Example / $F(X) = X^n - X$

For $n \geq 2$, the critical points satisfy $nc^{n-1} = 1$. They are nonzero and distinct, and

$$F(c) = -\frac{n-1}{n}c.$$

The critical values are therefore distinct. It follows that $X^n - X - T$ has geometric and arithmetic group S_n in characteristic zero. Over $\mathbb{Q}$, Hilbert irreducibility then supplies infinitely many rational parameters t for which $X^n - X - t$ has group S_n.

Derivatives and inertia determine the generic group, and Hilbert irreducibility gives full-group fibers. For a specified t, factorization, discriminants, or resolvents can determine its group.

References

[Serre2007Topics] / §4.4 and §3.3. The Morse theorem and specialization framework; the derivative calculation is supplied here. Source.

30.5 Specializations of a cubic

For $F = X^3 - 3X$, the critical points are ± 1, with values ∓ 2. Thus $F(X) - T$ has generic group S_3. Its discriminant is

$$\Delta(X^3 - 3X - T) = 27(4 - T^2).$$

Examples over $\mathbb{Q}$

- $t = 0$**:** $X(X^2 - 3)$ is reducible; its splitting-field group is C_2, fixing the rational root.
- $t = 1$**:** $X^3 - 3X - 1$ has no rational root and discriminant 81, so its group is $A_3 = C_3$.
- $t = 2$**:** $(X - 2)(X + 1)^2$ has a repeated root. This is a branch value, outside the separable-fiber setting.
- $t = 3$**:** $X^3 - 3X - 3$ is Eisenstein at 3, and its discriminant is -135, a nonsquare. Its group is S_3.

Even away from the branch values, specialization can lower the group or destroy irreducibility. A generic monodromy calculation does not by itself determine the group of an individual fiber.

References

[ConradCubicQuartic] / Equation (1.1) and Theorem 2.1. Cubic discriminant formula and the irreducible cubic group test; specialized identities are checked directly. Source.

[Serre2007Topics] / §4.4. The generic Morse theorem. Source.

30.6 Positive characteristic

The characteristic-zero hypothesis avoids wild ramification. In characteristic p, the polynomial $X^p - X$ has derivative -1; its cover $X^p - X = T$ is unramified at finite points, and its geometric group is the additive cyclic group of order p, acting by $X \mapsto X + a$ for $a \in \mathbb{F}_p$. Ramification at infinity is wild. The tame Riemann-Hurwitz argument used above therefore does not apply.

There are positive-characteristic versions of the Morse theorem, but their hypotheses must be stated separately. The theorem above concerns characteristic zero.

References

[Serre2007Topics] / Theorem 4.4.5; remark after Corollary 4.4.7, pp. 41-42. Positive-characteristic scope and the Artin-Schreier boundary. Source.

Questions

Do separable specializations retain the generic group?

Settled in stated scope / The characteristic-zero Morse family X^3-3X-T over $\mathbb{Q}$

No. At $T = 0$ the polynomial is reducible, and at $T = 1$ it is irreducible with group C_3. Both fibers are separable. Hilbert irreducibility gives many full-group fibers, with possible exceptional specializations.

References

[Serre2007Topics] / §3.3 and §4.4. Specialization framework; the two counterexamples are calculated in this article. Source.

Related topics

Precedes. Specialization and Hilbert irreducibility. Specialization relates the generic group to the groups of rational fibers.

Uses. Ramification and inertia. Inertia at a simple branch value is generated by a transposition.

Uses. Transitivity, primitivity, and cycle criteria. A transitive group generated by transpositions is symmetric.

Contrasts with. Composition, iteration, and arboreal groups. Composition gives intermediate covers and block systems; critical values can coincide as a result.

Constructs examples for. Inverse Galois theory and generic polynomials. The family X^n−X−T gives regular symmetric-group extensions of Q(T).

CHAPTER 31

Inverse Galois theory and generic polynomials

foundation / Intermediate / Q and number fields / Research overview

The inverse Galois problem asks which finite groups occur over a given field. Regular realizations and generic polynomials impose further conditions on families of extensions.

Overview

- What does it mean to realize a finite group by a polynomial?
- How do regular realizations and generic polynomials differ?
- Which existence results are known?

Prerequisites: Fields, irreducibility, and Galois actions; Specialization and Hilbert irreducibility

31.1 Realizations and parametrizations

The inverse Galois problem over K asks whether a given finite group G is $\mathrm{Gal}(L/K)$ for some finite Galois extension. One may also ask for an explicit polynomial whose splitting field is L. A *regular realization* over $K(T)$ requires $L \cap \overline{K} = K$. Thus the constant field remains unchanged.

A polynomial $P(\mathbf{T}, x)$ is *parametric* for G over K if its generic splitting group is G and every G-extension of K is obtained by specialization. It is *generic* if this remains true over every extension field of K. A family with group G at infinitely many parameters need not parametrize all G-extensions.

References

[JensenLedetYui2002] / §0.1, Definition 0.1.1. Parametric versus generic; existence versus construction. Source.

31.2 Regular realizations

> **Regular realization and specialization**
>
> A regular finite Galois extension of $K(T)$ with nontrivial group G, for K a number field, gives infinitely many linearly disjoint G-extensions of K by Hilbert irreducibility.

Regular realizations can be constructed using invariant fields, branched covers and rigidity, embedding problems, or arithmetic families such as torsion on elliptic curves. The cover must have the required geometric group, descend to the chosen base field, and be regular there. Hilbert irreducibility then gives arithmetic specializations. A complex cover with group G does not by itself give a G-extension of $\mathbb{Q}$.

References

[Serre2007Topics] / Proposition 3.3.3, Corollary 3.3.4, and §4.1. Linear disjointness from a regular family. Source.
[MalleMatzat2018]. Rigidity, embedding problems, and geometric constructions. Source.

31.3 Examples

Examples

If $\operatorname{char} K \neq 2$, the polynomial $x^2 - T$ is generic for C_2: every quadratic extension of every field $E \supseteq K$ has the form $E(\sqrt{a})$ for a nonsquare $a \in E$. The exceptional specializations $T = a \in E^2$ split. A generic polynomial may therefore have specializations with a smaller group.

For a cyclic cubic extension of $\mathbb{Q}$, take

$$f = x^3 + x^2 - 2x - 1.$$

The rational-root candidates ± 1 are not roots, so f is irreducible. Its discriminant is 49, hence the transitive group lies in $A_3 = C_3$ and is exactly C_3. A root is $\zeta_7 + \zeta_7^{-1}$, which generates the real cubic subfield of $\mathbb{Q}(\zeta_7)$.

References

[ConradCubicQuartic]. Cubic discriminant test; the explicit calculations are included here. Source.
[Milne2022]. Cyclotomic extensions and quadratic extensions. Author's book page.

31.4 Existence results

Every finite solvable group occurs over $\mathbb{Q}$ by Shafarevich's theorem. Symmetric and alternating groups also occur over $\mathbb{Q}$, as Hilbert proved. The general assertion for every finite group remains open in the cited sources. The regular inverse Galois problem is stronger.

An explicit realization also specifies a faithful permutation action. A transitive action of degree n corresponds to a degree-n irreducible polynomial whose splitting group acts in that way. The regular action has degree $|G|$, often much larger than the smallest degree of a faithful transitive action. Databases of number fields therefore organize groups by degree and transitive action, not only abstract isomorphism type.

References

[JensenLedetYui2002] / Theorems 0.2.1 and 0.2.4. Hilbert and Shafarevich existence theorems. Source.
[KluenersMalle2000]. Explicit realizations of transitive permutation groups. Source.
[MalleMatzat2018]. The ordinary and regular inverse problems. Source.

31.5 Limitations

A finite list of examples realizes only the listed groups and actions. Even a complete computation up to a degree bound leaves the general inverse problem unresolved. Failure to find a polynomial does not establish nonexistence. A generic polynomial requires a proof that every extension in the stated class occurs by specialization.

For a polynomial family, symmetry often bounds the Galois group from above. Irreducibility, local cycle types, and resolvents can then determine the group. A separate argument is needed to determine which extensions occur among its specializations.

References

[JensenLedetYui2002] / §0.1. The separate universality requirement. Source.
[FiekerKlueners2014]. Group determination for explicit rational polynomials. Source.

Questions

Does every finite group occur as a Galois group over Q?

Open in cited source / Finite Galois extensions of Q; no fixed degree bound

This is the classical inverse Galois problem. The cited monographs describe known realizations. For a specified group, one can also ask for a realization in a prescribed transitive action.

References

[MalleMatzat2018]. Inverse Galois problem and known realizations. Source.

Does every finite group admit a regular realization over Q(T)?

Conjecture in cited source / Regular finite Galois extensions of Q(T)

This is Serre's Conjecture 4.1.1. A positive answer would imply the ordinary inverse Galois problem by specialization. The conjecture concerns finite groups, not arbitrary profinite groups.

References

[Serre2007Topics] / Conjecture 4.1.1. Source.

Related topics

Uses. Specialization and Hilbert irreducibility. Hilbert irreducibility transfers a regular function-field construction to number fields.

Uses. Algorithms for Galois groups. An explicit realization requires a proof of its Galois group.

Uses. Resolvents and subgroup tests. Resolvents distinguish candidate permutation groups.

Contrasts with. Irreducibility over finite fields. Over a finite base field, every finite Galois group is cyclic.

Applies to. Composition, iteration, and arboreal groups. Arboreal existence problems prescribe an entire compatible tower of groups.

Part VII

Composition and iteration

CHAPTER 32

Composition, iteration, and arboreal groups

method / Intermediate / general fields and number fields / Research overview

Polynomial composition gives extension towers and Galois actions on rooted trees. Irreducibility of all iterates is weaker than surjectivity of the arboreal representation.

Overview

- When is a composition irreducible?
- Why do iterated Galois groups lie in wreath products?
- Which arithmetic conditions control the group at every level?

Prerequisites: Fields, irreducibility, and Galois actions; Transitivity, primitivity, and cycle criteria

32.1 Factorization and composition

Multiplicative factorization $f = gh$ differs from composition $f = g \circ h$. In composition theory, the factors are nonconstant maps and the units are linear changes of variable. For example, $x^4 = (x^2)\circ(x^2)$, whereas every nonconstant polynomial over $\mathbb{C}$ factors multiplicatively into linear factors. Ritt's "prime" polynomials are indecomposable under composition.

For $f \in \mathbb{C}[x]$, composition is connected to intermediate fields of $\mathbb{C}(x)/\mathbb{C}(f(x))$ and to blocks in the monodromy action of $f(x) - T$. This is a function-field group, not the splitting group of the particular polynomial $f(x)$ over $\mathbb{Q}$.

References

[Ritt1922]. Functional decomposition; consult the recorded erratum. Source.

[Mueller1995Monodromy]. Primitive monodromy of complex polynomial maps. Source.

32.2 Capelli's lemma

Theorem (Capelli's composition lemma)

Let $F \in K[x]$ be irreducible of degree m, let $g \in K[x]$ have degree $d \geq 1$, and let α be a root of F. Then $F(g(x))$ is irreducible over K if and only if $g(x) - \alpha$ is irreducible over $K(\alpha)$.

For a root β of $g(x) - \alpha$, we have $\alpha = g(\beta)$, so $K(\alpha) \subseteq K(\beta)$. The tower law gives $[K(\beta) : K] = m[K(\beta) : K(\alpha)]$. This equals $md = \deg(F \circ g)$ exactly when $g(x) - \alpha$ is a scalar multiple of a minimal polynomial of degree d. This proves both implications.

The criterion reduces the problem to factorization over a root field. For quadratic iterates, this requires deciding whether certain elements are squares in that field.

References

[Schinzel2000]. Composition and reducibility; the field-degree proof is given in full here. Source.
[Odoni1985]. Composition as the starting point for iterated Galois theory. Source.

32.3 Rooted trees and wreath products

Let $f \in K[x]$ have degree $d \geq 2$, and suppose each iterate $f^{\circ n}$ is separable. Put a separate copy of its roots at level n, with an edge from β to $f(\beta)$. The resulting regular rooted d-ary tree is preserved by Galois, because every automorphism commutes with evaluation of f.

$$G_n \;\leq\; \mathrm{Aut}(T_{d,n}) \cong [S_d]^n, \qquad |\mathrm{Aut}(T_{d,n})| = (d!)^{1+d+\cdots+d^{n-1}}.$$

The right side is an iterated wreath product, generally much smaller than S_{d^n}. Irreducibility of every iterate, called *stability*, is equivalent to transitivity at every level. Surjectivity onto the tree automorphism group is stronger.

References

[Jones2013Arboreal] / §1. The preimage tree, arboreal representation, and natural maximal group. Source.

32.4 Example

Example

For $f = x^2 + 1$, the first iterate is irreducible over $\mathbb{Q}$, and

$$f^{\circ 2} = x^4 + 2x^2 + 2$$

is Eisenstein at 2. Its four roots fall into two pairs according to whether $f(\beta) = i$ or $-i$. Thus its group is contained in $C_2 \wr C_2$, of order eight.
The quadratic critical-orbit criterion uses $c_1 = -a$ and $c_n = f_a^{\circ n}(0)$ for $n \geq 2$, where $f_a = x^2 + a$ and $a \in \mathbb{Q} \setminus \{0, -1\}$. It says that the level-n group is full precisely when $c_1, \ldots, c_n$ are independent in $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$. Here $(c_1, c_2) = (-1, 2)$; none of $-1, 2, -2$ is a rational square. Consequently $G_2 = C_2 \wr C_2$, the dihedral group of order eight. This calculation determines only the first two levels.

References

[Stoll1992] / Theorem in §1, part 1, equivalence (a)-(b). Square-class criterion for quadratic iterates; restated in Stoll2026 Proposition1.7. Source.

32.5 Postcritically finite polynomials

A polynomial is *postcritically finite* (PCF) when every critical point has finite forward orbit. For $f = x^2 - 2$, the finite critical orbit is $0 \mapsto -2 \mapsto 2 \mapsto 2$. Every iterate is nevertheless irreducible over $\mathbb{Q}$.

Proof

The identity $f(z + z^{-1}) = z^2 + z^{-2}$ gives

$$f^{\circ n}(z + z^{-1}) = z^{2^n} + z^{-2^n}.$$

For a primitive 2^{n+2}-th root of unity ζ, the element $\zeta+\zeta^{-1}$ is therefore a root of $f^{\circ n}$. It has degree $\varphi(2^{n+2})/2 = 2^n$, the degree of the iterate. All its conjugates lie in the same real cyclotomic field, so

$$K_n = \mathbb{Q}(\zeta+\zeta^{-1}), \qquad G_n \cong C_{2^n}.$$

Thus the action is transitive at every level, whereas $|G_n| = 2^n$ is much smaller than $|\operatorname{Aut}(T_{2,n})| = 2^{2^n-1}$ for $n \geq 2$.

Benedetto-Ghioca-Juul-Tucker describe the ambient groups imposed by PCF relations and give criteria for attaining them: the 2025 paper treats periodic critical points; its 2026 companion treats strictly preperiodic ones. These groups depend on critical-orbit relations, the base field, and the target point. The latter paper treats the classical Chebyshev case above separately from its general constructions.

References

[BenedettoGhiocaJuulTucker2025Periodic] / §§1.2-1.6, Theorem 1.5. Periodic PCF case; maximality is inside the constrained group M, not the full binary-tree group. Source.

[BenedettoGhiocaJuulTucker2026Preperiodic] / §1.4; Theorems 1.2 and 7.6. §1.4 explicitly separates x^2-2 and records the Chebyshev identity; the cyclotomic calculation at target zero is supplied here. Source.

32.6 Generic and arithmetic groups

For the generic monic degree-d polynomial $F = x^d+s_{d-1}x^{d-1}+\cdots+s_0$ over a characteristic-zero field k, Odoni proves $\operatorname{Gal}(F^{\circ n}/k(\mathbf{s})) = [S_d]^n$ for every fixed n. Juul extends this to positive characteristic, excluding $(d, \operatorname{char} k) = (2,2)$ for polynomials. The induction uses disjoint ramification in the new branches to obtain independent permutations.

Over number fields, Benedetto-Juul prove all-level existence in every even degree and in odd degree when $[K:\mathbb{Q}]$ is odd. Specter's 2018 preprint states all degrees over every number field. In contrast, Dittmann-Kadets construct characteristic-zero Hilbertian fields admitting no finite-index arboreal image. Thus the existence theorem depends on the base field.

References

[Odoni1985]. Characteristic-zero generic theorem. Source.

[Juul2019] / Theorem 1.1. Positive-characteristic exception and generic full wreath products. Source.

[BenedettoJuul2019] / Abstract and main theorem. Published number-field cases. Source.

[Specter2018] / Version 2. All-degree number-field statement, labeled as a preprint. Source.

[DittmannKadets2022] / Theorem 1.2. General Hilbertian-field counterexample. Source.

32.7 Finite-level detection

Preprint submitted 10 November 2026. Stoll gives revised proofs and rational-parameter extensions of maximality criteria for x^2+a. Proposition 1.7 restates the square-class test. Theorem 6.1 gives explicit regions and congruence classes where maximality at level two implies maximality at every level. Its hypotheses do not include every rational a.

The paper also formulates a uniform finite-level detection conjecture and reports supporting computations. These finite calculations do not prove the conjecture for all parameters. Relations among critical-orbit square classes can cause the first failure of maximality at a later level.

Example

For $a = 2/3$, the first four adjusted critical values are

$$(c_1, c_2, c_3, c_4) = \left(-\frac{2}{3}, \frac{10}{9}, \frac{154}{81}, \frac{28090}{6561}\right).$$

The first three square classes are independent: the sign detects c_1, the prime 5 detects c_2, and the prime 7 detects c_3. But $c_2 c_4 = (530/243)^2$. Hence the level-three group is full and the level-four group is not. A universal detection level, if one exists, must therefore be at least four.

References

[Stoll2026Arboreal] / Proposition 1.7, Theorem 6.1, Proposition 8.2, and Remark 8.5. arXiv:2611.14014v1, 10 November 2026. The a=2/3 relation is checked directly; it is an exact obstruction, not evidence from a finite numerical search. Source.

Questions

Does a universal finite level detect maximality for x^2+a over Q?

Conjecture in cited source / a∈Q, base point 0, full binary preimage tree

Stoll's Conjecture 8.1 asks whether one integer N suffices: if the level-N representation is surjective, must the full arboreal representation be surjective? The 2026 preprint discusses evidence for $N = 4$, not a proof for all rational parameters. The exact example $a = 2/3$ above rules out every $N < 4$. Changing the base point gives a different question.

References

[Stoll2026Arboreal] / Conjecture 8.1 and Remark 8.5. Source.

Does every Hilbertian field admit a surjective arboreal representation?

Settled in stated scope / General Hilbertian fields of characteristic zero

No. Dittmann-Kadets give counterexample fields. These counterexamples are compatible with the positive existence results over number fields and with generic maximality at each fixed finite level.

References

[DittmannKadets2022] / Theorem 1.2. Source.

Related topics

Uses. Specialization and Hilbert irreducibility. Finite-level generic groups specialize by Hilbert irreducibility. A single parameter valid at every level requires an additional argument.

Uses. Transitivity, primitivity, and cycle criteria. Blocks and wreath products identify the ambient action.

Uses. Discriminants, parity, and ramification. Square classes and ramification control the quadratic extensions introduced at each level.

Generalizes. Inverse Galois theory and generic polynomials. The target is a compatible tower of prescribed finite groups.

Uses. Newton polygons and Galois groups. Valuations can prove irreducibility or produce new ramification at each level.

Part VIII

Finite fields

CHAPTER 33

Irreducibility over finite fields

method / Start here / finite fields / Classical theory

Frobenius powers give an exact irreducibility criterion and a formula for the number of irreducible polynomials. Primitive polynomials satisfy an additional multiplicative-order condition.

Overview

- How can irreducibility be certified without a full factorization?
- How many monic irreducibles of a given degree exist?
- Why is the Galois group cyclic, and what does primitive mean?

Prerequisites: Fields, irreducibility, and Galois actions

33.1 The Frobenius identity

Let q be a prime power. In $\mathbb{F}_q[x]$,

$$x^{q^m} - x = \prod_{\substack{P \text{ monic irreducible} \\ \deg P \mid m}} P(x).$$

Every element of $\mathbb{F}_{q^m}$ is a root, the derivative is -1, and an element of degree d over $\mathbb{F}_q$ belongs to $\mathbb{F}_{q^m}$ exactly when $d \mid m$. Thus each indicated factor occurs once.

If f is irreducible of degree n, adjoining one root gives $\mathbb{F}_{q^n}$, which already contains every conjugate. Its Galois group is C_n, generated by $z \mapsto z^q$. Noncyclic monodromy groups over finite fields arise over function fields such as $\mathbb{F}_q(T)$, not this constant-field extension.

References

[LidlNiederreiter1997]. Finite-field structure and irreducible polynomials. Source.

[GuralnickZieve2010]. A contrasting function-field monodromy setting. Source.

33.2 The irreducibility criterion

Theorem (Frobenius-power criterion)

A monic polynomial $f \in \mathbb{F}_q[x]$ of degree $n \geq 1$ is irreducible if and only if

$$x^{q^n} \equiv x \pmod{f}, \qquad \gcd(f, x^{q^{n/\ell}} - x) = 1 \quad \text{for every prime } \ell \mid n.$$

The first condition makes f squarefree and forces every factor degree to divide n. If a proper divisor $d \mid n$ occurred as a factor degree, some prime $\ell \mid n/d$ would give $d \mid n/\ell$, contradicting the corresponding gcd. The reverse implication follows from the same Frobenius identity. The criterion uses modular exponentiation and Euclidean gcds, without expanding the powers as polynomials.

References

[Rabin1980]. Irreducibility testing in finite-field algorithms. Source.

[FLINTFactorizationDocumentation] / Irreducibility testing over finite fields, Theorem 1. Official documentation states the criterion and its exact algorithmic use. Source.

33.3 Example

Example

Take $f = x^4 + x + 1 \in \mathbb{F}_2[x]$. Repeated squaring modulo f gives

$$x^4 \equiv x + 1, \qquad x^8 \equiv x^2 + 1, \qquad x^{16} \equiv x.$$

The only prime divisor of $n = 4$ is 2, and $x^4 - x \equiv 1 \pmod f$. Thus the required gcd is one, and f is irreducible. Its splitting field has 16 elements, and the Galois group over $\mathbb{F}_2$ has order four.

The congruence $x^{16} \equiv x \pmod f$ alone does not suffice: the reducible quartic $x^4 + x = x(x+1)(x^2+x+1)$ also divides $x^{16} - x$. The gcd conditions eliminate smaller-degree factors.

References

[Rabin1980]. Application of the exact criterion; the displayed remainders are direct calculations. Source.

33.4 Counting irreducible polynomials

If $N_q(n)$ counts monic irreducibles of degree n, taking degrees in the Frobenius identity and applying Möbius inversion gives

$$N_q(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) q^{n/d}.$$

For example, $N_2(4) = (16 - 4)/4 = 3$. For fixed q and large n, the proportion of monic degree-n polynomials that are irreducible is asymptotic to $1/n$. Random sampling followed by exact testing gives a construction algorithm. Shoup and others also study deterministic constructions and faster algorithms.

References

[LidlNiederreiter1997]. Counting formula from Möbius inversion. Source.

[Shoup1990Irreducible]. Algorithms for constructing irreducible polynomials. Source.

[Shoup1994]. Fast constructions. Source.

33.5 Primitive polynomials and prescribed coefficients

An irreducible polynomial of degree n is *primitive* if a root generates $\mathbb{F}_{q^n}^{\times}$, a group of order $q^n - 1$. Primitivity concerns the multiplicative order of a root; the Galois group remains cyclic, with order equal to the degree of the polynomial. For the quartic above, the root α has order 15: $\alpha^3 \neq 1$ and $\alpha^5 = \alpha^2 + \alpha \neq 1$, excluding the proper maximal divisors 5 and 3 of 15.

Prescribed coefficients, sparse support, and primitivity impose different restrictions on the counting problem. The Hansen-Mullen primitivity problem is stronger than its irreducibility problem. Wan's theorem and the finite checks of Ham-Mullen concern irreducibility; Tuxanidy-Wang give a later proof. Results on several prescribed coefficients have restrictions on the parameters. In particular, a zero constant coefficient precludes irreducibility in degree greater than one.

References

[HansenMullen1992]. Primitive polynomials and prescribed coefficients. Source.

[Wan1997]. Hansen-Mullen irreducibility theorem outside finitely many cases. Source.

[HamMullen1998]. Completion of the remaining finite cases. Source.

[TuxanidyWang2018]. A later proof of the irreducibility theorem. Source.

[Pollack2013Prescribed]. Several prescribed coefficients; the range is approximately a square root of the degree, not the full degree. Source.

33.6 Discriminants and factor parity

For a squarefree polynomial of degree n with r irreducible factors over a finite field of odd order, Frobenius has sign $(-1)^{n-r}$. Its discriminant is a square exactly when $n-r$ is even. This is the Stickelberger discriminant-parity theorem. Swan develops its characteristic-two lifting form and applies it to trinomials. The required parity may still allow several factors, so it does not prove irreducibility.

References

[Swan1962] / Theorem 1 and added-in-proof attribution, pp.1099-1100. The basic theorem is credited by Swan to Stickelberger, with Dalen's prior proof noted. Source.

Questions

Which prescribed supports admit irreducibles over a fixed finite field?

Research direction / A specified finite field, degree, and support

The counting formula proves existence without coefficient restrictions. A prescribed support or coefficient pattern may force factors, and existence must then be studied for that pattern. Exact certificates can establish individual cases; a general result requires a uniform argument.

References

[HansenMullen1992]. Source.

[Swan1962]. Source.

[MullenPanario2013]. Source.

Related topics

Supplies examples for. Reduction modulo primes. Finite-field certificates can prove rational irreducibility at good primes.

Supplies examples for. Frobenius and factorization modulo primes. Finite-field factor degrees record global Frobenius cycle types.

Uses. Polynomial factorization algorithms. Modular exponentiation and gcds are shared algorithmic components.

Contrasts with. Inverse Galois theory and generic polynomials. Only cyclic groups occur over a finite constant field.

Uses. Discriminants, parity, and ramification. Factor-count parity is encoded by the discriminant in odd characteristic.

CHAPTER 34

Irreducible polynomials with coefficient restrictions

method / Intermediate / finite fields / Published results

Frobenius orbits count irreducible polynomials over finite fields and give construction algorithms. Prescribed coefficients, allowed coefficient sets, and factorization statistics lead to distinct counting problems.

Overview

- How many monic irreducibles of degree n exist over a finite field?
- How does counting yield an exact construction algorithm?
- Which arbitrary coefficients can be prescribed simultaneously?
- How does restricting every coefficient differ from prescribing selected coefficients?
- Why do factor degrees resemble permutation cycles, and in which limit?

Prerequisites: Fields, irreducibility, and Galois actions; Irreducibility over finite fields

34.1 Counting irreducible polynomials

Let $N_q(n)$ count monic irreducibles of degree n over $\mathbb{F}_q$, where q is a prime power. Every element of $\mathbb{F}_{q^m}$ is a root of $x^{q^m} - x$, whose derivative is -1. A monic irreducible divides this polynomial exactly when its degree divides m.

Derivation

$$q^m = \sum_{d \mid m} d N_q(d), \qquad N_q(n) = \frac{1}{n} \sum_{d \mid n} \mu(d) q^{n/d}.$$

The second identity is Möbius inversion; μ is the Möbius function. Every proper divisor of n is at most $n/2$, giving $N_q(n) = q^n/n + O(q^{n/2}/n)$, uniformly for $q \geq 2$. Thus a uniform monic polynomial is irreducible with probability asymptotic to $1/n$ as $n \to \infty$, even for fixed q.

Example

$$N_2(6) = \frac{64 - 8 - 4 + 2}{6} = 9.$$

Thus nine of the 64 monic binary sextics are irreducible. All nine have constant coefficient one, so nine of the 32 constant-one sextics are irreducible.

References

[LidlNiederreiter1997]. Finite-field structure and irreducible-polynomial counting; the derivation is included explicitly. Source.

[Pollack2013Prescribed] / §1 and Lemma 4. Counting normalization and the standard estimate for degree-n irreducibles. Source.

34.2 Construction algorithms

Independent uniform sampling of monic polynomials, followed by the exact Frobenius-power irreducibility test, gives a Las Vegas construction. Sampling continues until the test proves irreducibility; the runtime is random and the output is certified. For binary sextics, sampling only constant-one candidates takes exactly $32/9$ trials on average.

If an extension $\mathbb{F}_{q^n}$ is already represented, choose α outside its maximal proper subfields. Equivalently, require $\alpha^{q^{n/\ell}} \neq \alpha$ for every prime $\ell \mid n$. Its minimal polynomial is then $\prod_{i=0}^{n-1}(x-\alpha^{q^i})$. This construction assumes a representation of the extension field.

Shoup develops deterministic constructions, including bounds polynomial for small characteristic, and faster construction methods. Polynomial dependence on the characteristic p need not be polynomial in its bit length $\log p$.

References

[Rabin1980]. Probabilistic finite-field algorithms, with exact irreducibility verification. Source.

[Shoup1990Irreducible] / Introduction. Deterministic constructions and dependence on the characteristic. Source.

[Shoup1994]. Faster construction of irreducible polynomials of prescribed degree. Source.

34.3 One prescribed coefficient

Theorem (Hansen-Mullen)

Given $n \geq 2$, $0 \leq j < n$, and $c \in \mathbb{F}_q$, a monic irreducible degree-n polynomial with coefficient $[x^j]f = c$ exists, except precisely when $j = 0, c = 0$, or when q is even and $(n, j, c) = (2, 1, 0)$.

The exceptional cases follow from elementary obstructions: a zero constant term supplies the factor x; in characteristic two, $x^2 + c$ is a square because finite fields are perfect. For $n \geq 3$, the constant-term obstruction is the only one.

Wan proved a large-parameter range and Ham-Mullen finished the remaining cases. Tuxanidy-Wang give another proof: sufficiently long periods of finite-field discrete Fourier transforms force elements of full degree in supports of functions encoding elementary symmetric coefficients. Irreducibility here does not require a root to generate the multiplicative group.

References

[Wan1997]. The large-parameter part of the Hansen-Mullen irreducibility theorem. Source.

[HamMullen1998]. Completion of the remaining cases. Source.

[TuxanidyWang2018] / Theorem 1.1 and Introduction. Exact single-coefficient statement, exceptions, and the Fourier-period proof mechanism. Source.

34.4 Several prescribed coefficients

Let $I \subset \{0, \dots, n-1\}$ have size r; prescribe arbitrary $a_i \in \mathbb{F}_q$, requiring $a_0 \neq 0$ if $0 \in I$. Pollack proves, for fixed $\varepsilon > 0$ and $r \leq (1-\varepsilon)\sqrt{n}$,

$$\#\{f \text{ irreducible with these coefficients}\} \sim \sigma N_q(n), \quad \sigma = \begin{cases} q^{-r}, & 0 \notin I, \\ q^{1-r}/(q-1), & 0 \in I, \end{cases}$$

as $n \to \infty$, uniformly in the field and prescriptions.

Theorem (Ha)

For every prime power q, if $n \geq 52$ and $r \leq n/10$, every such prescription is realized by a monic irreducible. Thus any six chosen coefficients of a degree-60 polynomial over $\mathbb{F}_2$ can be prescribed, subject to the constant-term condition.

Ha also obtains $\lfloor (1/4 - \varepsilon)n \rfloor$ prescriptions for $n \geq 8$, $0 < \varepsilon < 1/4$, and $q \geq q_0(\varepsilon)$. The proofs use exponential-sum estimates and the function-field circle method, with a positive main term larger than the error. The conclusions apply within the stated ranges for the number of prescribed coefficients.

References

[Pollack2013Prescribed] / Theorem 1 and Corollary 2. Uniform growing-degree asymptotic, including the constant-coefficient normalization. Source.

[Ha2016] / Theorem 1.3, Corollary 1.2(2), and Introduction. Explicit all-fields linear range, large-field quarter-degree range, and proof strategy. Source.

34.5 Forbidden coefficient values

Prescribing a few coefficients and restricting *all* coefficients to large sets are different problems. Porritt treats a common forbidden set using the function-field circle method. Cheng allows the forbidden set to depend on the coefficient position.

Theorem / Cheng, fixed field and growing degree

Fix a prime power $q \geq 300$. For each $0 \leq i < n$, let $R_i \subseteq \mathbb{F}_q$ have $|R_i| \leq \sqrt{q}/2$, and put $A_i = \mathbb{F}_q \setminus R_i$. The number of monic irreducibles $x^n + \sum_{i<n} a_i x^i$ with $a_i \in A_i$ satisfies

$$\#\mathcal{J} = \frac{q}{n(q-1)} \prod_{i=0}^{n-1} |A_i| \left(\Delta + O(q^{-\sqrt{n/45}})\right), \qquad \Delta = \frac{|A_0 \setminus \{0\}|}{|A_0|},$$

for sufficiently large n. The cited theorem gives an explicit threshold. The estimate applies to allowed sets satisfying the stated size condition.

The factor Δ removes the constant-zero obstruction. For example, if every coefficient must be nonzero, the count is asymptotic to $q(q-1)^{n-1}/n$. If each position must avoid a specified value c_i, multiply this main term by 1 when $c_0 = 0$, and by $(q-2)/(q-1)$ otherwise. These counts apply as the degree tends to infinity.

References

[Porritt2019Restricted] / Theorem 1 and §4. Common forbidden set; q≥500, |R|<$\sqrt{q}/2$, n≥100(log q)^2 in the displayed theorem. Source.

[Cheng2025Restricted] / Theorem 1.1 and Corollary 1.3. Position-dependent forbidden sets, exact constant-term factor, and explicit sufficient threshold. Source.

34.6 Square coefficients

Theorem / Bary-Soroker-Shmueli, published 2026

Fix $n \geq 1$. Choose the n nonleading coefficients of a monic degree-n polynomial independently and uniformly from the *set of square values* $S_q = \{u^2 : u \in \mathbb{F}_q\}$, including zero. Then

$$\Pr(f \text{ is irreducible over } \mathbb{F}_q) = \frac{1}{n} + O_n(q^{-1/2}), \qquad q \longrightarrow \infty.$$

For odd q, uniform sampling from S_q differs from squaring a uniform element of $\mathbb{F}_q$: the latter gives zero half the weight of each nonzero square. For even q, squaring is a bijection and this is the unrestricted model.

The proof uses finite-field Chebotarev and disjointness of suitable function-field extensions. The paper also handles coefficient sets of bounded algebraic complexity. Its fixed-degree error term does not establish a fixed-field, growing-degree analogue.

Example

Over $\mathbb{F}_5$, the square values are $0, 1, 4$. The polynomial $x^2 + x + 1$ has square coefficients and is irreducible because its discriminant is 2, a nonsquare. By contrast, $x^2 + 1 = (x - 2)(x + 2)$. Thus square coefficients alone do not imply irreducibility.

References

[BarySorokerShmueli2026Squares] / Theorem 1; Theorem 2 and §§3-5 for the broader method. Monic polynomial, independent uniform square values, fixed n, q→∞; the O-constant depends on n. Source.

34.7 Factorization statistics

For a partition $\lambda = 1^{m_1}2^{m_2}\cdots$ of n, unique factorization gives the exact number of squarefree monic polynomials of type λ:

$$\prod_{j=1}^{n} \binom{N_q(j)}{m_j} \sim \frac{q^n}{z_\lambda}, \qquad z_\lambda = \prod_j j^{m_j} m_j!, \qquad q \to \infty \quad (n \text{ fixed}).$$

Indeed, choose the distinct factors of each degree independently. The factor $1/z_\lambda$ is also the probability of cycle type λ in S_n.

Theorem (Bank-Bary-Soroker-Rosenzweig)

Let f be monic of degree n and $3 \leq m < n$. Bank-Bary-Soroker-Rosenzweig prove that among $f + \{h : \deg h \leq m\}$, the number with factorization type λ, counting factor degrees with multiplicity, equals

$$\frac{q^{m+1}}{z_\lambda} + O_n(q^{m+1/2}).$$

The proof identifies S_n as geometric monodromy of the parameter family and translates Frobenius cycle types into point counts. This group belongs to a function-field cover; an individual finite-field splitting extension still has cyclic Galois group.

Remark

The interval estimate has a constant depending on n; its stated asymptotic fixes n and lets $q \to \infty$. It does not establish the corresponding fixed-q, growing-degree assertion. The unrestricted $1/n$ probability in that regime follows separately from Möbius inversion.

References

[Cohen1970]. Foundational use of function-field Galois groups to study factorization distributions. Source.

[BankBarySorokerRosenzweig2015] / Theorem 2.3 and §4. Uniform interval estimate for 3≤m<n and its geometric-monodromy proof; smaller m requires extra hypotheses. Source.

Questions

Can any one coefficient be prescribed?

Settled in stated scope / Monic degree n≥2 polynomials over every finite field

Yes, with exactly the constant-zero and even-characteristic quadratic exceptions stated above. Primitive-polynomial prescriptions are a different problem with additional multiplicative conditions.

References

[TuxanidyWang2018] / Theorem 1.1. Source.

Which coefficient positions allow stronger bounds?

Research direction / Fixed finite field, growing degree, and a specified family of prescription patterns

For a specified pattern of positions and values, one can ask for a larger proportion of prescribed coefficients, an asymptotic count, or an efficient construction. The bounds above establish sufficient ranges; they need not be optimal or the strongest known for every pattern.

References

[Ha2016] / Theorem 1.1 and Corollary 1.2. Source.

[Pollack2013Prescribed] / Corollary 2. Source.

Can degree and coefficient restrictions vary simultaneously?

Research direction / A specified sequence of fields, degrees, and allowed coefficient sets

The problem depends on which of q, n and the sets A_i vary. Cheng's theorem allows increasing degree with large allowed sets, whereas the square-coefficient theorem fixes the degree. For a specified model, one can seek estimates uniform in both parameters or compare the ranges of these results. This formulation does not assert that every intermediate regime is open.

References

[Cheng2025Restricted] / Theorem 1.1. Source.

[BarySorokerShmueli2026Squares] / Theorems 1-2. Source.

Related topics

Uses. Irreducibility over finite fields. Frobenius-power tests certify the output of construction algorithms.

Generalizes. Frobenius and factorization modulo primes. Cycle types record factor degrees both for arithmetic reduction and for finite-field parameter families.

Applies to. Polynomial factorization algorithms. Extension-field representations and irreducibility testing are basic inputs to factorization algorithms.

Contrasts with. Random polynomials: models and methods. Finite-field models distinguish increasing field size from increasing degree; neither is the integer-coefficient model.

Supplies examples for. Specialization and Hilbert irreducibility. The geometric group of a finite-field family can differ from the groups of its specializations.

Part IX

Algorithms and certificates

CHAPTER 35

Polynomial factorization algorithms

algorithm / Intermediate / Finite fields, Q, and p-adic fields / Published results

Finite-field factorization uses Frobenius maps and linear algebra. Factorization over the rationals combines modular factorization, Hensel lifting, and lattice reduction.

Overview

- Which methods underlie exact polynomial factorization?
- Why is modular factorization followed by recombination?
- How can a randomized computation still produce exact factors?
- How do recent modular-composition advances relate to factorization?
- How are local factors found when reduction has repeated factors?

Prerequisites: Reduction modulo primes; Irreducibility over finite fields

35.1 Finite-field factorization

Let the polynomial have coefficients in $\mathbb{F}_q$. A squarefree decomposition separates repeated factors, using gcds with the derivative. If the derivative vanishes, the polynomial is a p-th power over the finite field and one first takes its p-th root.

For a squarefree polynomial, $\gcd(f, x^{q^d} - x)$ selects its irreducible factors whose degrees divide d. Removing the factors already found and repeating the gcd calculation gives a distinct-degree decomposition. Equal-degree splitting then separates factors of the same degree. Cantor-Zassenhaus uses random choices for this splitting step; the output consists of exact polynomial factors whose product and irreducibility can be checked.

References

[CantorZassenhaus1981]. Source.

[GathenGerhard2013]. Squarefree, distinct-degree, and equal-degree factorization. Source.

35.2 Berlekamp's algorithm

Proposition

If f is squarefree with r distinct irreducible factors, then

$$\dim_{\mathbb{F}_q} \{h \in \mathbb{F}_q[x]/(f) : h^q = h\} = r.$$

Indeed, the Chinese remainder theorem identifies the quotient with a product $\prod_{i=1}^{r} \mathbb{F}_{q^{d_i}}$. The fixed elements of Frobenius on each component are exactly $\mathbb{F}_q$. Finding the fixed subspace is linear algebra. A fixed element assumes constant values on the components; gcds $\gcd(f, h-c)$, for $c \in \mathbb{F}_q$, separate components with different values.

Example

Over $\mathbb{F}_2$, take $f = x^4 + x$. Since $f' = 1$, it is squarefree. In the quotient, $x^4 = x$, and solving $h^2 = h$ gives the basis $1, x^2 + x, x^3$. Thus there are three irreducible factors. The gcds from $h = x^3$ first split off x; those from $h = x^2 + x$ split the remaining cubic. The result is

$$x^4 + x = x(x + 1)(x^2 + x + 1).$$

The quadratic has no root in $\mathbb{F}_2$, which completes the factorization.

References

[Berlekamp1967]. Frobenius-fixed algebra and the linear-algebra factorization method. Source.

35.3 Modular composition

The operation *modular composition* takes $a, g, f \in K[x]$, with f monic of degree n and $\deg a, \deg g < n$, and computes $g(a(x)) \bmod f(x)$. It avoids storing a composition whose unreduced degree can be nearly n^2.

Frobenius as modular composition

In $A = \mathbb{F}_q[x]/(f)$, put $a = x^q \bmod f$. For every $h \in \mathbb{F}_q[x]$,

$$h(x)^q \equiv h(a(x)) \pmod{f}.$$

The coefficients are fixed by the q-power map. Thus one application of Frobenius is a modular composition. Iterates of this map supply the powers used in distinct-degree factorization and irreducibility tests.

Example

Over $\mathbb{F}_2$, take $f = x^3+x+1$. Since $x^3 = x+1$ in the quotient, applying Frobenius to $h = x^2+x$ gives

$$h^2 = x^4 + x^2 = (x^2 + x) + x^2 = x.$$

Equivalently, compose h with $a = x^2$ and reduce modulo f.

Brent-Kung's baby-step/giant-step method groups powers of a into blocks and combines them by matrix multiplication. Later methods encode small algebraic relations between x and a in polynomial matrices, or reduce the task to multivariate evaluation. Both approaches avoid forming the full intermediate composition.

References

[BrentKung1978] / Composition algorithms; see also Neiger-Salvy-Schost-Villard (2024), Section 1.1, for modular composition. Source.

[KedlayaUmans2011] / Sections 3 and 7. Source.

[NeigerSalvySchostVillard2024] / Section 1.1 and Theorem 1.1. Source.

35.4 Complexity bounds

A *field-operation count* charges unit cost for arithmetic in K. A *bit-operation count* includes the representation and size of coefficients. Over $\mathbb{F}_q$, these are not interchangeable uniformly in q. Modular composition is one step of a factorization algorithm.

Source	Precise scope	Contribution
Kedlaya-Umans (2011)	Modular composition over $\mathbb{F}_q$; bit complexity	For each fixed $\varepsilon > 0$, $n^{1+\varepsilon}\log^{1+o(1)} q$ bit operations in sufficiently large degree. This leads to a randomized finite-field factorization bound, displayed below.
Neiger-Salvy-Schost-Villard (2024)	Arbitrary fields of known characteristic; arithmetic complexity	A Las Vegas method with expected $O(n^{1.43})$ field operations. Small finite fields are handled through a suitably enlarged extension; a failed randomized trial is detected.
Neiger-Salvy-Schost-Villard (2026)	Fixed f, generic inner polynomial a; arithmetic complexity	Two relation matrices give $\widetilde{O}(n^{(\omega+3)/4})$ field operations. Here ω is a feasible square-matrix multiplication exponent. The genericity requirement is essential.
van der Hoeven-Lecerf (2026)	Modular composition over $\mathbb{Z}/r\mathbb{Z}$; bit complexity	A sharper nearly linear bound via spiroids and Fourier transforms with proved precision control. Prime finite fields are a special case.

The Kedlaya-Umans randomized *factorization* bound is

$$O\big(n^{3/2+o(1)}\log^{1+o(1)} q + n^{1+o(1)}\log^{2+o(1)} q\big).$$

A faster modular-composition subroutine does not by itself turn this exponent into 1. Guo-Narayanan-Umans (2016) organize related algebraic problems through reductions that would transfer an improvement below the factorization exponent 3/2. These reductions do not establish an optimal lower bound.

Conditional approach. Umans-Wang (2025) propose structured sets of integers with a divisibility-covering property as a route to faster factorization. Their proposed improvement depends on a number-theoretic conjecture; it is not an unconditional algorithm beating 3/2. The paper was listed for ISAAC in December 2026 when checked on 12 November; the reference below identifies the available preprint.

References

[KedlayaUmans2011] / Corollary 7.2 and finite-field factorization results. Source.

[NeigerSalvySchostVillard2024] / Theorem 1.1 and the following discussion of sampling and small finite fields. Source.

[NeigerSalvySchostVillard2026] / Theorem 1.3 and the paragraph immediately following it. Source.

[HoevenLecerf2026] / Theorem 1 and Sections 1.3-1.4; author version dated 12 August 2026. Source.

[GuoNarayananUmans2016] / Introduction and reduction theorems. Source.

[UmansWang2025] / Abstract and introduction; conditional strategy. Source.

35.5 Hensel lifting and recombination

Let $f \in \mathbb{Z}[x]$ be monic and squarefree. Choose a prime where its reduction remains squarefree, factor there, and lift the pairwise coprime factors modulo $p, p^2, p^4, \ldots$ by Hensel's lemma. Coprimality permits the correction equations to be solved at each stage.

A rational factor may be a product of several lifted factors. Recombination determines which products descend to $\mathbb{Z}[x]$. Bounds for the coefficients of a genuine factor turn sufficiently accurate modular coefficients into unique candidate integers. Exact division then accepts or rejects a candidate.

Reducibility modulo p alone gives no rational factor, since the local factors need not descend separately. An irreducible good reduction, however, proves rational irreducibility immediately.

References

[Zassenhaus1969]. Source.

[GathenGerhard2013]. Hensel lifting, coefficient bounds, and modular factorization. Source.

35.6 Local factorization and higher Newton polygons

Local factorization over $\mathbb{Q}_p$ remains possible when every factor of $\bar{f}$ is repeated. A residual polynomial separates roots that share a Newton slope; higher Newton polygons continue when that residual information is insufficient. The Montes method organizes successive slopes, residue-field extensions, and polynomial representatives into *types*. Complete types identify irreducible local factors and their ramification indices and residue degrees.

Guàrdia-Montes-Nart give the underlying theory and an algorithm for prime ideal decomposition and discriminants. Poteaux-Weimann's ISSAC 2022 paper improves local factorization through Hensel lifting adapted to these valuations and a divide-and-conquer strategy. Its sharper complexity bound assumes residue characteristic zero or greater than the polynomial degree; residual-field factorization costs are accounted for separately.

A local factor is represented to a stated p-adic precision together with exact structural data; its coefficients need not be rational. Recovering rational factors still requires global recombination and certification.

References

[GuardiaMontesNart2012Newton] / §§2-4. Higher-order polygon and residual-polynomial theory. Source.

[GuardiaMontesNart2011Montes] / §2.2, Theorem 2.2; §3. Complete types and the optimized Montes method. Source.

[PoteauxWeimann2022Local] / Assumption 1; §§4-5, Theorems 3-4. Published algorithmic improvements; assumptions and residual-field costs remain explicit. Source.

35.7 Lattice reduction

The number of subsets of lifted factors is exponential in the number of factors. The Lenstra-Lenstra-Lovász theorem gives a deterministic algorithm for factoring rational polynomials in time polynomial in their degree and coefficient bit size, using lattice reduction. The lattice encodes congruence information; short vectors reveal relations compatible with a genuine factor.

Van Hoeij recasts practical recombination through a structured knapsack problem. Modern variants, including FLINT's implementation, use logarithmic derivatives to turn products into sums, so a prospective factor corresponds to a vector of choices among local factors. Lattice reduction narrows the possible choice vectors before expensive exact tests. Belabas-van Hoeij-Klüners-Steel (2009) proved polynomial-time complexity for van Hoeij's original rational-polynomial algorith-

m and natural variants. This recombination method uses local information to find exact global factors.

References

[LLL1982]. Polynomial-time factorization over Q. Source.

[VanHoeij2002]. Recombination by knapsack and lattice methods. Source.

[BHKS2009]. Polynomial-time analysis of van Hoeij's original rational-polynomial factorization algorithm and natural variants. Source.

[FLINTFactorizationDocumentation] / Factoring polynomials over the integers: Van Hoeij's algorithm and Complexity. Implementation-oriented explanation of modular factorization. Source.

35.8 Factorization certificates

The identity $f = c \prod_i g_i^{e_i}$ proves a factorization, but not its completeness. Completeness additionally requires proof that each g_i is irreducible over the stated field. A modular irreducibility witness, a proved algorithm, or a suitable exact criterion can supply that step. A complete certificate specifies the coefficient field, normalization, multiplicities, and irreducibility proofs.

In a Las Vegas algorithm, random choices affect the runtime, while every returned factorization is correct. Numerical methods require exact verification before their output gives a factorization certificate.

References

[GathenGerhard2013]. Source.

[Rabin1980]. Finite-field algorithms and exact irreducibility tests. Source.

Questions

Is irreducibility over Q decidable in polynomial time?

Settled in stated scope / Dense univariate polynomials given by rational coefficients in binary

Yes. LLL gives polynomial-time factorization in the degree and coefficient bit size. The bound applies to dense binary input; it does not give the same complexity for every sparse encoding or extension-field representation.

References

[LLL1982]. Source.

Can the 2026 generic composition bound be extended to arbitrary input?

Research direction / Algebraic modular composition over a field; the two-relation-matrix method

For each degree-n modulus f, Theorem 1.3 supplies a nonempty Zariski-open set of coefficient vectors for a on which the faster bound holds. The algorithm can detect failure of its genericity conditions and fall back on another method. The authors explicitly leave a Las Vegas reduction for arbitrary input to future work. A nonempty Zariski-open set over an algebraic closure need not contain a point over a small finite base field; "generic" must not be silently replaced by "every."

References

[NeigerSalvySchostVillard2026] / Theorem 1.3 and following paragraph. Source.

Related topics

Uses. Irreducibility over finite fields. Frobenius tests certify the finite-field factors.

Generalizes. Reduction modulo primes. An irreducible good reduction can terminate a rational factorization algorithm.

Applies to. Algorithms for Galois groups. A group computation begins with exact factorization and separability checks.

Uses. Newton polygons and Galois groups. Residual polynomials and higher Newton polygons determine local factorization data.

CHAPTER 36

Algorithms for Galois groups

algorithm / Intermediate / Q, number fields, and p-adic fields / Published results

Exact factorization, modular cycle types, and relative resolvents determine Galois groups. A certified computation records the permutation action and the data for its decisive tests.

Overview

- What data certify a computed Galois group?
- How are resolvents combined with modular factorizations?
- What are the software limits and certification requirements?

Prerequisites: Frobenius and factorization modulo primes; Resolvents and subgroup tests; Polynomial factorization algorithms

36.1 Input and permutation action

The input is a polynomial and a ground field. For $f \in \mathbb{Q}[x]$, clear denominators, remove content, and compute a squarefree factorization. A transitive-group label describes the action for an irreducible factor. If f is reducible, its whole splitting field is a compositum; the group need not be the direct product of the groups of the factors because their splitting fields can intersect.

For an irreducible degree-n input, the output should specify the subgroup of S_n up to conjugacy, together with its action on the roots. An abstract isomorphism type alone may lose information needed for cycle types, resolvents, subfields, or iteration.

References

[Cox2012]. Source.
[FiekerKlueners2014] / Introduction and Section 2. Source.

36.2 Subgroup descent

1. **Irreducibility and separability.** A factorization or irreducibility certificate establishes the input assumptions.
2. **Structural restrictions.** The discriminant, reciprocity, composition, and known intermediate fields restrict the ambient permutation group.
3. **Local factorizations.** Exact irreducible factorizations at good primes determine cycle types.
4. **Subgroup tests.** Primitive-group theorems, relative resolvents, and subgroup descent eliminate incompatible candidates.
5. **Certification.** The decisive tests require exact verification and retained proof data.

Stauduhar's method makes the fourth step systematic. Starting from a known overgroup, it tests relative invariants for maximal subgroups and descends whenever a certified resolvent condition permits it.

References

[Stauduhar1973]. Source.
[FiekerKlueners2014] / Sections 3 and 7. Source.

36.3 Solvability by radicals

Landau-Miller (1985) give a polynomial-time decision procedure for solvability by radicals over $\mathbb{Q}$, in the dense binary input model. Their paper also constructs radical expressions in a compact straight-line-program representation. The size bound applies to this compact encoding and need not survive expansion of repeated subexpressions.

The method exploits blocks of imprimitivity and the restricted size of primitive solvable permutation groups. Thus one can answer a structural question about a Galois group without first listing all its elements. This is different from constructing the splitting field in an explicit basis: its degree may be $n!$ for a degree-n polynomial with group S_n.

References

[LandauMiller1985] / Introduction and Sections 2-4. Source.

36.4 Software

Tool	Scope	Verification data
Magma	Native `GaloisGroup` accepts integer, rational, and number-field polynomials without a fixed degree cap, but does not automatically prove its conditional steps. The documented `GaloisProof(f,S)` function covers integer polynomials; an absolute number-field overload is also available.	Input, options, `GaloisData`, successful proof result, and version.
PARI/GP 2.19	`polgalois` requires irreducible input over $\mathbb{Q}$, of degree at most seven, or eleven with `galdata`. Its resolvent method can rarely return a wrong answer when initial numerical precision is insufficient.	Polynomial, precision, complete output, numbering convention, and version.
SageMath 10.10	The rational-polynomial `galois_group` interface requires irreducible input. It uses PARI through degree eleven and attempts GAP above that; GAP's documented algorithm range is through degree fifteen, with some higher cases possible. The Magma wrapper identifies groups only through degree thirty.	Routine, algorithm, backend version, permutation action, and separate certification outcome.

Native Magma's degree-independent algorithm and Sage's wrapper have different limits. A database containing transitive groups of a given degree does not guarantee that the group-

computation algorithm handles every polynomial of that degree. Runtime and memory can be decisive even without a formal degree cap.

For Sage number fields, `gc_numbering` controls whether permutations act on roots of the original defining polynomial or of the Galois closure's defining polynomial. The documentation also warns that calling `galois_group` on a relative extension currently computes its absolute Galois closure over $\mathbb{Q}$; the intended ground field must therefore be specified explicitly.

Sage also documents `galois_group_davenport_smith_test`: return values 1 and 2 certify S_n and A_n, while 0 means inconclusive. Its default checks irreducibility; setting `assume_irreducible=True` requires an independently established hypothesis. This specialized test is different from unrestricted group determination.

References

[MagmaGaloisDocumentation] / GaloisGroup and GaloisProof. Input fields, degree independence, conditional output, and documented certification scope. Source.

[PARIDocumentation] / polgalois, PARI/GP 2.19.0. Degree limits, output conventions, and explicit precision warning. Source.

[SageGaloisDocumentation] / Polynomial_rational_flint.galois_group; galois_group_davenport_smith_test; NumberField.galois_group, SageMath 10.10. Backend limits, certified specialized test, root-action conventions, and relative-field warning. Source.

36.5 Example

Example

For $f = x^4 + 2x + 2$, the following data determine the group:

1. Eisenstein at 2, proving irreducibility.
2. The resolvent $R = T^3 - 8T - 4$, with no rational root among $\pm 1, \pm 2, \pm 4$.
3. The discriminant $\Delta(f) = 1616 = 16 \cdot 101$, a nonsquare.

The quartic classification proves the group is S_4. The companion script verifies these identities and root tests by exact arithmetic.

A corresponding Magma command sequence is:

```
Zx<x> := PolynomialRing(Integers());
f := x^4 + 2*x + 2;
G, roots, S := GaloisGroup(f);
assert GaloisProof(f, S);
Order(G);
```

These commands are a template; no Magma execution is claimed here. The algebraic proof above establishes $|G| = 24$ without relying on that execution.

References

[ConradCubicQuartic] / Table 5 and Theorem 3.6. Source.

[MagmaGaloisDocumentation] / GaloisGroup and GaloisProof. Source.

36.6 Local Galois groups

For a rational polynomial, computing its group over $\mathbb{Q}_p$ is distinct from computing its group over $\mathbb{Q}$. If $L/\mathbb{Q}$ is the global splitting field and $\mathfrak{P} \mid p$, the local splitting-field group identifies with the decomposition subgroup $D_{\mathfrak{P}} \subseteq \mathrm{Gal}(L/\mathbb{Q})$. The embedding depends on the choice of $\mathfrak{P}$, up to conjugacy.

Doris (2020) develops local algorithms using ramification towers, global models, and resolvents. A global model supplies a tractable overgroup; local factorization of resolvents distinguishes the desired subgroup. The base field therefore remains part of the result even when global auxiliary calculations are used.

Certification also depends on the implementation. Remark 3.5 of the paper identifies a heuristic complex-root bound in the then-current implementation. The author's package documentation later describes an exact p-adic input mode that produces proved results. The mode and verification outcome are part of the computation record; an algorithm does not by itself certify a finite-precision implementation.

References

[Neukirch1999] / Decomposition groups and completions. Source.

[Doris2020] / Sections 1.1-4 and Remark 3.5; linked author package documentation for the exact-input mode. Source.

36.7 Computations in families

A computation for a parameterized family specifies the parameter range, normalization, exceptions, and input polynomials. A finite verified range does not establish a result for all parameters; numerical experiments require a separate certification step.

Factor degrees usually identify the Frobenius cycle type rather than its precise conjugacy class in the group. Dokchitser and Dokchitser give finer invariants for distinguishing classes when cycle type is insufficient. This matters when arithmetic applications require more than the abstract group or its order.

A reproducible record includes exact coefficients, ground field, source theorem or algorithm, witness primes, discriminant, group action, certification outcome, and the date of the run. Databases supply comparison examples; the original computation or theorem supplies the proof.

References

[DokchitserDokchitser2013]. Distinguishing Frobenius classes beyond their cycle types. Source.

[LMFDBFieldsGroups]. Comparison data for number fields and transitive groups. Source.

Questions

How can a computed group be certified?

Research direction / Reproducible computations for explicit polynomial families

Short certificates and verifiers for subgroup containment and noncontainment can simplify computations in a specified family. The quartic example illustrates such a certificate. The problem is to find suitable certificates for the chosen family, rather than to establish that Galois-group certification is possible.

References

[FiekerKlueners2014] / Section 7. Source.

[MagmaGaloisDocumentation] / GaloisProof. Source.

Related topics

Uses. Resolvents and subgroup tests. Relative invariants certify containment in smaller subgroups.

Uses. Frobenius and factorization modulo primes. Factorization at small primes restricts the candidate groups.

Uses. Polynomial factorization algorithms. Exact factors and irreducibility checks establish the input assumptions.

Applies to. Eulerian polynomials. Computed tables certify finite ranges; an all-degree result requires a separate argument.

Part X

Several variables

CHAPTER 37

Absolute irreducibility and Newton polytopes

method / Intermediate / arbitrary fields / Published results

An indecomposable Newton polytope can prove absolute irreducibility. When the polytope decomposes, factorization depends on the coefficients.

Overview

- How does absolute irreducibility differ from irreducibility over the ground field?
- When does the Newton polytope certify absolute irreducibility?
- Why does a decomposable polytope leave irreducibility undecided?

Prerequisites: Fields, irreducibility, and Galois actions

37.1 Absolute irreducibility

A nonconstant $f \in K[x_1, \dots, x_r]$ is *absolutely irreducible* if it remains irreducible over $\overline{K}$. For a univariate polynomial this forces degree one. In several variables, this means that the coordinate ring of the hypersurface defined by f remains a domain after extension to $\overline{K}$. Equivalently, the hypersurface is geometrically integral.

For example, x^2+y^2 is irreducible over $\mathbb{Q}$ but factors over $\mathbb{Q}(i)$ as $(x+iy)(x-iy)$. Irreducibility over the ground field therefore does not imply absolute irreducibility. Irreducibility and smoothness of the hypersurface are also distinct properties.

References

[Gao2001Polytopes]. Absolute irreducibility of multivariate polynomials. Source.

[Lecerf2013]. Rational and absolute factorization in several variables. Source.

37.2 Newton polytopes and products

For $f = \sum_{\mathbf{a}} c_{\mathbf{a}} x^{\mathbf{a}}$, its Newton polytope is

$$\operatorname{Newt}(f) = \operatorname{conv}\{\mathbf{a} \in \mathbb{Z}^r_{\geq 0} : c_{\mathbf{a}} \neq 0\}.$$

For a product, $\operatorname{Newt}(gh) = \operatorname{Newt}(g) + \operatorname{Newt}(h)$, where the right side is the Minkowski sum. For a linear weight with unique maxima on the two factor polytopes, the product of the highest-weight terms is nonzero and is the unique highest-weight term of the product. Varying the weight determines the polytope and proves the identity.

A lattice polytope is integrally indecomposable if every expression as a sum of two lattice polytopes has a point as one summand. A point polytope corresponds to a monomial, not necessarily to a constant.

References

[GaoLauder2001] / §2, Lemma 1 and Corollary 2. Product formula, with its origin attributed to Ostrowski, and the monomial caveat. Source.

37.3 The indecomposability criterion

Theorem (Newton-polytope criterion)

Let $f \in K[x_1, \dots, x_r]$ be nonconstant and divisible by no variable x_i. If $\mathrm{Newt}(f)$ is integrally indecomposable, then f is absolutely irreducible.

Any factorization over $\overline{K}$ decomposes the same lattice polytope. One factor must therefore have point support and be a monomial. Since no variable divides f, that factor is a nonzero constant. This excludes every nontrivial factorization.

The criterion depends only on the support. It remains valid after changing the nonzero coefficients, over any field, provided the support is unchanged.

References

[Gao2001Polytopes]. Integral indecomposability criterion; the elementary proof is included. Source.

[GaoLauder2001] / Corollary 2. Explicit statement excluding divisibility by any variable. Source.

37.4 Example

Example

Consider $f = 1 + x^2y + xy^3$ over any field. Its Newton polygon has vertices $(0,0), (2,1), (1,3)$. The oriented edge vectors are $(2,1), (-1,2), (-1,-3)$, all primitive integer vectors.

A nontrivial Minkowski summand of a triangle must itself be a triangle with parallel corresponding edges; a segment summand would introduce another edge direction. The two triangular summands would therefore divide each edge into two positive integral lengths. A primitive edge has lattice length one, so this is impossible. The polygon is integrally indecomposable. The nonzero constant term excludes nonconstant monomial factors. The criterion therefore proves absolute irreducibility in every characteristic.

References

[Gao2001Polytopes]. Triangle indecomposability criteria; this particular support is checked explicitly here. Source.

37.5 Limitations

Monomial factors. The polynomial $x(1 + x + y)$ has a translated indecomposable triangle as its Newton polygon, but has a nonconstant monomial factor. The hypothesis excluding variable divisors is essential.

The converse is false. Over a field of characteristic different from two, x^2+y^2+1 is absolutely irreducible, although its polygon is twice the standard simplex and hence decomposable. Its projective homogenization $X^2 + Y^2 + Z^2$ is nonsingular; a reducible projective conic over an algebraically closed field is a pair of lines, possibly repeated, and is singular. In characteristic two this example becomes $(x + y + 1)^2$, so the characteristic restriction is necessary.

References

[GaoLauder2001] / §5. A decomposable polytope makes the shape test inconclusive; the examples are direct checks. Source.

37.6 Coefficient criteria and differential methods

If the polytope decomposes, the coefficients determine whether a factorization exists. Effective Noether forms describe the reducibility locus. Fix a total-degree bound d. There are finitely many integer polynomials in the coefficient variables such that all of them vanish exactly when the specialized polynomial is reducible over the algebraic closure or its total degree drops below d. Thus, for a polynomial of *exact* degree d, absolute irreducibility is certified by at least one nonzero Noether form. In positive characteristic, reduce the integer coefficients of the forms modulo that characteristic. Differential methods instead recover factor information through linear systems. For a squarefree bivariate $f = \prod_i f_i$ in characteristic zero, the identity

$$\frac{df}{f} = \sum_i \frac{df_i}{f_i}$$

suggests looking for closed rational one-forms with denominator f. Writing such a form as $(g/f)\,dx + (h/f)\,dy$ gives the polynomial linear equation

$$f\,(g_y - h_x) = g f_y - h f_x.$$

An algorithm based on this equation requires degree bounds and hypotheses on the characteristic. Gao's paper develops the method in characteristic zero or sufficiently large characteristic; small characteristic requires separate arguments.

The polytope criterion can be applied before methods that use the coefficients. A reducible specialization $f(x, t)$ does not establish that $f(x, y)$ itself is reducible.

References

[Kaltofen1995] / §5, Theorem 7. Vanishing describes absolute reducibility together with degree drop; the fixed-degree qualification is essential. Source.

[Gao2003PDE]. Differential-equation factorization with characteristic restrictions. Source.

[Lecerf2013]. Multivariate factorization methods and their hypotheses. Source.

[FLINTFactorizationDocumentation] / Multivariate factorisation: irreducibility from the Newton polytope. The polytope test is used as a certificate/pretest, not as an absolute factorization routine. Source.

Questions

Which coefficient conditions imply absolute irreducibility when the Newton polytope decomposes?

Research direction / A fixed multivariate support and a specified coefficient field

A decomposition of the polytope gives possible supports of factors, but does not determine whether those factors exist. For a specified family, one can study the reducibility locus in coefficient space or combine the support conditions with differential or lifting methods. The question concerns criteria adapted to that family, rather than a general factorization algorithm.

References

[GaoLauder2001] / §5. Source.

[Kaltofen1995]. Source.

[Gao2003PDE]. Source.

Related topics

Contrasts with. Newton polygons and Galois groups. Valuation polygons use coefficient valuations; Newton polytopes use exponent support.

Generalizes. Sparse polynomials and factor reversal. Sparse support can constrain factors through convex geometry.

Applies to. Polynomial factorization algorithms. The polytope criterion can precede factorization methods that use the coefficients.

Contrasts with. Specialization and Hilbert irreducibility. Factorization of a specialization need not imply factorization of the multivariate polynomial.

Part XI

Bibliography

232 references

The reference keys are used throughout the handbook. This list includes the original 188-source collection and later additions. Publication status is recorded for each entry.

Foundations and reference books

[Serre1979] Local Fields

Serre, Jean-Pierre. 1979. Springer. 67. Graduate Texts in Mathematics.

Book / Published book

Treats valuations, ramification, the different, and discriminants. These provide the local field framework for Newton polygons and for constructing inertia elements in polynomial Galois groups.

English translation of Corps locaux by Marvin Jay Greenberg.

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Lidl, Rudolf and Niederreiter, Harald. 1997. Cambridge University Press. 20. Edition 2. Encyclopedia of Mathematics and its Applications.

Book / Published book

Treats irreducible and primitive polynomials over finite fields, factorization, and constructions. An irreducible polynomial of degree n over a finite field has cyclic splitting-field group of order n; Galois groups over the rationals need not be cyclic.

The standard bibliographic year is 1997 (BNF and AMS review); the Cambridge product page lists a 24 October 1996 release date.

https://www.cambridge.org/core/books/finite-fields/75BDAA74ABAE713196E718392B9E5E72

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[Neukirch1999] Algebraic Number Theory

Neukirch, Jürgen. 1999. Springer. 322. Grundlehren der mathematischen Wissenschaften.

Book / Published book

Treats prime decomposition, ramification, and the global arithmetic of splitting fields, relating reduction modulo primes to arithmetic properties of number fields.

Translated from German by Norbert Schappacher.

https://link.springer.com/book/10.1007/978-3-662-03983-0

https://doi.org/10.1007/978-3-662-03983-0

[Schinzel2000] Polynomials with Special Regard to Reducibility

Schinzel, Andrzej. 2000. Cambridge University Press. 77. Encyclopedia of Mathematics and its Applications.

Book / Published book

Treats arbitrary and lacunary polynomials over several classes of fields. Its bibliography documents the classical literature on reducibility.

The monograph treats several broad field classes. Its publisher explicitly excludes results valid only over finite fields, local fields, or the rational field; use it together with field-specific references.

https://doi.org/10.1017/CBO9780511542916

https://www.cambridge.org/core/books/polynomials-with-special-regard-to-reducibility/FBC37D46462D46A6E121FFC9A09391C5

[Lang2002] Algebra

Lang, Serge. 2002. Springer. 211. Edition 3. Graduate Texts in Mathematics.

Book / Published book

Develops the algebra of irreducibility, field extensions, and Galois theory over general fields, including the distinction between separability and irreducibility.

https://link.springer.com/book/10.1007/978-1-4613-0041-0

https://doi.org/10.1007/978-1-4613-0041-0

[Prasolov2004] Polynomials

Prasolov, Victor V. 2004. Springer. 11. Algorithms and Computation in Mathematics.

Book / Published book

Treats irreducibility on pp. 47-76, special polynomial forms on pp. 77-131, and Galois theory on pp. 181-218, relating algebraic arguments to root location and explicit constructions.

The English first edition is ©2004 (hardcover published 9 July 2004); the linked eBook was released 23 September 2009. These are editions/formats of the same book.

https://link.springer.com/book/10.1007/978-3-642-03980-5

https://doi.org/10.1007/978-3-642-03980-5

[Cox2012] Galois Theory

Cox, David A. 2012. John Wiley & Sons. Edition 2.

Book / Published book

Develops field extensions and explicit Galois-group calculations through worked examples. Chapters 13 and 14 treat resolvents, computation, and solvable permutation groups.

https://onlinelibrary.wiley.com/doi/book/10.1002/9781118218457

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[Milne2022] Fields and Galois Theory

Milne, James S. 2022. Kea Books.

Book / Published book; author errata available

Develops field and Galois theory through proofs and examples; Chapter 4 treats Galois-group computation. The linked file is the 2022 book. The author's errata address, among other points, irreducibility in Q[X] versus Z[X] and the degree-two exception in polynomial counting. The frozen course notes v5.10 have different pagination.

Author errata: https://www.jmilne.org/math/Books/fterrata.html. Course-note version 5.10 is frozen; the author directs readers to the book for corrections and updates.

https://www.jmilne.org/math/Books/index.html

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[Milne2020ANT] Algebraic Number Theory (v3.08)

Milne, James S. 2020. Author course notes, 166 pages.

Exposition / Author course notes; version 3.08, 19 July 2020

Free notes supplying precise proofs of the discriminant-index identity, prime decomposition, Hensel lifting, unramified extensions, decomposition groups, and tame/wild inertia. Version 3.08 is dated 19 July 2020; this is a different work from Fields and Galois Theory.

https://www.jmilne.org/math/CourseNotes/ANT.pdf

Classical criteria and sparse-polynomial methods

[schoenemann1846] Von denjenigen Moduln, welche Potenzen von Primzahlen sind. (Fortsetzung)

Schönemann, Theodor. 1846. Journal für die reine und angewandte Mathematik. 32. pp. 93-105.

Published work / published

Precedes Eisenstein's treatment of the criterion now often called Schönemann-Eisenstein. Cox's historical account explains the role of higher congruences and translates the statement into modern polynomial language.

https://www.degruyterbrill.com/document/doi/10.1515/crll.1846.32.93/html

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[eisenstein1850] Über die Irreductibilität und einige andere Eigenschaften der Gleichung, von welcher die Theilung der ganzen Lemniscate abhängt

Eisenstein, Gotthold. 1850. Journal für die reine und angewandte Mathematik. 39. pp. 160-179.

Published work / published

Develops Eisenstein's criterion in connection with division of the lemniscate and Gaussian integers. For an integer polynomial, the familiar criterion requires a prime dividing every nonleading coefficient but not the leading coefficient, with its square not dividing the constant coefficient.

https://doi.org/10.1515/crll.1850.39.160

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https://people.math.ethz.ch/~halorenz/4students/Algebra/Schoenemann_Eisenstein.pdf

[capelli1901binomials] Sulla riduttibilità della funzione x^n-A in un campo qualunque di razionalità

Capelli, Alfredo. 1901. Mathematische Annalen. 54. pp. 602-603.

Published work / published

Classical reference for the binomial irreducibility criterion, including the exceptional fourth-power obstruction when the exponent is divisible by four. Distinguish this binomial criterion from the related Capelli composition lemma; the note itself points back to Capelli's 1897 and 1898 work.

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[dumas1906] Sur quelques cas d'irréductibilité des polynômes à coefficients rationnels

Dumas, Gustave. 1906. Journal de Mathématiques Pures et Appliquées, 6e série. 2. pp. 191-258. 6.

Published work / published

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Perron, Oskar. 1907. Journal für die reine und angewandte Mathematik. 132. pp. 288-307.

Published work / published

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Schur, Issai. 1908. Archiv der Mathematik und Physik. 13. pp. 367. 3.

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Original question on products of distinct integral linear factors perturbed by ± 1. The link is to the later historical account that verifies this citation, rather than to a scan of the original problem; that account also records Schur's Problem275 (1909) on power compositions.

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Pólya, George. 1919. Jahresbericht der Deutschen Mathematiker-Vereinigung. 28. pp. 31-40.

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Dorwart, H. L. 1935. The American Mathematical Monthly. 42(6). pp. 369-381.

Published work / published

Surveys classical criteria based on coefficient divisibility, coefficient size, and values at integers. Its bibliography records the literature available at the time.

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[maclane1936values] A construction for absolute values in polynomial rings

MacLane, Saunders. 1936. Transactions of the American Mathematical Society. 40(3). pp. 363-395.

Published work / published

Constructs polynomial valuations through successive augmentations and key polynomials. This framework supports irreducibility tests beyond a first Newton polygon and underlies modern local factorization methods.

https://users.encs.concordia.ca/~ford/UNCG/smvals.pdf

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Schinzel, Andrzej. 1965. Acta Arithmetica. 11(1). pp. 1-34.

Published work / published; errata available

Studies structural reducibility, especially for sparse polynomials and trinomials. Specific statements should be read with the 1966 errata.

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/11/1/95825/on-the-reducibility-of-polynomials-and-in-particular-of-trinomials

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[schinzel1966errata] Errata to the paper "On the reducibility of polynomials and in particular of trinomials"

Schinzel, Andrzej. 1966. Acta Arithmetica. 11(4). pp. 491.

Published work / published erratum

Corrects statements in Schinzel's 1965 paper; the correction and original paper should be consulted together.

https://doi.org/10.4064/aa-11-4-491

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/en/publishing-house/journals-and-series/acta-arithmetica/all/11/4/96020/errata-to-the-paper-on-the-reducibility-of-polynomials-and-in-particular-of-trinomials

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Tverberg, Helge. 1973. Mathematica Scandinavica. 32. pp. 5-21.

Published work / published

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Brillhart, John and Filaseta, Michael and Odlyzko, Andrew. 1981. Canadian Journal of Mathematics. 33(5). pp. 1055-1059.

Published work / published

Generalizes the criterion that a prime's digit expansion gives an irreducible polynomial when its digits are used as coefficients. It identifies coefficient restrictions under which a single prime evaluation proves irreducibility.

https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/on-an-irreducibility-theorem-of-a-cohn/F1644BDFD66CFC166841810C29CBA91A

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https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/S0025579300011931

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[filaseta1999smallnorm] On the factorization of polynomials with small Euclidean norm

Filaseta, Michael. 1999. Number Theory in Progress, Vol. 1. pp. 143-163. de Gruyter.

Published work / published

Uses the Euclidean norm of the coefficient vector to constrain factorization, including that of sparse polynomials and their nonreciprocal parts. Irreducibility of the nonreciprocal part does not imply irreducibility of the whole polynomial.

Pages143-163 follow the author's bibliography and MathSciNet citation reproduced in Sawin-Shusterman-Stoll; the publisher chapter wrapper gives143-164, apparently including a trailing page.

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Filaseta, Michael and Ford, Kevin and Konyagin, Sergei. 2000. Illinois Journal of Mathematics. 44(3). pp. 633-643.

Published work / published

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https://www.ford126.web.illinois.edu/wwwpapers/ffk.pdf

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Murty, M. Ram. 2002. The American Mathematical Monthly. 109(5). pp. 452-458.

Published work / published

Relates prime values, coefficient bounds, and irreducibility, with Cohn's theorem as an example. A single prime value requires additional size or coefficient hypotheses; the paper states such hypotheses.

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Thangadurai, R. 2007. RMS Mathematical Newsletter. 17(2). pp. 29-37.

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The author's bibliography confirms RMS Mathematical Newsletter 17(2) (2007), 29-37. A distinct 2009 chapter with the same title is coauthored with N. Saradha.

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https://www.hri.res.in/~thanga/expository.html

[cox2011eisenstein] Why Eisenstein Proved the Eisenstein Criterion and Why Schönemann Discovered It First

Cox, David A. 2011. The American Mathematical Monthly. 118(1). pp. 3-21.

Published work / published

Explains the history and mathematics of Eisenstein's criterion through Gauss's lemma, finite fields, Hensel lifting, Gaussian integers, and lemniscatic division. It distinguishes the arguments and contributions of Schönemann and Eisenstein.

The cited Monthly version (2011), pp. 3-21, is a reprint with minor changes of the article in Normat 57(2) (2009), pp. 49-73; the Monthly first-page footnote explicitly acknowledges the earlier publication. Publisher online migration date 13 December 2017 is not the publication year.

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Győry, Kálmán and Hajdu, Lajos and Tijdeman, Robert. 2011. Monatshefte für Mathematik. 163(4). pp. 415-443.

Published work / published

Research paper with a substantial historical introduction and bibliography on small-value criteria, perturbed products and compositions. Its extensions use interpolation, algebraic number theory and Diophantine approximation, including results over imaginary quadratic rings and compositions of CM type.

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Bonciocat, Nicolae Ciprian. 2015. Communications in Algebra. 43(8). pp. 3102-3122.

Published work / published

Combines coefficient divisibility at several primes through Newton polygons. Intersecting the permitted factor degrees can exclude every proper factor even when each prime alone is inconclusive.

Published in 2015; open author preprint v2 dated 21 March 2014. Theorem A gives the several-prime factor-degree intersection criterion. The former secondary verification link DOI 10.1090/proc/14856 is replaced by this paper's own primary text.

https://www.tandfonline.com/doi/abs/10.1080/00927872.2014.910800

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Published work / published

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Filaseta, Michael and Li, Huixi and Patane, Frank and Skabelund, Dane. 2020. Acta Arithmetica. 196(2). pp. 187-201.

Published work / published

Improves the sufficient exponent threshold in the robust-pair method of Sawin, Shusterman and Stoll. The conclusion concerns the nonreciprocal part, so reciprocal factors still require separate analysis when proving full irreducibility.

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https://doi.org/10.4064/aa190907-11-2

https://huixil.wordpress.com/wp-content/uploads/2020/06/on-the-irreducibility-of-the-non-reciprocal-part-of-polynomials-of-the-form-fx-x5en-gx.pdf

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Sawin, Will and Shusterman, Mark and Stoll, Michael. 2020. Acta Arithmetica. 192(2). pp. 111-139.

Published work / published

Develops Ljunggren's factor-reversal method into irreducibility algorithms for x^N c(x^-1)+d(x). It treats cyclotomic obstructions separately, improves effective bounds under additional robustness hypotheses, and reviews earlier large-gap results.

https://arxiv.org/abs/1803.10811

https://doi.org/10.4064/aa180526-12-6

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/192/2/113301/irreducibility-of-polynomials-with-a-large-gap

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[GuardiaMontesNart2012Newton] Newton polygons of higher order in algebraic number theory

Guàrdia, Jordi and Montes, Jesús and Nart, Enric. 2012. Transactions of the American Mathematical Society. 364(1). pp. 361-416.

Published work / Published article; open author preprint v2 (2008)

Develops Ore's higher Newton polygons for local factorization. Product and residual-polynomial theorems refine slope restrictions to irreducibility certificates, and higher-order indices control termination. These methods address repeated factors in reduction and cases where a first Newton polygon is inconclusive. Theorem numbers cited in this atlas refer to arXiv v2 where indicated.

https://doi.org/10.1090/S0002-9947-2011-05442-5

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Galois group detection and effective prime methods

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Jordan, Camille. 1871. Journal de Mathématiques Pures et Appliquées (2). 16. pp. 383-408.

Published work / Published article

Establishes primitive-group results used to prove containment of A_n in polynomial Galois groups. Modern expositions give the prime-cycle criterion in current terminology, including its primitivity hypothesis.

https://www.numdam.org/item/JMPA_1871_2_16__383_0/

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Dixon, John D. and Mortimer, Brian. 1996. Graduate Texts in Mathematics 163. Springer.

Book / Published monograph

Treats transitivity, block systems, primitive groups, and Jordan-type criteria for large permutation groups. These criteria apply once modular factorization or local ramification has supplied cycle information.

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Stevenhagen, Peter and Lenstra, Jr., Hendrik W. 1996. The Mathematical Intelligencer. 18(2). pp. 26-37.

Exposition / Published expository article

Explains Frobenius conjugacy classes and their relation to splitting modulo primes, with the history and proof of Chebotarev's theorem. It describes what modular factorization data can establish and what remains undetermined.

https://pub.math.leidenuniv.nl/~lenstrahw/PUBLICATIONS/1996d/art.pdf

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https://pub.math.leidenuniv.nl/~lenstrahw/PUBLICATIONS/pub.html

[Jones2014Cycles] Primitive permutation groups containing a cycle

Jones, Gareth A. 2014. Bulletin of the Australian Mathematical Society. 89(1). pp. 159-165.

Published work / Published article

Classifies primitive finite permutation groups containing a cycle and strengthens Jordan's prime-cycle theorem. Theorems 1.1-1.2 and Corollary 1.3 justify the cycle criteria used here. Prefer the published version: Remark 1.9 corrects the older preprint's distinction between polynomial composition and primitivity of a fixed polynomial's root action.

Published online 18 July 2013; journal issue February 2014. Use the published text: preprint Comments 1 and 3 became Remarks 1.4 and 1.6, and the inaccurate fixed-polynomial indecomposability claim in arXiv v1 Comment 6 was corrected to a divisibility criterion in published Remark 1.9.

https://doi.org/10.1017/S000497271300049X

https://www.cambridge.org/core/services/aop-cambridge-core/content/view/BBF3021D8434AD2C0BB0D3D6CE637008/S000497271300049Xa.pdf/primitive-permutation-groups-containing-a-cycle.pdf

https://arxiv.org/abs/1209.5169

https://www.cambridge.org/core/journals/bulletin-of-the-australian-mathematical-society/article/primitive-permutation-groups-containing-a-cycle/BBF3021D8434AD2C0BB0D3D6CE637008

[LagariasOdlyzko1977] Effective versions of the Chebotarev density theorem

Lagarias, Jeffrey C. and Odlyzko, Andrew M. 1977. Algebraic Number Fields: L-functions and Galois Properties (Proc. Sympos., Univ. Durham, Durham, 1975). pp. 409-464. Academic Press. Edited by Fröhlich, Albrecht.

Published work / Published proceedings chapter

Gives conditional and unconditional effective Chebotarev estimates. These bound the primes with specified Frobenius conjugacy classes.

The main link is Lagarias's bibliography, which verifies the 1977 proceedings chapter (pp. 409-464); it is not a full-text link.

https://websites.umich.edu/~lagarias/zeta.html

https://www-users.cse.umn.edu/~odlyzko/doc/complete.html

https://doi.org/10.1090/mcom/3665

[Cohen1981Hilbert] The Distribution of Galois Groups and Hilbert's Irreducibility Theorem

Cohen, Stephen D. 1981. Proceedings of the London Mathematical Society (3). 43(2). pp. 227-250.

Published work / Published article

Gives quantitative bounds for exceptional specializations and Galois-group distributions in polynomial families by combining Hilbert irreducibility with sieve estimates.

https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/plms/s3-43.2.227

https://doi.org/10.1112/plms/s3-43.2.227

https://academic.oup.com/plms/article-abstract/s3-43/2/227/1545903

[Serre1981Chebotarev] Quelques applications du théorème de densité de Chebotarev

Serre, Jean-Pierre. 1981. Publications Mathématiques de l'IHÉS. 54. pp. 123-201.

Published work / Published article

Develops effective Chebotarev estimates and arithmetic applications, specifying the dependence on analytic hypotheses, extension degrees, and discriminants.

https://numdam.org/articles/10.1007/BF02698692/

https://doi.org/10.1007/BF02698692

https://www.college-de-france.fr/media/jean-pierre-serre/UPL7902704499476033975_UPL61373_Serre_Chebotarev.pdf

[ThornerZaman2019Chebotarev] A unified and improved Chebotarev density theorem

Thorner, Jesse and Zaman, Asif. 2019. Algebra & Number Theory. 13(5). pp. 1039-1068.

Published work / Published article

Gives an unconditional effective Chebotarev theorem with improved uniformity, including a possible exceptional zero. Its parameter dependence determines when an unconditional estimate can replace a GRH-based estimate in a probabilistic or computational argument.

https://msp.org/ant/2019/13-5/ant-v13-n5-p02-s.pdf

https://doi.org/10.2140/ant.2019.13.1039

https://arxiv.org/abs/1803.02823

https://experts.illinois.edu/en/publications/a-unified-and-improved-chebotarev-density-theorem/

[KappeWarren1989Quartic] An Elementary Test for the Galois Group of a Quartic Polynomial

Kappe, Luise-Charlotte and Warren, Bette. 1989. The American Mathematical Monthly. 96(2). pp. 133-137.

Published work / Published article

Original source of the elementary quartic-group test that separates cyclic and dihedral cases after the cubic resolvent and discriminant tests. Publisher metadata verified; the full formulation and proof used in this atlas are checked in Conrad's author-hosted exposition, Theorem 4.1. The publisher also assigns DOI 10.1080/00029890.1989.11972158.

https://www.jstor.org/stable/2323198

https://doi.org/10.2307/2323198

https://www.jstor.org/stable/i315188

https://www.tandfonline.com/doi/abs/10.1080/00029890.1989.11972158

https://kconrad.math.uconn.edu/blurbs/galoistheory/cubicquartic.pdf

Special polynomial families

[Selmer1956] On the irreducibility of certain trinomials

Selmer, Ernst S. 1956. Mathematica Scandinavica. 4. pp. 287-302.

Published work / Published

Proves irreducibility of x^n-x-1 over Q for every n at least 2. The later full symmetric-group conclusion has a separate attribution.

https://doi.org/10.7146/math.scand.a-10478

[Ljunggren1960] On the irreducibility of certain trinomials and quadrinomials

Ljunggren, Wilhelm. 1960. Mathematica Scandinavica. 8. pp. 65-70.

Published work / Published

Studies factorization of sparse polynomials through the distinction between reciprocal and nonreciprocal factors. Its factor-reversal method extends to later large-gap problems.

https://doi.org/10.7146/math.scand.a-10593

[NartVila1979] Equations of the type X^n+aX+b with absolute Galois group S_n

Nart, Enric and Vila, Núria. 1979. Revista de la Universidad de Santander. 2. pp. 821-825.

Published work / Published

An early source for symmetric Galois groups of trinomials; in particular, it precedes Osada for x^n-x-1. The institutional catalogue supplies an English title although the original paper is in Spanish.

https://portalrecerca.uab.cat/en/publications/equations-of-the-type-xnaxb-with-absolute-galois-group-sn/

https://kconrad.math.uconn.edu/blurbs/gradnumthy/galoisselmerpoly.pdf

[Osada1987] The Galois groups of the polynomials X^n+aX^l+b

Osada, Hiroyuki. 1987. Journal of Number Theory. 25(2). pp. 230-238.

Published work / Published

Gives arithmetic conditions forcing the full symmetric group for trinomials, using ramification to obtain transpositions. It includes an independent proof for x^n-x-1.

https://doi.org/10.1016/0022-314X(87)90029-1

https://kconrad.math.uconn.edu/blurbs/gradnumthy/galoisselmerpoly.pdf

[CohenMovahhediSalinier1997] Double transitivity of Galois groups of trinomials

Cohen, Stephen D. and Movahhedi, A. and Salinier, Alain. 1997. Acta Arithmetica. 82(1). pp. 1-15.

Published work / Published

Studies how arithmetic hypotheses on a trinomial force double transitivity. This is an intermediate group-theoretic step that can be combined with cycle or ramification information to obtain a large Galois group.

https://doi.org/10.4064/aa-82-1-1-15

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/en/publishing-house/journals-and-series/acta-arithmetica/all/82/1/109606/double-transitivity-of-galois-groups-of-trinomials

[CohenMovahhediSalinier1999] Galois groups of trinomials

Cohen, Stephen D. and Movahhedi, A. and Salinier, Alain. 1999. Journal of Algebra. 222(2). pp. 561-573.

Published work / Published

Extends earlier results of Osada using primitivity, cycle structure, and arithmetic restrictions. Apply the precise irreducibility and coprimality assumptions in the paper rather than treating sparsity alone as a sufficient condition.

https://doi.org/10.1006/jabr.1999.8033

https://www.sciencedirect.com/science/article/pii/S002186939998033X

https://www.unilim.fr/pages_perso/alain.salinier/publications.html

[Schur1929I] Einige Sätze über Primzahlen mit Anwendungen auf Irreduzibilitätsfragen, I

Schur, Issai. 1929. Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse. pp. 125-136.

Published work / Published; citation verified through later literature

Classical prime-divisor arguments applied to irreducibility. Read alongside Part II and the 1930/1931 papers to trace the origin of factorial-coefficient and orthogonal-polynomial methods.

The linked source is Rabarison-Pazuki-Molin (2024), bibliography entry [20], which verifies the original citation. It is not a scan of Schur's 1929 paper.

https://www.numdam.org/articles/10.5802/pmb.57/

[Schur1930] Gleichungen ohne Affekt

Schur, Issai. 1930. Sitzungsberichte der Preußischen Akademie der Wissenschaften, Physikalisch-Mathematische Klasse. pp. 443-449.

Published work / Published; citation verified through later literature

Proves the Galois-group theorem for exponential Taylor polynomials. Coleman later gives a proof using Newton polygons.

The linked source is Rabarison-Pazuki-Molin (2024), bibliography entry [22], which verifies the original citation. It is not a scan of Schur's 1930 paper.

https://www.numdam.org/articles/10.5802/pmb.57/

https://web.math.princeton.edu/~nmk/kGalois_11bis.pdf

[Schur1931] Affektlose Gleichungen in der Theorie der Laguerreschen und Hermiteschen Polynome

Schur, Issai. 1931. Journal für die reine und angewandte Mathematik. 165. pp. 52-58.

Published work / Published

Studies irreducibility and Galois groups of Laguerre and Hermite polynomials. Statements for Hermite polynomials must account for parity and the forced factor at zero.

https://doi.org/10.1515/crll.1931.165.52

https://gdz.sub.uni-goettingen.de/id/PPN243919689_0165?tify=%7B%22view%22%3A%22info%22%2C%22pages%22%3A%5B57%5D%7D

https://www.numdam.org/articles/10.5802/pmb.57/

[Coleman1987] On the Galois groups of the exponential Taylor polynomials

Coleman, Robert F. 1987. L'Enseignement Mathématique. 33. pp. 183-189.

Published work / Published

Proves Schur's theorem for sum_{j=0}^n x^j/j!: irreducible over Q, with group A_n when 4 divides n and S_n otherwise (n at least 2). The proof combines Newton-polygon slopes, long cycles, and the discriminant square test.

https://doi.org/10.5169/seals-87891

[Martin2004] The Galois group of x^n-x^{n-1}-...-x-1

Martin, Paulo A. 2004. Journal of Pure and Applied Algebra. 190(1–3). pp. 213-223.

Published work / Published

Proves that the generalized Fibonacci polynomial has group S_n when n is even or prime. The prime-degree argument includes irreducibility of its reduction modulo that prime.

https://doi.org/10.1016/j.jpaa.2003.10.028

https://core.ac.uk/download/pdf/82404251.pdf

https://web.math.princeton.edu/~nmk/kGalois_11bis.pdf

[FilasetaKumchevPasechnik2007] On the irreducibility of a truncated binomial expansion

Filaseta, Michael and Kumchev, Angel and Pasechnik, Dmitrii. 2007. Rocky Mountain Journal of Mathematics. 37(2). pp. 455-464.

Published work / Published

Studies irreducibility of the first k+1 terms of (1+x)^n and related coefficient families through factorial-coefficient criteria.

https://doi.org/10.1216/rmjm/1181068761

https://ora.ox.ac.uk/objects/uuid%3A28aa8966-f92a-4ae1-818b-661a3ced8861

https://people.math.sc.edu/filaseta/papers/FKPpaper120304.pdf

[ChambertLoir2011] The theorem of Jentzsch–Szegő on an analytic curve: application to the irreducibility of truncations of power series

Chambert-Loir, Antoine. 2011. International Journal of Number Theory. 7(7). pp. 1807-1823.

Published work / Published

Relates zero-distribution on analytic curves, including nonarchimedean curves, to irreducibility information for truncations. Use the corrected published version or arXiv v3; the arXiv history specifically records a correction to the application.

https://arxiv.org/abs/1004.3813

https://doi.org/10.1142/S1793042111004691

[FilasetaMoy2018] On the Galois group over Q of a truncated binomial expansion

Filaseta, Michael and Moy, Richard. 2018. Colloquium Mathematicum. 154(2). pp. 295-308.

Published work / Published

For a fixed truncation degree r other than 6, proves that sufficiently large n gives S_r for the first r+1 terms of (1+x)^n. At r=6 it bounds the number of exceptions n up to N by O(log N).

https://doi.org/10.4064/cm7474-3-2018

[ShokriShaffafTaleb2019] Galois groups of Taylor polynomials of some elementary functions

Shokri, Khosro Monsef and Shaffaf, Jafar and Taleb, Reza. 2019. International Journal of Number Theory. 15(6). pp. 1127-1141.

Published work / Published

Studies truncations associated with logarithmic and cosine functions. The cosine examples naturally involve a signed-permutation group because of their even symmetry.

https://doi.org/10.1142/S1793042119500623

https://www.numdam.org/articles/10.5802/pmb.57/

https://www.worldscientific.com/doi/10.1142/S1793042119500623

[RabarisonPazukiMolin2024] Exponentielle tronquée et autres contes galoisiens

Rabarison, Patrick and Pazuki, Fabien and Molin, Pascal. 2024. Publications mathématiques de Besançon. Algèbre et théorie des nombres. pp. 105-117.

Published work / Published

Surveys exponential truncations and gives explicit investigations of Padé approximants, relating Schur's classical results to later computational questions.

https://www.numdam.org/articles/10.5802/pmb.57/

https://doi.org/10.5802/pmb.57

[KatzRivinSpeculations] Speculations in Galois Theory

Katz, Nicholas M. and Rivin, Igor. Undated. Author-hosted note.

Exposition / Author-hosted contribution to the Laumon volume; publication year not verified

Poses limiting-frequency questions for Galois groups of Taylor truncations and reports computations in many families. The note explicitly separates experimental evidence from theorems; its posted copy has no date on the title page.

https://web.math.princeton.edu/~nmk/kGalois_11bis.pdf

https://www.laumonvolume.fr/articles

[Grosswald1978] Bessel Polynomials

Grosswald, Emil. 1978. Springer. 698. Lecture Notes in Mathematics.

Book / Published

Treats Bessel polynomials, including their algebraic irreducibility and Galois groups. Its irreducibility conjecture was later resolved by Filaseta and Trifonov.

https://link.springer.com/book/10.1007/BFb0063135

https://doi.org/10.1007/BFb0063135

[FilasetaLam2002] On the irreducibility of the generalized Laguerre polynomials

Filaseta, Michael and Lam, Tsit-Yuen. 2002. Acta Arithmetica. 105(2). pp. 177-182.

Published work / Published

For each fixed rational parameter outside the negative integers, proves irreducibility over Q in every sufficiently large degree. The threshold depends on the parameter, so this is not a uniform theorem when the parameter varies with degree.

https://doi.org/10.4064/aa105-2-4

https://people.math.sc.edu/filaseta/papers/laguerrepap0202.pdf

[FilasetaTrifonov2002] The irreducibility of the Bessel polynomials

Filaseta, Michael and Trifonov, Ognian. 2002. Journal für die reine und angewandte Mathematik. 550. pp. 125-140.

Published work / Published

Resolves Grosswald's irreducibility conjecture for Bessel polynomials in every positive degree. Proves a stronger coefficient-perturbation statement using Newton polygons and prime-divisor estimates.

https://doi.org/10.1515/crll.2002.069

https://people.math.sc.edu/filaseta/papers/besselconj0701.pdf

https://people.math.sc.edu/filaseta/paperindex.html

[Hajir2005] On the Galois group of generalized Laguerre polynomials

Hajir, Farshid. 2005. Journal de théorie des nombres de Bordeaux. 17(2). pp. 517-525.

Published work / Published

Introduces a Newton-index criterion for containing A_n and applies it, together with irreducibility results, to generalized Laguerre polynomials with fixed rational parameter outside the negative integers in sufficiently large degree. Irreducibility is an explicit input to the group criterion.

https://jtnb.centre-mersenne.org/item/JTNB_2005__17_2_517_0/

https://doi.org/10.5802/jtnb.505

https://arxiv.org/abs/math/0406308

https://numdam.org/articles/10.5802/jtnb.505/

[Hajir2009] Algebraic properties of a family of generalized Laguerre polynomials

Hajir, Farshid. 2009. Canadian Journal of Mathematics. 61(3). pp. 583-603.

Published work / Published

Studies negative integral parameters linked to degree, a regime different from the fixed-parameter theorem. Newton polygons organize irreducibility and large-group results, together with precise remaining conjectures.

https://doi.org/10.4153/CJM-2009-031-6

https://arxiv.org/abs/math/0406307

https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/algebraic-properties-of-a-family-of-generalized-laguerre-polynomials/0E2DDA7A33FA3B483679E4693D093D36

[FilasetaKiddTrifonov2012] Laguerre polynomials with Galois group A_m for each m

Filaseta, Michael and Kidd, Travis and Trifonov, Ognian. 2012. Journal of Number Theory. 132. pp. 776-805.

Published work / Published

Completes Schur's generalized Laguerre realizations of A_m by treating m congruent to 2 modulo 4. The constructed polynomials are irreducible for $m > 2$, while $m = 2$ is the split case.

https://doi.org/10.1016/j.jnt.2011.09.012

https://www.sciencedirect.com/science/article/pii/S0022314X11002502

https://people.math.sc.edu/filaseta/papers/FKTpap07.pdf

[BanerjeeFilasetaFinchLeidy2013] On classifying Laguerre polynomials which have Galois group the alternating group

Banerjee, Pradipto and Filaseta, Michael and Finch, Carrie E. and Leidy, J. Russell. 2013. Journal de théorie des nombres de Bordeaux. 25(1). pp. 1-30.

Published work / Published

Classifies the integer parameter pairs with nonzero square discriminant into explicit infinite families and a finite remainder. A square discriminant constrains the group to A_n but does not by itself prove irreducibility or equality with A_n.

https://www.numdam.org/articles/10.5802/jtnb.822/

https://doi.org/10.5802/jtnb.822

[JindalLaishram2022] Families of Laguerre polynomials with alternating group as Galois group

Jindal, Ankita and Laishram, Shanta. 2022. Journal of Number Theory. 241. pp. 387-429.

Published work / Published

Proves irreducibility and group results for integer parameters in the interval [-2n-4,-2n], including Bessel-related cases. Verifies parts of the alternating-group conjectural classification in that range.

https://doi.org/10.1016/j.jnt.2022.04.001

https://digitalcommons.isical.ac.in/journal-articles/2870/

https://www.sciencedirect.com/science/article/abs/pii/S0022314X22001068

[CullinanHajirSell2009] Algebraic properties of a family of Jacobi polynomials

Cullinan, John and Hajir, Farshid and Sell, Elizabeth. 2009. Journal de théorie des nombres de Bordeaux. 21(1). pp. 97-108.

Published work / Published

For fixed n at least 6, obtains irreducibility for all but finitely many rational parameters in a particular one-parameter Jacobi subfamily. The Galois-group exceptional set is finite for odd n and thin for even n.

https://www.numdam.org/articles/10.5802/jtnb.659/

https://doi.org/10.5802/jtnb.659

[CullinanHajir2014] On the Galois groups of Legendre polynomials

Cullinan, John and Hajir, Farshid. 2014. Indagationes Mathematicae. 25(3). pp. 534-552.

Published work / Published

Studies the groups of parity-reduced Legendre polynomials through discriminants and Newton polygons. Its results and conjectures must be stated for the correct reduced polynomial, with irreducibility assumptions where required.

https://doi.org/10.1016/j.indag.2014.01.004

https://www.numdam.org/articles/10.5802/pmb.57/

[CullinanHajir2017] Algebraic properties of Kaneko-Zagier lifts of supersingular polynomials

Cullinan, John and Hajir, Farshid. 2017. Proceedings of the American Mathematical Society. 145(6). pp. 2291-2304.

Published work / Published

Connects rational lifts of supersingular polynomials with specialized Jacobi polynomials. Gives new irreducibility cases and investigates the Mahlburg-Ono maximal-Galois-group conjecture after removing trivial factors.

https://doi.org/10.1090/proc/13212

[Cullinan2024] On the irreducibility of the Krawtchouck polynomials

Cullinan, John. 2024. Journal of the Ramanujan Mathematical Society. 39(4). pp. 323-335.

Published work / Published

Determines Newton polygons for certain Krawtchouck polynomials and uses them to prove new irreducibility cases.

https://www.mathjournals.org/jrms/2024-039-004/2024-039-004-002.html

https://arxiv.org/abs/2212.07539

[Carlitz1952BernoulliEuler] Note on irreducibility of the Bernoulli and Euler polynomials

Carlitz, Leonard. 1952. Duke Mathematical Journal. 19. pp. 475-481.

Published work / Published

An early arithmetic source for irreducibility in Bernoulli and Euler families. These are the Appell polynomials defined by exponential generating functions, and should not be conflated with descent-enumerating Eulerian polynomials.

https://doi.org/10.1215/S0012-7094-52-01948-0

https://www.mathstat.dal.ca/~dilcher/bernc.html

https://www.impan.pl/shop/publication/transaction/download/product/83168

https://pmc.ncbi.nlm.nih.gov/articles/PMC7114963/

[Brillhart1969] On the Euler and Bernoulli polynomials

Brillhart, John. 1969. Journal für die reine und angewandte Mathematik. 234. pp. 45-64.

Published work / Published

Studies factorization and roots of Euler and Bernoulli polynomials, including irreducibility results and normalizations after removing forced factors.

https://doi.org/10.1515/crll.1969.234.45

https://www.mathstat.dal.ca/~dilcher/bernb.html

[FoataSchutzenberger1970] Théorie géométrique des polynômes eulériens

Foata, Dominique and Schützenberger, Marcel-Paul. 1970. Springer. 138. Lecture Notes in Mathematics.

Book / Published

Develops the conventions, generating functions, and symmetries of Eulerian polynomials used in their arithmetic study.

The linked December 2005 TeX retypesetting is authorized; it reproduces the 1970 monograph with different pagination.

https://www.mat.univie.ac.at/~slc/books/foaschuetz.html

[SharmaStraus1974] On the irreducibility of a class of Euler Frobenius polynomials

Sharma, A. and Straus, Ernst G. 1974. Canadian Mathematical Bulletin. 17(2). pp. 265-273.

Published work / Published

Proves irreducibility results for Euler-Frobenius polynomials. Theorem 6 at parameter r=1 includes the classical Eulerian polynomial at an odd prime index, where a shift gives Eisenstein's criterion.

https://doi.org/10.4153/CMB-1974-052-x

https://www.cambridge.org/core/services/aop-cambridge-core/content/view/C8F1019355B9DAEDF0FC0F60677DEAC7/S0008439500070818a.pdf/on_the_irreducibility_of_a_class_of_euler_frobenius_polynomials.pdf

[MinacNguyenTan2023] Fekete polynomials, quadratic residues, and arithmetic

Mináč, Ján and Nguyen, Tung T. and Tân, Nguyen Duy. 2023. Journal of Number Theory. 242. pp. 532-575.

Published work / Published

Introduces normalized reciprocal and trace polynomials associated with prime-conductor Fekete polynomials, relates values to class numbers, and studies their splitting fields. The proposed full wreath-product group is a conjecture in this source, not an all-prime theorem.

https://arxiv.org/abs/2111.05256

https://doi.org/10.1016/j.jnt.2022.05.001

https://www.sciencedirect.com/science/article/pii/S0022314X22001251

[MinacNguyenTan2024] On the arithmetic of generalized Fekete polynomials

Mináč, Ján and Nguyen, Tung T. and Tân, Nguyen Duy. 2024. Experimental Mathematics. 33(4). pp. 723-754.

Published work / Published

Extends the arithmetic investigation to quadratic characters with possibly composite conductor. Cyclotomic factors, trace-polynomial reductions, and extra symmetries constrain the possible Galois groups.

https://arxiv.org/abs/2206.11778

https://doi.org/10.1080/10586458.2023.2293283

https://www.tandfonline.com/doi/abs/10.1080/10586458.2023.2293283

[AndersonBertelliODorney2026] Galois groups of reciprocal polynomials and the van der Waerden-Bhargava theorem

Anderson, Theresa C. and Bertelli, Adam and O'Dorney, Evan M. 2026. Algebra & Number Theory. 20(3). pp. 603-628.

Published work / Published

For separable reciprocal polynomials of degree 2n and growing coefficient height H, gives exceptional proportion of order H^(-1) log H for nonmonic n $\geq$ 1 and monic n $\geq$ 3. The corrected version leaves the monic quartic case open; the expected group is the full signed-permutation group.

https://arxiv.org/abs/2406.18970v2

https://doi.org/10.2140/ant.2026.20.603

https://arxiv.org/abs/2406.18970

https://arxiv.org/html/2406.18970v2

https://msp.org/ant/2026/20-3/ant-v20-n3-s.pdf

https://msp.org/index/ail.php?jpath=ant&l=0

[DLMFBernoulliEuler] Bernoulli and Euler Polynomials (Chapter 24)

NIST Digital Library of Mathematical Functions. Undated. NIST Digital Library of Mathematical Functions.

Resource / Maintained reference resource

Gives conventions, generating identities, Bernoulli denominator arithmetic, and complete multiple-root classifications: every B_n is squarefree; E_5 is the unique Euler exception. Appell Euler polynomials are distinct from Eulerian descent polynomials.

https://dlmf.nist.gov/24

https://dlmf.nist.gov/24.2

https://dlmf.nist.gov/24.4

https://dlmf.nist.gov/24.12

https://dlmf.nist.gov/24.10

[Deng2026Legendre] A proof of the irreducibility conjecture for Legendre polynomials

Deng, Zikang. 2026. arXiv preprint.

Preprint / Preprint claim; version 2, 27 September 2026; no peer-reviewed publication verified

Preprint claiming a proof of Stieltjes' irreducibility conjecture for parity-normalized Legendre polynomials in every degree, using upper and lower bounds for a resultant. Included to keep the historical conjecture record current; the full proof has not been independently verified in this atlas.

https://arxiv.org/abs/2609.22336v2

https://arxiv.org/html/2609.22336v2

https://arxiv.org/abs/2609.22336

[Heidrich1984Eulerian] On the factorization of Eulerian polynomials

Heidrich, Arnošt J. J. 1984. Journal of Number Theory. 18(2). pp. 157-168.

Published work / Published article; original full text not accessed in this review

Studies arithmetic factorization of Eulerian polynomials, including the lower bound p−1 for the degree of a large irreducible factor when p is the largest prime not exceeding the index. The atlas gives an elementary Newton-polygon derivation and the prime-plus-one consequence.

Publication metadata cross-checked in the University of Utah journal bibliography; the factor-degree result is attributed to Heidrich in the introduction of Ta Thi Hoai An and Ngo Quoc Hoan, Multiple Eulerian polynomials. No original-paper theorem number is claimed.

https://doi.org/10.1016/0022-314X(84)90050-7

https://ftp.math.utah.edu/public_html/pub/tex/bib/jnumbertheory1980.pdf

https://math.ac.vn/uploads/files/IMH20230604.pdf

[ChidambaramMinacNguyenTan2025] Fekete polynomials of principal Dirichlet characters

Chidambaram, Shiva and Mináč, Ján and Nguyen, Tung T. and Tân, Nguyen Duy. 2025. Journal of Experimental Mathematics. 1(1). pp. 51-93.

Published work / Published article

Studies the 0/1 polynomials associated with principal Dirichlet characters: cyclotomic factors, the remaining polynomial, separability, and Galois groups. Distinct from the prime-conductor quadratic-character family; records conjectures suggested by calculations.

Use publisher year2025, volume1(1), pp51-93. Some discovery indexes display2026, but the journal page and PDF state2025.

https://jexpmath.org/index.php/jem/article/view/Vol-1Issue-1Paper-4

https://arxiv.org/abs/2307.14896

https://doi.org/10.56994/JXM.001.001.004

https://jexpmath.org/index.php/jem/article/download/Vol-1Issue-1Paper-4/Vol-1Issue-1Paper-4/54417

[Dilcher2008MultipleZeros] On multiple zeros of Bernoulli polynomials

Dilcher, Karl. 2008. Acta Arithmetica. 134(2). pp. 149-155.

Published work / Published work

Completes the proof that every Bernoulli polynomial is squarefree, using Brillhart's restriction on possible multiple roots and congruences for centered expansions. Section 4 recalls that E_5 is the unique Euler polynomial with multiple roots.

https://doi.org/10.4064/aa134-2-6

https://www.impan.pl/shop/publication/transaction/download/product/83519

https://dlmf.nist.gov/24.12

[Kimura2003Bernoulli] On universal higher order Bernoulli numbers and polynomials

Kimura, Noriaki. 2003. Report of the Research Institute of Industrial Technology, Nihon University. pp. 1-22. Report no. 70.

Published work / Published work

Research report on Bernoulli families and generalizations. Section 2 states Carlitz's even-index irreducibility criterion and its relation to McCarthy's base-p digit-sum condition; later sections discuss higher-order and universal versions.

Report number 70 (2003), ISSN 0386-1678. The report's definition of p-Eisenstein indexes coefficients in descending order; the atlas uses the equivalent usual Eisenstein criterion on the reciprocal polynomial.

https://www.cit.nihon-u.ac.jp/laboratorydata/kenkyu/publication/report/70.pdf

[PatelSiksek2017Powers] On powers that are sums of consecutive like powers

Patel, Vandita and Siksek, Samir. 2017. Research in Number Theory. 3. Article 2.

Published work / Published work

Proves an unconditional derangement theorem for even Bernoulli Galois groups, using 2-adic Newton polygons and decomposition groups. Applies Chebotarev to equations involving sums of powers; records the even-index irreducibility conjecture.

https://doi.org/10.1007/s40993-016-0068-0

https://link.springer.com/article/10.1007/s40993-016-0068-0

https://pmc.ncbi.nlm.nih.gov/articles/PMC7114963/

[ODorney2026Twisted] Galois groups of twisted reciprocal polynomials: a uniform asymptotic

O'Dorney, Evan M. 2026. arXiv preprint.

Preprint / Preprint; v1 (25 September 2026)

For fixed twist $b\neq 0$ and half-degree $m\geq 4$, reports an asymptotic for exceptional hyperoctahedral Galois groups in an adapted Euclidean height. Primary arXiv indexed abstract checked; theorem details and constants require the full version.

The primary arXiv abstract and v1 endpoint could not be reopened on 12 November 2026. The earlier abstract-only catalogue entry is retained; full theorem statements, constants and any later version remain unverified in this pass.

https://arxiv.org/abs/2609.31613v1

https://arxiv.org/abs/2609.31613

[LaishramShorey2016III] Irreducibility of generalized Hermite-Laguerre polynomials III

Laishram, Shanta and Shorey, Tarlok N. 2016. Journal of Number Theory. 164. pp. 303-322.

Published work / Published

All-degree irreducibility for eight third- and quarter-integer Laguerre parameters, with $(q,n)=(1/4,2)$ exceptional. Only the ordinary assertion is used: the printed power-substitution wording needs qualification because $L_1^{(-3/4)}(x^4)=1/4-x^4$ is reducible.

https://doi.org/10.1016/j.jnt.2016.01.003

https://arxiv.org/abs/1306.0736v2

https://www.isid.ac.in/~statmath/index.php?Action=ViewAbs&PreprintId=311&module=Preprint

[GrosuGrosu2021] The irreducibility of some Wronskian Hermite polynomials

Grosu, Codruţ and Grosu, Corina. 2021. Indagationes Mathematicae. 32(2). pp. 456-497.

Published work / Published

Studies arithmetic and zeros of two-part Wronskian Hermite families using Newton polygons and Schur congruences. The atlas's restatement of Theorem 2 excludes the unit remainder R_(2,1)=1, since irreducibility here requires a nonconstant polynomial.

https://doi.org/10.1016/j.indag.2020.12.004

https://arxiv.org/abs/2007.00065

https://arxiv.org/html/2007.00065v1

https://www.sciencedirect.com/science/article/pii/S0019357720301233

[BanerjeeBera2022] Classifying families of orthogonal polynomials having Galois group the alternating group

Banerjee, Pradipto and Bera, Ranjan. 2022. Journal of Number Theory. 240. pp. 31-49.

Published work / Published

Extends the classification program from square discriminants to alternating Galois groups of generalized Laguerre polynomials, for integer $\alpha \neq n$, up to finitely many exceptions. The diagonal $\alpha = n$ remains outside the stated scope.

https://doi.org/10.1016/j.jnt.2021.12.009

https://www.sciencedirect.com/science/article/abs/pii/S0022314X22000087

[Melotti2025Eulerian] Distribution of the roots of Eulerian polynomials

Melotti, Paul. 2025. Electronic Communications in Probability (accepted on author page); author manuscript arXiv:2507.15908v2.

Accepted manuscript / Accepted; author manuscript checked

A combinatorial proof of the known log-Cauchy zero distribution, using moments that become exactly constant. The paper uses x times the Eulerian polynomial normalized here. The author records acceptance; final journal metadata was not independently retrieved.

https://arxiv.org/abs/2507.15908v2

https://arxiv.org/html/2507.15908v2

https://www.imo.universite-paris-saclay.fr/~paul.melotti/recherche.html

[Chen2026Dodecic] Elementary characterization of the Galois groups of x^12+ax^6+b

Chen, Malcolm Hoong Wai. 2026. Mathematica Slovaca.

Published work / Published

Classifies the 16 possible groups under irreducibility over Q. Lower-degree power compositions, square and cube tests, and a cubic resolvent give an explicit decision algorithm. Published online 2 September 2026.

https://doi.org/10.1515/ms-2026-0503

https://www.degruyterbrill.com/document/doi/10.1515/ms-2026-0503/html

[AwtreyPataneToone2025] On Galois groups of power compositional nonic polynomials

Awtrey, Chad and Patane, Frank and Toone, Brian. 2025. Bulletin of the Australian Mathematical Society. 112(3). pp. 458-472.

Published work / Published

Classifies the 11 possible transitive groups for irreducible $g(x^3)$, with g a rational monic cubic, using two discriminants and two resolvents. Also gives an algorithm to recognize number fields with such a defining polynomial. Online 10 February 2025; issue dated December 2025.

https://doi.org/10.1017/S0004972725000048

https://www.cambridge.org/core/journals/bulletin-of-the-australian-mathematical-society/article/abs/on-galois-groups-of-power-compositional-nonic-polynomials/D88BEDEF37A18AA15F6EAF653E4B6200

https://math.samford.edu/~cawtrey/power-comp-nonics.pdf

Counting, random polynomials, and random matrices

[vanderWaerden1936] Die Seltenheit der reduziblen Gleichungen und der Gleichungen mit Affekt

van der Waerden, B. L. 1936. Monatshefte für Mathematik und Physik. 43. pp. 133-147.

Published work / Published

Studies the rarity of reducible polynomials and exceptional Galois groups in coefficient boxes as coefficient height grows with degree fixed.

https://link.springer.com/article/10.1007/BF01707594

https://doi.org/10.1007/BF01707594

[Chela1963] Reducible polynomials

Chela, R. 1963. Journal of the London Mathematical Society. 38(1). pp. 183-188.

Published work / Published

Gives asymptotic counts of reducible integer polynomials in fixed-degree coefficient boxes, including the main contribution from linear factors. Reducibility is distinct from having an exceptional transitive Galois group.

https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-38.1.183

https://doi.org/10.1112/jlms/s1-38.1.183

https://arxiv.org/pdf/2111.06507

[Gallagher1973] The large sieve and probabilistic Galois theory

Gallagher, P. X. 1973. Analytic Number Theory. 24. pp. 91-101. American Mathematical Society. Proceedings of Symposia in Pure Mathematics.

Published work / Published

Uses the large sieve to combine factorization information at many primes and bound exceptional Galois groups in fixed-degree integer coefficient boxes.

https://doi.org/10.1090/pspum/024/0332694

https://doi.org/10.1090/pspum/024

https://www.cecm.sfu.ca/organics/papers/odlyzko/paper/html/node7.html

https://doi.org/10.1017/S1474748023000154

[Dietmann2012] On the distribution of Galois groups

Dietmann, Rainer. 2012. Mathematika. 58(1). pp. 35-44.

Published work / Published

Bounds the number of monic integer polynomials of fixed degree and bounded height with a prescribed Galois group, using resolvents and integral-point estimates.

https://arxiv.org/abs/1010.5341

https://doi.org/10.1112/S0025579311002105

https://pure.royalholloway.ac.uk/en/publications/on-the-distribution-of-galois-groups/

https://www.cambridge.org/core/journals/mathematika/article/abs/on-the-distribution-of-galois-groups/389955B071D757E6F61BDF5776AC9AE5

[ChowDietmann2020] Enumerative Galois theory for cubics and quartics

Chow, Sam and Dietmann, Rainer. 2020. Advances in Mathematics. 372. Article 107282.

Published work / Published

Studies counts for individual Galois groups in degrees three and four, refining the question of how often the full symmetric group occurs.

https://arxiv.org/abs/1807.05820

https://doi.org/10.1016/j.aim.2020.107282

https://wrap.warwick.ac.uk/id/eprint/138618/

https://www.sciencedirect.com/science/article/pii/S000187082030308X

[BarySorokerBenPorathMatei2024] Probabilistic Galois Theory: The square discriminant case

Bary-Soroker, Lior and Ben-Porath, Or and Matei, Vlad. 2024. Bulletin of the London Mathematical Society. 56(6). pp. 2162-2177.

Published work / Published

Studies the frequency of alternating Galois groups and nonzero square discriminants in the fixed-degree large-box model. Constructions and lower bounds show why a naive random-integer model for the discriminant can be misleading.

https://londmathsoc.onlinelibrary.wiley.com/doi/full/10.1112/blms.13049

https://doi.org/10.1112/blms.13049

https://arxiv.org/abs/2207.12493

[Bhargava2025] Galois groups of random integer polynomials and van der Waerden's Conjecture

Bhargava, Manjul. 2025. Annals of Mathematics. 201(2). pp. 339-377.

Published work / Published

Proves the order-of-magnitude bound O(H^(n-1)) for monic integer polynomials of each fixed degree $n \geq 3$ and height at most H whose Galois group is not S_n. Degree is fixed; this does not settle the fixed-coefficient-law, growing-degree problem.

https://annals.math.princeton.edu/2025/201-2/p01

https://doi.org/10.4007/annals.2025.201.2.1

https://arxiv.org/abs/2111.06507

https://arxiv.org/html/2609.40123v1

[OdlyzkoPoonen1993] Zeros of polynomials with 0,1 coefficients

Odlyzko, A. M. and Poonen, Bjorn. 1993. L'Enseignement Mathématique. 39(3–4). pp. 317-348.

Published work / Published

Poses the asymptotic irreducibility question for monic 0-1 polynomials with constant coefficient 1. Its principal results concern the geometry and topology of their zero sets.

https://www.cecm.sfu.ca/organics/papers/odlyzko/paper/html/paper.html

https://www-users.cse.umn.edu/~odlyzko/doc/sequence.html

https://www.cecm.sfu.ca/organics/papers/odlyzko/paper/html/node1.html

[Konyagin1999] On the number of irreducible polynomials with 0,1 coefficients

Konyagin, S. V. 1999. Acta Arithmetica. 88(4). pp. 333-350.

Published work / Published

For the 0-1 model, unconditionally excludes small-degree factors with high probability and obtains a lower bound of order 1/log n for irreducibility, using finite-field and exponential-sum methods.

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/88/44/110736/on-the-number-of-irreducible-polynomials-with-0-1-coefficients

https://doi.org/10.4064/aa-88-4-333-350

https://arxiv.org/pdf/1810.13360

[BarySorokerKozma2017Bivariate] Is a bivariate polynomial with ± 1 coefficients irreducible? Very likely!

Bary-Soroker, Lior and Kozma, Gady. 2017. International Journal of Number Theory. 13(4). pp. 933-936.

Published work / Published

Studies irreducibility of random bivariate sign polynomials. The additional variable permits arguments unavailable in the univariate fixed-coefficient model.

https://doi.org/10.1142/S1793042117500488

https://weizmann.elsevierpure.com/en/publications/is-a-bivariate-polynomial-with-1-coefficients-irreducible-very-li/

https://arxiv.org/pdf/1810.13360

[BorstEtAl2018] Irreducibility of random polynomials

Borst, Christian and Boyd, Evan and Brekken, Claire and Solberg, Samantha and Wood, Melanie Matchett and Wood, Philip Matchett. 2018. Experimental Mathematics. 27(4). pp. 498-506.

Published work / Published

Computational study of several sampling models and a universality heuristic relating reducibility to low-degree factors. Treat its numerical evidence and conjectures separately from its rigorous statements.

https://arxiv.org/abs/1705.03709

https://doi.org/10.1080/10586458.2017.1325790

https://www.tandfonline.com/doi/abs/10.1080/10586458.2017.1325790

[BreuillardVarju2019] Irreducibility of random polynomials of large degree

Breuillard, Emmanuel and Varjú, Péter P. 2019. Acta Mathematica. 223(2). pp. 195-249.

Published work / Published; main general results conditional on RH

Under Riemann hypotheses for suitable Dedekind zeta functions, proves irreducibility and alternating-group containment for broad independent-coefficient models, with careful treatment of zero and cyclotomic factors. The method connects finite-field random walks, root counts and effective prime-ideal estimates.

https://arxiv.org/abs/1810.13360

https://doi.org/10.4310/ACTA.2019.v223.n2.a1

[ORourkeWood2019] Low-degree factors of random polynomials

O'Rourke, Sean and Wood, Philip Matchett. 2019. Journal of Theoretical Probability. 32(2). pp. 1076-1104.

Published work / Published

Converts pointwise delocalization of roots into exclusion of factors of low algebraic degree. Covers several random-matrix characteristic-polynomial models as well as random polynomials; absence of low-degree factors alone is weaker than irreducibility.

https://arxiv.org/abs/1608.01938

https://doi.org/10.1007/s10959-018-0839-8

https://math.colorado.edu/~seor3821/research/

[BarySorokerKozma2020] Irreducible polynomials of bounded height

Bary-Soroker, Lior and Kozma, Gady. 2020. Duke Mathematical Journal. 169(4). pp. 579-598.

Published work / Published

Proves high-probability irreducibility and containment of A_n for suitable fixed coefficient distributions, including uniform coefficients in {1,...,210}. Combines reductions at several primes with probabilistic permutation-group arguments.

https://arxiv.org/abs/1710.05165

https://doi.org/10.1215/00127094-2019-0047

[BarySorokerKoukoulopoulosKozma2023] Irreducibility of random polynomials: general measures

Bary-Soroker, Lior and Koukoulopoulos, Dimitris and Kozma, Gady. 2023. Inventiones Mathematicae. 233(3). pp. 1041-1120.

Published work / Published

Develops Fourier equidistribution, a sieve for factor degrees and permutation-group methods for independent integer coefficients, conditioning on nonzero constant term. Proves high-probability irreducibility and Galois group A_n or S_n under explicit distributional hypotheses, including a uniform interval of at least 35 integers; much more general laws yield exclusion of factors up to a positive fraction of the degree.

https://arxiv.org/abs/2007.14567

https://doi.org/10.1007/s00222-023-01193-6

https://cris.tau.ac.il/en/publications/irreducibility-of-random-polynomials-general-measures/

[BarySorokerGoldgraber2025] Full Galois groups of polynomials with slowly growing coefficients

Bary-Soroker, Lior and Goldgraber, Noam. 2025. Bulletin of the London Mathematical Society. 57(3). pp. 941-955.

Published work / Published

For monic degree-n polynomials with coefficients uniform in [-L(n),L(n)], proves full symmetric Galois group with high probability whenever L(n) tends to infinity. The interval may grow arbitrarily slowly; the theorem does not assert the same conclusion for a fixed interval.

https://arxiv.org/abs/2404.13559

https://doi.org/10.1112/blms.70008

https://cris.tau.ac.il/en/publications/full-galois-groups-of-polynomials-with-slowly-growing-coefficient-2/

[BarySorokerHokkenKozmaPoonen2025] Irreducibility of Littlewood polynomials of special degrees

Bary-Soroker, Lior and Hokken, David and Kozma, Gady and Poonen, Bjorn. 2025. International Mathematics Research Notices. 2025(21). pp. rnaf326.

Published work / Published

Proves unconditionally that the limsup of the irreducibility probability for uniformly random Littlewood polynomials is 1, using factorization constraints in special degrees. This does not establish an all-degree limit.

https://arxiv.org/abs/2308.04878

https://doi.org/10.1093/imrn/rnaf326

https://academic.oup.com/imrn/article/2025/21/rnaf326/8300532

[Bazin2025] Irreducibility of random polynomials of $\mathbb{Z}$[X]

Bazin, Pierre-Alexandre. 2025. International Mathematics Research Notices. 2025(11). pp. rnaf136.

Published work / Published

Improves rates in the Bary-Soroker-Koukoulopoulos-Kozma framework when near-uniformity holds modulo more primes. The zero-constant-term obstruction remains: for example, uniform intervals of length at least 10^8 give P(reducible) $\leq$ P(a_0=0)+O(n^(-1/2)), rather than an unconditioned bound tending to zero.

https://arxiv.org/abs/2402.06643

https://doi.org/10.1093/imrn/rnaf136

[HokkenKoukoulopoulos2025] Irreducibility and Galois groups of random reciprocal polynomials of large degree

Hokken, David and Koukoulopoulos, Dimitris. 2025.

Preprint / Preprint; version 2 dated 23 September 2026

For reciprocal polynomials of degree 2m and coefficient laws satisfying a Fourier hypothesis, proves high-probability irreducibility and a Galois group equal to the hyperoctahedral group or one of two index-two subgroups. The paired-root symmetry changes both the natural ambient group and the factorization arguments.

Version 2, 23 September 2026

https://arxiv.org/abs/2510.18857v2

https://arxiv.org/abs/2510.18857

[BlacharBreuillardKozma2026] Galois groups of random polynomials of large degree

Blachar, Guy and Breuillard, Emmanuel and Kozma, Gady. 2026.

Preprint / Preprint; version 1 dated 30 September 2026; scope issue noted

Preprint reporting an O(1/log n) square-discriminant probability bound in a high-entropy bounded-coefficient model (Rényi entropy of order 2 greater than 12). The atlas reports the constant-term-one subclass and the uniform-interval full-S_n corollary conditioned on a nonzero constant term. The unqualified v1 statement permits a zero constant term and misses the persistent discriminant-zero event when the first two coefficients vanish; the full proof has not been independently verified.

Version 1, 30 September 2026

Theorem 1.1 in v1 includes square discriminant without excluding zero and permits an arbitrary independent bounded constant-term law. If the constant and linear coefficients are iid with law μ and $\mu(0)>0$, discriminant zero has probability at least $\mu(0)^2$, contradicting the unrestricted bound. Section 8 invokes an O(1/n) repeated-root bound without removing this event. The website records only the a0=1 reported subclass and the explicitly conditioned interval corollary; it does not certify or repair the preprint proof.

https://arxiv.org/abs/2609.40123v1

https://arxiv.org/abs/2609.40123

https://arxiv.org/html/2609.40123v1

[Kogler2026] The Odlyzko-Poonen Conjecture on Irreducibility of Random Polynomials

Kogler, Constantin. 2026.

Preprint / Preprint; version 2 dated 14 October 2026

Preprint presenting an unconditional proof of the irreducibility limit for monic polynomials with constant coefficient 1 and all remaining coefficients independent uniform in {0,1}. The claim concerns this coefficient model and remains a preprint result.

Version 2, 14 October 2026

https://arxiv.org/abs/2609.26771v2

https://arxiv.org/abs/2609.26771

[VarjuXu2026] A random polynomial with multiplicative coefficients is almost surely irreducible

Varjú, Péter P. and Xu, Max Wenqiang. 2026. International Mathematics Research Notices. 2026(16). pp. rnag178.

Published work / Published; conditional on RH

Under RH for Dedekind zeta functions, proves irreducibility with probability 1-O(d^(-1/2+epsilon)) for specified random multiplicative-coefficient models. The theorem concerns P_d(x)/x, after removal of the obligatory factor x; these coefficients are dependent.

https://arxiv.org/abs/2511.04240

https://doi.org/10.1093/imrn/rnag178

https://academic.oup.com/imrn/article/2026/16/rnag178/8761102

[Rivin2008] Walks on groups, counting reducible matrices, polynomials, and surface and free group automorphisms

Rivin, Igor. 2008. Duke Mathematical Journal. 142(2). pp. 353-379.

Published work / Published

Relates random walks on arithmetic groups and finite quotients to irreducible characteristic polynomials. Its random-walk and height models differ from models with independent bounded matrix entries.

https://arxiv.org/abs/math/0703532

https://doi.org/10.1215/00127094-2008-009

https://ww2.newschool.edu/WorkArea/DownloadAsset.aspx?id=54362

[Eberhard2022] The characteristic polynomial of a random matrix

Eberhard, Sean. 2022. Combinatorica. 42(4). pp. 491-527.

Published work / Published; conditional on ERH

Assuming ERH, proves high-probability irreducibility and containment of A_n for characteristic polynomials of matrices with independent entries from a fixed nontrivial finitely supported integer law. The dependent polynomial coefficients require matrix-specific input.

https://arxiv.org/abs/2008.01223

https://doi.org/10.1007/s00493-020-4657-0

https://pure.qub.ac.uk/en/publications/the-characteristic-polynomial-of-a-random-matrix/

[FerberJainSahSawhney2023] Random symmetric matrices: rank distribution and irreducibility of the characteristic polynomial

Ferber, Asaf and Jain, Vishesh and Sah, Ashwin and Sawhney, Mehtaab. 2023. Mathematical Proceedings of the Cambridge Philosophical Society. 174(2). pp. 233-246.

Published work / Published; irreducibility application conditional on ERH

Under ERH, proves high-probability irreducibility for characteristic polynomials of random symmetric sign matrices, using sharp finite-field rank-distribution estimates and inverse Littlewood-Offord methods.

https://arxiv.org/abs/2106.04049

https://doi.org/10.1017/S0305004122000226

https://www.cambridge.org/core/journals/mathematical-proceedings-of-the-cambridge-philosophical-society/article/abs/random-symmetric-matrices-rank-distribution-and-irreducibility-of-the-characteristic-polynomial/99A806811892A282D875B22DF443BC19

[AndersonODorney2026Matrices] Galois groups of random integer matrices

Anderson, Theresa C. and O'Dorney, Evan M. 2026. Mathematical Proceedings of the Cambridge Philosophical Society. pp. 1-16.

Published work / Published online 12 August 2026; First View

Studies fixed-size integer matrices with entries bounded by a height parameter and counts characteristic polynomials with non-symmetric Galois group. Uses the large sieve, Fourier analysis and geometric sieving; this is a different asymptotic regime from Eberhard's growing-dimension theorem.

https://www.cambridge.org/core/journals/mathematical-proceedings-of-the-cambridge-philosophical-society/article/galois-groups-of-random-integer-matrices/B87E2B0D71138A94EF7BB1179891456A

https://doi.org/10.1017/S0305004126102126

[BarySorokerGarzoniSodin2026] Irreducibility of the characteristic polynomials of random tridiagonal matrices

Bary-Soroker, Lior and Garzoni, Daniele and Sodin, Sasha. 2026. Journal of Number Theory. 280. pp. 973-998.

Published work / Published; conditional on RH

Under RH for certain Dedekind zeta functions, proves irreducibility with exponentially high probability and alternating-or-symmetric Galois group for a specified tridiagonal model. A related model has a smaller Galois group.

https://arxiv.org/abs/2502.17218

https://doi.org/10.1016/j.jnt.2025.09.008

[LuczakPyber1993] On random generation of the symmetric group

Łuczak, Tomasz and Pyber, László. 1993. Combinatorics, Probability and Computing. 2(4). pp. 505-512.

Published work / Published

Supplies the permutation-group principle behind several probabilistic Galois arguments: a random permutation rarely belongs to a proper transitive subgroup other than A_n. Invariable generation is relevant because modular factorization determines conjugacy classes, rather than labeled permutations.

https://www.cambridge.org/core/journals/combinatorics-probability-and-computing/article/abs/on-random-generation-of-the-symmetric-group/C6FE584196D2BE56BB85CAE23BEDF59D

https://doi.org/10.1017/S0963548300000869

[EberhardFordKoukoulopoulos2016] Permutations contained in transitive subgroups

Eberhard, Sean and Ford, Kevin and Koukoulopoulos, Dimitris. 2016. Discrete Analysis. 2016. Article 12.

Published work / Published

Sharpens quantitative bounds for permutations lying in transitive subgroups other than S_n or A_n. It treats imprimitive and primitive obstructions separately, supplying estimates for large-group arguments after irreducibility has been established.

Paper No. 12

https://discreteanalysisjournal.com/article/849-permutations-contained-in-transitive-subgroups

https://doi.org/10.19086/da.849

https://arxiv.org/abs/1605.01068

[EberhardFordGreen2017] Invariable generation of the symmetric group

Eberhard, Sean and Ford, Kevin and Green, Ben. 2017. Duke Mathematical Journal. 166(8). pp. 1573-1590.

Published work / Published

Shows that three uniform random permutations fail to invariably generate S_n with probability tending to one, whereas four succeed with probability bounded away from zero. Explains why information from a small fixed collection of modular factorizations has a nontrivial combinatorial threshold.

https://arxiv.org/abs/1508.01870

https://doi.org/10.1215/00127094-0000007X

https://pure.qub.ac.uk/en/publications/invariable-generation-of-the-symmetric-group/

[PeledSenZeitouni2016] Double roots of random Littlewood polynomials

Peled, Ron and Sen, Arnab and Zeitouni, Ofer. 2016. Israel Journal of Mathematics. 213(1). pp. 55-77.

Published work / Published

Gives sharp parity-sensitive double-root asymptotics for fair-sign Littlewood polynomials. Separates the easier squarefreeness question from irreducibility and explains the dominant repeated roots at ± 1.

Theorem 1.2, specialized to fair signs; publication metadata confirmed by the publisher, arXiv and the author's institutional record. Publisher and institutional abstract rendering interchanges formulas, so mathematical statements were checked against the full paper.

https://arxiv.org/abs/1409.2034

https://doi.org/10.1007/s11856-016-1328-3

https://arxiv.org/html/1409.2034v2

https://experts.umn.edu/en/publications/double-roots-of-random-littlewood-polynomials/

https://link.springer.com/article/10.1007/s11856-016-1328-3

[FeldheimSen2017] Double roots of random polynomials with integer coefficients

Feldheim, Ohad N. and Sen, Arnab. 2017. Electronic Journal of Probability. 22. pp. 1-23. Article 10.

Published work / Published

For bounded iid integer coefficients with largest atom at most $1/2$, locates repeated-root probability at 0 and ± 1 up to o(n^(-2)); when zero is excluded, obtains O(n^(-2)). Allowing zero coefficients can leave a persistent discriminant-zero event.

Theorem 1.1. The model randomizes every coefficient, including the leading one; it is not stated as a fixed-monic theorem.

https://arxiv.org/abs/1603.03811

https://doi.org/10.1214/17-EJP24

https://arxiv.org/html/1603.03811v1

https://experts.umn.edu/en/publications/double-roots-of-random-polynomials-with-integer-coefficients/

[Kuba2009] On the distribution of reducible polynomials

Kuba, Gerald. 2009. Mathematica Slovaca. 59(3). pp. 349-356.

Published work / Published

Establishes sharp orders for reducible, not necessarily monic integer polynomials of fixed degree and bounded coefficient height: H^n in degree $n \geq 3$, and H^2 log H in degree two. Complements monic counting theorems and isolates the contribution of linear factors.

Published in 2009; the author's arXiv copy was deposited in 2020. Theorems 1-4 in that copy.

https://www.sav.sk/index.php?doc=journal-list&journal_article_no=5434&lang=en&part=article_response_page

https://doi.org/10.2478/s12175-009-0131-6

https://arxiv.org/abs/2008.05815

https://arxiv.org/pdf/2008.05815

[BenPorath2026RootSubsets] Counting Polynomials via Galois Actions on Root Subsets

Ben-Porath, Or. 2026.

Preprint / Preprint; arXiv version 1

Obtains upper bounds for monic bounded-height polynomials with specified permutation representations, including actions on subsets, ordered tuples, and regular actions. The method recovers splitting fields from carefully chosen root-subset stabilizers and controls heights of their generators.

arXiv:2603.14617v1, submitted 15 March 2026

Theorems 1.1-1.4; Lemma 3.1 and Proposition 3.3 explain the root-subset mechanism. The HTML title page displays 24 August 2026 although the arXiv record lists only v1 submitted 15 March 2026; version identifier is retained explicitly.

https://arxiv.org/abs/2603.14617

https://arxiv.org/html/2603.14617v1

[EntinPopov2024] Probabilistic Galois theory in function fields

Entin, Alexei and Popov, Alexander. 2024. Finite Fields and Their Applications. 98. Article 102466.

Published work / Published; full-group conclusion in the fixed-coefficient-degree model conditional

Studies monic polynomials in y with independent uniformly bounded-degree coefficients in F_q[x]. The two limiting regimes differ: growing coefficient degree gives S_n with probability tending to one; growing y-degree gives irreducibility limit 1−q^(−d), and conditioned on irreducibility, the group is A_n or S_n with probability tending to one. The full-S_n refinement in odd characteristic uses a uniform polynomial Chowla conjecture.

Theorems 1.1, 1.2, 1.4 and 1.6 checked in arXiv v3 (4 July 2024); the precise uniformity required for the conditional result is Conjecture 1.5. The publisher article endpoint could not be reopened during the 12 November 2026 second pass; publication metadata was retained from the earlier primary-publisher check.

https://www.sciencedirect.com/science/article/pii/S1071579724001059

https://doi.org/10.1016/j.ffa.2024.102466

https://arxiv.org/abs/2311.14862

https://arxiv.org/abs/2311.14862v3

https://arxiv.org/html/2311.14862v3

[MichelenYakir2025Discriminant] Law of large numbers for the discriminant of random polynomials

Michelen, Marcus and Yakir, Oren. 2025.

Preprint / Preprint; version 1 dated 13 June 2025

Proves a universal first-order law for log absolute discriminant of iid mean-zero, variance-one sub-Gaussian Kac polynomials with no zero coefficient atom. Connects Mahler measure, reciprocal symmetry in distribution and root separation. A size asymptotic does not determine the discriminant's square class.

arXiv:2506.12206v1

Theorem 1, equation (10), and §1.3. The abstract omits hypotheses that are explicit in Theorem 1; the atlas preserves them.

https://arxiv.org/abs/2506.12206

https://arxiv.org/html/2506.12206v1

https://marcusmichelen.org/research.php

[MichelenYakir2026RootSeparation] Limit law for root separation in random polynomials

Michelen, Marcus and Yakir, Oren. 2026. Advances in Mathematics. 496. Article 110990.

Published work / Published

Establishes a limiting law for the smallest root separation at scale n^(-5/4) for mean-zero sub-Gaussian iid coefficients with no atom at zero. Corollary 1.2 implies squarefreeness with probability tending to one without the older integer-law largest-atom restriction.

Theorem 1.1 and Corollary 1.2; published in June 2026. The sharp Littlewood repeated-root asymptotic of Peled-Sen-Zeitouni supplies a quantitative conclusion not contained in the limit statement alone.

https://www.sciencedirect.com/science/article/pii/S0001870826002124

https://doi.org/10.1016/j.aim.2026.110990

https://arxiv.org/abs/2505.02723

https://arxiv.org/html/2505.02723v1

https://www.sciencedirect.com/journal/advances-in-mathematics/vol/496/suppl/C

Hilbert irreducibility and inverse Galois theory

[Hilbert1892] Ueber die Irreducibilität ganzer rationaler Functionen mit ganzzahligen Coefficienten

Hilbert, David. 1892. Journal für die reine und angewandte Mathematik. 110. pp. 104-129.

Published work / Published article

The original source for Hilbert's irreducibility theorem. It is the historical starting point for passing from irreducible polynomial families over a rational function field to irreducible arithmetic specializations; use a modern treatment for the general language of Hilbertian fields.

https://www.degruyterbrill.com/document/doi/10.1515/crll.1892.110.104/html

https://doi.org/10.1515/crll.1892.110.104

[DvornicichZannier1994] Fields containing values of algebraic functions

Dvornicich, Roberto and Zannier, Umberto. 1994. Annali della Scuola Normale Superiore di Pisa, Classe di Scienze (4). 21(3). pp. 421-443.

Published work / Published article

Studies the number fields generated by specialized values of algebraic functions, addressing the diversity of fields obtained from irreducible specializations.

https://www.numdam.org/item/ASNSP_1994_4_21_3_421_0/

[HajirWong2006] Specializations of one-parameter families of polynomials

Hajir, Farshid and Wong, Siman. 2006. Annales de l'Institut Fourier. 56(4). pp. 1127-1163.

Published work / Published article

Develops geometric and group-theoretic conditions that make the exceptional set of reducible specializations over a number field finite. For fixed n $\geq$ 10, all but finitely many parameter values in any fixed number field give irreducible generalized Laguerre polynomials with Galois group S_ n.

https://www.numdam.org/item/AIF_2006__56_4_1127_0/

https://doi.org/10.5802/aif.2208

https://arxiv.org/abs/math/0405139

https://www.numdam.org/item/10.5802/aif.2208.pdf

[DvornicichZannier2007] Cyclotomic Diophantine problems (Hilbert irreducibility and invariant sets for polynomial maps)

Dvornicich, Roberto and Zannier, Umberto. 2007. Duke Mathematical Journal. 139(3). pp. 527-554.

Published work / Published article

Develops Hilbert irreducibility in cyclotomic extensions through the arithmetic of torsion points on tori. Applications to roots-of-unity parameters or dynamics over cyclotomic fields require the stated structural hypotheses.

https://arpi.unipi.it/handle/11568/112812

https://doi.org/10.1215/S0012-7094-07-13934-6

https://ror.circle-u.eu/items/44e0b4f8-aef4-40f1-8da3-2a2f2391946c/full

https://www.ae-info.org/ae/Member/Zannier_Umberto/Publications

[Serre2007Topics] Topics in Galois Theory

Serre, Jean-Pierre. 2007. A K Peters/CRC Press, Research Notes in Mathematics 1. Edition 2. A K Peters/CRC Press.

Book / Published monograph

Treats Hilbert irreducibility, thin sets, explicit Galois realizations, rigidity, and sieve methods. Chapter 3 and Section 4.6 explain preservation of generic Galois groups under specialization. Section 4.4 proves the Morse criterion using inertia and Riemann-Hurwitz to obtain full symmetric monodromy.

Notes written by Henri Darmon. Publisher electronic edition copyright 2007; the physical second edition is also commonly catalogued and cited as 2008.

https://www.routledge.com/Topics-in-Galois-Theory-2nd-Edition/Serre/p/book/9780429064401

https://doi.org/10.1201/b10588

https://api.pageplace.de/preview/DT0400.9781439865255_A38251687/preview-9781439865255_A38251687.pdf

https://indico.math.cnrs.fr/event/11410/attachments/4760/7351/SerreTopicsGaloisTheory.pdf

[Zannier2010] Hilbert irreducibility above algebraic groups

Zannier, Umberto. 2010. Duke Mathematical Journal. 153(2). pp. 397-425.

Published work / Published article

Studies irreducible fibers above Zariski-dense cyclic subgroups of algebraic tori and powers of an elliptic curve without complex multiplication, under a pullback irreducibility condition. The method combines torsion specialization with descent using Chebotarev, and yields a Bertini-type result for tori.

https://arxiv.org/abs/1001.3079

https://doi.org/10.1215/00127094-2010-027

https://ricerca.sns.it/handle/11384/64

[DebesLegrand2013] Specialization results in Galois theory

Dèbes, Pierre and Legrand, François. 2013. Transactions of the American Mathematical Society. 365(10). pp. 5259-5275.

Published work / Published article

Uses twisting to turn prescribed specialization questions into rational-point questions. One application combines irreducible specialization with prescribed residue-degree partitions at finitely many sufficiently large primes for covers over Q with geometric monodromy S_n.

https://arxiv.org/abs/1106.6151

https://doi.org/10.1090/S0002-9947-2013-05800-X

https://cris.technion.ac.il/en/publications/specialization-results-in-galois-theory/

https://pro.univ-lille.fr/fileadmin/user_upload/pages_pros/pierre_debes/A52-DebesLegrand1.pdf

[FriedJarden2023] Field Arithmetic

Fried, Michael D. and Jarden, Moshe. 2023. Springer, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, 11. Edition 4. Springer.

Book / Published monograph

Treats Hilbertian fields, absolute Galois groups, specialization, and PAC fields. The fourth edition updates the earlier editions and adds a chapter on Hilbertian subfields of Galois extensions.

https://link.springer.com/book/10.1007/978-3-031-28020-7

https://doi.org/10.1007/978-3-031-28020-7

[Volklein1996] Groups as Galois Groups: An Introduction

Völklein, Helmut. 1996. Cambridge Studies in Advanced Mathematics 53. Cambridge University Press.

Book / Published monograph

Develops the geometric approach to inverse Galois theory through covers, Riemann's existence theorem, and rigidity, constructing polynomial families from prescribed permutation data.

https://www.cambridge.org/core/books/groups-as-galois-groups/A2233CBEE2CD250DDE2B867502C0A4DE

https://doi.org/10.1017/CBO9780511471117

https://assets.cambridge.org/97805215/62805/frontmatter/9780521562805_frontmatter.pdf

[JensenLedetYui2002] Generic Polynomials: Constructive Aspects of the Inverse Galois Problem

Jensen, Christian U. and Ledet, Arne and Yui, Noriko. 2002. Mathematical Sciences Research Institute Publications 45. Cambridge University Press.

Book / Published monograph

Constructs generic polynomials using invariant-theoretic methods. The MSRI/SLMath page provides free downloadable chapters and explicit families for specified finite groups.

https://library.slmath.org/books/Book45/index.html

https://library.slmath.org/books/Book45/desc.html

[MalleMatzat2018] Inverse Galois Theory

Malle, Gunter and Matzat, B. Heinrich. 2018. Springer Monographs in Mathematics. Edition 2. Springer.

Book / Published monograph

A systematic reference for constructing Galois extensions through rigidity, braid actions, embedding problems, additive polynomials, and rigid analytic methods. It also includes tables of example polynomials for transitive and primitive groups of small degree.

https://link.springer.com/book/10.1007/978-3-662-55420-3

https://doi.org/10.1007/978-3-662-55420-3

[SiksekStoll2026] Hilbert's Irreducibility for $\mathbb{G}_m$

Siksek, Samir and Stoll, Michael. 2026. arXiv preprint.

Preprint / Preprint; version 1, 3 September 2026

Describes exceptional S-unit fibers of a curve cover as a finite set together with finitely many power cosets. Theorem 3 makes the cosets effective but leaves the finite remainder ineffective; Corollary 1 characterizes infinitely many reducible f(x)−p_1^{e_1}···p_s^{e_s}.

Source title uses the notation G_m for the multiplicative group; author order in the PDF is Samir Siksek and Michael Stoll.

https://arxiv.org/abs/2609.04551v1

https://arxiv.org/html/2609.04551v1

https://arxiv.org/abs/2609.04551

https://www.mathe2.uni-bayreuth.de/stoll/schrift.html

[BarySorokerGarzoniMatei2025Powers] On the irreducibility of f(2^n,3^m,X) and other such polynomials

Bary-Soroker, Lior and Garzoni, Daniele and Matei, Vlad. 2025. Mathematische Zeitschrift. 310. pp. Article 11.

Published work / Published article

Under GRH and the geometric pullback condition (PB), Theorem 1.1 proves density-one irreducibility along independent power parameters. The power-pullback condition is stronger than generic irreducibility and excludes X^2-t. Theorem 1.1 counts exponent vectors in symmetric boxes, including negative exponents.

https://doi.org/10.1007/s00209-025-03717-7

https://link.springer.com/article/10.1007/s00209-025-03717-7

https://arxiv.org/abs/2405.04058

[Debes1992Powers] On the irreducibility of the polynomials P(t^m, Y)

Dèbes, Pierre. 1992. Journal of Number Theory. 42(2). pp. 141-157.

Published work / Published article

Characterizes infinite reducibility along a geometric progression by a reducible scaled power pullback (Theorem 1.2). Under the geometric pullback condition (PB), one obtains irreducibility for all but finitely many integer exponents, unconditionally. The argument uses Siegel's theorem; finiteness is distinct from effective enumeration of exceptional exponents.

Original scanned author-hosted paper inspected visually: Theorem 1.2 is on printed page 144. Publication metadata verified against the publisher and author list.

https://doi.org/10.1016/0022-314X(92)90018-K

https://pro.univ-lille.fr/fileadmin/user_upload/pages_pros/pierre_debes/A12-JNT1.pdf

https://pro.univ-lille.fr/pierre-debes/publications

https://www.sciencedirect.com/science/article/pii/0022314X9290018K

https://link.springer.com/article/10.1007/s00209-025-03717-7

Composition, iteration, and arboreal groups

[Odoni1985] The Galois Theory of Iterates and Composites of Polynomials

Odoni, R. W. K. 1985. Proceedings of the London Mathematical Society (3). 51(3). pp. 385-414.

Published work / Published article

Studies Galois groups under composition and iteration. In characteristic zero, the iterates of the generic degree-d polynomial have the full iterated wreath-product group, giving finite-level specialization results and motivating arboreal questions.

https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/plms/s3-51.3.385

https://doi.org/10.1112/plms/s3-51.3.385

https://arxiv.org/pdf/1410.3814

[Stoll1992] Galois groups over Q of some iterated polynomials

Stoll, Michael. 1992. Archiv der Mathematik. 59. pp. 239-244.

Published work / Published article

Characterizes maximality of iterated quadratic Galois groups through arithmetic square-class conditions, relating critical orbits to the growth of splitting fields.

https://link.springer.com/article/10.1007/BF01197321

https://doi.org/10.1007/BF01197321

https://www.mathe2.uni-bayreuth.de/stoll/schrift.html

[AyadMcQuillan2000] Irreducibility of the iterates of a quadratic polynomial over a field

Ayad, Mohamed and McQuillan, Donald L. 2000. Acta Arithmetica. 93(1). pp. 87-97.

Published work / Published article; correction available

Develops criteria for irreducibility of successive quadratic iterates. Read it together with the authors' 2001 correction before using the stated tests in a proof or computation.

https://doi.org/10.4064/aa-93-1-87-97

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/93/1

https://geodesic.mathdoc.fr/articles/10.4064/aa99-1-9/

[AyadMcQuillan2001Correction] Corrections to "Irreducibility of the iterates of a quadratic polynomial over a field" (Acta Arith. 93 (2000), 87–97)

Ayad, Mohamed and McQuillan, Donald L. 2001. Acta Arithmetica. 99(1). pp. 97.

Published work / Published correction

Corrects Ayad and McQuillan's quadratic-iteration paper and should be read with the original article.

https://geodesic.mathdoc.fr/articles/10.4064/aa99-1-9/

https://doi.org/10.4064/aa99-1-9

[Jones2012EveryPrime] An iterative construction of irreducible polynomials reducible modulo every prime

Jones, Rafe. 2012. Journal of Algebra. 369. pp. 114-128.

Published work / Published article

Constructs polynomials of arbitrarily large degree that are irreducible over Q, and over suitable other global fields, but reducible modulo every prime. Failure to find an irreducible reduction is therefore not evidence of reducibility over the ground field.

https://arxiv.org/abs/1012.2857

https://doi.org/10.1016/j.jalgebra.2012.05.020

https://www.sciencedirect.com/science/article/pii/S0021869312002918

https://doi.org/10.1090/bull/1665

[Jones2013Arboreal] Galois representations from pre-image trees: an arboreal survey

Jones, Rafe. 2013. Publications mathématiques de Besançon. Algèbre et théorie des nombres. pp. 107-136.

Published work / Published survey

Surveys Galois actions on trees of iterated preimages, stability, and density questions. Its bibliography covers the foundational literature; its open-problem discussion predates subsequent results.

https://www.numdam.org/articles/10.5802/pmb.a-154/

https://doi.org/10.5802/pmb.a-154

https://arxiv.org/abs/1402.6018

[Specter2018] Polynomials with Surjective Arboreal Galois Representations Exist in Every Degree

Specter, Joel. 2018. arXiv preprint.

Preprint / Preprint; version 2, 12 March 2018

Gives the number-field form of Odoni's existence conjecture in all degrees. The source checked is the 2018 preprint, cited as such in subsequent literature; this must be distinguished from the false assertion over all Hilbertian fields.

https://arxiv.org/abs/1803.00434v2

https://arxiv.org/abs/1803.00434

https://math.jhu.edu/~jspecter/Odoni.pdf

https://arxiv.org/pdf/2012.03076

[BenedettoJuul2019] Odoni's conjecture for number fields

Benedetto, Robert L. and Juul, Jamie. 2019. Bulletin of the London Mathematical Society. 51(2). pp. 237-250.

Published work / Published article

Proves existence of full arboreal Galois groups for even degree over every number field and for odd degree when the number field has odd degree over Q. The paper does not cover every number-field case suggested by the title.

https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/blms.12225

https://doi.org/10.1112/blms.12225

https://arxiv.org/abs/1803.01987

[Juul2019] Iterates of Generic Polynomials and Generic Rational Functions

Juul, Jamie. 2019. Transactions of the American Mathematical Society. 371(2). pp. 809-831.

Published work / Published article

Extends the generic full wreath-product result to arbitrary characteristic for rational functions and to polynomials except the degree-2, characteristic-2 case. The paper also records the contributions of Odoni's unpublished work to these results.

https://arxiv.org/abs/1410.3814

https://arxiv.org/abs/1410.3814v6

https://doi.org/10.1090/tran/7229

https://arxiv.org/pdf/1410.3814

https://sites.google.com/view/jamiejuul/research

[Looper2019] Dynamical Galois groups of trinomials and Odoni's conjecture

Looper, Nicole R. 2019. Bulletin of the London Mathematical Society. 51(2). pp. 278-292.

Published work / Published article

Constructs integral polynomials with surjective arboreal representations in every prime degree. It also gives conditional results in certain composite degrees using Vojta's conjecture, so the unconditional and conditional statements should be separated.

https://arxiv.org/abs/1609.03398

https://doi.org/10.1112/blms.12227

https://londmathsoc.onlinelibrary.wiley.com/doi/full/10.1112/blms.12227

https://sites.google.com/view/nicole-looper/research

[DittmannKadets2022] Odoni's conjecture on arboreal Galois representations is false

Dittmann, Philip and Kadets, Borys. 2022. Proceedings of the American Mathematical Society. 150(8). pp. 3335-3343.

Published work / Published article

Disproves Odoni's conjecture over general Hilbertian fields of characteristic zero. The counterexamples do not contradict positive existence results over number fields.

https://arxiv.org/abs/2012.03076

https://arxiv.org/abs/2012.03076v2

https://doi.org/10.1090/proc/15920

https://arxiv.org/pdf/2012.03076

https://cris.huji.ac.il/en/publications/odonis-conjecture-on-arboreal-galois-representations-is-false-2/

[Ritt1922] Prime and composite polynomials

Ritt, Joseph F. 1922. Transactions of the American Mathematical Society. 23(1). pp. 51-66.

Published work / Published article with erratum

The classical source for functional decomposition of complex polynomials into indecomposable components. Here prime means indecomposable under composition, not irreducible under multiplication; read together with the erratum in the same volume, p. 431.

https://www.jstor.org/stable/1988911

https://doi.org/10.1090/S0002-9947-1922-1501189-9

https://www.ams.org/journals/tran/1922-023-01/S0002-9947-1922-1501189-9/

https://doi.org/10.1090/S0002-9947-1922-1501205-4

[Mueller1995Monodromy] Primitive monodromy groups of polynomials

Müller, Peter. 1995. Recent Developments in the Inverse Galois Problem. 186. pp. 385-401. American Mathematical Society. Contemporary Mathematics.

Published work / Published book chapter; author-hosted version

Classifies geometric monodromy groups of indecomposable complex polynomials and discusses restrictions for polynomials with rational coefficients. The group is that of f(X)−t over a function field, rather than the splitting-field group of f(X) over the coefficient field.

https://ypfmde.github.io/Papers/mon.pdf

https://ypfmde.github.io/publications.html

https://doi.org/10.1090/conm/186

[ZieveMueller2008] On Ritt's polynomial decomposition theorems

Zieve, Michael E. and Müller, Peter. 2008. arXiv preprint.

Preprint / Preprint; no journal publication verified

Describes all complete decompositions of a complex polynomial, refining Ritt's theory. Both authors' publication pages list the work as a preprint; no journal publication is asserted here.

https://arxiv.org/abs/0807.3578

https://websites.umich.edu/~zieve/papers/peter.html

https://ypfmde.github.io/publications.html

[MuzychukPakovich2011] Jordan-Hölder theorem for imprimitivity systems and maximal decompositions of rational functions

Muzychuk, Mikhail and Pakovich, Fedor. 2011. Proceedings of the London Mathematical Society. 102(1). pp. 1-24.

Published work / Published article

Studies imprimitivity lattices for permutation groups with a cyclic subgroup having at most two orbits, with applications to decomposition. It extends aspects of Ritt's first theorem and explains why unrestricted rational-function analogues can fail.

https://doi.org/10.1112/plms/pdq009

https://academic.oup.com/plms/article-abstract/102/1/1/1478963

https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/plms/pdq009

[Stoll2026Arboreal] When do all iterates of x^2+a have maximal Galois group?

Stoll, Michael. 2026. arXiv preprint.

Preprint / Preprint; version 1, 10 November 2026

Reworks quadratic critical-orbit criteria and extends explicit sufficient conditions to rational parameters. Theorem 6.1 imposes region and congruence restrictions; Conjecture 8.1 asks for uniform finite-level detection. Neither the computational evidence nor the reported formalization establishes that conjecture in general.

https://arxiv.org/abs/2611.14014v1

https://arxiv.org/html/2611.14014v1

https://arxiv.org/abs/2611.14014

https://www.mathe2.uni-bayreuth.de/stoll/papers/arboreal-surjective.pdf

https://www.mathe2.uni-bayreuth.de/stoll/schrift.html

https://arxiv.org/pdf/2611.14014v1

[BenedettoGhiocaJuulTucker2025Periodic] Arboreal Galois groups of postcritically finite quadratic polynomials: The periodic case

Benedetto, Robert L. and Ghioca, Dragos and Juul, Jamie and Tucker, Thomas J. 2025. Journal of the London Mathematical Society. 112(2). pp. e70257.

Published work / Published article

Describes groups for x^2+c with periodic critical point in characteristic different from two. Theorem 1.5 gives equivalent square-root, finite-level, and all-level conditions for attaining the specified constrained group, which is not the full binary-tree automorphism group. The target point and cyclotomic constants enter the hypotheses.

https://doi.org/10.1112/jlms.70257

https://arxiv.org/html/2411.06745v2

https://arxiv.org/abs/2411.06745

https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms.70257

[BenedettoGhiocaJuulTucker2026Preperiodic] Arboreal Galois groups of postcritically finite quadratic polynomials: The strictly preperiodic case

Benedetto, Robert L. and Ghioca, Dragos and Juul, Jamie and Tucker, Thomas J. 2026. Canadian Journal of Mathematics.

Published work / Published article

Complements the periodic case by treating strictly preperiodic critical orbits of x^2+c. Separates short-tail and long-tail cases and gives criteria for the resulting constrained groups. Section 1.4 excludes the already-classical Chebyshev map x^2-2, whose towers admit a Kummer description.

Published online 30 March 2026; no unverified final volume or page range assigned. Mathematical locators refer to author preprint v1.

https://doi.org/10.4153/S0008414X2610217X

https://arxiv.org/html/2507.08347v1

https://arxiv.org/abs/2507.08347

Finite fields: construction and distribution

[Shoup1990Irreducible] New algorithms for finding irreducible polynomials over finite fields

Shoup, Victor. 1990. Mathematics of Computation. 54. pp. 435-447.

Published work / Published article; author-hosted version

Constructs irreducible polynomials of a prescribed degree, including deterministic polynomial-time results for small characteristic. Distinguish dependence polynomial in the characteristic from dependence polynomial in its bit length.

https://www.shoup.net/papers/detirred.pdf

https://doi.org/10.1090/S0025-5718-1990-0993933-0

https://www.shoup.net/papers/

[HansenMullen1992] Primitive polynomials over finite fields

Hansen, Tom and Mullen, Gary L. 1992. Mathematics of Computation. 59(200). pp. 639-643.

Published work / Published article with supplementary tables

Provides explicit primitive-polynomial tables and formulates influential existence conjectures with one prescribed coefficient. The primitivity question and the irreducibility question are distinct; a primitive polynomial has a root generating the multiplicative group of the extension field.

https://doi.org/10.1090/S0025-5718-1992-1134730-7

https://www.ams.org/journals/mcom/1992-59-200/S0025-5718-1992-1134730-7/S0025-5718-1992-1134730-7.pdf

https://pure.psu.edu/en/publications/primitive-polynomials-over-finite-fields/

[Shoup1994] Fast construction of irreducible polynomials over finite fields

Shoup, Victor. 1994. Journal of Symbolic Computation. 17. pp. 371-391.

Published work / Published article

Gives faster algorithms for constructing an irreducible polynomial of specified degree over a finite field, thereby producing an explicit representation of the corresponding extension field.

https://doi.org/10.1006/jsco.1994.1025

https://www.shoup.net/papers/fastirred.pdf

https://www.shoup.net/papers/

https://www.sciencedirect.com/science/article/pii/S074771718471025X

[Wan1997] Generators and irreducible polynomials over finite fields

Wan, Daqing. 1997. Mathematics of Computation. 66(219). pp. 1195-1212.

Published work / Published article

Uses character-sum estimates to study generators and irreducible polynomials with prescribed coefficients. Establishes the Hansen-Mullen irreducibility assertion outside finitely many cases; Ham and Mullen settle the remaining cases.

https://doi.org/10.1090/S0025-5718-97-00835-1

https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/new-proof-of-the-hansenmullen-irreducibility-conjecture/189A3A4627AC3912AC76E77F6E5221E8

https://www.pollack-math.net/prescribed.pdf

[HamMullen1998] Distribution of irreducible polynomials of small degrees over finite fields

Ham, Kie H. and Mullen, Gary L. 1998. Mathematics of Computation. 67(221). pp. 337-341.

Published work / Published article

Completes the Hansen-Mullen irreducibility conjecture by verifying the finite set of cases left by Wan's theorem. The statement includes necessary exceptions, notably a prescribed zero constant coefficient and the quadratic zero-linear-coefficient case in characteristic two.

https://doi.org/10.1090/S0025-5718-98-00904-1

https://pure.psu.edu/en/publications/distribution-of-irreducible-polynomials-of-small-degrees-over-fin/

https://citeseerx.ist.psu.edu/document?doi=a767d4e1ec243bd2a44b517e4a0961a31a90255f&repid=rep1&type=pdf

[Cohen2006Primitive] Primitive polynomials with a prescribed coefficient

Cohen, Stephen D. 2006. Finite Fields and Their Applications. 12(3). pp. 425-491.

Published work / Published article

Establishes the Hansen-Mullen primitivity assertion for degrees at least nine by character-sum and sieve methods. This is a stronger condition than irreducibility and this paper alone is not the full treatment of the small-degree cases.

https://doi.org/10.1016/j.ffa.2005.08.001

https://eprints.gla.ac.uk/12863/

https://www.sciencedirect.com/science/article/pii/S1071579705000663

[MullenPanario2013] Handbook of Finite Fields

Mullen, Gary L. and Panario, Daniel (editors). 2013. CRC Press.

Book / Published edited handbook

Treats irreducible and primitive polynomials, prescribed coefficients, factorization algorithms, and finite-field applications, with an extensive bibliography.

https://doi.org/10.1201/b15006

https://pure.psu.edu/en/publications/handbook-of-finite-fields/

[Pollack2013Prescribed] Irreducible polynomials with several prescribed coefficients

Pollack, Paul. 2013. Finite Fields and Their Applications. 22. pp. 70-78.

Published work / Published article; author-hosted version

Gives an explicit estimate for monic irreducibles with coefficients prescribed at arbitrary positions, with a nonzero constant term when it is prescribed. For sufficiently large degree n, it allows about $(1-\varepsilon)\sqrt{n}$ prescribed coefficients uniformly over the ground finite field.

https://www.pollack-math.net/prescribed.pdf

https://doi.org/10.1016/j.ffa.2013.03.001

https://www.sciencedirect.com/science/article/pii/S1071579713000336

[Ha2016] Irreducible polynomials with several prescribed coefficients

Ha, Junsoo. 2016. Finite Fields and Their Applications. 40. pp. 10-25.

Published work / Published article

Improves prescribed-coefficient results to a positive proportion of the degree by adapting Bourgain's method. Its large-field range approaches one quarter of the coefficients; the characteristic, size, and nonzero-constant-term conditions must be retained when applying the theorem.

https://arxiv.org/abs/1601.06867

https://arxiv.org/html/1601.06867v1

https://doi.org/10.1016/j.ffa.2016.02.006

https://www.sciencedirect.com/science/article/pii/S1071579716000277

https://arxiv.org/abs/1601.06867v1

[TuxanidyWang2018] A new proof of the Hansen-Mullen irreducibility conjecture

Tuxanidy, Aleksandr and Wang, Qiang. 2018. Canadian Journal of Mathematics. 70(6). pp. 1373-1389.

Published work / Published article

Proves the one-prescribed-coefficient existence theorem using least periods of discrete Fourier transforms over finite fields. This replaces the character-sum approach while retaining the necessary exceptional cases.

https://arxiv.org/abs/1604.04023

https://arxiv.org/abs/1604.04023v1

https://doi.org/10.4153/CJM-2017-022-1

https://www.cambridge.org/core/journals/canadian-journal-of-mathematics/article/new-proof-of-the-hansenmullen-irreducibility-conjecture/189A3A4627AC3912AC76E77F6E5221E8

https://arxiv.org/html/1604.04023v1

[Swan1962] Factorization of polynomials over finite fields

Swan, Richard G. 1962. Pacific Journal of Mathematics. 12(3). pp. 1099-1106.

Published work / Published article

Presents the Stickelberger discriminant-parity theorem, including characteristic two via lifting, and derives reducibility obstructions for binary trinomials. Swan explicitly credits the basic parity theorem to Stickelberger and notes Dalen's earlier proof.

https://msp.org/pjm/1962/12-3/pjm-v12-n3-p27-p.pdf

https://doi.org/10.2140/pjm.1962.12.1099

https://www.math.uchicago.edu/~swan/bibliography.html

[Cohen1970] The distribution of polynomials over finite fields

Cohen, Stephen D. 1970. Acta Arithmetica. 17(3). pp. 255-271.

Published work / Published article

Studies factorization patterns of f(X)-a g(X) as a varies over a finite field, relating factor degrees to cycle types in a function-field Galois group under the stated arithmetic hypotheses.

https://doi.org/10.4064/aa-17-3-255-271

https://www.impan.pl/en/publishing-house/journals-and-series/acta-arithmetica/all/17/3/97248/the-distribution-of-polynomials-over-finite-fields

https://bibliotekanauki.pl/articles/1394036.pdf

[GuralnickZieve2010] Polynomials with PSL(2) monodromy

Guralnick, Robert M. and Zieve, Michael E. 2010. Annals of Mathematics. 172(2). pp. 1315-1359.

Published work / Published article

Classifies positive-characteristic polynomials with specified PSL(2) monodromy under a ramification hypothesis, with applications to exceptional polynomials and decomposition after extending constants. This concerns function-field Galois groups; splitting fields of individual polynomials over finite fields have cyclic Galois groups.

https://annals.math.princeton.edu/2010/172-2/p11

https://doi.org/10.4007/annals.2010.172.1315

https://arxiv.org/abs/0707.1835

[BankBarySorokerRosenzweig2015] Prime polynomials in short intervals and in arithmetic progressions

Bank, Efrat and Bary-Soroker, Lior and Rosenzweig, Lior. 2015. Duke Mathematical Journal. 164(2). pp. 277-295.

Published work / Published article

Proves asymptotic counts for irreducibles in short polynomial intervals and arithmetic progressions, and treats other factorization types. The asymptotic is fixed degree with the field size tending to infinity, with explicit small-characteristic and interval-length restrictions.

https://arxiv.org/abs/1302.0625

https://arxiv.org/html/1302.0625v3

https://doi.org/10.1215/00127094-2856728

https://arxiv.org/abs/1302.0625v3

[Entin2021Restricted] Factorization statistics of restricted polynomial specializations over large finite fields

Entin, Alexei. 2021. Israel Journal of Mathematics. 242(1). pp. 37-53.

Published work / Published article

Studies factorization patterns when specialization parameters are restricted to sufficiently large, regular subsets of a prime field. The degree stays fixed as the characteristic tends to infinity; the result is not a growing-degree statement over one fixed finite field.

https://arxiv.org/abs/1810.07512

https://doi.org/10.1007/s11856-021-2101-9

https://cris.tau.ac.il/en/publications/factorization-statistics-of-restricted-polynomial-specializations/

https://www.math.tau.ac.il/~aentin/pub.html

[Porritt2019Restricted] Irreducible polynomials over a finite field with restricted coefficients

Porritt, Sam. 2019. Canadian Mathematical Bulletin. 62. pp. 429-439.

Published work / Published article

Counts monic irreducibles when every nonleading coefficient avoids a common small forbidden set. Theorem 1 gives explicit q,n,|R| conditions and a constant-term correction; Theorem 2 uses additional interval structure. The field-size and degree ranges matter.

https://doi.org/10.4153/CMB-2018-027-x

https://arxiv.org/html/1711.06243v1

https://arxiv.org/abs/1711.06243

[Cheng2025Restricted] Counting irreducible polynomials with restricted coefficients

Cheng, Kaimin. 2025. Finite Fields and Their Applications. 108. pp. Article 102691.

Published work / Published article

Extends the common-forbidden-set model to position-dependent forbidden sets. Theorem 1.1 allows at most $\sqrt{q}/2$ exclusions per coefficient for q≥300 and sufficiently large degree; Corollary 1.3 treats avoiding one specified value at every position. Includes the necessary constant-term density factor.

Issue date December 2025. The publisher-indexed article preview supplies Theorem 1.1 and Corollary 1.3; direct page fetching intermittently returns HTTP 403.

https://doi.org/10.1016/j.ffa.2025.102691

https://www.sciencedirect.com/science/article/pii/S1071579725001212

https://www.sciencedirect.com/science/article/abs/pii/S1071579725001212

[BarySorokerShmueli2026Squares] Irreducibility of polynomials with square coefficients over finite fields

Bary-Soroker, Lior and Shmueli, Roy. 2026. Finite Fields and Their Applications. 109. pp. Article 102714.

Published work / Published article

For fixed degree n, independent nonleading coefficients uniform on the set of square values give irreducibility probability 1/n+O_n(q^(−1/2)) as q grows. Theorem 2 covers more general sets of bounded algebraic complexity. Uniform square values differ from squaring uniform field elements in odd characteristic.

Published January 2026; DOI contains 2025. Mathematical theorem locators refer to the public author preprint v1, while publication metadata is verified from the author institution and publisher.

https://doi.org/10.1016/j.ffa.2025.102714

https://arxiv.org/html/2410.16814v1

https://arxiv.org/abs/2410.16814

https://cris.tau.ac.il/en/publications/irreducibility-of-polynomials-with-square-coefficients-over-finit/

Algorithms for factorization and Galois groups

[Berlekamp1967] Factoring polynomials over finite fields

Berlekamp, Elwyn R. 1967. Bell System Technical Journal. 46(8). pp. 1853-1859.

Published work / Published article

Factors polynomials over finite fields by linear algebra in the Frobenius-fixed subalgebra, which detects the irreducible factors.

https://doi.org/10.1002/j.1538-7305.1967.tb03174.x

https://onlinelibrary.wiley.com/doi/abs/10.1002/j.1538-7305.1967.tb03174.x

https://www.nokia.com/bell-labs/publications-and-media/publications/factoring-polynomials-over-finite-fields/

[Zassenhaus1969] On Hensel factorization, I

Zassenhaus, Hans. 1969. Journal of Number Theory. 1(3). pp. 291-311.

Published work / Published article

The classical modular route to integer-polynomial factorization: factor modulo a prime, lift p-adically, then recombine the lifted factors. It explains why local factors must be grouped before recovering rational factors.

https://doi.org/10.1016/0022-314X(69)90047-X

https://www.sciencedirect.com/science/article/pii/0022314X6990047X

https://www.numdam.org/item/JTNB_2009__21_1_15_0/

[Rabin1980] Probabilistic algorithms in finite fields

Rabin, Michael O. 1980. SIAM Journal on Computing. 9(2). pp. 273-280.

Published work / Published article

Develops finite-field algorithms for irreducible-polynomial construction, root finding, and factorization, alongside Frobenius-power tests for irreducibility of monic polynomials.

The MIT repository link describes the earlier January 1979 report Probabilistic Algorithm in Finite Fields (MIT-LCS-TR-213). This entry cites the 1980 SIAM journal article; the repository report and journal version have different publication dates.

https://doi.org/10.1137/0209024

https://epubs.siam.org/doi/10.1137/0209024

https://dspace.mit.edu/entities/publication/0c9f4e53-8b19-4a3c-8d9e-0a53238d2eb3

[CantorZassenhaus1981] A new algorithm for factoring polynomials over finite fields

Cantor, David G. and Zassenhaus, Hans. 1981. Mathematics of Computation. 36(154). pp. 587-592.

Published work / Published article

Gives a randomized finite-field factorization algorithm, including the splitting of products whose irreducible factors have equal degree. The search is randomized; the resulting factors can be checked exactly.

https://doi.org/10.1090/S0025-5718-1981-0606517-5

https://flintlib.org/doc/gr_poly/gr_poly_factor_equal_deg.html

https://www.ams.org/journals/mcom/1981-36-154/S0025-5718-1981-0606517-5/

[LLL1982] Factoring polynomials with rational coefficients

Lenstra, A. K. and Lenstra, Jr., H. W. and Lovász, L. 1982. Mathematische Annalen. 261(4). pp. 515-534.

Published work / Published article

Proves polynomial-time factorization over the rationals using lattice basis reduction. The lattice algorithm also supports algebraic-relation searches and certificate construction.

The CWI record is the 1982 preprint report IW 195/82; the UC Davis link supplies scanned full text. The Numdam link is the 2009 Belabas-van Hoeij-Klüners-Steel follow-up, not another edition of this 1982 article.

https://doi.org/10.1007/BF01457454

https://www.math.ucdavis.edu/~deloera/MISC/LA-BIBLIO/trunk/Lovasz/LovaszLenstraLenstrafactor.pdf

https://ir.cwi.nl/pub/9304

https://www.numdam.org/item/JTNB_2009__21_1_15_0/

[Shoup1995] A new polynomial factorization algorithm and its implementation

Shoup, Victor. 1995. Journal of Symbolic Computation. 20(4). pp. 363-397.

Published work / Published article

Develops and implements baby-step/giant-step finite-field factorization, examining its computational performance.

The open-text link is the author's version. The published article appeared in Journal of Symbolic Computation 20(4), October 1995, pages 363-397.

https://doi.org/10.1006/jsco.1995.1055

https://www.shoup.net/papers/factorimpl.pdf

https://www.sciencedirect.com/science/article/pii/S0747717185710553

https://www.shoup.net/papers/

[KaltofenShoup1998] Subquadratic-time factoring of polynomials over finite fields

Kaltofen, Erich and Shoup, Victor. 1998. Mathematics of Computation. 67(223). pp. 1179-1197.

Published work / Published article

Improves the complexity of randomized polynomial factorization over finite fields through a baby-step/giant-step framework used in later factorization algorithms.

The open-text link is the authors' version of the journal work. The 1998 Mathematics of Computation article expands an extended abstract presented at STOC 1995; the two publication dates identify different versions.

https://doi.org/10.1090/S0025-5718-98-00944-2

https://www.shoup.net/papers/subquadratic.pdf

https://www.shoup.net/papers/

https://kaltofen.math.ncsu.edu/bibliography/index.html

https://flintlib.org/doc/algorithms/polynomials/factoring-finite-fields.html

[VanHoeij2002] Factoring polynomials and the knapsack problem

van Hoeij, Mark. 2002. Journal of Number Theory. 95(2). pp. 167-189.

Published work / Published article

Replaces the recombination search in modular factorization with a structured knapsack problem solved by lattice reduction.

The open-text link is the author's version. The journal publication is August 2002; the year 2001 embedded in its DOI is not the publication year.

https://doi.org/10.1006/jnth.2001.2763

https://www.math.fsu.edu/~hoeij/knapsack/paper/knapsack.pdf

https://www.math.fsu.edu/~hoeij/papers.html

https://www.sciencedirect.com/science/article/pii/S0022314X01927635

https://flintlib.org/doc/algorithms/polynomials/factoring-integers.html

[BHKS2009] Factoring polynomials over global fields

Belabas, Karim and van Hoeij, Mark and Klüners, Jürgen and Steel, Allan. 2009. Journal de théorie des nombres de Bordeaux. 21(1). pp. 15-39.

Published work / Published article

Establishes polynomial-time complexity for van Hoeij's original rational factorization algorithm and natural variants. It also treats global-field extensions of the approach and bivariate factorization over finite fields.

https://www.numdam.org/articles/10.5802/jtnb.655/

https://doi.org/10.5802/jtnb.655

https://www.numdam.org/item/JTNB_2009__21_1_15_0/

https://ris.uni-paderborn.de/record/34889

[Stauduhar1973] The determination of Galois groups

Stauduhar, R. P. 1973. Mathematics of Computation. 27(124). pp. 981-996.

Published work / Published article

Introduces subgroup descent using relative invariants and resolvents, a method underlying later Galois-group algorithms.

https://doi.org/10.1090/S0025-5718-1973-0327712-4

https://docs.magma-maths.org/GlobalFields/GaloisGroupsAndAutomorphisms/bibliography.html

https://www.cambridge.org/core/journals/lms-journal-of-computation-and-mathematics/article/computation-of-galois-groups-of-rational-polynomials/70FFFC0B1BBA91C846342618C93D1E08

[LandauMiller1985] Solvability by radicals is in polynomial time

Landau, Susan and Miller, Gary Lee. 1985. Journal of Computer and System Sciences. 30(2). pp. 179-208.

Published work / Published article

Shows that deciding solvability by radicals for rational polynomials is possible in polynomial time. This decision problem should be distinguished from constructing an entire splitting field or listing every element of a large Galois group.

https://doi.org/10.1016/0022-0000(85)90013-3

https://www.cs.cmu.edu/~glmiller/Publications/Papers/LaMi85.pdf

https://www.cs.cmu.edu/~glmiller/Publications/b2hd-LaMi85.html

https://www.sciencedirect.com/science/article/pii/0022000085900133

[SoicherMcKay1985] Computing Galois groups over the rationals

Soicher, Leonard and McKay, John. 1985. Journal of Number Theory. 20(3). pp. 273-281.

Published work / Published article

Develops practical computational methods using resolvents and low-degree transitive groups. The paper also gives rational realizations of the transitive permutation groups in degrees three through seven.

https://doi.org/10.1016/0022-314X(85)90022-8

https://www.sciencedirect.com/science/article/pii/0022314X85900228

https://webspace.maths.qmul.ac.uk/l.h.soicher/pubs.html

https://docs.magma-maths.org/GlobalFields/GaloisGroupsAndAutomorphisms/bibliography.html

[Dummit1991] Solving solvable quintics

Dummit, David S. 1991. Mathematics of Computation. 57(195). pp. 387-401.

Published work / Published article; corrections available

A concrete sextic-resolvent treatment of solvable irreducible quintics and radical expressions. Use it with Robin Chapman's 1992 correction to the proof of Theorem 1 and the numerical errata on the author's webpage.

https://site.uvm.edu/ddummit/

https://site.uvm.edu/ddummit/files/2021/04/Solving_Solvable_Quintics__Math_Comp_57_no195_1991__pp_387_401.pdf

https://doi.org/10.1090/S0025-5718-1991-1079014-X

https://site.uvm.edu/ddummit/files/2021/04/Solving_Solvable_Quintics_note_by_Robin_Chapman.pdf

[GeisslerKlueners2000] Galois group computation for rational polynomials

Geissler, Katharina and Klüners, Jürgen. 2000. Journal of Symbolic Computation. 30(6). pp. 653-674.

Published work / Published article

Develops Galois-group computation using p-adic roots and refinements of Stauduhar's method, preceding Fieker and Klüners's approach without an a priori degree bound.

https://doi.org/10.1006/jsco.2000.0377

https://www.sciencedirect.com/science/article/pii/S0747717100903778

https://www.cambridge.org/core/journals/lms-journal-of-computation-and-mathematics/article/computation-of-galois-groups-of-rational-polynomials/70FFFC0B1BBA91C846342618C93D1E08

[KluenersMalle2000] Explicit Galois realization of transitive groups of degree up to 15

Klüners, Jürgen and Malle, Gunter. 2000. Journal of Symbolic Computation. 30(6). pp. 675-716.

Published work / Published article

Constructs explicit rational polynomials realizing all transitive groups of degree at most fifteen and organizes the constructions by permutation group.

https://doi.org/10.1006/jsco.2000.0378

https://ris.uni-paderborn.de/record/34899

https://www.sciencedirect.com/science/article/pii/S074771710090378X

[DokchitserDokchitser2013] Identifying Frobenius elements in Galois groups

Dokchitser, Tim and Dokchitser, Vladimir. 2013. Algebra & Number Theory. 7(6). pp. 1325-1352.

Published work / Published article

Identifies Frobenius conjugacy classes beyond the factor-degree data of reduction modulo a prime, distinguishing classes that share a cycle type in the chosen permutation representation.

https://doi.org/10.2140/ant.2013.7.1325

https://msp.org/ant/2013/7-6/ant-v7-n6-p02-s.pdf

https://arxiv.org/abs/1009.5388

[FiekerKlueners2014] Computation of Galois groups of rational polynomials

Fieker, Claus and Klüners, Jürgen. 2014. LMS Journal of Computation and Mathematics. 17(1). pp. 141-158.

Published work / Published article

Constructs relative invariants for Galois-group computation without an a priori degree bound. The method underlies current Magma computations, subject to practical time and memory limits.

https://doi.org/10.1112/S1461157013000302

https://arxiv.org/abs/1211.3588

https://www.cambridge.org/core/journals/lms-journal-of-computation-and-mathematics/article/computation-of-galois-groups-of-rational-polynomials/70FFFC0B1BBA91C846342618C93D1E08

[Cohen1993] A Course in Computational Algebraic Number Theory

Cohen, Henri. 1993. Springer. 138. Graduate Texts in Mathematics.

Book / Published book

Develops algorithms for polynomial factorization, number fields, and Galois-group computation, with the arithmetic needed for their implementation.

https://doi.org/10.1007/978-3-662-02945-9

https://link.springer.com/book/10.1007/978-3-662-02945-9

[GathenGerhard2013] Modern Computer Algebra

von zur Gathen, Joachim and Gerhard, Jürgen. 2013. Cambridge University Press. Edition 3.

Book / Published book

Treats polynomial arithmetic, finite-field factorization, Hensel lifting, and factorization over the rationals, with complexity models and algorithmic proofs.

https://doi.org/10.1017/CBO9781139856065

https://www.cambridge.org/core/books/modern-computer-algebra/DB3563D4013401734851CF683D2F03F0

https://cosec.bit.uni-bonn.de/science/mca/

[GuardiaMontesNart2011Montes] Higher Newton polygons in the computation of discriminants and prime ideal decomposition in number fields

Guàrdia, Jordi and Montes, Jesús and Nart, Enric. 2011. Journal de théorie des nombres de Bordeaux. 23(3). pp. 667-696.

Published work / Published article

Presents an optimized Montes method using types and their representatives. Complete types identify irreducible local factors and reveal ramification indices and residue degrees; the algorithm computes prime ideal decomposition and local index information without requiring a global integral basis first.

https://www.numdam.org/articles/10.5802/jtnb.782/

https://www.numdam.org/item/10.5802/jtnb.782.pdf

https://doi.org/10.5802/jtnb.782

[PoteauxWeimann2022Local] Local Polynomial Factorisation: Improving the Montes Algorithm

Poteaux, Adrien and Weimann, Martin. 2022. Proceedings of the 2022 International Symposium on Symbolic and Algebraic Computation (ISSAC '22). pp. 149-157.

Published work / Published conference paper (5 July 2022); related 2026 arXiv deposit

Improves local irreducibility and factorization using generalized Hensel lifting, approximate roots, and divide-and-conquer. The sharper complexity bounds require residue characteristic zero or sufficiently large; the source separately includes residual-field factorization costs. A same-title arXiv record, 2607.02153 (2 July 2026), is linked as a related deposit, not a separate discovery; its full text could not be retrieved in this audit.

https://doi.org/10.1145/3476446.3535487

https://weimann.users.lmno.cnrs.fr/issac.pdf

https://www.issac-conference.org/2022/papers.php

https://arxiv.org/abs/2607.02153

[BrentKung1978] Fast algorithms for manipulating formal power series

Brent, Richard P. and Kung, Hsiang-Tsung. 1978. Journal of the ACM. 25(4). pp. 581-595.

Published work / Published article

Classical fast composition and reversion algorithms. The baby-step/giant-step matrix method also underlies modular composition; distinguish the arbitrary-modulus version from techniques that use the special modulus x^n. The author page records an erratum on page 590.

https://doi.org/10.1145/322092.322099

https://maths-people.anu.edu.au/~brent/pub/pub045.html

https://www.eecs.harvard.edu/~htk/publication/1978-jacm-brent-kung

[KedlayaUmans2011] Fast polynomial factorization and modular composition

Kedlaya, Kiran S. and Umans, Christopher. 2011. SIAM Journal on Computing. 40(6). pp. 1767-1802.

Published work / Published article

Proves a near-linear bit-complexity bound for modular composition over finite fields, yielding the degree-3/2 randomized factorization benchmark. The method uses multivariate evaluation and supports fast Frobenius iteration; its bit-complexity bounds differ from field-operation bounds. Corollary 7.2 states the modular-composition bound for each fixed positive epsilon.

https://doi.org/10.1137/08073408X

https://epubs.siam.org/doi/10.1137/08073408X

https://users.cms.caltech.edu/~umans/papers/KU08-final.pdf

https://dspace.mit.edu/entities/publication/fd268b79-5301-4239-8ce3-4b964b6b72d6

[GuoNarayananUmans2016] Algebraic problems equivalent to beating exponent 3/2 for polynomial factorization over finite fields

Guo, Zeyu and Narayanan, Anand Kumar and Umans, Chris. 2016. 41st International Symposium on Mathematical Foundations of Computer Science (MFCS 2016), LIPIcs. 58. pp. 47:1-47:14.

Published work / Published conference paper

Relates finite-field factorization to other algebraic problems by reductions, identifying improvements that would lower its degree exponent below 3/2. These equivalences do not establish optimality of the exponent 3/2.

https://doi.org/10.4230/LIPIcs.MFCS.2016.47

https://arxiv.org/abs/1606.04592

https://drops.dagstuhl.de/entities/document/10.4230/LIPIcs.MFCS.2016.47

[NeigerSalvySchostVillard2024] Faster modular composition

Neiger, Vincent and Salvy, Bruno and Schost, Éric and Villard, Gilles. 2024. Journal of the ACM. 71(2). pp. Article 11.

Published work / Published article

A Las Vegas algorithm for modular composition over arbitrary fields of known characteristic, with expected arithmetic complexity O(n^1.43). Polynomial matrices encode algebraic relations. Theorem 1.1 detects failed trials; the discussion handles small finite fields by extension. This counts field operations, unlike finite-field bit-complexity bounds.

https://doi.org/10.1145/3638349

https://arxiv.org/abs/2110.08354

https://cs.uwaterloo.ca/~eschost/publications/NSSV24.pdf

https://perso.ens-lyon.fr/gilles.villard/BIBLIOGRAPHIE/biblio.html

[NeigerSalvySchostVillard2026] Faster modular composition using two relation matrices

Neiger, Vincent and Salvy, Bruno and Schost, Éric and Villard, Gilles. 2026. Proceedings of the 51st International Symposium on Symbolic and Algebraic Computation (ISSAC 2026).

Published work / Published conference paper

Theorem 1.3 improves algebraic modular composition to soft-O(n^((omega+3)/4)) for generic inner polynomials relative to a fixed modulus. Genericity failure is detected; an arbitrary-input Las Vegas reduction is left as a research question. Published at ISSAC 2026, with the published version on arXiv v2.

https://doi.org/10.1145/3815436.3815451

https://arxiv.org/abs/2601.17422

https://arxiv.org/html/2601.17422v2

https://perso.ens-lyon.fr/gilles.villard/BIBLIOGRAPHIE/biblio.html

[HoevenLecerf2026] Fast modular composition using spiroids

van der Hoeven, Joris and Lecerf, Grégoire. 2026. Journal of Complexity. 96. pp. 102073.

Published work / Published article

Gives a sharper nearly linear bit bound for modular composition over Z/rZ using structured complex evaluation points and controlled-precision Fourier transforms. Prime finite fields are included. The authors distinguish asymptotic improvement from practical competitiveness. Journal publication is October 2026.

https://doi.org/10.1016/j.jco.2026.102073

https://researchportal.ip-paris.fr/en/publications/fast-modular-composition-using-spiroids/

https://www.texmacs.org/joris/fftcomp/fftcomp.pdf

https://www.texmacs.org/joris/fftcomp/fftcomp.html

[UmansWang2025] A number-theoretic conjecture implying faster algorithms for polynomial factorization and integer factorization

Umans, Chris and Wang, Siki. 2025. arXiv:2511.10851; listed for ISAAC 2026.

Preprint / Preprint; listed for ISAAC 2026

Proposes a structured divisibility-covering conjecture which, under the paper's hypotheses, would improve the degree exponent for finite-field factorization below 3/2. The improvement is conditional. The work is available as a 2025 preprint and was listed in the December 2026 ISAAC program at the verification date.

https://arxiv.org/abs/2511.10851

https://doi.org/10.48550/arXiv.2511.10851

https://www.algo-door.com/isaac2026/program.html

[Doris2020] Computing the Galois group of a polynomial over a p-adic field

Doris, Christopher. 2020. International Journal of Number Theory.

Published work / Published article

Develops local Galois-group algorithms using resolvents, ramification, and global models; a local group occurs as a decomposition subgroup in a global computation. Remark 3.5 records a heuristic precision bound in the original implementation. The linked package separately documents an exact p-adic input mode for proved results.

https://arxiv.org/abs/2003.05834

https://doi.org/10.1142/S179304212050092X

https://arxiv.org/html/2003.05834v1

https://cjdoris.github.io/

https://github.com/cjdoris/pAdicGaloisGroup

https://research-information.bris.ac.uk/en/publications/computing-the-galois-group-of-a-polynomial-over-a-p-adic-field/

Databases and software

[KluenersMalle2001] A database for field extensions of the rationals

Klüners, Jürgen and Malle, Gunter. 2001. LMS Journal of Computation and Mathematics. 4. pp. 182-196.

Published work / Published article

Explains the arithmetic database, its construction methods, and its emphasis on Galois groups and signatures. Consult the live database separately, since its coverage has grown beyond the original paper.

https://doi.org/10.1112/S1461157000000851

https://arxiv.org/abs/math/0102232

https://ris.uni-paderborn.de/record/34897

https://www.cambridge.org/core/journals/lms-journal-of-computation-and-mathematics/article/database-for-field-extensions-of-the-rationals/75F226702791F692154D9BE954D08F0C

[FLINTFactorizationDocumentation] FLINT documentation: polynomial factorization algorithms

The FLINT Team. Undated.

Resource / Maintained software documentation; revision in progress

Official explanations of finite-field and integer-polynomial factorization, connecting Berlekamp, Cantor-Zassenhaus, Hensel lifting, and lattice recombination with library routines. These are implementation guides; use the cited original papers for theorem statements and complexity proofs.

The documentation was undergoing a major revision when checked. Its October 2026 banner states that parts of the text were generated by an LLM and asks readers to report inaccuracies. Mathematical claims in this atlas are independently supported by the original research papers.

https://flintlib.org/doc/algorithms/polynomials/factoring-finite-fields.html

https://flintlib.org/doc/algorithms/polynomials/factoring-integers.html

https://flintlib.org/doc/algorithms/polynomials/irreducibility.html

[KluenersMalleDatabase] A database for number fields

Klüners, Jürgen and Malle, Gunter. Undated.

Resource / Maintained mathematical database

A source of defining polynomials indexed by Galois group and signature. At access, the site reports coverage through degree twenty-three for all transitive groups except M23, with stated limitations on signatures and discriminant minima.

https://galoisdb.math.upb.de/

[LMFDBFieldsGroups] The L-functions and Modular Forms Database: number fields and Galois groups

The LMFDB Collaboration. Undated.

Resource / Maintained mathematical database

Searchable number-field records and transitive-group data, with labels, invariants, and links between objects. Consult the completeness and reliability pages for each collection; a group appearing in a transitive-group catalogue does not by itself prove its realization over the rationals.

Number-field completeness is bounded by degree, signature, discriminant, or ramification conditions. The reliability page marks certain class-group and unit data that depend on GRH. Transitive-group catalogue completeness is a separate statement and does not establish realization over the rationals.

https://www.lmfdb.org/NumberField/

https://www.lmfdb.org/GaloisGroup/

https://www.lmfdb.org/NumberField/Completeness

https://www.lmfdb.org/NumberField/Reliability

https://www.lmfdb.org/GaloisGroup/Completeness

https://www.lmfdb.org/GaloisGroup/Reliability

[MagmaGaloisDocumentation] Magma Handbook: Galois groups

Magma Computational Algebra System. Undated.

Resource / Maintained software documentation

Official documentation for the Stauduhar-based GaloisGroup implementation, with no fixed degree cap for the documented integer, rational and number-field polynomial inputs. GaloisGroup does not itself certify its conditional steps; the documented GaloisProof overloads attempt certification for integer polynomials or absolute number fields. Record a successful result and the exact input/options.

https://docs.magma-maths.org/GlobalFields/GaloisGroupsAndAutomorphisms/galois.html

[PARIDocumentation] PARI/GP reference manual: general number fields and polgalois

The PARI Group. Undated.

Resource / Maintained software documentation

Official reference for polgalois and number-field computations. PARI/GP 2.19 requires irreducible rational input of degree at most seven, or eleven with galdata. The manual explicitly warns that insufficient initial precision can rarely produce a wrong result; record precision, output convention, and version.

https://pari.math.u-bordeaux.fr/dochtml/ref/General_number_fields.html

[SageGaloisDocumentation] SageMath reference manual: polynomial and number-field Galois groups

The Sage Developers. Undated.

Resource / Maintained software documentation

Official interfaces for polynomial and number-field Galois groups. SageMath 10.10 documents PARI through degree 11, GAP fallback with a documented range through degree 15 and some higher examples, and a Magma wrapper limited by group identification through degree 30. The number-field interface distinguishes original-root and Galois-closure actions and warns about absolute versus relative ground fields. A wrapper is not an independent backend proof.

https://doc.sagemath.org/html/en/reference/polynomial_rings/sage/rings/polynomial/polynomial_rational_flint.html

https://doc.sagemath.org/html/en/reference/number_fields/sage/rings/number_field/number_field.html

https://doc.sagemath.org/html/en/reference/number_fields/sage/rings/number_field/galois_group.html

Multivariate absolute irreducibility

[Ruppert1986] Reduzibilität ebener Kurven

Ruppert, Wolfgang M. 1986. Journal für die reine und angewandte Mathematik. 369. pp. 167-191.

Published work / Published article

A classical source on reducibility of plane curves and algebraic criteria for absolute irreducibility. This is a multivariate question: a univariate polynomial of degree greater than one always splits over an algebraic closure.

Bibliographic details checked in Lecerf's survey and an AMS bibliography; the publisher full text was not retrieved.

https://doi.org/10.1515/crll.1986.369.167

https://www.numdam.org/item/CCIRM_2013__3_1_A2_0/

https://doi.org/10.1090/MCOM/3007

[Kaltofen1995] Effective Noether irreducibility forms and applications

Kaltofen, Erich. 1995. Journal of Computer and System Sciences. 50(2). pp. 274-295.

Published work / Published article

Gives effective bounds for algebraic irreducibility criteria, with applications to exceptional specializations in several variables and to factorization complexity.

https://www.sciencedirect.com/science/article/pii/S0022000085710239

https://users.cs.duke.edu/~elk27/bibliography/95/Ka95_jcss.pdf

https://doi.org/10.1006/jcss.1995.1023

[Gao2001Polytopes] Absolute irreducibility of polynomials via Newton polytopes

Gao, Shuhong. 2001. Journal of Algebra. 237(2). pp. 501-520.

Published work / Published article

Uses integral indecomposability of Newton polytopes to obtain absolute irreducibility criteria for sparse multivariate polynomials. Monomial factors must be excluded when applying a polytope criterion to a polynomial.

https://www.sciencedirect.com/science/article/pii/S0021869300985867

https://doi.org/10.1006/jabr.2000.8586

https://www.math.clemson.edu/~sgao/pub.html

[Gao2003PDE] Factoring multivariate polynomials via partial differential equations

Gao, Shuhong. 2003. Mathematics of Computation. 72(242). pp. 801-822.

Published work / Published article

A factorization method based on a differential equation and linear algebra, in characteristic zero or sufficiently large characteristic. Connects rational and absolute factorization and provides an effective reduction from multivariate to bivariate problems.

https://doi.org/10.1090/S0025-5718-02-01428-X

https://www.math.clemson.edu/~sgao/papers/fac_bipoly.pdf

https://www.math.clemson.edu/~sgao/WEB/research.html

https://dblp.org/rec/journals/moc/Gao03.html

[Lecerf2013] Factorisation des polynômes à plusieurs variables

Lecerf, Grégoire. 2013. Journées Nationales de Calcul Formel (13-17 May 2013). pp. 1-85. CIRM. Les cours du CIRM.

Published work / Published lecture notes

A French-language course on multivariate factorization, including absolute factorization and computational algebraic geometry, with an extensive bibliography.

Journées Nationales de Calcul Formel, 13-17 May 2013, talk 2.

https://www.numdam.org/articles/10.5802/ccirm.18/

https://doi.org/10.5802/ccirm.18

https://www.numdam.org/item/CCIRM_2013__3_1_A2_0/

[GaoLauder2001] Decomposition of Polytopes and Polynomials

Gao, Shuhong and Lauder, Alan G. B. 2001. Discrete & Computational Geometry. 26(1). pp. 89-104.

Published work / Published article

Relates Minkowski decompositions of lattice polytopes to possible polynomial factors, gives polygon-decomposition algorithms, and develops projection tests in higher dimension. Corollary 2 gives an absolute-irreducibility certificate with the necessary exclusion of monomial factors. A decomposable polytope is inconclusive.

https://link.springer.com/article/10.1007/s00454-001-0024-0

https://arxiv.org/html/math/0012099v1

https://doi.org/10.1007/s00454-001-0024-0

https://arxiv.org/abs/math/0012099

https://arxiv.org/pdf/math/0012099

Free expositions and course materials

[FilasetaCourse2016] Math 788F: The Theory of Irreducible Polynomials

Filaseta, Michael. 2016. University of South Carolina course materials.

Exposition / Expository notes

Provides a syllabus, problems, and an introduction to Newton polygons for a course on irreducibility methods. The page does not currently link openly to every chapter of the proposed book.

https://people.math.sc.edu/filaseta/gradcourses/Math788F/Math788F.html

[ConradAllPrimes] An irreducible that factors modulo all primes

Conrad, Keith. Undated. Author-hosted expository note.

Exposition / Expository notes

Explains why irreducibility over the rationals need not be detectable by one irreducible reduction. The obstruction is visible in the cycle types available in the Galois group.

Undated; accessed 11 November 2026.

https://kconrad.math.uconn.edu/blurbs/ringtheory/reducibleallp.pdf

https://kconrad.math.uconn.edu/blurbs/

[ConradCubicQuartic] Galois groups of cubics and quartics (not in characteristic 2)

Conrad, Keith. Undated. Author-hosted expository note.

Exposition / Expository notes

Worked low-degree classifications using discriminants and resolvents. The characteristic restriction matters; the author's collection also contains a separate all-characteristics treatment.

Undated; accessed 11 November 2026.

https://kconrad.math.uconn.edu/blurbs/galoistheory/cubicquartic.pdf

https://kconrad.math.uconn.edu/blurbs/

[ConradFiniteTests] Irreducibility tests in F_p[T]

Conrad, Keith. Undated. Author-hosted expository note.

Exposition / Expository notes

Explains finite-field irreducibility tests and the congruence and gcd calculations that give modular certificates.

Undated; accessed 11 November 2026.

https://kconrad.math.uconn.edu/blurbs/ugradnumthy/irredtestFpT.pdf

https://kconrad.math.uconn.edu/blurbs/

[ConradIrredQ] Irreducibility tests in Q[T]

Conrad, Keith. Undated. Author-hosted expository note.

Exposition / Expository notes

Proves Gauss's lemma, the reduction-modulo-a-prime criterion, and Eisenstein's criterion, establishing the prerequisites for more specialized irreducibility tests.

Undated; accessed 11 November 2026.

https://kconrad.math.uconn.edu/blurbs/ringtheory/irredtestsoverQ.pdf

https://kconrad.math.uconn.edu/blurbs/

[ConradModP] Galois groups over Q and factorizations mod p

Conrad, Keith. Undated. Author-hosted expository note.

Exposition / Expository notes

Explains how squarefree factorizations modulo good primes produce cycle types in a Galois group, with an explicit discriminant condition.

Undated; accessed 11 November 2026.

https://kconrad.math.uconn.edu/blurbs/gradnumthy/galois-Q-factor-mod-p.pdf

https://kconrad.math.uconn.edu/blurbs/

[ConradSnAn] Recognizing Galois groups S_n and A_n

Conrad, Keith. Undated. Author-hosted expository note.

Exposition / Expository notes

Gives permutation-group recognition criteria and examples that use modular factorization and real roots to identify S_n or A_n.

Undated; accessed 11 November 2026.

https://kconrad.math.uconn.edu/blurbs/galoistheory/galoisSnAn.pdf

https://kconrad.math.uconn.edu/blurbs/